

CORRECTED CUMULATIVE READER

Collatz

Research continuations

Split-Zero comparisons, original arithmetic, and retained-source reductions

Complete recovered mathematical notes, with an additive all-even-exponent continuation

16 September 2026

Public collaboration name: Kokuno Yumeto. Research and development with ChatGPT 5.6 Sol and GPT-6 Astra. Original source authors retain their attribution.

Corrected reader edition

This reader corrects the presentation of the supplied 16 September 2026 cumulative archive. The original ZIP, original PDF, and historical receipts remain preserved as the original delivery. Its Markdown converter had treated pairs of literal caret characters in some unmarked inline arithmetic as superscript markup. The corrected converter disables that interpretation; unmarked formulas retain their literal source notation, while explicit mathematical displays remain mathematical displays. Source-specific repairs to unintended emphasis are recorded separately and do not rewrite the Markdown.

Chapter 20 also states explicitly that its unrestricted support and coefficient bounds apply to positive odd $n > 1$. At the fixed root, the conventions are $R_*(1) = 1$, $K_*(1) = 0$, and $H(V_1) = 0$. This is the reviewed domain clarification; it does not assert that the residual root set has been exhausted.

The source map records both original and edition SHA-256 identities. The first nineteen Markdown sources are unchanged. Historical source-time publication and execution statements remain historical, and the anthology is not a new independent proof audit of all twenty chapters.

Public collaboration name: Kokuno Yumeto. The research and its continuation were developed with ChatGPT 5.6 Sol and GPT-6 Astra. The models are identified as tools in the research process; the historical mathematical literature keeps its own named authors and citations.

Archive scope and how to read it

This volume transcribes the full mathematical notes available in the fourteen supplied continuation archives, adds four complete historical notes retrieved from the pinned GitHub source, and adds the present all-even-exponent note. The original Markdown, code, manifests, proof ledgers, and historical receipts remain separately preserved in the accompanying ZIP. This is an anthology of source versions, not a newly unified proof of Collatz.

The source-time publication and verification statements in each chapter are retained. Some are now historical: PR 2 was merged on 16 September 2026, and the main branch was read at commit 9453b5306b3a897389dece309f9c119dffbcc489. No new GitHub publication is implied by this edition. Historical finite test counts are not added into a claim of a new repository-wide mathematical audit.

The bundle is a cumulative archive of the supplied artifacts, not a complete clone of every GitHub workbench file. Three additional historical directories are pinned in `provenance/GITHUB_ONLY.json`: the history-law, anchored-period, and ternary-join checkpoints. Their four complete mathematical notes were retrieved through the GitHub connection and checked byte-for-byte against their Git blob identities. They are included in this volume and the source overlay. Their other companion files were not supplied locally. The accompanying `tools/fetch_github_sources.py` restores any missing files in those exact pinned directories on a networked machine, without overwriting an existing differing file. No placeholder is presented as missing source code or evidence.

Every included chapter has a source locator and SHA-256 in `latex/SOURCE_MAP.json`. Formula text is preserved in per-chapter math inventories. The original delivery type-set equation tags and scaled overwide displays to the available line width. The corrected reader also preserves unmarked inline notation as explained above. One malformed source subscript is grouped as $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ in the presentation; the unchanged original Markdown and the explicit typography correction record remain in the archive. Original equation numbers are local to their source chapter. Original files, not the presentation conversion, remain the archival record.

The current continuation is the final chapter. It proves complete common-future families for every positive even interior exponent, corrects their ternary layer through the exact identity $\nu_3((2^e + 2)/3) = \nu_3(e - 1)$, and makes template membership a finite test at each original source. Its budget-preserving version retains the predecessor's support bound. Unresolved roots remain represented.

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Chapter 1

Actual Collatz history laws: exact counting and a Gaussian total-variation threshold

Source: repository/collatz_reconstruction/research_program/history_law_cutoff_20260914/RESEARCH_NOTE.md.

Date: 14 September 2026. **Status:** written mathematical continuation with independently implemented exact finite checks; not an independent external review or a proof of Collatz convergence.

Working repository: KokunoYumeto/collatz-workbench. **Source revision:** 63407034387466870f9e092dcdf39fb5a1ec0399.

This note continues the *Collatz* manuscript `collatz_reconstruction/research_program/split_zero_history_20260913/note.tex`, especially its exact odd-return cylinder lemma, finite spectral reconstruction theorem, and closing distinction between encoding a distribution and estimating it. No source in the zeta repository is changed or needed to run this contribution. The separate earlier web-session integral-extension package is not incorporated or certified here.

The contribution is an explicit finite-sample law comparison, including a matching information-theoretic obstruction and its Gaussian transition, followed by an offset-preserving first-descent counter. The underlying cylinder coding and geometric comparison belong to the established literature and the existing workbench. In particular, Tao's Proposition 1.9 and Section 4 already use the same cylinder/binomial-count mechanism. This note makes no global priority claim for its deductions.

1.1 1. The same map, with an actual sampling measure

Use the odd-return map

$$T(n) = \frac{3n+1}{2^{\nu_2(3n+1)}} \quad (n \text{ positive and odd}).$$

Fix integers $N \geq 1$, $b \geq 0$, and $m \geq 1$. Choose the starting integer uniformly from

$$I_{N,b} = \{2b+1, 2b+3, \dots, 2(b+N)-1\}.$$

The interval contains exactly N odd integers; N is not its upper endpoint. All state-

ments below are uniform in its location b , even when b varies arbitrarily with N .

Let $P_{N,b,m}$ be the probability law of the **whole chronological word**

$$(a_1, \dots, a_m), \quad a_i = \nu_2(3T^{i-1}(n) + 1).$$

The comparison law on $\mathbb{N}_{>0}^m$ is

$$G_m(w) = 2^{-A(w)}, \quad A(w) = a_1 + \dots + a_m.$$

It is the product of m geometric laws with $\Pr(a = j) = 2^{-j}$, mean 2 and variance 2. This is a *comparison* law: independence is not imposed on the actual finite-integer sample.

Throughout,

$$\Delta(P, G) = \frac{1}{2} \sum_w |P(w) - G(w)| = \sup_E |P(E) - G(E)|.$$

Tao's V7 equation (1.9) uses the full sum, twice this convention. The Gaussian limit below is therefore $2\Phi(2c)$ in that convention.

1.2 2. Exact cylinders and exact counts

Write $A_j = a_1 + \dots + a_j$, $A_0 = 0$, and retain the original affine numerator

$$B_w = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}.$$

The chronological branch product is

$$T_w(x) = \frac{3^m x + B_w}{2^{A_m}}.$$

Let r_w be the unique odd integer in $[1, 2^{A_m+1})$ satisfying

$$3^m r_w + B_w \equiv 2^{A_m} \pmod{2^{A_m+1}}, \quad h_w = (r_w - 1)/2.$$

Lemma 2.1 (original exact cylinder, including the last oddness bit). A positive odd integer $n = 2j + 1$ has exponent word w if and only if

$$j \equiv h_w \pmod{2^{A_m}}.$$

Proof. For one exponent the congruence says precisely that $(3n + 1)/2^{a_1}$ is an odd integer, so the valuation is exactly a_1 . For a longer word, $B_w = 3^{m-1} + 2^{a_1} B_{w'}$. The congruence first forces $2^{a_1} \mid 3n + 1$. Divide by this power and put $y = (3n + 1)/2^{a_1}$. The resulting suffix congruence is

$$3^{m-1} y + B_{w'} \equiv 2^{A_m - a_1} \pmod{2^{A_m - a_1 + 1}}.$$

Because the suffix has positive length, its right side is even and $B_{w'}$ is odd, forcing y odd. Induction establishes all exact exponents, including the last. Reversing these steps proves necessity. The original coefficient 3^m is odd, hence invertible modulo the displayed power of two. Positivity is preserved by every actual step. Substituting $n = 2j + 1$ gives the stated modulus in j . This is the same cylinder lemma as in the

predecessor, rederived to fix the sampling convention. $\square$

It follows that the number of occurrences of w in $I_{N,b}$ is exactly

$$C_{N,b}(w) = \left\lfloor \frac{b + N - 1 - h_w}{2^{A_m}} \right\rfloor - \left\lfloor \frac{b - 1 - h_w}{2^{A_m}} \right\rfloor. \quad (2.1)$$

In particular it is either $\lfloor N2^{-A_m} \rfloor$ or $\lceil N2^{-A_m} \rceil$, and

$$|P_{N,b,m}(w) - 2^{-A_m}| \leq \frac{1}{N}. \quad (2.2)$$

No equidistribution theorem with a fixed modulus is being reused for a moving modulus: (2.1) is an exact interval count for every modulus and every interval location.

1.3 3. A finite, fully explicit total-variation envelope

For $H \geq 0$, define

$$K_m(H) = \begin{cases} \binom{H}{m}, & H \geq m, \\ 0, & H < m, \end{cases} \quad Q_m(H) = 2^{-H} \sum_{j=0}^{\min(m-1, H)} \binom{H}{j}.$$

There are exactly $K_m(H)$ positive words with $A_m \leq H$: sum $\binom{A-1}{m-1}$ over $m \leq A \leq H$. Under G_m ,

$$G_m(A_m > H) = Q_m(H). \quad (3.1)$$

Indeed, independent fair Bernoulli trials have geometric waiting times between successes; the m -th success occurs after time H precisely when the first H trials have at most $m-1$ successes. Equivalently (3.1) follows from the finite negative-binomial sum. This also handles $H < m$, where the answer is 1.

Theorem 3.1 (uniform finite-sample sandwich). For all N, b, m, H as above,

$$\boxed{\max\{0, Q_m(H) - N2^{-(H+1)}\} \leq \Delta(P_{N,b,m}, G_m) \leq \min\{1, Q_m(H) + K_m(H)/N\}.} \quad (3.2)$$

Different values of H may be chosen for the two bounds, or each side may be optimized over all H .

Proof. For any two discrete probabilities, $\Delta(P, G) = 1 - \sum_w \min(P(w), G(w))$. On the $K_m(H)$ words of total exponent at most H , (2.2) gives

$$\sum_w \min(P(w), G(w)) \geq 1 - Q_m(H) - K_m(H)/N.$$

This proves the upper bound. Let F be the actual support of P . There are at most N words in F , because there are only N sampled starting integers. Each atom with $A_m > H$ has G -mass at most $2^{-(H+1)}$. Thus

$$G(F) \leq 1 - Q_m(H) + N2^{-(H+1)}.$$

The event F^c has P -mass zero, and yields the lower bound. This step concerns the actual labelled word, not a projection to a spectrum or an affine multiplier. $\square$

1.3.1 Sharper finite versions

Rounding rather than absolute discrepancy gives

$$\Delta(P_{N,b,m}, G_m) \leq \min \left\{ 1, Q_m(H) + \frac{1}{N} \sum_{A=m}^H \binom{A-1}{m-1} \{N2^{-A}\} \right\}, \quad (3.3)$$

where braces denote fractional part. To prove it use $P(w) \geq \lfloor N2^{-A} \rfloor / N$ in the overlap formula. Every term is an explicit rational number; an empty sum is zero.

There is also a sharp bound from support cardinality alone. Let $h_* \geq m-1$ be the largest integer with $K_m(h_*) \leq N$. Then

$$\Delta(P_{N,b,m}, G_m) \geq Q_m(h_*) - (N - K_m(h_*))2^{-(h_*+1)}. \quad (3.4)$$

The N largest atoms of G_m consist of every word of exponent sum at most h_* , followed by $N - K_m(h_*)$ words of sum $h_* + 1$. No set of at most N atoms has larger mass. Taking its complement proves (3.4); thus it is optimal when only that support-size information is used, not a claim of optimality among Collatz samples.

For a dyadic sample size $N = 2^L$, every cylinder with $A_m \leq L$ has exactly its geometric probability. A cylinder with $A_m > L$ contains at most one sampled starting integer. Hence

$$\Delta(P_{2^L,b,m}, G_m) = Q_m(L) - \sum_{\substack{w \in \text{supp } P_{2^L,b,m} \\ A(w) > L}} 2^{-A(w)}. \quad (3.5)$$

In particular $\Delta \leq Q_m(L)$, uniformly in b . This is an exact formula with an actual finite observed-support sum, not a normal approximation.

1.4 4. The sharp growing-history transition

Let $L_N = \log_2 N$, and let Φ denote the standard normal cumulative distribution function.

Theorem 4.1 (Gaussian critical window). For every fixed real c , put

$$m_N = \left\lfloor \frac{L_N}{2} + c\sqrt{L_N} \right\rfloor.$$

Then, as $N \rightarrow \infty$,

$$\sup_{b \geq 0} |\Delta(P_{N,b,m_N}, G_{m_N}) - \Phi(2c)| \rightarrow 0. \quad (4.1)$$

Thus the whole-word geometric model tends to total variation 0 below $(1/2 - \varepsilon) \log_2 N$, and to total variation 1 above $(1/2 + \varepsilon) \log_2 N$. The latter statement obstructs full-history approximation, not the usefulness of any particular coarser statistic.

Proof of the critical window. Choose $a_N = L_N^{1/4}$, $H_- = \lfloor L_N - a_N \rfloor$, and $H_+ = \lceil L_N + a_N \rceil$. For large N , these are nonnegative. Since $K_m(H) \leq 2^H$, (3.2) gives

$$Q_{m_N}(H_+) - 2^{-a_N-1} \leq \Delta(P_{N,b,m_N}, G_{m_N}) \leq Q_{m_N}(H_-) + 2^{-a_N}. \quad (4.2)$$

The errors are independent of b . If Y_H is binomial with parameters $(H, 1/2)$, then

$$Q_m(H) = \Pr\left(\frac{2Y_H - H}{\sqrt{H}} \leq \frac{2m - 2 - H}{\sqrt{H}}\right).$$

The centered normalized variable converges in distribution to a standard normal. In this special case its characteristic function is exactly $(\cos(t/\sqrt{H}))^H$, which tends to $e^{-t^2/2}$, since $\log \cos u = -u^2/2 + O(u^4)$. The continuity theorem for characteristic functions gives the asserted distributional limit. For both $H_{\pm}$, the threshold on the right tends to $2c$, because $a_N = o(\sqrt{L_N})$. Continuity of Φ and (4.2) complete the proof, uniformly in b . $\square$

Explicit off-critical estimates. If $0 < \varepsilon < 1/2$ and $1 \leq m \leq (1/2 - \varepsilon)L_N$, then, for sufficiently large N ,

$$\Delta(P_{N,b,m}, G_m) \leq N^{-\varepsilon} + N^{-\varepsilon^2/(2 \log 2)}. \quad (4.3)$$

Here $\log 2$ is the natural logarithm of 2. Take $H = \lfloor (1 - \varepsilon)L_N \rfloor$. The counting term is at most $N^{-\varepsilon}$, and $H/2 - (m - 1) \geq \varepsilon L_N/2 + 1/2$. For a fair binomial variable, $\Pr(Y_H - H/2 \leq -d) \leq e^{-2d^2/H}$; since $H \leq L_N$, the tail is at most $e^{-\varepsilon^2 L_N/2}$.

For $m \geq (1/2 + \varepsilon)L_N$, set $H = \lceil (1 + \varepsilon)L_N \rceil$. When $\varepsilon L_N > 1$,

$$1 - \Delta(P_{N,b,m}, G_m) \leq \frac{1}{2}N^{-\varepsilon} + \exp\left(-\frac{2(\varepsilon L_N/2 - 1/2)^2}{H}\right). \quad (4.4)$$

This follows from the lower half of (3.2) and the upper binomial tail $\Pr(Y_H \geq m)$. If $m > H$, that tail is simply zero, and the displayed bound remains valid. The binomial tail inequality used in both estimates follows directly from Markov's inequality and $\mathbb{E}e^{t(Y_H - H/2)} = (\cosh(t/2))^H \leq e^{Ht^2/8}$, optimizing at $t = 4d/H$. No untracked large-deviation constant is needed.

The clocks remain explicit: m counts odd returns; A_m counts divisions by two, or shortened-map steps; the corresponding full-map clock is $m + A_m$. The threshold is stated in the first clock, not silently transported to another.

1.5 5. What the predecessor's spectral reconstruction now receives

Fix m, H . Retain all words $W_{m, \leq H}$ and add one **overflow label** ∂ for $A_m > H$. The actual finite probability vector is

$$p_w = C_{N,b}(w)/N, \quad p_{\partial} = 1 - \sum_{w \in W_{m, \leq H}} p_w.$$

The comparison vector has masses $g_w = 2^{-A(w)}$, $g_{\partial} = Q_m(H)$. Overflow is not discarded or renormalized away. The coarse-graining is deterministic, so its total variation is at most (3.2) or (3.3). The same bound controls the expectation error for **every** function f on words with $0 \leq f \leq 1$, including simultaneous congruences or prime-power divisibility tests on the same B_w . Those tests need not be independent, and their moduli may vary. Positivity $2^{A(w)} - 3^m > 0$ must still accompany the positive integral-cycle predicate.

For the finite spectral map choose the original integer node r_w for each retained word and node 0 for overflow. Nodes remain distinct even across different sums A : an equality $r_w = r_v > 0$ would give that very positive integer two different actual

length- m histories, contradicting uniqueness of iteration. Also $r_w < 2^{H+1}$, and no such node is zero.

There are $K_m(H) + 1$ nodes, including overflow. The Hurwitz family is

$$Z_p(s, t) = p_\partial \zeta(s, 1) + \sum_{w \in W_{m, \leq H}} p_w \zeta(s, 1 + tr_w).$$

Its mass is one and its unchanged displacement coordinate obeys $|t|(2^{H+1} - 1) < 1$. The predecessor's full-cluster inverse, retaining every multiplicity of its chosen nontrivial zeta zero, determines moments through order $K_m(H)$ from that cluster jet. Lagrange interpolation on these distinct nodes then reconstructs every mass, including overflow. In the degenerate case $H < m$, only overflow remains, with weight one and no jet needed. This is an application of the proved finite construction, not a claim that a zero branch can be computed without the original distribution.

Supported zero weights and absent labels remain different: all displayed labels are retained even when their sample count is zero. A separate retained-label mask cannot be reconstructed from their zero masses alone. No analytic conditioning estimate for the inverse jet map follows from the total-variation estimate. For an individual polynomial observable, a moment-error bound is the sum of the absolute interpolation coefficients times the corresponding moment errors; those potentially large coefficients cannot be suppressed.

The new arithmetic input is (2.1), (3.2), and (4.1), rather than another encoding of unspecified weights. Conversely, invertible spectral encoding cannot repair the full-word obstruction above the threshold. Noninvertible observations may behave much better and must be investigated using their actual fibres.

1.6 6. Exact first-descent certificates with the additive term retained

The statistical theorem concerns distributions of words. First descent additionally depends on the starting integer within its cylinder. For a prefix $w_{[1,j]}$, set

$$B_j = \sum_{i=0}^{j-1} 3^{j-1-i} 2^{A_i}, \quad D_j = 2^{A_j} - 3^j.$$

On the actual cylinder,

$$T^j(n) < n \iff D_j > 0 \text{ and } n > B_j/D_j.$$

Define

$$\tau(w) = \min_{1 \leq j \leq m: D_j > 0} \lfloor B_j/D_j \rfloor,$$

with $\min \emptyset = +\infty$. Then **no strict descent below the starting value during the first m odd returns is equivalent to $n \leq \tau(w)$.**

This follows by taking the conjunction of the exact inequalities for every prefix. A nonpositive D_j cannot give descent because $n, B_j > 0$. Equality is not descent. In particular, the word (2) at $n = 1$ has multiplier $3/4 < 1$ but returns to 1; the additive term cannot be deleted.

Let

$$U_w = \min\{b + N - 1, \lfloor (\tau(w) - 1)/2 \rfloor\},$$

with the infinite threshold interpreted as $U_w = b + N - 1$. The number of no-descent starts in the word's cylinder is

$$Z_w = \begin{cases} \left\lfloor \frac{U_w - h_w}{2^{A(w)}} \right\rfloor - \left\lfloor \frac{b - 1 - h_w}{2^{A(w)}} \right\rfloor, & U_w \geq b, \\ 0, & U_w < b. \end{cases}$$

Hence a sum-cap certificate computes

$$D_{\text{low}} = \sum_{w \in W_{m, \leq H}} (C_{N,b}(w) - Z_w), \quad R = N - \sum_{w \in W_{m, \leq H}} C_{N,b}(w).$$

If D_{actual} is the total number of sampled starts that descend by the m -th odd return, then

$$\boxed{D_{\text{low}} \leq D_{\text{actual}} \leq D_{\text{low}} + R.} \quad (6.1)$$

The error R is the *exact number of sample starts in the retained overflow state*. It is not an assumed geometric tail. This is an executable finite certificate for actual starts, independent of any global convergence assertion.

For $N = 4096, b = 0, m = 5, H = 16$, the checker obtains 3,929 accounted starts and 167 overflow starts. Formula (6.1) gives **3,320 to 3,487** strict-descending starts. A separate direct iteration of all 4,096 starts gives **3,487**. Agreement with the upper endpoint is a result of this particular fixture, not a general theorem that every overflow start descends.

1.7 7. Verification, reproduction, and remaining work

`check_history_law.py` uses the Python standard library and exact integer/Fraction arithmetic. It checks 55 capped families, 4,422 positive integer lifts, 72 actual interval laws, 6,600 cylinder counts, 1,224 TV envelopes (including their rounding refinement), 3,006 descent equivalences, 216 capped descent certificates, 21 moment-weight recoveries including zero weights and overflow, and three synchronized arithmetic-predicate recoveries. Interval fixtures include a location greater than 10^{30} , emphasizing the uniform-in-location statement.

A separate table examines 332,032 starting integers across sample sizes $2^8, 2^{12}, 2^{16}, 2^{18}$, at three horizons for each size. These are exact finite TV calculations with rational lower/upper bounds, including (3.4) and the dyadic identity (3.5). Decimal renderings are display only. They do not certify the Gaussian limit; its proof is Section 4. In particular the modest-size values at $m = L/2$ are not misreported as already equal to their limiting value $1/2$.

Ordinary and optimized Python runs have identical JSON output:

```
python -B check_history_law.py > verification.normal.json
python -O -B check_history_law.py > verification.optimized.json
```

Five negative controls reject an even start, a zero exponent, a negative interval location, lost probability mass, and the incorrect assertion that 1 strictly descends in one return. Assertions are explicit and remain active under optimization. No zeta zero

was sampled, no Lean was run, no existing workbench-wide checker was rerun, and no claim about all infinite trajectories is made.

For the next local session, the useful boundary is now precise: full histories cannot remain approximately geometric after their information threshold, but an actual arithmetic pushforward can discard much of that information. The next calculation should specify a fixed jointly retained observable (for example the offset modulo powers of three together with an affine descent barrier), compute its fibres or recurrence, and bound its distribution without asserting full-word approximation beyond (4.1). The overflow/descent counter already provides finite fixtures against which such a proposal can be tested.

1.8 References and source-use record

Kokuno Yumeto. (2026, September 13). *Finite Collatz histories through support and Hurwitz zero clusters* [Working manuscript]. In *Collatz research workbench*, revision 63407034387466870f9e092dcdf39fb5a1ec0399, `collatz_reconstruction/research_program/split_zero_history_20260913/note.tex`. The mathematical text was read through its complete end, including exact odd-return coordinates, cluster-to-moment inversion, reconstruction, sharpness, arithmetic predicates, and the distinction between encoding and descent. Its Git blob is 5fec479a48c4253eadffb9ed3c78da81b58a0000. The accompanying audit was read as historical provenance, not treated as a newly executed independent audit.

Tao, T. (2026, July 16). *Almost all orbits of the Collatz map attain almost bounded values* (arXiv:1909.03562, Version 7). <https://arxiv.org/html/1909.03562v7>. Specifically Definition 1.7, equation (1.9), Proposition 1.9, and Section 4. The older journal publication is in *Forum of Mathematics, Pi*, 10, e12 (2022). The finite residue/binomial comparison here is explicitly credited to this established mechanism. Proposition 1.9 is not invoked by assuming its strong residue-uniformity hypothesis (1.11) for arbitrary intervals: formula (2.1) supplies the actual discrepancy instead. No strengthening of Tao's orbit-minimum theorem is claimed.

National Institute of Standards and Technology. (n.d.). *Digital Library of Mathematical Functions*, Section 25.11, equation 25.11.10. <https://dlmf.nist.gov/25.11.E10>. Retrieved September 14, 2026. Used only as the primary reference for the Hurwitz Taylor identity already employed in the predecessor, not as a source of Collatz trajectory estimates.

Chapter 2

Stopped Collatz transport on the original arithmetic lattices

Source: repository/collatz_reconstruction/research_program/stopped_affine_transport_20260914/note.md.

2.1 Exact restart charts, image cohomology, synchronized marks, and a completed stopping law

Date: 14 September 2026

Status: self-contained mathematical continuation with exact-arithmetic replay; not externally reviewed.

Intended repository: KokunoYumeto/collatz-workbench

Publication status: local add-only contribution. No repository checkout, commit, branch, pull request, or CI result is asserted.

2.1.1 What this contribution changes

The source problem is to continue Collatz histories without replacing an actual arithmetic output by an unmarked, uniformly sampled odd interval. The output of a prescribed word retains a power-of-three lattice step. This note constructs the original source and target lattices, their exact inverse charts, their affine image cosets, and their integral two-term complexes. It then restarts on the **actual target progression**. Neither the affine offset nor the target lattice is discarded.

The stopping observation retains every exponent until an actual universal affine descent certificate has been obtained. It retains the original first-descent index, all synchronized residue readouts, congruence incompatibility masks, and declared labels of weight zero. A full countable stopped law has an explicit finite-sample error bound, proved below; this law is not a law for the discarded post-stop history. Its relation to the full history is the explicit stopping map and its fibers.

The square-root estimate developed in Section 7 retains the translation through the invertible coordinate map $x \rightarrow x-1$. The two conjugated affine branches and the inverse coordinate map are displayed. No hypothesis about random deterministic Collatz trajectories is used.

2.1.2 Source status

The author-supplied conversation contains the earlier affine-packet and full-history-cutoff statements and claimed GitHub locators. Those claims were not independently recovered from the current Collatz repository. The current GitHub pages and raw Collatz files did not return readable content in this session. A cached Zeta README was readable, but not its requested methodology section or the current Collatz hand-off.

Historical records in the connected Drive were readable: the original instructions about exact morphisms, retained coefficients, and constructive continuation after an obstruction, and an older Collatz note defining the original maps. The detailed private locators are in the local intake record outside the repository patch. These historical records are not represented as the current GitHub methodology section. The Collatz, lattice, counting, and chain-map results used here are proved in this note; the standard Hurwitz special-value identity is explicitly cited in [R3]. No mathematical dependency is silently imported from the Zeta arithmetic-weight programme.

This is a self-contained continuation, not a priority claim. Stopping-time methods have a classical literature, including Terras [R1]. Tao's orbit-minimum result and its first-passage/3-adic transport formulation are a major existing reference [R2]. This note makes no claim to strengthen that result.

2.2 1. Original words, affine numerators, and legal cylinders

Write the positive odd-return map as

$$T(n) = \frac{3n+1}{2^{\nu_2(3n+1)}} \quad (n \in 2\mathbb{Z}_{\geq 0} + 1).$$

For a finite exponent word $p = (a_1, \dots, a_m)$, every $a_i \geq 1$, retain

$$A_j = \sum_{i=1}^j a_i, \quad A_0 = 0, \quad L_j = 3^j, \quad C_j = \sum_{i=1}^j 3^{j-i} 2^{A_{i-1}}. \quad (1.1)$$

Set $A = A_m$, $L = L_m$, $C = C_m$, and $U = 2^A$. Empty words have $(A, L, C) = (0, 1, 0)$. The formal branch is

$$F_p(x) = \frac{Lx + C}{U}, \quad F_j(x) = \frac{L_j x + C_j}{2^{A_j}}. \quad (1.2)$$

2.2.1 Proposition 1.1 — Literal append and composition maps

Appending an exponent a gives

$$(A, L, C) \mapsto (A + a, 3L, 3C + 2^A). \quad (1.3)$$

For words p, v , in chronological order,

$$\begin{aligned} A(pv) &= A(p) + A(v), & L(pv) &= L(v)L(p), \\ C(pv) &= L(v)C(p) + 2^{A(p)}C(v), & F_{pv} &= F_v \circ F_p. \end{aligned} \quad (1.4)$$

Proof. Substituting $F_p(x)$ into $(3x+1)/2^a$ gives $(3Lx+3C+2^A)/2^{A+a}$. Substitution into the full map F_v gives (1.4), with its original denominator $2^{A(p)+A(v)}$. Induction gives (1.1). ■

The integral matrix

$$\mathcal{M}_p = \begin{pmatrix} L & C \\ 0 & U \end{pmatrix} \quad (1.5)$$

satisfies $\mathcal{M}_{pv} = \mathcal{M}_v \mathcal{M}_p$. The equation

$$\mathcal{M}_p \begin{pmatrix} n \\ 1 \end{pmatrix} = U \begin{pmatrix} F_p(n) \\ 1 \end{pmatrix} \quad (1.6)$$

records the denominator; it does not replace the integral matrix by its diagonal spectrum.

2.2.2 Proposition 1.2 — Exact positive-integer cylinder

There is exactly one odd residue ρ_p modulo 2^{A+1} whose positive representatives have first exponent word p . In the explicitly selected section $1 \leq \rho_p < 2^{A+1}$, it is

$$\rho_p \equiv (2^A - C)L^{-1} \pmod{2^{A+1}}. \quad (1.7)$$

All positive odd starts with that word are exactly

$$n = \rho_p + 2^{A+1}k, \quad k \in \mathbb{Z}, \quad n > 0. \quad (1.8)$$

For every prefix, including $j = 0$,

$$T^j(n) = F_j(\rho_p) + 2^{A+1-A_j}L_jk. \quad (1.9)$$

Proof. First prove cylinder uniqueness by extension, rather than infer every intermediate valuation merely from final integrality. The empty word is the odd class modulo 2. On a legal word cylinder of exponent sum B , its odd terminal value, as the integer lift varies, has the form $u + 2 \cdot 3^j k$, with u odd. Requiring the next exponent to equal a is the congruence

$$3(u + 2 \cdot 3^j k) + 1 \equiv 2^a \pmod{2^{a+1}}.$$

After division of this congruence by 2, the coefficient of k is odd. There is exactly one solution for $k \pmod{2^a}$. This refines the original cylinder to one odd class modulo 2^{B+a+1} , and proves every intermediate exponent is exact.

For a legal final word, its odd terminal value implies $Ln + C \equiv 2^A \pmod{2^{A+1}}$. The odd integer L is invertible modulo this power of two, so the unique class already constructed is (1.7). Its representative is odd. Formula (1.9) is direct substitution in (1.2). ■

2.2.3 Section changes are explicit

Another section $\rho'_p = \rho_p + 2^{A+1}h$ has lift $k' = k - h$. In (1.9), the base value changes by $2^{A+1-A_j}L_jh$, and the lift term changes by its negative. All original values are identical. For a modulus M , the mark changes by $k' \equiv k - h \pmod{M}$; the integer quotient

changes by the corresponding division algorithm. The section is a coordinate choice with this transition map, not a discarded arithmetic datum.

2.3 2. A stopping observation certified by the entire affine expression

For a nonempty prefix define

$$D_j = 2^{A_j} - L_j.$$

For every positive real n , exact subtraction gives

$$F_j(n) < n \iff D_j > 0 \text{ and } D_j n > C_j. \quad (2.1)$$

In particular, a multiplier $L_j/2^{A_j} < 1$ alone does not decide descent. For $p = (1, 1, 3)$, $(L, U, C) = (27, 32, 19)$: the multiplier is below one, but $F_p(3) = 100/32 > 3$.

Fix the literal reference value **3**. Let $\mathcal{L}$ be the following prefix-free set of words:

$$p \in \mathcal{L} \iff F_j(3) \geq 3 \ (0 \leq j < m), \quad F_m(3) < 3. \quad (2.2)$$

Here $F_j(3)$ is evaluation of the original formal affine prefix. An arbitrary prescribed word is not asserted to be an actual exponent word of the integer 3.

2.3.1 Theorem 2.1 — Universal actual descent on every supported leaf

For every $p \in \mathcal{L}$,

$$3D_m > C_m > 0. \quad (2.3)$$

Every actual positive odd start $n \geq 3$ in the cylinder (1.8) satisfies

$$0 < T^m(n) < n. \quad (2.4)$$

Proof. Equation (2.2) says $3L + C < 3U$, which is (2.3). Then $D_m n \geq 3D_m > C_m$ for every $n \geq 3$. Equation (2.1) and legality of the cylinder give (2.4). Positivity follows from the original positive numerator and denominator. ■

This stopping time is the first **reference-certified** return, not necessarily the actual first return below the starting integer. Section 5 retains the genuine first-descent index separately.

2.3.2 The history map and its fibers

Let $\Omega = \mathbb{N}_{\geq 1}^{\mathbb{N}}$. The stopping map

$$\sigma : \Omega \longrightarrow \mathcal{L} \sqcup \{\infty\} \quad (2.5)$$

returns the first prefix satisfying (2.2), or ∞ when no such prefix occurs. Its fiber above a finite leaf is precisely

$$\sigma^{-1}(p) = \{p\} \times \Omega \quad (2.6)$$

under concatenation. Thus the post-stop tail is exactly the data forgotten by this map. No claim is made that the stopped observation recovers that tail.

On actual integer starts the source is the image of the actual itinerary map $n \mapsto (a_1(n), a_2(n), \dots)$. Its fiber is the literal intersection with (2.6), namely (1.8). The symbol ∞ remains a declared symbol for the actual source as well. No probability-zero argument deletes it.

For finite $H \geq 0$, let

$$\mathcal{L}_H = \{p \in \mathcal{L} : A(p) \leq H\}. \quad (2.7)$$

The finite observer returns these leaves or an explicit **cutoff overflow** symbol. Overflow includes both later stopping and never stopping; those possibilities have not been identified.

2.4 3. Exact restart on arbitrary original arithmetic progressions

Retain the complete source chart

$$n(t) = a + 2dt, \quad t = b, b+1, \dots, b+N-1, \quad (3.1)$$

where a is odd, d, N are positive integers, $b \in \mathbb{Z}$, and $a + 2db \geq 3$. The odd factor of d , its power of two, the source interval, and the offset are all part of the state.

2.4.1 Theorem 3.1 — Source chart, target chart, and inverse

For a nonempty word p , set

$$g = \gcd(d, 2^A), \quad c = (\rho_p - a)/2. \quad (3.2)$$

The compatibility mask is

$$\kappa_p(a, d) = \mathbf{1}_{g|c}. \quad (3.3)$$

An incompatible word has empty intersection with this source progression. For a compatible word set

$$P = 2^A/g, \quad s \equiv (c/g)(d/g)^{-1} \pmod{P}, \quad 0 \leq s < P, \quad (3.4)$$

with $s = 0$ for $P = 1$. Then the actual source points with word p are exactly

$$t = s + Pv, \quad (3.5)$$

and their terminal values are exactly

$$T^m(n(s + Pv)) = u + 2d'v, \quad (3.6)$$

where

$$u = \frac{L(a + 2ds) + C}{2^A} \in 2\mathbb{Z} + 1, \quad d' = \frac{d}{g}L. \quad (3.7)$$

The source and target parametrizations have inverse parameter maps

$$v = (t - s)/P, \quad v = (y - u)/(2d'). \quad (3.8)$$

Their common parameter range is exactly

$$\left\lceil \frac{b - s}{P} \right\rceil \leq v \leq \left\lfloor \frac{b + N - 1 - s}{P} \right\rfloor. \quad (3.9)$$

Proof. The original cylinder equation is $a + 2dt \equiv \rho_p \pmod{2^{A+1}}$, or $dt \equiv c \pmod{2^A}$. Elementary divisibility gives (3.3). Division by the recorded common factor g , with that factor retained in the chart, gives the invertible coefficient d/g modulo P . This proves (3.4) and (3.5). Substitution in the original numerator gives

$$\frac{L(a + 2d(s + Pv)) + C}{2^A} = \frac{L(a + 2ds) + C}{2^A} + 2\frac{d}{g}Lv.$$

The base numerator gives an odd integer by Proposition 1.2, even when the chosen section lies outside the sampled interval. Equations (3.8) are literal inverse maps on their stated fibers. Restricting $b \leq t \leq b + N - 1$ proves (3.9). ■

2.4.2 The commuting integral matrix diagram

Define the source and target lattice embeddings

$$J_p = \begin{pmatrix} 2dP & a + 2ds \\ 0 & 1 \end{pmatrix}, \quad K_p = \begin{pmatrix} 2d' & u \\ 0 & 1 \end{pmatrix}.$$

The exact relation is

$$\boxed{\mathcal{M}_p J_p = 2^A K_p.} \quad (3.10)$$

Both columns are equal as integer vectors. The first column is the identity $LdP = 2^A d'$; the second is $L(a + 2ds) + C = 2^A u$. Thus the diagram includes the original integral lattices and the denominator map.

2.4.3 Marked restart, with every factor retained

For any positive integer M and $0 \leq r < M$, write $v = r + M\ell$. The exact restriction and target are

$$t = s + Pr + PM\ell, \quad y = u + 2d'r + 2Md'\ell. \quad (3.11)$$

The bounds in (3.9) become the corresponding ceiling/floor bounds on ℓ . This is an exact restart family with parameters

$$(a_{\text{new}}, d_{\text{new}}) = (u + 2d'r, Md'). \quad (3.12)$$

The original full-support index set and the compatibility mask remain recorded even for empty sampled fibers.

2.4.4 Exactly how dyadic information is consumed

Write $s_0 = \nu_2(d)$. Equation (3.7) gives

$$\nu_2(d') = \max\{s_0 - A, 0\}. \quad (3.13)$$

An additional mark modulus M adds $\nu_2(M)$ to this exponent. For odd d , the unmarked target step parameter is $d' = d3^m$, still odd. For even d , the original fixed dyadic digits are consumed according to (3.13), with every incompatible branch still marked. These are exact relationships between the two source descriptions, not an assumption that conditional branches become unconditioned.

2.5 4. The arithmetic image complex and its exact localizations

The target of (3.6), in the invertible odd-index coordinate

$$z = (y - 1)/2, \quad y = 2z + 1, \quad (4.1)$$

is the affine lattice

$$z = z_0 + d'v, \quad z_0 = (u - 1)/2. \quad (4.2)$$

The division in (4.1) is an explicitly inverted bijection between odd integers and integers. The original value y and its step $2d'$ are recoverable by its displayed inverse.

2.5.1 Theorem 4.1 — Image cohomology with the affine coset retained

The two-term integral complex

$$K_{p,a,d}^{\text{image}} = \left[\mathbb{Z} \xrightarrow{d'} \mathbb{Z} \right] \quad \text{in degrees } 0, 1 \quad (4.3)$$

has

$$H^0 = 0, \quad H^1 = \mathbb{Z}/d'\mathbb{Z}. \quad (4.4)$$

The image progression is the distinguished affine coset

$$[z_0] \in H^1; \quad z \text{ is in that image exactly when } [z] = [z_0]. \quad (4.5)$$

The order of this distinguished class is $d'/\gcd(d', z_0)$. A zero class still specifies a present image lattice; it does not remove the source word, the ambient group, or the source map.

Proof. Multiplication by the positive integer d' is injective on $\mathbb{Z}$. Its cokernel is the indicated quotient. Equation (4.2) is equivalent to $z - z_0 \in d'\mathbb{Z}$, proving (4.5). The least positive integer k with $d' \mid kz_0$ is $d'/\gcd(d', z_0)$. ■

This complex records the arithmetic image lattice. It is not declared to equal an unread workbench complex. The exact map to the source dynamics is (3.10), and the affine image condition is (4.5).

Changing the lift section by $v' = v - h$ changes u to $u + 2d'h$ and z_0 to $z_0 + d'h$. The class in (4.5) is unchanged by this displayed coordinate transition.

2.5.2 Theorem 4.2 — Exact coefficient comparison and homotopy

Let ℓ be a prime. Write, retaining both factors,

$$d' = \ell^e w, \quad \ell \nmid w.$$

The coefficient map $\mathbb{Z} \hookrightarrow \mathbb{Z}_\ell$ gives the commutative square with multiplication by the same original d' on both rows. Over $\mathbb{Z}_\ell$, the complex (4.3) has

$$H^0 = 0, \quad H^1 = \mathbb{Z}_\ell / d' \mathbb{Z}_\ell. \quad (4.6)$$

The explicit comparison with $[\mathbb{Z}_\ell \xrightarrow{\ell^e} \mathbb{Z}_\ell]$ has degree-zero map multiplication by w and degree-one map the identity. Its inverse has degree-zero map multiplication by w^{-1} and degree-one map the identity. Both compositions are identities, and the distinguished degree-one coset is unchanged.

For $e = 0$, the complex is contractible with homotopy

$$h^1(x) = (d')^{-1}x. \quad (4.7)$$

Proof. A nonzero integer acts injectively on $\mathbb{Z}_\ell$, proving degree-zero vanishing. The cokernel is (4.6). The chain-map equation is $\ell^e w = d'$. The unit inverse w^{-1} exists in $\mathbb{Z}_\ell$, and gives the displayed inverse chain map. When $e = 0$, both $d'h^1$ and h^1d' are the identity. ■

For odd d , the image complex therefore contracts over $\mathbb{Z}_2$, while its original integral quotient has order $d3^m$. At the prime 3 its quotient has order $3^{m+\nu_3(d)}$. These statements are related by the coefficient embeddings and explicit chain maps above; they are not assertions of unrelated categories.

2.5.3 Finite coefficients retain both image and kernel

For every $N \geq 1$, put $g_N = \gcd(d', N)$. The equation

$$d'v = z - z_0 \pmod{N} \quad (4.8)$$

has exactly g_N solutions for $g_N \mid z - z_0$, and zero otherwise. Its homogeneous kernel has g_N elements. The explicit connecting isomorphism is

$$\beta : H^0(K^{\text{image}} \otimes \mathbb{Z}/N) \longrightarrow (\mathbb{Z}/d'\mathbb{Z})[N], \quad [v] \mapsto \left[\frac{d'v}{N} \right]. \quad (4.9)$$

The fraction in (4.9) is an integer on its domain. Changing the lift of v by Nq changes it by $d'q$. Injectivity follows because $d'v/N \in d'\mathbb{Z}$ forces $v \in N\mathbb{Z}$; both groups have g_N elements. Thus no finite-coefficient kernel is silently discarded.

2.5.4 The dyadic and triadic images of a word

For the unrestricted odd cylinder $n = \rho_p + 2^{A+1}k$, write its terminal value as $y = u + 2Lk$. Over the odd 2-adic integers, the branch inverse is

$$n = \frac{2^A y - C}{L}. \quad (4.10)$$

The odd integer L is a 2-adic unit, so every odd 2-adic y has this inverse, lying in the original cylinder. On ordinary integers the same inverse is integral precisely when

$$2^A y \equiv C \pmod{3^m}; \quad (4.11)$$

with the required positivity and source bound imposed literally. Consequently the square formed by the integer embeddings into $\mathbb{Z}_2$, the branch map, and (4.10) commutes. Its integer target is the affine lattice described by (4.11), whose step is $2 \cdot 3^m$. This is the exact arithmetic information lost by replacing the integer target with all odd integers.

2.5.5 Worked image defect

For $p = (1, 1, 3, 2)$,

$$(A, L, C, \rho_p) = (7, 81, 89, 247),$$

$$247 + 256k \mapsto 157 + 162k. \quad (4.12)$$

In the target coordinate (4.1), this is

$$z = 78 + 81k.$$

Thus the image complex has differential 81, $H^1 = \mathbb{Z}/81\mathbb{Z}$, and distinguished coset [78], of order 27. Over $\mathbb{Z}_2$, the homotopy is multiplication by 81^{-1} . The original integral lattice and class remain recorded alongside that contraction. The identity (4.12), not the 2-adic contraction alone, is the restart input.

2.6 5. Exact synchronized residues and the genuine first descent

Fix a compatible source chart from Section 3. Let $n_* = a + 2ds$. For a prefix of length j , substitution gives

$$T^j(n(s + Pv)) = F_j(n_*) + 2^{A+1-A_j} \frac{d}{g} L_j v. \quad (5.1)$$

Every displayed base and coefficient is integral. The coefficient follows from $2dP/2^{A_j}$, and the base is integral by cylinder legality.

For every modulus M , the mark $r = v \bmod M$ therefore determines, jointly,

$$(n \bmod M, T(n) \bmod M, \dots, T^m(n) \bmod M) \quad (5.2)$$

by the explicit readout (5.1). This assertion includes even M , powers of 3, and mixed moduli. No coprimality between M and the displayed coefficients is needed. Repeated or constant readouts are retained as such.

2.6.1 The first-descent cells

For $D_j > 0$, define the exact source-index threshold

$$\theta_j = \frac{C_j/D_j - a}{2d}. \quad (5.3)$$

The source starts with actual first descent at j lie in

$$E_j(p) = \left(\theta_j, \min_{i < j, D_i > 0} \theta_i \right], \quad (5.4)$$

with upper endpoint $+\infty$ when there is no preceding positive D_i . Set $E_j = \emptyset$ for $D_j \leq 0$, or when the lower endpoint is at least the upper endpoint. Each declared index $j = 1, \dots, m$ is retained even for an empty cell.

Proof. By (2.1), actual descent at j is equivalent to $D_j > 0$ and $t > \theta_j$. Absence of descent at each earlier index with $D_i > 0$ is $t \leq \theta_i$; earlier nonpositive D_i never produce positive-source descent. Intersecting these exact inequalities gives (5.4). ■

For $p \in \mathcal{L}$, these cells partition every real source value t with $n(t) \geq 3$, because the terminal certificate guarantees at least one descent. Their boundaries keep the strict/non-strict distinction: equality at a threshold is not strict descent.

2.6.2 The joint observation and its complete fiber

For a source integer that stops at p , define

$$\pi(n) = (p, r, J), \quad r = v \bmod M, \quad (5.5)$$

where J is its actual first descent index. The exact fiber is

$$\{a + 2d(s + Pr + PM\ell) : \ell \in \mathbb{Z}, b \leq s + Pr + PM\ell < b + N, s + Pr + PM\ell \in E_J(p)\}. \quad (5.6)$$

Equation (5.6) supplies an inverse parametrization of the fiber, not merely its cardinality. Together, (5.1) and (5.6) recover the synchronized residue vector, the genuine first descent, and the target restart lattice. Forgetting r, J , or a selected residue read-out is a specified projection with fibers obtained by union over exactly the forgotten labels.

2.6.3 Example: reference stopping and actual first descent both retained

For the source 247, the word $(1, 1, 3, 2)$ gives

$$247 \rightarrow 371 \rightarrow 557 \rightarrow 209 \rightarrow 157.$$

The actual first descent is $J = 3$. The reference values at 3 are

$$3 \rightarrow 5 \rightarrow 8 \rightarrow 25/8 \rightarrow 83/32.$$

The reference certificate first occurs at $m = 4$. At prefix 3, $D_3 = 5$ and $C_3 = 19$, so the exact actual descent threshold is $19/5$. The source 247 exceeds it; the reference 3 does not. Both indices and their map (5.4) are retained.

2.7 6. Exact counts, including empty labels and overflow

For each declared label $\lambda = (p, r, j)$, let c_λ be the number of points in (5.6). This is computed by integer ceiling and floor operations on that original progression and its exact open/closed interval. Explicitly, for a compatible chart let

$$o = s + Pr, \quad h = PM,$$

and write $E_j = (\theta_j, \eta_j]$. Put

$$l = \max\{b, \lfloor \theta_j \rfloor + 1\}, \quad u = \min\{b + N - 1, \lfloor \eta_j \rfloor\},$$

where an infinite upper endpoint contributes no extra restriction. For $l \leq u$,

$$c_\lambda = \max\left\{0, \left\lfloor \frac{u - o}{h} \right\rfloor - \left\lceil \frac{l - o}{h} \right\rceil + 1\right\}. \quad (6.1)$$

Otherwise the count is zero. An incompatible chart or empty first-descent cell also has count zero, with its reason separately recorded. These labels are not removed from the declared target set.

Define the comparison weight on the **same threshold interval** by

$$q_\lambda = \frac{|[b, b + N) \cap E_j(p)|}{NMP} \quad (6.2)$$

for compatible charts, and zero for incompatible charts. The length in (6.2) is ordinary interval length in the original source parameter t . This is an explicitly defined lattice-density comparison. It depends on the source interval when the first-descent mark is included. It is not an assumption of independence between residue marks and actual deterministic descent.

2.7.1 Proposition 6.1 — Exact cell discrepancy

For every declared finite leaf label,

$$\left| \frac{c_\lambda}{N} - q_\lambda \right| \leq \frac{1}{N}. \quad (6.3)$$

For a fixed compatible leaf, $\sum_{r,j} q_{p,r,j} = 1/P$.

Proof. Intersecting (5.6) with the source range gives one interval, possibly empty and with its specified endpoint conventions. The number of points of $o + h\mathbb{Z}$ in any such bounded interval differs from its length divided by h by at most one. This follows directly from the two endpoint floor/ceiling formulas, including the cases in which endpoints themselves are lattice points. Division by N gives (6.3). The first-descent intervals partition the source real interval; their lengths sum to N . Summing over M marks in (6.2) gives $1/P$. ■

The reference mass of a compatible unmarked leaf is therefore

$$\mu_{a,d}(p) = \frac{1}{P} = \frac{g}{2^A}, \quad \mu_{a,d}(p) = 0 \text{ for an incompatible leaf.} \quad (6.4)$$

For odd d , $g = 1$, every leaf is compatible, and this is exactly 2^{-A} . For even d , equation (6.4), rather than the unconditioned weight, is retained.

At cutoff H , set

$$t_H^{a,d} = 1 - \sum_{p \in \mathcal{L}_H} \mu_{a,d}(p). \quad (6.5)$$

The actual overflow count is $N - \sum_{p \in \mathcal{L}_H, r, j} c_{p,r,j}$. Both overflow labels are included, even at value zero. The next two sections prove that (6.5) is nonnegative, tends to zero, and bounds the full countable comparison law.

2.8 7. A retained-translation square-root estimate and the complete-word tail

This section constructs the reference probability space and proves its tail bound. Nothing in this construction assigns independent random exponents to one fixed integer trajectory.

2.8.1 7.1 Literal binary factorization and its cylinder law

Define the half-step affine maps

$$f_0(x) = x/2, \quad f_1(x) = (3x + 1)/2. \quad (7.1)$$

The original odd branch is exactly

$$g_a = f_0^{a-1} \circ f_1. \quad (7.2)$$

A word $(a_1, \dots, a_m)$ corresponds to the binary input string

$$1 \, 0^{a_1-1} \, 1 \, 0^{a_2-1} \dots 1 \, 0^{a_m-1},$$

followed by the next odd input bit 1. It uses A executed half-steps and $A + 1$ known input bits. Its initial bit is fixed to 1.

For completeness, parity-prefix lifting gives one odd residue modulo 2^h for every prescribed binary input prefix of length h with initial bit 1. Indeed the affine output after h half-steps has odd numerator slope 3^j and denominator 2^h . The two lifts of a legal input class modulo 2^{h+1} change that output by the odd integer 3^j , so they realize the two possible next parities exactly once. Induction proves the assertion.

Consequently uniform odd residues give probability $2^{-(h-1)}$ to each such length- h prefix. These consistent finite laws are the fair binary product law after the forced initial odd bit. By (7.2), its exponent-word cylinder probabilities are exactly

$$\mu(p) = 2^{-A(p)} = \prod_{i=1}^m 2^{-a_i}. \quad (7.3)$$

2.8.2 7.2 Exact coordinate morphism for the potential

Use the bijection

$$\phi : x \mapsto z = x - 1, \quad \phi^{-1}(z) = z + 1. \quad (7.4)$$

The conjugated maps, retaining their translations, are

$$\phi f_0 \phi^{-1}(z) = \frac{z-1}{2}, \quad \phi f_1 \phi^{-1}(z) = \frac{3z+2}{2}. \quad (7.5)$$

The potential is $V(x) = \sqrt{x-1}$ on $x \geq 3$. The source domain becomes $z \geq 2$; it is not changed to a different positivity chamber.

2.8.3 Theorem 7.1 — Exact square-root contraction

For every real $x \geq 3$,

$$\frac{\sqrt{f_0(x)-1} + \sqrt{f_1(x)-1}}{2} \leq \frac{\sqrt{15}}{4} \sqrt{x-1}. \quad (7.6)$$

Equality holds at $x = 7$. In particular, the same bound holds for the killed expectation in which branches with new value below 3 contribute zero.

Proof. Put

$$A_x = f_0(x) - 1 = (x-2)/2, \quad B_x = f_1(x) - 1 = (3x-1)/2.$$

Both are positive on the stated domain, and the original translations give the exact identity

$$2A_x + B_x = \frac{5}{2}(x-1). \quad (7.7)$$

The Cauchy-Schwarz inequality, with its displayed weights, gives

$$(\sqrt{A_x} + \sqrt{B_x})^2 \leq (1/2 + 1)(2A_x + B_x) = \frac{15}{4}(x-1).$$

Division by 4 and taking nonnegative square roots gives (7.6). Its equality condition $B_x = 4A_x$ is exactly $x = 7$. An entirely polynomial certificate is

$$R = \frac{15}{4}(x-1) - A_x - B_x = \frac{7x-9}{4} > 0, \quad R^2 - 4A_x B_x = \frac{(x-7)^2}{16}. \quad (7.8)$$

Removing positive contributions from branches killed below 3 preserves the bound. ■

2.8.4 7.3 Survival of the binary reference process

Apply (7.1) to the literal reference $X_0 = 3$, with the first bit forced odd and subsequent bits fair. Let

$$\tau = \inf\{h \geq 1 : X_h < 3\}, \quad S_h = \Pr(\tau > h).$$

Before killing, $V(X_h) \geq \sqrt{2}$. The forced first step gives $X_1 = 5$, $V(X_1) = 2$. Iterating the killed estimate gives, for $h \geq 1$,

$$S_h \leq \sqrt{2} \left(\frac{\sqrt{15}}{4} \right)^{h-1}. \quad (7.9)$$

The entirely rational constants used by the checker are

$$\rho = \frac{31}{32}, \quad C_0 = \frac{3}{2}. \quad (7.10)$$

They satisfy $15/16 < \rho^2$ and $2 < (C_0\rho)^2$. Thus, including $h = 0$,

$$S_h \leq C_0\rho^h. \quad (7.11)$$

This proves $\Pr(\tau = \infty) = 0$ in the specified binary reference probability space.

2.8.5 7.4 Completion to the original odd-return word

The first crossing below 3 occurs on an even half-step: $f_1(x) > x$ for $x \geq 3$. After that crossing, finish the remaining even run before the next odd input. Let R be the number of additional zero input bits. The completed original exponent word has sum

$$A_{\text{stop}} = \tau + R. \quad (7.12)$$

For every finite stopping prefix, the unused bits are fair, and

$$\Pr(R = r \mid \text{that prefix}) = 2^{-r-1}, \quad r \geq 0. \quad (7.13)$$

This follows by counting the specified next r zeros and the following 1. It proves independence from the stopping prefix directly on the original binary cylinders. Earlier completed odd returns were at least 3; the completed return in (7.12) is below 3. Thus its leaf is exactly the one in (2.2).

Let

$$t_H = \Pr(A_{\text{stop}} > H) = 1 - \sum_{p \in \mathcal{L}_H} 2^{-A(p)}.$$

Then $t_0 = 1$, and

$$t_H = \frac{t_{H-1} + S_H}{2} \quad (H \geq 1). \quad (7.14)$$

Proof. Decompose by τ . For a crossing at $\tau \leq H$, equation (7.13) gives $\Pr(R > H - \tau) = 2^{-(H-\tau+1)}$; crossings after H have total mass S_H . Subtracting the expressions at H and $H - 1$, and using $\Pr(\tau = H) = S_{H-1} - S_H$, gives (7.14). ■

Since

$$K = \frac{31}{20}, \quad K(2\rho - 1) = C_0\rho,$$

induction in (7.14) proves

$$\boxed{t_H \leq K\rho^H}. \quad (7.15)$$

In particular, the leaf masses sum to one. All complete exponents, rather than a prematurely cut even run, are retained.

2.8.6 7.5 Exact leaf count and a quantitative growth bound

The number of binary prefixes first killed at length h is

$$b_h = 2^{h-1}(S_{h-1} - S_h).$$

Each gives exactly one completed odd leaf for each additional even-run length $R = 0, \dots, H - h$. The decomposition by the first crossing and additional zeros is unique. Therefore

$$|\mathcal{L}_H| = \sum_{h=1}^H b_h (H - h + 1). \quad (7.16)$$

Equation (7.11) gives $b_h \leq C_0(2\rho)^{h-1}$. Summing the geometric derivative series yields

$$|\mathcal{L}_H| \leq C_0 \frac{(2\rho)^{H+1}}{(2\rho - 1)^2}, \quad \sum_{p \in \mathcal{L}_H} m(p) \leq H |\mathcal{L}_H|. \quad (7.17)$$

These are explicit bounds, not fitted decay rates. The checker separately enumerates binary survival and complete odd leaves and verifies the exact identities (7.14) and (7.16) through exponent sum 20.

2.9 8. The full stopped law, with joint arithmetic marks

2.9.1 8.1 Reference measure for the original source progression

Uniform residues of t modulo growing powers of two in (3.1) induce a measure on odd dyadic source classes. Its word-cylinder mass is exactly the congruence count (6.4). This also gives a direct finite-level construction of that measure.

Write $d = 2^{s_0} d_o$ with d_o odd. The affine map $t \mapsto a + 2dt$ is a bijection from uniform dyadic parameters onto the class $n \equiv a \pmod{2^{s_0+1}}$, with inverse parameter obtained by division by the retained 2^{s_0+1} and the dyadic unit d_o . That class has mass 2^{-s_0} among all odd dyadic classes. Thus every event E satisfies the explicit conditional-measure comparison

$$\mu_{a,d}(E) = 2^{s_0} \mu(E \cap \{n \equiv a \pmod{2^{s_0+1}}\}). \quad (8.1)$$

Applied to the event of no completed leaf by exponent sum H , it proves

$$0 \leq t_H^{a,d} \leq 2^{s_0} t_H \leq 2^{s_0} K \rho^H. \quad (8.2)$$

For odd d , $t_H^{a,d} = t_H$. In particular, $\sum_{p \in \mathcal{L}} \mu_{a,d}(p) = 1$, for every fixed retained progression.

2.9.2 8.2 The two countable probability laws

Let $P_{a,d,b,N,M}$ be the actual pushforward of the uniform law on the N integer parameters in (3.1) by (5.5), with the additional never-certified symbol ∞ .

Let $Q_{a,d,b,N,M}$ assign every finite label its weight (6.2), and assign $Q(\infty) = 0$. Equations (6.4) and (8.2) prove that its finite-label weights sum to one. Its declared zero labels, including ∞ , remain in the target set. This is a precise definition of the comparison law, not an assertion that it equals actual arithmetic data.

2.9.3 Theorem 8.1 — All-horizon joint-observation bound

For every admissible source progression, every $N, M \geq 1$, and every integer $H \geq 0$,

$$\Delta(P_{a,d,b,N,M}, Q_{a,d,b,N,M}) \leq \min \left\{ 1, t_H^{a,d} + \frac{M}{N} \sum_{p \in \mathcal{L}_H} m(p) \right\}. \quad (8.3)$$

The left side concerns the full countable stopped observation with no prescribed final return horizon. For odd d , the bound is uniform in the original location a, b and the retained odd step d .

Proof. Let S be the finite declared labels lying over $\mathcal{L}_H$, and let $e = |S|/N = M \sum m(p)/N$. Equation (6.3) gives $\sum_{\lambda \in S} |P(\lambda) - Q(\lambda)| \leq e$. Its consequence for total mass is $P(S^c) \leq Q(S^c) + e = t_H^{a,d} + e$. Therefore

$$2\Delta(P, Q) \leq e + P(S^c) + Q(S^c) \leq 2e + 2t_H^{a,d}.$$

This inequality includes every tail leaf and the actual never-certified atom, so it has not discarded either. The bound by one is the defining maximum of total variation. ■

For the stopped word alone, the identical proof with one cell per leaf gives

$$\Delta(P^{\text{word}}, \mu_{a,d}) \leq \min\{1, t_H^{a,d} + |\mathcal{L}_H|/N\}. \quad (8.4)$$

2.9.4 Explicit convergence rate for odd source step

Set $\eta = -\log_2(31/32) > 0$. For parameters $N \geq M(1 + \log_2 N)$, choose the explicit nonnegative integer

$$H = \left\lfloor \log_2 \frac{N}{M(1 + \log_2 N)} \right\rfloor. \quad (8.5)$$

Equations (7.15), (7.17), and (8.3) yield

$$\Delta(P, Q) \leq \left(K + C_0 \frac{2\rho}{(2\rho - 1)^2} \right) \rho^H \leq \frac{K + C_0(2\rho)/(2\rho - 1)^2}{\rho} \left(\frac{M(1 + \log_2 N)}{N} \right)^\eta. \quad (8.6)$$

Indeed $MH2^H/N \leq H/(1 + \log_2 N) \leq 1$, and the floor in (8.5) loses at most one factor of ρ . For fixed M , this tends to zero uniformly over the stated odd-step source families. This explicit rate is not optimized.

For general even step, the finite inequality remains (8.3) with the exact conditional tail (6.5), or its bound (8.2). The dyadic factor is not omitted to assert uniformity in an unrestricted growing even step.

2.9.5 Exact projection to other observations

For any declared readout map ψ of the retained labels, define the pushforward by summing its literal fibers. Then

$$\Delta(\psi_*P, \psi_*Q) = \frac{1}{2} \sum_y \left| \sum_{\lambda: \psi(\lambda)=y} (P(\lambda) - Q(\lambda)) \right| \leq \Delta(P, Q). \quad (8.7)$$

Thus (8.3) applies to the synchronized residue vector, the actual first-descent index, or their joint readout, by their explicit maps in Section 5. The proof shows exactly which fibers are combined.

The observation in (8.3) forgets the post-stop tail by (2.6). It does not recover the predecessor's entire fixed-length history law. No equivalence between those two distributions is asserted; their exact relation is the pushforward (2.5) and its displayed fibers.

2.10 9. Restart composition and the retained dyadic/triadic correspondence

2.10.1 9.1 Exact ancestral state

A restart node consists of a current family

$$x(t) = a + 2dt, \quad b \leq t < b + N,$$

an original source family on the same parameter

$$n_0(t) = a_0 + 2d_0t,$$

and the complete concatenated history w . They satisfy the coefficient identities

$$L(w)a_0 + C(w) = 2^{A(w)}a, \quad L(w)d_0 = 2^{A(w)}d. \quad (9.1)$$

For a compatible next leaf p and mark restriction (3.11), set

$$o = s + Pr, \quad h = PM, \quad t = o + h\ell.$$

The new original coefficients are

$$a'_0 = a_0 + 2d_0o, \quad d'_0 = d_0h, \quad (9.2)$$

and the new current coefficients are

$$a' = u + 2d'r, \quad d'_{\text{current}} = Md'. \quad (9.3)$$

The new history is wp , with composition law (1.4). Direct substitution in (9.1) proves its two new coefficient identities. This is an inductive commuting diagram at every restart, not just endpoint agreement on a sampled orbit.

Within each node, distinct leaf cylinders are disjoint by prefix-freeness. Congruence marks partition each compatible cylinder. Exact first-descent cells further partition

by (5.4). Empty fibers, cutoff overflow, and the singleton value 1 are separately retained. The singleton 1 is an absorbing boundary state, because $T(1) = 1$; Theorem 2.1 is used only at current sources at least 3.

2.10.2 9.2 The explicit dyadic itinerary map

The infinite regular exponent code has the 2-adic realization

$$\Xi(a_1, a_2, \dots) = - \sum_{i=1}^{\infty} \frac{2^{A_{i-1}}}{3^i}. \quad (9.4)$$

The series converges in $\mathbb{Z}_2$ because $A_{i-1} \rightarrow \infty$; its first term is odd and later terms are even. Splitting the sum after m terms gives the exact branch equation

$$\Xi(p\omega) = \frac{2^{A(p)}\Xi(\omega) - C(p)}{L(p)}. \quad (9.5)$$

It follows, using (1.7), that its prefix cylinder has the original residue and modulus. Conversely, the finite cylinders of any infinite exponent code are nested and have moduli tending to infinity, so they specify the unique dyadic integer (9.4). The points where a dyadic trajectory reaches $-1/3$, and hence an infinite valuation at a finite stage, are excluded from the regular exponent domain by their explicit defining equation. They form a countable set of finite-prefix inverse images under (9.5). No positive integer lies in that set: every finite actual iterate of a positive integer is positive. A separate exceptional symbol may be assigned to those dyadic points; its dyadic measure is zero.

This records the exact domain and exceptional set of the coding. Formula (9.5) is the same inverse as (4.10), with its integer integrality criterion (4.11) still retained.

2.10.3 Theorem 9.1 — Product law of completed blocks in the reference code

Successive words in $\mathcal{L}$, obtained by restarting the reference stopping rule on the unused exponent code, have the cylinder law

$$\Pr(p_1, \dots, p_r) = \prod_{i=1}^r 2^{-A(p_i)}. \quad (9.6)$$

Proof. Prefix-free stopping parses a concatenation of r leaf words uniquely. Prescribing that stopped tuple is exactly prescribing the concatenated exponent prefix $p_1 \cdots p_r$. Its cylinder mass from (7.3) is $2^{-\sum_i A(p_i)}$, which is (9.6). Section 7 proves that each unused reference tail completes a leaf with probability one, so these masses have total one for each fixed r . ■

This is a theorem about the specified reference code. For actual finite integer samples, the corresponding concatenated cylinder is counted by its original floor formula and restricted through the restart charts. No independence of actual finite-sample block outcomes is asserted.

The dyadic branch (9.5) maps onto all regular odd dyadic tail values, while its ordinary integer target is the triadic congruence class (4.11). The two assertions coexist through the explicit embedding and inverse diagram of Section 4. In an integer

restart the step $d3^m$, or its fully retained even-step counterpart (3.7), remains part of the source. Equation (9.6) does not authorize replacing that source with consecutive odd integers.

At the actual integer 1, the infinite code $(2, 2, \dots)$ realizes 1 by (9.4). The unabsorbed reference parser would keep returning (2). The positive-integer engine instead maps that exact source point to the absorbing label 1. This is a specified change on a literal singleton of reference measure zero, not a claim that the singleton is absent from a finite integer sample.

2.11 10. Support-aware chain maps and finite Hurwitz reconstruction

The arithmetic image complex in Section 4 and the actual support-incidence complex are linked by an affine incidence complex carrying both the original mass and the actual terminal odd-index. The comparison chain maps, their cohomology sequence, and a retained-anchor homotopy are given first.

2.11.1 10.0 Exact chain-level bridge through the actual terminal moment

The link between support incidence and the arithmetic image complex can be made at chain level, not merely by citing the same source chart. Fix a compatible declared label $\lambda = (p, r, j)$. In (3.11), let F_λ be the finite set of actual lift parameters ℓ in the fiber (5.6). It is a consecutive integer interval, possibly empty. Retain

$$D_\lambda = Md', \quad z_\lambda = \frac{u + 2d'r - 1}{2}, \quad z(\ell) = z_\lambda + D_\lambda \ell. \quad (10.0a)$$

Here $z(\ell)$ is the **actual terminal odd-index**, not an arbitrary numerical label. Form the affine incidence complex

$$\mathcal{A}_\lambda = \left[\mathbb{Z}[F_\lambda] \xrightarrow{d_{\mathcal{A}}} \mathbb{Z}^2 \right], \quad d_{\mathcal{A}}(e_\ell) = (1, z(\ell)). \quad (10.0b)$$

Its first coordinate is the original support incidence. Its second coordinate is the actual terminal moment. Write the one-label support complex and marked image complex as

$$\mathcal{J}_\lambda = [\mathbb{Z}[F_\lambda] \xrightarrow{\delta} \mathbb{Z}], \quad \delta(e_\ell) = 1, \quad \mathcal{K}_\lambda = [\mathbb{Z} \xrightarrow{D_\lambda} \mathbb{Z}].$$

There is the following explicit span of chain maps:

$$\mathcal{J}_\lambda \xleftarrow{q} \mathcal{A}_\lambda \xrightarrow{f} \mathcal{K}_\lambda, \quad (10.0c)$$

with

$$q^0 = \text{id}, \quad q^1(a, b) = a, \quad f^0(e_\ell) = \ell, \quad f^1(a, b) = b - z_\lambda a. \quad (10.0d)$$

The chain equations are exactly

$$q^1 d_{\mathcal{A}}(e_\ell) = 1 = \delta(e_\ell), \quad f^1 d_{\mathcal{A}}(e_\ell) = z(\ell) - z_\lambda = D_\lambda f^0(e_\ell). \quad (10.0e)$$

The coefficient z_λ is retained in the map; it has not been set to zero without its inverse data.

The exact sequence exposing the image torsion

The kernel complex of q has only $\mathbb{Z}$ in degree one, embedded by $b \mapsto (0, b)$. Its resulting exact cohomology sequence is

$$0 \longrightarrow H^0(\mathcal{A}_\lambda) \longrightarrow H^0(\mathcal{J}_\lambda) \xrightarrow{\theta} \mathbb{Z} \longrightarrow H^1(\mathcal{A}_\lambda) \longrightarrow H^1(\mathcal{J}_\lambda) \longrightarrow 0, \quad (10.0f)$$

where the connecting map, with its sign fixed by the displayed embedding, is

$$\theta\left(\sum_{\ell} c_{\ell} e_{\ell}\right) = \sum_{\ell} c_{\ell} z(\ell) = D_{\lambda} \sum_{\ell} c_{\ell} \ell, \quad \sum_{\ell} c_{\ell} = 0. \quad (10.0g)$$

Direct proof of exactness. The kernel of θ consists exactly of source vectors of both mass zero and terminal moment zero, which is $H^0(\mathcal{A}_\lambda)$. The next map sends b to the class of $(0, b)$. Its kernel consists of moments of mass-zero source vectors, exactly the image of θ . The next map is induced by the first-coordinate projection, and is surjective. Its kernel is represented by the second-coordinate vectors after subtracting a source lift of the first coordinate whenever that coordinate is in the support image. This proves the remaining exactness, including the empty source case. ■

Let $n_\lambda = |F_\lambda|$. For an empty fiber, $H^1(\mathcal{A}_\lambda) = \mathbb{Z}^2$. For a singleton, its sole image vector $(1, z)$ is primitive, so $H^1(\mathcal{A}_\lambda) \cong \mathbb{Z}$. For at least two points, their consecutive parameters give

$$\theta(H^0(\mathcal{J}_\lambda)) = D_\lambda \mathbb{Z}, \quad H^1(\mathcal{A}_\lambda) \cong \mathbb{Z}/D_\lambda \mathbb{Z}, \quad \text{rank } H^0(\mathcal{A}_\lambda) = n_\lambda - 2. \quad (10.0h)$$

Indeed the first two columns have determinant D_λ , and every column difference is $(0, D_\lambda k)$. Thus the original terminal lattice appears as the cokernel of the **actual terminal-moment map on the support kernel**. Empty and singleton fibers retain their different free cohomology instead of being silently forced into the at-least-two-point calculation.

Explicit homotopy retaining the anchor

For $n_\lambda \geq 2$, choose the first two actual parameters $\ell_0, \ell_0 + 1$, and retain $z_* = z(\ell_0)$. The anchored comparison has

$$\bar{f}^0(e_\ell) = \ell - \ell_0, \quad \bar{f}^1(a, b) = b - z_* a.$$

Its right inverse from $\mathcal{K}_\lambda$ is

$$i^0(1) = e_{\ell_0+1} - e_{\ell_0}, \quad i^1(1) = (0, 1).$$

Let

$$R_\ell = e_\ell - e_{\ell_0} - (\ell - \ell_0)(e_{\ell_0+1} - e_{\ell_0}),$$

let $P^0(e_\ell) = R_\ell$, $P^1 = 0$, and put $h^1(a, b) = a e_{\ell_0}$. Direct substitution gives

$$\bar{f}i = 1, \quad 1 - i\bar{f} - P = d_{\mathcal{A}}h + hd_{\mathcal{A}}. \quad (10.0i)$$

Each R_ℓ has both zero mass and zero original terminal moment. The vectors outside the two anchors give a free basis of the kernel. Hence (10.0i) is a complete chain homotopy onto the marked image complex plus the actual affine-relation module in degree zero.

The original and anchored comparison maps themselves differ by the explicit chain homotopy $s^1(a, b) = \ell_0 a$: $f - \bar{f} = d_{\mathcal{K}}s + sd_{\mathcal{A}}$. The original coset is retained because $z_* - z_\lambda = D_\lambda \ell_0$. Thus neither the anchor nor its translation has been silently discarded.

Mark refinement of the image complex

The map from $\mathcal{K}_\lambda$ back to the unmarked complex in Section 4 is multiplication by M in degree zero and the identity in degree one. Its chain equation is $d'M = D_\lambda$. The cohomology maps fit the exact sequence

$$0 \longrightarrow \mathbb{Z}/M\mathbb{Z} \xrightarrow{[c] \mapsto [d'c]} \mathbb{Z}/(Md')\mathbb{Z} \longrightarrow \mathbb{Z}/d'\mathbb{Z} \longrightarrow 0. \quad (10.0j)$$

Its middle-to-right map is literal reduction. The marked coset $[z_\lambda]$ maps to the original $[(u-1)/2]$, because their difference is $d'r$. This is the exact lattice data carried by forgetting a mark.

Seven-point instance

On the actual points $247 + 256\ell$, $\ell = 0, \dots, 6$, use $p = (1, 1, 3, 2)$, $M = 1$, and actual first descent $j = 3$. The affine incidence matrix has columns

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 78 & 159 & 240 & 321 & 402 & 483 & 564 \end{pmatrix}. \quad (10.0k)$$

The support map is its first row. The terminal-moment map on the support kernel has image $81\mathbb{Z}$, giving the exact sequence

$$0 \longrightarrow \mathbb{Z}^5 \longrightarrow \mathbb{Z}^6 \xrightarrow{\theta} \mathbb{Z} \longrightarrow \mathbb{Z}/81\mathbb{Z} \longrightarrow 0. \quad (10.0l)$$

In particular, the image torsion in Section 4 is recovered from these original finite arithmetic values and explicit source maps, not inserted as an unrelated auxiliary group.

2.11.2 10.1 Incidence complex of the actual observation

Fix a finite source sample I and the declared finite label set Λ_H of all (p, r, j) with $p \in \mathcal{L}_H$, together with overflow. Incompatible labels and empty first-descent cells remain in Λ_H , with their masks recorded. The actual map is $\pi_H : I \rightarrow \Lambda_H$.

Form

$$K_{\pi_H}^\bullet = \left[\mathbb{Z}[I] \xrightarrow{\delta} \mathbb{Z}[\Lambda_H] \right], \quad \delta(e_n) = e_{\pi_H(n)}. \quad (10.1)$$

For each occupied label λ , select one actual source point n_λ in its fiber and retain that selection. Define

$$h^1(e_\lambda) = \begin{cases} e_{n_\lambda}, & \lambda \text{ occupied,} \\ 0, & \lambda \text{ empty.} \end{cases} \quad (10.2)$$

Let $P^0 = 1 - h^1\delta$ and $P^1 = 1 - \delta h^1$. These formulas give

$$1 - P^0 = h^1\delta, \quad 1 - P^1 = \delta h^1, \quad \delta P^0 = 0, \quad P^1\delta = 0. \quad (10.3)$$

They are explicit projections and a chain homotopy onto the following zero-differential cohomology representatives:

$$\begin{aligned} H^0 &\cong \bigoplus_{\lambda \text{ occupied}} \langle e_n - e_{n_\lambda} : n \in \pi_H^{-1}(\lambda), n \neq n_\lambda \rangle, \\ H^1 &\cong \mathbb{Z}[\{\lambda \in \Lambda_H : \pi_H^{-1}(\lambda) = \emptyset\}]. \end{aligned} \quad (10.4)$$

Proof. In a source fiber, δ sums its coefficients into its target basis vector. Its kernel is freely generated by the displayed differences. Its image contains exactly the occupied target basis vectors. The other target basis vectors form a free cokernel basis. Equations (10.3) follow on each source and target basis vector from (10.2). ■

Changing the selected source representative in a fiber changes $h^1(e_\lambda)$ by the explicitly known kernel vector $e_{n'_\lambda} - e_{n_\lambda}$. This retains the section-change map as well.

Thus a declared label with zero actual weight remains a basis element; it is not removed because a numerical weight vanishes. The distinct reasons for zero—source congruence incompatibility, empty threshold interval, or no point in this particular finite sample—remain metadata on those exact labels.

2.11.3 10.2 Pullback, pushforward, and a retained count vector

The function pullback is

$$\pi_H^* f = f \circ \pi_H. \quad (10.5)$$

On formal source measures it is dual to the exact pushforward δ , and

$$\delta \left(\sum_{n \in I} e_n \right) = \sum_{\lambda \in \Lambda_H} c_\lambda e_\lambda. \quad (10.6)$$

For a further declared projection $\psi : \Lambda_H \rightarrow \Gamma$, the chain map from the incidence complex of π_H to that of $\psi\pi_H$ is the identity in degree zero and $e_\lambda \mapsto e_{\psi(\lambda)}$ in degree one. Its chain equation is literal composition of the source maps. In degree one its kernel is generated by label contrasts $e_\lambda - e_{\lambda_0}$ inside the fibers of ψ . This exhibits exactly the information lost by a further observation.

2.11.4 10.3 Finite spectral input with its label dictionary retained

Choose and retain a bijective enumeration

$$\iota : \Lambda_H \longrightarrow \{1, \dots, d_\Lambda\}.$$

The associated diagonal label operator has eigenvalues $\lambda_i = i$, with an idempotent e_i for **every** declared label. The dictionary ι^{-1} is part of the object; discarding it would

discard the connection to (5.6).

Put $w_i = c_i/N$, including zero weights, and

$$M_k = \sum_{i=1}^{d_\Lambda} w_i \lambda_i^k, \quad Z(s) = \sum_{i=1}^{d_\Lambda} w_i \zeta(s, \lambda_i). \quad (10.7)$$

The finite weights are the actual counts (10.6), not an unspecified probability vector. The equality $M_0 = 1$ includes overflow.

The standard Hurwitz-Bernoulli identity [R3] is

$$\zeta(-k, x) = -\frac{B_{k+1}(x)}{k+1}, \quad B_{k+1}(x) = \sum_{r=0}^{k+1} \binom{k+1}{r} B_{k+1-r} x^r. \quad (10.8)$$

For this finite source, it gives the literal triangular relation

$$Z(-k) = -\frac{1}{k+1} \sum_{r=0}^{k+1} \binom{k+1}{r} B_{k+1-r} M_r. \quad (10.9)$$

Starting from $M_0 = 1$, the explicit inverse is

$$M_{k+1} = -(k+1)Z(-k) - \sum_{r=0}^k \binom{k+1}{r} B_{k+1-r} M_r. \quad (10.10)$$

For each label form the Lagrange polynomial

$$L_i(X) = \prod_{j \neq i} \frac{X - \lambda_j}{\lambda_i - \lambda_j} = \sum_{k=0}^{d_\Lambda-1} l_{ik} X^k.$$

Its exact weight reconstruction is

$$w_i = \sum_{k=0}^{d_\Lambda-1} l_{ik} M_k. \quad (10.11)$$

Proof. Equation (10.9) follows by inserting the finite sum (10.7) into (10.8). The coefficient of M_{k+1} inside the sum is one, which gives (10.10). Finally $\sum_k l_{ik} M_k = \sum_j w_j L_i(\lambda_j) = w_i$. All finite sums and their label maps are explicit. In particular (10.11) reconstructs zero weights without deleting their idempotents. ■

The checker uses exact Bernoulli fractions and polynomial arithmetic to test (10.9)–(10.11). It does not claim numerical analytic continuation of zeta functions or identification with an inaccessible predecessor implementation.

2.12 11. A complete worked restart family

Start with the compatible family in (4.12). On the target $157 + 162t$, the next leaf (3) selects $t = 8u$. The exact two-block diagram is

$$247 + 2048u \mapsto 157 + 1296u \mapsto 59 + 486u, \quad u \in \mathbb{Z}_{\geq 0}. \quad (11.1)$$

The first map uses four odd returns; the second uses one. Each displayed block begins at a value at least 3 and strictly descends. The step 486 is $2 \cdot 3^5$, not 2.

For the anchor 247 the successive reference-certified blocks are

$$(1, 1, 3, 2), \quad (3), \quad (1, 2, 1, 4), \quad (1, 3), \quad (1, 2, 3), \quad (4). \quad (11.2)$$

Their endpoint values are respectively

$$157, \quad 59, \quad 19, \quad 11, \quad 5, \quad 1.$$

The complete prefix data are:

Completed block	Total odd returns	Total exponent sum	Original source step	Terminal step	Anchor endpoint
1	4	7	256	162	157
2	5	10	2,048	486	59
3	9	18	524,288	39,366	19
4	11	22	8,388,608	354,294	11
5	14	28	536,870,912	9,565,938	5
6	15	32	8,589,934,592	28,697,814	1

Each row is an infinite-family identity

$$247 + 2^{A+1}t \mapsto x_{\text{anchor}} + 2 \cdot 3^m t, \quad t \geq 0, \quad (11.3)$$

with the row's original word, A , and m . In particular,

$$T^{15}(247 + 2^{33}t) = 1 + 28\,697\,814t \quad (11.4)$$

on precisely the concatenated word in (11.2). The assertion that the terminal value equals 1 applies to the anchor $t = 0$; equation (11.4) states the other values exactly and does not assert that the whole infinite family has already reached 1.

2.13 12. Executed verification and precise scope

The accompanying `verify.py` uses only the Python standard library, integers, and exact rational fractions. Every proof check is an explicit exception check rather than an `assert`, so optimization does not disable the checks.

The verified families include 340 original words of lengths 1 through 4 with exponents 1 through 4; independent binary and odd-leaf enumerations through exponent sum 20; 180 marked finite-source laws with odd and even progression steps, large source offsets, and odd/even mark moduli; synchronized residue readouts modulo 3, 4, 9, and 12; integral image complexes and finite coefficient kernels at powers of 2 and 3; enriched-incidence chain maps and homotopies, including empty, singleton, and occupied-fiber cases; exact product-cylinder counts for one, two, and three stopped blocks; the worked infinite-family identities; and a three-round original-source restart replay.

For the 17 source points $3, 5, \dots, 35$, exponent cutoff 4 and mark modulus 3, the finite incidence/Hurwitz fixture has 16 declared labels, 8 occupied labels, rank $H^0 = 9$, and rank $H^1 = 8$. Every zero-weight label is reconstructed exactly.

The three-round finite restart starts with the 1,024 odd integers 3, 5, ..., 2049 and uses exponent cutoff 10 at each block:

Round	Input points	Accepted descent points	Already at 1, absorbed	Explicit cutoff overflow	Output progressions
1	1,024	809	0	215	71
2	809	637	5	167	302
3	637	502	14	121	423

At each round the counts in the three outcome columns sum to the input count. Every accepted point is checked against actual valuations, terminal values, original-source ancestry, and strict descent. Overflow points are retained as unresolved outcomes of that finite cutoff and are not silently sent through the restart. Output points reaching 1 are absorbed at the next input boundary; the table does not count every final output at 1 as already absorbed.

The exact stopped profiles include

Exponent cutoff	Number of leaves	Sum of leaf lengths	Reference overflow
4	4	5	1/2
8	28	75	69/256
12	212	1,102	339/2048
16	1,601	12,116	1901/16384
20	17,396	181,297	10845/131072

These are finite exact checks of the displayed formulas. The infinite-series, coefficient-comparison, and all-horizon statements rest on the proofs above, not on extrapolation from these numbers. Ordinary and optimized execution receipts are supplied separately. No Lean execution, external mathematical audit, current-workbench replay, GitHub CI success, or remote publication is claimed.

2.13.1 What remains an actual integer question

For a finite source sample, the never-certified symbol has actual weight

$$P(\infty) = \frac{1}{N} \# \{n \in I : \sigma(\text{actual itinerary of } n) = \infty\}. \quad (12.1)$$

Its reference weight is zero, and the map from source points to that label is still present. A positive actual point contributes $1/N$, regardless of its zero reference measure. Equations (8.3) and (12.1) state the relationship explicitly. They do not prove the count in (12.1) is zero for every sample, nor that every individual positive start enters an accepted restart branch.

The concrete deterministic continuation object is the restart system (3.10), (4.3)–(4.11), and (9.1)–(9.3), with original source ancestry attached. It supports exact further work on the unresolved integer fibers without substituting a uniform odd source or deleting a zero-reference label. The present contribution constructs that system and executes several levels of it; it does not replace the remaining integer-fiber question by an unproved assumption.

2.14 References

[R1] Riho Terras, *A stopping time problem on the positive integers*, Acta Arithmetica **30** (1976), 241–252, DOI 10.4064/aa-30-3-241-252. Publisher bibliographic record inspected on 14 September 2026. The full paper was not used as a proof dependency.

[R2] Terence Tao, *Almost all orbits of the Collatz map attain almost bounded values*, arXiv 1909.03562; Forum of Mathematics, Pi **10** (2022), e12. The current abstract and metadata inspected on 14 September 2026 identify version 7, revised 16 July 2026. The abstract states the logarithmic-density orbit-minimum theorem and the first-passage/3-adic transport method. No assertion of having audited its complete current proof is made here. Source: <https://arxiv.org/abs/1909.03562>.

[R3] NIST Digital Library of Mathematical Functions, equations 25.11.14 and 24.2.1–24.2.3, Hurwitz values at nonpositive integers and the Bernoulli generating functions/polynomials. Inspected on 14 September 2026. Sources: <https://dlmf.nist.gov/25.11.E14> and <https://dlmf.nist.gov/24.2>.

[R4] Author-supplied conversation and readable historical author workbench records, September 2026 intake: original affine Collatz definitions, exact-morphism methodology, and the claimed predecessor path. Private record locators and the distinction between readable historical material and inaccessible current repository files are preserved in the local source-intake record, outside the repository patch.

Chapter 3

Collatz cycles, relative cohomology, and counterexample-preserving descent compression

Source: repository/collatz_reconstruction/research_program/cycle_relative_cohomology_20260914/note.md.

Date: 14 September 2026

Target: KokunoYumeto/collatz-workbench

Status: proofs and exact local replay; no external mathematical review, Lean certification, or remote publication claimed.

3.1 0. Source, question, and contribution

The preceding contribution is retained byte-for-byte at `../stopped_affine_transport_20260914/`. Its exact chart sends a legal word cylinder to its actual affine image, and its finite source-incidence construction records the image lattice. This continuation adds the return-to-source condition, its integral obstruction, the actual orbit graph relative to the fixed point 1, and explicit chain-homotopy compression by certified descent paths.

The central constructive result is Section 6: every checked collection of actual strict-descent paths gives a chain retraction with a displayed homotopy, and **every closed cycle chain is fixed literally by that retraction**. Unlisted vertices remain separate roots. This is the precise mechanism by which the preceding descent work can be used without deleting a possible counterexample.

The relevant cycle information is not a bare Betti number or an unmarked torsion group. A proposed word retains its forcing class, rational primitive, integral/positive membership, actual valuation word, primitive period with its reconstruction map, and its induced map into the relative orbit graph.

Rational cycle formulas have established precedents, notably Lagarias [R1]. Operator formulations of Collatz also have an existing literature, for example Mori [R2]. No priority claim is made for those underlying ideas. All specific algebraic and graph statements used below are proved here. The predecessor's probability estimates are not used to decide whether an individual integer is a counterexample.

Current GitHub files were not readable through an authenticated connection in this session. Consequently no current remote base commit, methodology inspection, PR status, or remote integration is asserted. The source actually used is the attached predecessor archive, whose manifest and executable results were replayed locally.

3.2 1. Original affine word and the closed-cycle obstruction

Write

$$T(n) = \frac{3n+1}{2^{\nu_2(3n+1)}} \quad (n \in X := \{1, 3, 5, \dots\}).$$

For a nonempty word $p = (a_1, \dots, a_m)$, retain

$$\begin{aligned} A_i &= \sum_{j=1}^i a_j, \quad A_0 = 0, \quad U = 2^{A_m}, \quad L = 3^m, \quad D = U - L, \\ C &= \sum_{i=1}^m 3^{m-i} 2^{A_{i-1}}, \quad F_p(x) = \frac{Lx + C}{U}. \end{aligned} \quad (1.1)$$

Every exponent is a positive integer. Thus $D \neq 0$ and $\gcd(D, 6) = 1$. No denominator or affine term is removed.

Define the two-term cochain complex, in degrees 0 and 1,

$$K_p = [\mathbb{Z}^m \xrightarrow{B_p} \mathbb{Z}^m], \quad (B_p x)_i = 2^{a_i} x_{i+1} - 3x_i, \quad x_{m+1} = x_1. \quad (1.2)$$

For $m = 1$, the single matrix entry is $2^{a_1} - 3$, with both contributions added in the same entry.

3.2.1 Theorem 1.1. Integral forcing class and rational primitive

Put

$$\mathcal{F}_p(y) = \sum_{i=1}^m 3^{m-i} 2^{A_{i-1}} y_i.$$

Then

$$\mathcal{F}_p B_p = D e_1^t, \quad \det B_p = (-1)^{m-1} D, \quad (1.3)$$

$$H^0(K_p) = 0, \quad H^1(K_p) \xrightarrow[\mathcal{F}_p]{\simeq} \mathbb{Z}/D\mathbb{Z}, \quad [\mathbf{1}] \mapsto [C]. \quad (1.4)$$

The unique rational solution of $B_p x = \mathbf{1}$ has coordinates

$$x_i = \frac{C(p_i)}{D}, \quad (1.5)$$

where p_i starts at a_i and follows the original cyclic order. Its common reduced denominator is

$$q(p) = \frac{|D|}{\gcd(|D|, C)} = \text{ord}([C]). \quad (1.6)$$

Proof. In $\mathcal{F}_p(B_p x)$, each intermediate x_i cancels against the preceding row. The two remaining terms are $Ux_1 - Lx_1$. For $m \geq 2$, the determinant has the diagonal product and the cyclic permutation product only; their signed sum is $(-1)^{m-1}(U - L)$. The case $m = 1$ is direct.

The first and last coefficients of $\mathcal{F}_p$ are 3^{m-1} and $2^{A_{m-1}}$; they are coprime. For $m = 1$, the sole coefficient is 1. Therefore $\mathcal{F}_p$ is onto $\mathbb{Z}$. Equation (1.3) induces an onto map from the cokernel to $\mathbb{Z}/D\mathbb{Z}$. Both groups have order $|D|$, proving (1.4). This also proves $H^0 = 0$.

Solving the affine fixed-point equation gives $x_1 = C/D$. Applying the same telescoping calculation after cyclic relabelling gives (1.5). The exact transport is

$$2^{a_i} C(p_{i+1}) = 3C(p_i) + D. \quad (1.7)$$

Both 2 and 3 are units modulo every divisor of D . Hence the gcd with D is unchanged by cyclic transport, proving (1.6). ■

3.2.2 Explicit integral chain equivalence

Choose an integer column u with $\mathcal{F}_p u = 1$, using Bezout on the two endpoint coefficients. Define

$$\begin{aligned} f^0 &= e_1^t, & f^1 &= \mathcal{F}_p, & s^0 &= DB_p^{-1}u, & s^1 &= u, \\ h &= B_p^{-1}(I - u\mathcal{F}_p). \end{aligned} \quad (1.8)$$

These give maps between K_p and $[\mathbb{Z} \xrightarrow{D} \mathbb{Z}]$. They are integral: the cokernel calculation shows that Du and every column of $I - u\mathcal{F}_p$ lie in the image of B_p . Their identities are

$$fs = I, \quad B_p h = I - u\mathcal{F}_p, \quad hB_p = I - s^0 e_1^t. \quad (1.9)$$

Thus the original cycle equations have a specified chain equivalence, not just an abstract cokernel identification. The checker independently constructs (1.8) by rational Gaussian elimination and verifies the integrality and both homotopy identities.

3.2.3 Integral extension with its distinguished maps

The original matrix

$$\begin{pmatrix} L & C \\ 0 & U \end{pmatrix}$$

defines an extension of $\mathbb{Z}[t]$ -modules with end actions L and U . The resolution by multiplication by $t - U$ gives

$$\text{Ext}_{\mathbb{Z}[t]}^1(\mathbb{Z}_U, \mathbb{Z}_L) \simeq \mathbb{Z}/D\mathbb{Z}.$$

Changing the lift of the quotient generator by n times the submodule generator changes the original defect to $C - Dn$. Its extension class is therefore the same marked residue as in (1.4). The rational section is $n = C/D$. The coefficient em-

bedding into $\mathbb{Q}$ and this section are explicit; they do not replace the original integral extension.

3.2.4 Exact coefficient tests

For every positive modulus N , the equation $B_p x = \mathbf{1} \pmod{N}$ has

$$\begin{cases} \gcd(D, N) & \text{when } \gcd(D, N) \mid C, \\ 0 & \text{otherwise} \end{cases} \quad (1.10)$$

solutions. Tensor the integral chain equivalence (1.8) with $\mathbb{Z}/N\mathbb{Z}$: it leaves the single congruence $Dz = C$, which has exactly the stated number of solutions. In particular, all prime powers, not only the radical of D , participate.

For $p = (1, 1, 2, 4, 3)$,

$$D = 1805 = 5 \cdot 19^2, \quad C = 475 = 5^2 \cdot 19, \quad q = 19.$$

The rational orbit is $(5, 17, 35, 31, 7)/19$. There are 19 solutions modulo 19 and none modulo 361. The lift residual at $(0, 10, 6, 0, 6)$ is $(1, -1, -1, 5, -1)$, and its pairing with $\mathcal{F}_p$ is $-25 \not\equiv 0 \pmod{19}$.

3.3 2. The return obstruction comes from the preceding image chart

The predecessor's exact legal cylinder is

$$n(v) = r + 2Uv, \quad 1 \leq r < 2U, \quad r \equiv (U - C)L^{-1} \pmod{2U}.$$

Its actual image, with the same parameter, is

$$y(v) = u + 2Lv, \quad u = (Lr + C)/U. \quad (2.1)$$

The positive source is $v \geq 0$. Both r and u are odd. Put

$$z_0 = (r - 1)/2, \quad z_1 = (u - 1)/2, \quad k = z_1 - z_0 = (u - r)/2.$$

The coordinate map is explicitly $x \mapsto (x - 1)/2$ on odd integers, with inverse $z \mapsto 2z + 1$. In these coordinates the joint source/image point is

$$(z_0 + Uv, z_1 + Lv). \quad (2.2)$$

3.3.1 Theorem 2.1. Joint lattice and its image/return morphisms

Retain the parallel lattice complex

$$P_p = [\mathbb{Z} \xrightarrow{v \mapsto (Uv, Lv)} \mathbb{Z}^2].$$

It has the following integral chain maps, all with degree-zero map the identity:

$$\begin{aligned}
P_p &\longrightarrow [\mathbb{Z} \xrightarrow{U} \mathbb{Z}], & (s, t) &\mapsto s, \\
P_p &\longrightarrow I_p := [\mathbb{Z} \xrightarrow{L} \mathbb{Z}], & (s, t) &\mapsto t, \\
P_p &\longrightarrow R_p := [\mathbb{Z} \xrightarrow{D} \mathbb{Z}], & (s, t) &\mapsto s - t.
\end{aligned} \tag{2.3}$$

The first is the original source lattice, the second is the predecessor's image complex, and the third carries the return equation

$$n(v) = y(v) \iff Dv = k. \tag{2.4}$$

The original forcing and return marks obey

$$C = Dr + 2Uk, \quad [C] = 2U[k] \text{ in } \mathbb{Z}/D\mathbb{Z}. \tag{2.5}$$

Multiplication by $2U$ is an automorphism of that quotient. The two integral solvability tests therefore agree by a specified map. Their rational primitives satisfy

$$r + 2U(k/D) = C/D. \tag{2.6}$$

Proof. Each map in (2.3) sends the differential (U, L) to the stated target differential. Subtracting the two original formulas (2.1) gives $(n - y)/2 = Dv - k$. Rearranging $Uu = Lr + C$ gives (2.5). Since D is odd and coprime to U , the asserted unit property follows. Equation (2.6) is (2.5) divided by D . ■

For clarity, the image torsion $\mathbb{Z}/L\mathbb{Z}$ and return torsion $\mathbb{Z}/D\mathbb{Z}$ both arise from the same retained parallel lattice. Its cokernel is $\mathbb{Z}$, via the primitive row

$$\lambda(s, t) = Ls - Ut.$$

On cokernels, the image map is multiplication by $-U^{-1} \pmod{L}$, and the return map is multiplication by $L^{-1} \pmod{D}$, because

$$\lambda(s, t) \equiv -Ut \pmod{L}, \quad \lambda(s, t) \equiv L(s - t) \pmod{D}. \tag{2.7}$$

These are the exact comparison maps; the lattice moduli are not equated.

3.3.2 Actual source incidence remains attached

For any finite set of original parameters v_i , keep the source basis e_i and define

$$E_p = [\mathbb{Z}^s \longrightarrow \mathbb{Z}^3], \quad e_i \mapsto (1, z_0 + Uv_i, z_1 + Lv_i). \tag{2.8}$$

Projection to coordinates $(1, z_1 + Lv_i)$ is the preceding affine incidence map. Projection to the first coordinate is support incidence. There is also the chain map

$$E_p \longrightarrow P_p, \quad e_i \mapsto v_i, \quad (a, s, t) \mapsto (s - z_0 a, t - z_1 a). \tag{2.9}$$

The chain equation is $(Uv_i, Lv_i) = (Uv_i, Lv_i)$. Thus the new return detector is attached to the same actual source columns, rather than inferred from spectral similarity.

All maps exist for empty and singleton source sets as well. No occupied-fiber rank or torsion conclusion is imposed on those cases. A solution of (2.4) belongs to a finite supplied interval only when its exact parameter lies in that interval; this mask is

retained separately.

3.4 3. Trivial orbit, nontrivial orbit, and the marked primitive

3.4.1 Theorem 3.1. Positive integer cycles and the anchor at 1

The original word has a positive integer closed orbit exactly when

$$D > 0 \quad \text{and} \quad [C] = 0 \text{ in } \mathbb{Z}/D\mathbb{Z}. \quad (3.1)$$

For such an orbit, every exponent in the word is the actual valuation. Define the literal displacement

$$y_i = x_i - 1.$$

It satisfies

$$B_p y = b_p, \quad (b_p)_i = 4 - 2^{a_i}. \quad (3.2)$$

The trivial positive orbit is characterized by each of the equivalent statements

$$x_i = 1 \text{ (all } i), \quad y = 0, \quad b_p = 0, \quad a_i = 2 \text{ (all } i). \quad (3.3)$$

Every other positive integer closed orbit has $x_i \geq 3$, hence $y_i \geq 2$, at every position.

Proof. Every cyclic numerator is positive. Equation (1.5) therefore gives positivity exactly when $D > 0$; the common denominator calculation gives integrality exactly when $[C] = 0$. From the integer equation in row i , reduction modulo 2 makes x_i odd. The next coordinate is also odd, so $3x_i + 1 = 2^{a_i}x_{i+1}$ has valuation exactly a_i .

Subtract $B_p \mathbf{1}$, whose row is $2^{a_i} - 3$, from the original equation to obtain (3.2). Since B_p is injective, $b_p = 0$ implies $y = 0$, and the other equivalences are immediate. An orbit containing 1 remains at 1 under the actual deterministic map. Every positive odd integer other than 1 is at least 3. ■

The anchored forcing has

$$\mathcal{F}_p(b_p) = C - D, \quad (3.4)$$

which has the same cohomology class as C . Thus marking this same class again cannot distinguish trivial from nontrivial integral cycles: both are exact forcings. The distinguishing object at this stage is the **unique integral primitive together with its anchor and positivity**, followed by its actual orbit map in Section 4.

For example, (1,3) and (2,2) both have $D = 7$ and $H^1(K_p) = \mathbb{Z}/7\mathbb{Z}$. Their marked forcings are respectively [5] and [0]. The latter gives (1,1). The group itself is nonzero even for this repeated trivial orbit.

3.4.2 Repetition has two explicit maps

Let $p = q^r$, retaining the original word and the uniquely shortest repeating word q . With U_q, L_q, C_q, D_q unchanged, put

$$S_r = \sum_{j=0}^{r-1} U_q^{r-1-j} L_q^j.$$

Direct affine composition proves

$$D_p = S_r D_q, \quad C_p = S_r C_q. \quad (3.5)$$

The chain map

$$[\mathbb{Z} \xrightarrow{D_q} \mathbb{Z}] \longrightarrow [\mathbb{Z} \xrightarrow{D_p} \mathbb{Z}]$$

has degree maps $(1, S_r)$. It induces the injective map $[z] \mapsto [S_r z]$, and preserves the order of the forcing class. Meanwhile the graph map from the repeated labelled circle to the primitive orbit sends the fundamental cycle to r times the primitive cycle. Both factors, S_r and r , are retained; their source maps and meanings are specified rather than interchanged.

3.4.3 An arbitrary-length family excluded exactly

There is no nontrivial positive integer cycle with zero or one occurrence of the exponent 1 in its **primitive** word.

Proof. On every $x > 1$, a branch with $a \geq 2$ strictly decreases. This rules out a closed orbit with no exponent 1.

With exactly one exponent 1, put its source at the minimum $n \geq 3$. Its next value is $(3n+1)/2$, the maximum, since every remaining step decreases. A later exponent at least 3 would produce a value at most

$$\frac{3((3n+1)/2) + 1}{8} = \frac{9n+5}{16} < n,$$

contradicting minimality. All other exponents must therefore be 2. For the retained rotated word $p = (1, 2, \dots, 2)$,

$$D = 2 \cdot 4^{m-1} - 3^m, \quad C - D = 2 \cdot 3^{m-1}. \quad (3.6)$$

Since $\gcd(D, 6) = 1$, integrality forces $|D| = 1$. For $m = 1, 2$, $D = -1$ and the rational cycle is negative. For $m = 3$, $D = 5$; thereafter $D_{m+1} = 4D_m + 3^m > D_m$. No positive case remains. ■

This is a small, fully proved family exclusion, not a claim to supersede the much stronger cycle literature or to resolve the all-word problem.

3.5 4. The actual orbit graph relative to the trivial orbit

The graph has one vertex v_n and one edge e_n for every $n \in X$, with actual successor $T(n)$. Use the signed boundary convention

$$\partial e_n = v_n - v_{T(n)}.$$

The base subcomplex is the vertex v_1 and its loop e_1 . The **relative chain complex** is the actual quotient by this subcomplex:

$$\begin{aligned} C_1^{\text{rel}} &= \bigoplus_{n \in X \setminus \{1\}} \mathbb{Z}e_n, & C_0^{\text{rel}} &= \bigoplus_{n \in X \setminus \{1\}} \mathbb{Z}v_n, \\ \partial e_n &= v_n - v_{T(n)}, & v_1 &= 0. \end{aligned} \quad (4.1)$$

The coefficient modules here are direct sums: every chain has finite support. The quotient map sends exactly e_1, v_1 to zero and leaves every other original basis element unchanged.

3.5.1 The actual morphism from a solved word

A positive integral solution x of (1.2) defines a graph map from the labelled circle with m vertices:

$$i \mapsto x_i, \quad (i \rightarrow i+1) \mapsto e_{x_i}. \quad (4.2)$$

The proof of exact valuations in Theorem 3.1 makes this an actual map into the Collatz graph. On chains it sends $v_i \mapsto v_{x_i}$, $e_i \mapsto e_{x_i}$; the boundary equation is literal. Composing with the relative quotient kills the entire map for the trivial word. A nontrivial primitive word sends its fundamental cycle to

$$c_p = \sum_{i=1}^m e_{x_i} \neq 0, \quad \partial c_p = 0. \quad (4.3)$$

There are no degree-two chains, so this is a nonzero relative homology class. The cochain which is 1 on one chosen orbit edge and 0 on all others has pairing 1 with a primitive cycle and cannot be a coboundary. For a repeated word the pairing is its repetition count.

A primitive word has distinct orbit vertices: equality of two vertices forces equality of all following exponents under deterministic iteration and therefore a proper period of the word. This also proves the repetition assertion without changing the original word.

For completeness, the weighted equation matrix and the ordinary circle coboundary act on the same retained coordinates. Let $R(x)_i = x_{i+1}$, and put $\delta = I - R$. For the solved orbit,

$$B_p \text{diag}(x_i) = I - \text{diag}(3x_i + 1)\delta. \quad (4.4)$$

Indeed, the shifted coefficient is $2^{a_i}x_{i+1} = 3x_i + 1$. This is an exact operator identity. The positive integer diagonal map is retained with its integral cokernel; it is not asserted to be unimodular. Passage from the forced linear system to (4.2) uses the specified positive integral solution, not a purported isomorphism from a torsion group to a free graph group.

3.5.2 Theorem 4.1. Full relative groups, including nonperiodic orbits

Declare $a \sim b$ precisely when their forward orbits meet. This is an equivalence relation and exactly the connected-component relation of the actual graph. Let $\mathcal{B}$ be the components not containing 1 and $\mathcal{C}$ their periodic cycles. Then

$$H_0(C^{\text{rel}}) \simeq \bigoplus_{B \in \mathcal{B}} \mathbb{Z}, \quad H_1(C^{\text{rel}}) \simeq \bigoplus_{C \in \mathcal{C}} \mathbb{Z}. \quad (4.5)$$

Each component in $\mathcal{B}$ contains either a single periodic cycle or an injective forward ray. On the positive integers such an injective forward orbit tends to infinity.

Proof. Meeting of forward orbits is transitive: from a meeting of the first two and a meeting of the second and third, advance sufficiently along the second orbit. Every graph edge joins equivalent vertices; a meeting supplies a finite undirected path. Thus this is exactly connectivity.

Modulo the boundaries $v_n - v_{T(n)}$, the vertices of one component are equal. The sum-of-coefficients functional on any component in $\mathcal{B}$ annihilates every boundary and maps its common vertex class to 1. A finite combination of vertices with total coefficient zero in each component is a boundary: join its finitely many vertices by finitely many graph paths and telescope their differences. Components meeting 1 vanish in the relative quotient. This proves the first formula and gives both the quotient and its explicit finite relations.

A repeated vertex in a forward orbit produces a cycle. All vertices in its component eventually meet that cycle, so a component contains at most one cycle. Without a repeated vertex, the forward orbit is injective. An injective positive-integer orbit visits each finite set only finitely often and therefore tends to infinity.

Every finite-support closed edge chain is supported on cycles. To see this directly, remove leaves from the finite support graph; the coefficient of an edge incident to a terminal leaf is forced to zero by its boundary. The remaining finite components have all indegrees and outdegrees balanced. Since each vertex has at most one outgoing edge, their nonzero parts are directed cycles with a constant coefficient around each. The base loop was removed. The actual cycle sums are independent because their edges are disjoint. This proves the second formula. ■

3.5.3 Relative cohomology, with its coefficient space specified

Take the algebraic dual cochains

$$C_{\text{rel}}^0 = \{f : X \rightarrow \mathbb{Z} : f(1) = 0\}, \quad C_{\text{rel}}^1 = \{g : X \setminus \{1\} \rightarrow \mathbb{Z}\},$$

$$(\delta f)(n) = f(n) - f(T(n)). \quad (4.6)$$

These are products of copies of $\mathbb{Z}$, not finite-support cochains. The natural evaluation pairing with the finite-support chains (4.1) is always a finite sum. One has

$$H_{\text{rel}}^0 \simeq \prod_{B \in \mathcal{B}} \mathbb{Z}, \quad H_{\text{rel}}^1 \simeq \prod_{C \in \mathcal{C}} \mathbb{Z}. \quad (4.7)$$

The first map records the constant value of an invariant function on each component. The second records the period

$$\text{per}_C(g) = \sum_{n \in C} g(n). \quad (4.8)$$

Here is an explicit proof of the second identification. Every coboundary has zero period. Conversely, for a cochain with zero period on every cycle, choose one cycle vertex and prescribe its primitive value to be zero; successive equations determine the remaining values consistently around the cycle. On an acyclic component choose

a forward ray $r_0, r_1, \dots$, prescribe $f(r_0) = 0$, and set $f(r_{j+1}) = f(r_j) - g(r_j)$. Every other vertex eventually meets the selected ray or cycle; extend backward by the finite sum of g along that path. In the component of 1 use $f(1) = 0$ and the same finite sums. This constructs a primitive on every component. Finally any integer period assignment is realized by a cochain supported on one selected edge per cycle. This proves (4.7).

The relative removal of the trivial orbit is consequently exact: its own period has been removed by the displayed subcomplex quotient. An actual nontrivial positive cycle supplies a remaining period-1 detector. An acyclic component not reaching 1 contributes to H^0 but no cycle period in H^1 . These statements are proved by the component maps and explicit primitives above.

3.6 5. What resolves the full conjecture in this complex

Let $\eta(n) = 1$ on every relative edge. The following are equivalent, with no conjectural analytic estimate assumed:

every $n \in X$ reaches 1;

$$H_0(C^{\text{rel}}) = 0;$$

for each $n > 1$, v_n is the boundary of a finite integer edge chain;

$$\text{there exists } h : X \rightarrow \mathbb{Z}_{\geq 0}, \quad h(1) = 0, \quad \delta h = \eta. \quad (5.1)$$

Proof. A path from n to 1 gives the finite chain

$$H(v_n) = \sum_{j=0}^{\tau(n)-1} e_{T^j(n)}, \quad \partial H(v_n) = v_n. \quad (5.2)$$

Conversely the component augmentation in Theorem 4.1 makes such a boundary impossible in a component not containing 1. This proves the first three equivalences.

On the basin of 1, $h(n) = \tau(n)$, the exact number of odd returns to 1, satisfies the fourth statement. A nonnegative solution gives $h(T^j(n)) = h(n) - j$ for as long as 1 has not been reached. At $j > h(n)$ that would be negative, so every orbit reaches 1. Moreover iteration to 1 proves that any such solution equals τ . ■

This pinpoints the extra requirement beyond eliminating cycle periods. On an acyclic component the signed integer primitive of η constructed in Section 4 takes values $f(r_j) = f(r_0) - j$ on a forward ray. It is unbounded below. The original positive source and the nonnegative codomain in (5.1) detect precisely what that signed primitive fails to supply.

No global nonnegative primitive has been constructed in this contribution. The implemented certificates construct the actual columns (5.2) on proved source sets, while retaining the remaining roots and boundaries.

3.6.1 Formal inversion has a specified specialization problem

Identify edge and vertex bases by their source indices and define

$$P(v_n) = v_{T(n)}, \quad v_1 = 0.$$

Then $\partial = I - P$. On formal power series in t ,

$$(I - tP)^{-1}v_n = \sum_{j \geq 0} t^j P^j v_n. \quad (5.3)$$

The equality follows coefficient by coefficient. For a source reaching 1 it is a polynomial and its evaluation at $t = 1$ is the finite chain (5.2). For a periodic source away from 1, coefficients recur on the cycle and their proposed sum is not a finite integer coefficient. For an injective forward orbit, infinitely many distinct basis vectors occur, outside the direct sum. Equation (5.3) therefore has its exact source module and specialization behavior recorded; no formal inverse is passed off as an integral finite-support contraction.

3.7 6. Certified descent gives a counterexample-preserving chain retraction

A **checked descent record** consists of an actual positive odd source $n > 1$, its full actual exponent word p_n , and the terminal value $d(n) = T^{|p_n|}(n) < n$. Its retained path chain is

$$b_n = \sum_{j=0}^{|p_n|-1} e_{T^j(n)}, \quad \partial b_n = v_n - v_{d(n)}. \quad (6.1)$$

A path ending at 1 stops there, so no base loop is added. The preceding workbench's supported stopping leaves produce precisely such records using their original affine descent inequalities and actual source charts.

Take the actual finite collection of checked records stored by the algorithm. Unlisted vertices remain roots. Define recursively in increasing integer order:

$$\begin{aligned} r(1) &= 1, & H(v_1) &= 0; \\ r(n) &= n, & H(v_n) &= 0 \quad \text{for an unlisted } n; \\ r(n) &= r(d(n)), & H(v_n) &= b_n + H(v_{d(n)}) \quad \text{for a listed } n. \end{aligned} \quad (6.2)$$

Every recursive endpoint is strictly smaller than its source, so this is a finite, well-defined construction. It does not require knowledge of an unlisted orbit's future.

Define linear maps on the original relative modules by

$$Q(v_n) = v_{r(n)}, \quad F = I - H\partial \quad \text{on } C_1^{\text{rel}}. \quad (6.3)$$

3.7.1 Theorem 6.1. Exact compression and preservation of every cycle

The maps satisfy

$$\partial H = I - Q, \quad HQ = 0, \quad Q^2 = Q,$$

$$\partial F = Q\partial, \quad F^2 = F, \quad I - F = H\partial. \quad (6.4)$$

Consequently the image complex

$$[\operatorname{im} F \xrightarrow{\partial} \operatorname{im} Q]$$

is a chain-homotopy retract of the original relative complex, with inclusion, retraction (F, Q) , and homotopy H explicitly specified. Its homology and algebraic dual cohomology are preserved by those maps.

Most importantly,

$$\partial c = 0 \implies Fc = c. \quad (6.5)$$

Every closed cycle chain is retained **as the same chain on the original edges**.

Proof. Telescoping (6.1) along the finite recursion gives $\partial H(v_n) = v_n - v_{r(n)}$. A root has no descent record, so $H(v_{r(n)}) = 0$ and $r(r(n)) = r(n)$. This gives the first row of (6.4). Further,

$$\partial F = \partial - \partial H\partial = \partial - (I - Q)\partial = Q\partial.$$

From $H\partial H = H(I - Q) = H$,

$$F^2 = I - 2H\partial + H\partial H\partial = I - H\partial = F.$$

The remaining homotopy identity is the definition. The retraction is the identity on its image; its inclusion followed by retraction is the displayed projection, chain homotopic to the identity. Applying $\operatorname{Hom}(-, \mathbb{Z})$ to these actual identities gives the dual cochain homotopy without assuming an exactness theorem for unspecified coefficient spaces. Finally (6.5) follows by substituting $\partial c = 0$ into (6.3). ■

This does not contract every unresolved root to 1. It constructs a smaller, explicitly embedded complex whose nontrivial cycle and component obstructions are retained. Extending the checked descent collection extends the certificate; it does not silently supply missing records.

3.7.2 Executed positive control with an actual nontrivial cycle

For detector validation only, retain the explicitly changed map

$$T_-(n) = \frac{3n-1}{2^{\nu_2(3n-1)}}.$$

It has the positive cycle

$$5 \xrightarrow{a=1} 7 \xrightarrow{a=2} 5. \quad (6.6)$$

The corresponding word has $D = -1$. Its forcing is $-\mathbf{1}$, not $\mathbf{1}$, and its positive solution is $(5, 7)$. The relation to the original signed map is the literal conjugacy

$$T_-(n) = -T_+(-n),$$

with the same valuation since $3(-n) + 1 = -(3n - 1)$. The forcing change is tracked in every control certificate; it is not a positive Collatz counterexample.

Use only the checked descent record $7 \rightarrow 5$. Then

$$\begin{aligned} H(v_7) &= e_7, & H(v_5) &= 0, & Q(v_7) &= Q(v_5) = v_5, \\ F(e_7) &= 0, & F(e_5) &= e_5 + e_7, \\ F(e_5 + e_7) &= e_5 + e_7. \end{aligned} \tag{6.7}$$

Thus compression exposes a surviving loop at root 5 while preserving its actual two-edge cycle. Its detecting cochain has value 1 on e_5 and 0 on e_7 , and pairs to 1.

For the original map, the predecessor's block

$$247 \rightarrow 371 \rightarrow 557 \rightarrow 209 \rightarrow 157$$

has boundary $v_{247} - v_{157}$. Adding the independently computed actual path from 157 to 1 gives boundary exactly v_{247} . Both that bridge and the compression identities on the checked source families are executed in `verify.py`.

3.7.3 Theorem 6.2. Whole-source cutoff retractions and their compatible limit

The implementation also uses the predecessor's exact observer directly, without first replacing the infinite source by a finite list. For each integer cutoff $H \geq 0$, the observer accepts precisely the original reference-certified stopping words with total exponent sum at most H . Its check on any given positive integer terminates. Apply the recursion (6.2) to these actual records; write the resulting maps as

$$r_H, \quad h_H : C_0^{\text{rel}} \rightarrow C_1^{\text{rel}}, \quad Q_H, \quad F_H = I - h_H \partial.$$

This defines maps on **all** original positive odd vertices: every accepted block lowers its positive source, so a queried vertex needs only finitely many accepted blocks before reaching 1 or an actual cutoff root. No decision of never-stopping is made.

For $K \geq H$, the exact comparison identities are

$$\begin{aligned} r_K(r_H(n)) &= r_K(n), & r_H(r_K(n)) &= r_K(n), \\ h_K &= h_H + h_K Q_H, & h_H Q_K &= 0, \\ Q_K Q_H &= Q_H Q_K = Q_K, & F_K F_H &= F_H F_K = F_K. \end{aligned} \tag{6.8}$$

For every fixed vertex n , $r_H(n)$ is a nonincreasing positive integer sequence and is eventually constant. Its path chain $h_H(v_n)$ is also eventually constant. Denote the actual eventual values by $r_\infty(n)$ and $h_\infty(v_n)$. These are well-defined mathematical maps with finite-support columns, even though an algorithm is not supplied that recognizes the final cutoff for an unresolved root. They give the chain-homotopy retraction

$$Q_\infty(v_n) = v_{r_\infty(n)}, \quad F_\infty = I - h_\infty \partial, \tag{6.9}$$

onto a complex whose vertex basis is exactly the retained never-reference-certified positive integers, with 1 removed relatively. Every actual cycle chain remains unchanged by F_∞ .

Furthermore,

$$r_\infty(n) = 1 \iff r_H(n) = 1 \text{ for some finite } H \iff n \text{ reaches } 1. \quad (6.10)$$

Proof. A stopping word already accepted at H is the same first reference-certified word at K ; raising the cutoff does not change an accepted original word. Thus the K -path follows the entire H -path and then continues from its actual root. This proves the first root identity and $h_K = h_H + h_K Q_H$. A K -root other than 1 has no accepted word at K , hence none at H ; it is already an H -root. This proves the other root identity and $h_H Q_K = 0$.

The Q -identities follow on each original basis vertex. For the edge maps use $\partial h_H = I - Q_H$, the path extension identity, and $h_H Q_K = 0$ in direct multiplication of $(I - h_K \partial)(I - h_H \partial)$ and the reversed product. Both are $I - h_K \partial$, proving (6.8).

Since a new accepted block strictly lowers the old root, the root sequence is non-increasing in the ordinary positive integers. It is eventually constant. At any stage after that stabilization, a nonempty further descent from the root would produce a smaller root, which is impossible. The path extension identity then shows that the finite path column has stabilized as well. The finite chain identities of Theorem 6.1 pass to these eventual values on each basis column, and hence on every finite-support chain. This proves the retraction (6.9).

A final non-base root has no reference-certified word at any finite exponent cutoff, so it is exactly a never-reference-certified source. Conversely such a source is fixed at every cutoff. The vertex-basis assertion follows. Closed cycles are preserved by the same identity $F_\infty c = c - h_\infty \partial c = c$.

The first equivalence in (6.10) follows from eventual constancy. A finite cutoff reaching root 1 supplies the actual finite path, giving convergence. Conversely a converging integer $n \geq 3$ has some actual full word p with $F_p(n) = 1$. The slope of F_p is positive, so $F_p(3) \leq F_p(n) = 1 < 3$. Its first reference-certified stopping word therefore exists. Its terminal source also converges and is strictly smaller. Repeating this argument terminates in the positive integers and uses only finitely many finite stopping words. A cutoff exceeding all their exponent sums reaches root 1. ■

This is an exact localization of the remaining obstruction to the never-certified source, with a full chain map, inverse-on-image, and homotopy. The residual vertex basis is not declared empty. In particular, evaluating a finite cutoff does not recognize the eventual non-base roots in (6.9).

For every finite $H \geq 1$, the explicit positive integer

$$n_H = 2^{H+1} - 1$$

is still a cutoff root: its first H exponents are 1, and none of the formal prefixes sends the reference 3 below 3. It has no accepted leaf with exponent sum at most H . Thus the finite cutoff maps genuinely have retained positive roots. This is not an assertion that their **homology classes** survive to the full graph; Section 7 gives the actual transition maps and finite killing relations.

The replay on the 512 positive odd starts at most 1023 records:

Exponent cutoff	Sources with root 1	Distinct other roots in that finite source
4	6	256
8	159	138
12	259	88
16	287	61
32	342	23

Exponent cutoff	Sources with root 1	Distinct other roots in that finite source
64	509	2
128	512	0

The counts of distinct roots are not counts of non-base components or of divergent orbits. The unchanged observer leaves 27 as a root through cutoff 64 and gives an actual path to 1 at cutoff 128. Every entry is accompanied by original edge-boundary checks; no tail probability is used to kill a root.

3.8 7. Finite frontiers and integer support at infinite depth

3.8.1 Finite graph exhaustion with original boundaries

For a finite edge-source set $S \subset X \setminus \{1\}$, retain every vertex in

$$V_S = \{1\} \cup S \cup T(S),$$

but only the actual edges whose sources are in S . A target outside $S \cup \{1\}$ remains a named **frontier vertex**; no outgoing edge is invented and no frontier is sent to 1.

Every component has a basepoint, a unique frontier root, or an actual cycle. Relative H_1 consists precisely of the actual cycles already contained in the observed edges. Frontier components contribute to finite H_0 , but their eventual component type has not thereby been decided.

For the original map, observing only $3 \rightarrow 5$ gives relative $H_0 = \mathbb{Z}$. Adding the actual edge $5 \rightarrow 1$ kills that class. The explicit inclusion is the original vertex/edge inclusion; the class is killed by the new path, not by a relabelling of the old frontier.

Under increasing actual edge sets exhausting all positive odd sources, every finite chain and every finite boundary relation appears at a finite stage. Hence the direct limit of the finite homology groups is the full homology (4.5). A vertex class vanishes in the limit exactly when one of its finite-stage actual path relations kills it. This follows directly from the finite-support definition; no uniform finite stopping bound is inferred.

3.8.2 The residue tower and ordinary positive integers

For an infinite word, write r_m for the selected positive residue of its first m exponents modulo 2^{A_m+1} . Cylinder containment gives

$$r_{m+1} = r_m + 2^{A_m+1}b_m, \quad b_m \in \mathbb{Z}_{\geq 0}. \quad (7.1)$$

The exact digit b_m is retained; it is bounded above by the refinement modulus minus one. These compatible residues define a unique 2-adic point. The precise ordinary-integer criterion is

$$\text{the infinite word is realized by a positive integer} \iff (r_m) \text{ is bounded in } \mathbb{R} \iff r_m \text{ is eventually constant.} \quad (7.2)$$

Proof. The residues are nondecreasing positive integers by (7.1), so boundedness is equivalent to eventual constancy. An eventual constant n satisfies every cylinder congruence; cylinder legality then gives every finite prefix of the actual orbit of n . Conversely, an actual positive n has $r_m = n$ as soon as the modulus exceeds n . Since $A_m \geq m$, that occurs. ■

This is the exact map connecting all compatible dyadic data to ordinary positive support. No measure-zero set of codes is removed. For the word $(1, 1, \dots)$,

$$r_m = 2^{m+1} - 1,$$

so its 2-adic point is -1 , with no positive integer realization. For the word $(2, 2, \dots)$, $r_m = 1$ identically.

For a periodic code p^∞ , the same inverse-cylinder maps give the unique 2-adic fixed point C/D . It is a positive integer exactly by Theorem 3.1. For a nonperiodic code, (7.2) remains the exact integer-membership test; this contribution does not prove that every bounded-residue code eventually has exponent 2 forever.

3.8.3 The unbounded initial ascent is discharged with the original coordinates

For every positive odd integer write, using the unique valuation and its inverse chart,

$$n = 2^k u - 1, \quad k = \nu_2(n + 1) \geq 1, \quad u \geq 1 \text{ odd}.$$

For $0 \leq j \leq k - 1$,

$$T^j(n) = 3^j 2^{k-j} u - 1. \quad (7.3)$$

The first $k - 1$ exponents are exactly 1. With

$$r = \nu_2(3^k u - 1),$$

the next exponent is $r + 1$, and the completed terminal value is

$$y = \frac{3^k u - 1}{2^r}. \quad (7.4)$$

These identities follow by substituting the original branch and retaining its minus-one offset at every step. They show directly that the infinite all-1 code is absent from the positive source, without discarding the arbitrarily long finite all-1 runs. The exact descent condition for this whole block is

$$y < n \iff (2^{k+r} - 3^k)u > 2^r - 1. \quad (7.5)$$

No sign of a logarithmic average replaces the right-hand affine constant.

3.8.4 Localization of an actual unresolved component

Every actual component not containing 1 has a least positive integer $s \geq 3$. Every point on its forward orbit is at least s . For every prefix with original data (D_j, C_j) ,

$$D_j s \leq C_j. \quad (7.6)$$

For $D_j > 0$ this is the original no-descent inequality; for $D_j \leq 0$ it holds because $C_j > 0$. It implies $F_j(3) \geq 3$: for $D_j > 0$, $C_j \geq D_j s \geq 3D_j$, and for $D_j \leq 0$ the same conclusion is immediate. Therefore this minimum belongs to the predecessor's retained never-reference-certified support, not to a supported descent leaf.

This applies to a periodic component and to an acyclic component by the same original-source inequality. It identifies where a counterexample would survive the exact descent process. It does not infer that this support is empty from its reference weight.

3.9 8. Executed evidence and the remaining mathematical calculation

The new replay contains 42,507 named exact checks, 344 independently solved matrix fixtures, and 468 exhaustive modular fixtures. It visits every nonempty exponent word with total exponent sum at most 20: exactly 1,048,575 words. The only positive integral words in that finite domain are the ten retained repetitions $(2)^r$, $1 \leq r \leq 10$. This is an exhaustive bounded check, not an all-word estimate and not a new record in Collatz verification.

The checker also constructs finite-support path certificates for all 2,048 positive odd starts at most 4095. It verifies the relative graph ranks by independent rational matrix elimination on the small graph fixtures, checks the primitive and repeated cycle maps, preserves a real control cycle under descent compression, and rejects ten deliberately false or malformed inputs. Finite frontier observations are always labelled open, not divergent.

Both ordinary and optimized Python must reproduce `verification.json` byte-for-byte. The inherited 175,269-check replay and 142-check support certificate are run separately against the retained predecessor bytes. These counts are not merged into a claim of independent proof of an infinite assertion.

The remaining exact construction is a source-complete finite-support contraction, or equivalently the nonnegative integral primitive in (5.1), with all original orbit edges retained. Section 6 provides the verified extension operation for adding descent certificates without losing cycle or component obstructions. Section 7 supplies the exact positive-integer support criterion for a surviving infinite code. Neither universal source coverage nor emptiness of the remaining non-base components has been proved here.

The available work now consists of executable return and period detectors, a counterexample-preserving compression, explicit infinite-family exclusions, retained boundary/source criteria, and a replayable integration package. A positive nontrivial cycle would yield a finite certificate accepted by the original-forcing detector and a nonzero relative period. A nonperiodic failure would survive as a non-base component with the bounded-residue integer support in (7.2), even though it supplies no finite cycle period. No such original positive counterexample is asserted by this contribution.

3.10 References and inspection scope

[P] ../stopped_affine_transport_20260914/note.md, Sections 1–4, 9, and 10.0, with its original executable, manifest, and support certificate. The supplied files are retained unchanged and their finite outputs were replayed. This is not a claim of a fresh independent audit of every preceding infinite proof.

[R1] Jeffrey C. Lagarias, *The set of rational cycles for the $3x+1$ problem*, Acta Arithmetica 56 (1990), 33–53. Bibliographic record: <https://eudml.org/doc/206298>; author’s publication listing: <https://dept.math.lsa.umich.edu/~lagarias/3x%2B1.html>. The meta-data and author’s listing were inspected; no uninspected detailed result of that paper is used as a proof dependency.

[R2] Takehiko Mori, *Application of Operator Theory for the Collatz Conjecture*, arXiv:2411.08084. Abstract inspected at <https://arxiv.org/abs/2411.08084>. Cited to acknowledge existing operator/irreducibility formulations, not as a dependency for (4.5)–(6.5).

All proofs needed for the new computations are included above. Their status is the written argument plus exact finite replay, not an external acceptance or a claim to have settled Collatz.

Chapter 4

Residual-source descent, unbounded exponent tails, and period-driven cycle exclusion

Source: repository/collatz_reconstruction/research_program/residual_splice_20260914/note.md.

14 September 2026 — continuation in KokunoYumeto/collatz-workbench.

Source checkpoint: 32b8f3b9cc6dcfc4387ff783c4f4a6f6a365f717, PR #2. The original stopped-affine, cycle-relative, and integration files remain unchanged. This note and integral_blocks.md supply the new arguments. The verification records finite execution, not independent external mathematical review or a global Collatz proof.

4.1 1. Actual source and first-descent cells

Use the original map

$$T(n) = (3n + 1)/2^{\nu_2(3n+1)}, \quad n \in X = \{1, 3, 5, \dots\}.$$

For a nonempty exponent word $p = (a_1, \dots, a_m)$, keep

$$A_j = \sum_{i \leq j} a_i, \quad L_j = 3^j, \quad U_j = 2^{A_j}, \quad C_j = \sum_{i=1}^j 3^{j-i} 2^{A_{i-1}}, \quad D_j = U_j - L_j.$$

Write (A, L, U, C, D) for the final data. The predecessor's original legal cylinder is

$$n = r + 2Ut, \quad t \geq 0, \quad r \equiv (U - C)L^{-1} \pmod{2U}, \quad 1 \leq r < 2U,$$

with the exact image

$$T^m(n) = u + 2Lt, \quad Uu = Lr + C. \tag{1}$$

The comparison with the repository's September 13 notation is the identity on the labelled word, $S_j = A_j$, and $B_w = C_m$; the index substitution is $i = j + 1$. The original coefficients, residue representative, and affine map are identical under this comparison.

4.1.1 Proposition 1. Complete first-descent fiber

The set of original positive sources whose **first** strict odd-return descent is at the end of exactly p is empty for $D \leq 0$. For $D > 0$ it is

$$\{r + 2Ut : t \in \mathbb{Z}, \ell \leq t \leq h\}, \quad (2)$$

where

$$\ell = \max\left(0, \left\lfloor \frac{C - Dr}{2UD} \right\rfloor + 1\right), \quad h = \min_{\substack{j < m \\ D_j > 0}} \left\lfloor \frac{C_j - D_j r}{2UD_j} \right\rfloor. \quad (3)$$

The minimum of the empty set here is $+\infty$. A label with $h < \ell$ remains a declared empty first-descent fiber; its legal cylinder and coefficients are retained.

Proof. On the original cylinder,

$$T^j(n) < n \iff D_j n > C_j.$$

For $D_j \leq 0$, positivity of n, C_j rules out strict descent. For positive D_j , substituting $n = r + 2Ut$ and using integer floors gives (3): strict descent at the last step and non-descent at every earlier step. These are all the conditions. The inverse source parameter is $(n - r)/(2U)$, and the inverse target parameter is $(y - u)/(2L)$. Both are applied only on their specified images. ■

Distinct first-descent words have disjoint occupied fibers: the deterministic actual orbit has one first-descent index and one exponent list at that index. In particular, the mask for $(2, 2)$ is empty, whereas the mask for (2) has $r = 1, \ell = 1$; the fixed point 1 has not been treated as a strict descent.

For every supported source in (2), retain the original edge chain

$$b_p(n) = \sum_{j=0}^{m-1} e_{T^j(n)}, \quad \partial b_p(n) = v_n - v_{T^m(n)}. \quad (4)$$

The relative convention is exactly the predecessor's: only v_1, e_1 are sent to zero.

4.2 2. Whole exponent tails, without an overflow class

Let q be an original word all of whose nonempty prefixes have $D_j \leq 0$. The empty word is included, with $L = U = 1, C = 0, r = u = 1$. Every positive source in its cylinder has no earlier descent. Retain its chart

$$n = r + 2Ut, \quad x = T^{|q|}(n) = u + 2Lt, \quad t \geq 0.$$

Define $B \geq 1$ to be the least integer satisfying the **two original coefficient inequalities**

$$2^B r > 3u + 1, \quad 2^B U \geq 3L. \quad (5)$$

This integer exists. In this setting the second inequality is strict as well: $2^B U$ is a power of 2 and $3L$ a positive power of 3.

4.2.1 Theorem 2. Unbounded last-exponent descent family

All sources in the prefix cylinder with actual next exponent $a \geq B$ have their first strict descent at that next return. Their full source set is the single original progression

$$n = R + 2U 2^{B-1}v, \quad v \geq 0, \quad (6)$$

where

$$t_0 \equiv -\frac{3u+1}{2}(3L)^{-1} \pmod{2^{B-1}}, \quad 0 \leq t_0 < 2^{B-1}, \quad R = r + 2Ut_0. \quad (7)$$

No upper bound on a is imposed.

The full exponent label is recovered by

$$t = t_0 + 2^{B-1}v, \quad a = \nu_2(3u + 1 + 6Lt). \quad (8)$$

Its exact a -stratum, for each $a \geq B$, is

$$t \equiv \left(2^{a-1} - \frac{3u+1}{2}\right)(3L)^{-1} \pmod{2^a}. \quad (9)$$

Equation (9) gives exactly the predecessor's complete cylinder for q followed by a , with source step 2^{A+a+1} , target step $2 \cdot 3^{|q|+1}$, and the original affine numerator. Its first-descent interval is the entire positive cylinder.

Proof. Divisibility of $3x+1 = 3u+1+6Lt$ by 2^B is equivalent to (7), because $3L$ is odd. This proves (6), with the inverse parameter $v = (n-R)/(2U2^{B-1})$. Specifying valuation exactly a means congruence to 2^a modulo 2^{a+1} ; division of that congruence by 2 gives (9). It also proves (8) and the disjoint, complete stratification by actual exponents.

For all these sources,

$$\frac{3u+1+6Lt}{2^a} \leq \frac{3u+1+6Lt}{2^B} < r + 2Ut, \quad (10)$$

where the final strict inequality uses both inequalities in (5), including the strict constant term. Every previous $D_j \leq 0$ precludes an earlier descent. Each exact stratum has the original word chart (1), so its intermediate values and edge chain are retained. ■

4.2.2 A full symbolic family at the old residual frontier

For $q = (1, 2)$, its unchanged chart is

$$11 + 16t \mapsto 13 + 18t.$$

Equations (5)–(7) give $B = 2, t_0 = 0$. Thus every $v \geq 0$ has the actual path

$$11 + 32v \mapsto 17 + 48v \mapsto 13 + 36v \mapsto \frac{10 + 27v}{2^{\nu_2(10+27v)}}. \quad (11)$$

The full word is

$$(1, 2, 2 + \nu_2(10 + 27v)).$$

The terminal value is at most $10 + 27v < 11 + 32v$; the first two displayed values are

larger than the source. The original three-edge chain has boundary source minus the exact final value in (11). The residue class $11 \bmod 16$ is now split exactly: $11 \bmod 32$ has this descent, while $27 \bmod 32$ follows the noncontracting child $(1, 2, 1)$. No unspecified probability or changed target interval is used.

For the empty prefix, the tail is $n = 5 + 8v$, with $a \geq 3$. Together with the (2) cell it supplies every one-step descent source $n > 1, n \equiv 1 \pmod{4}$. In particular,

$$n_j = (4^j - 1)/3 \mapsto 1, \quad j \geq 2,$$

has actual exponent $2j$. For $j \geq 33$, these sources are unresolved by the original exponent cutoff 64 but are discharged by this one actual edge, without increasing that cutoff. The verifier executes $j = 33, 64, 128, 512$; the displayed identity proves the whole family.

4.2.3 Finite complete next-step decomposition

For any prefix q above, enumerate the finite list $1 \leq a < B$. A child with $D_{qa} \leq 0$ remains an original prefix frontier. A child with $D_{qa} > 0$ has its exact first-descent interval (2) and its finite complementary non-descent box from Section 5. All $a \geq B$ belong to the proved tail (6). These disjoint actual-exponent cases exhaust the full prefix cylinder. `tail_descent.py` implements this decomposition; it never adds an exponent-overflow atom.

4.3 3. Attaching new paths at actual retained roots

Let (h, Q, F) be the predecessor's actual path retraction, with

$$\partial h = I - Q, \quad hQ = 0, \quad Q^2 = Q, \quad F = I - h\partial.$$

Its root $r(n)$ is at most n ; $h(v_n)$ is the original finite path to that root. These properties hold for every whole-source cutoff retraction in the imported code, without declaring its roots to be convergent.

A new descent rule is applied **only at an old root** r . A cell from Section 1 or tail from Section 2 supplies an actual block $b : r \rightarrow d < r$. Attach the old path $h(v_d)$, giving

$$\tilde{b}_r = b + h(v_d), \quad \partial \tilde{b}_r = v_r - Q(v_d). \quad (12)$$

Its endpoint is an old root no greater than $d < r$. Iterate these root-to-root blocks. The recursion terminates by strict decrease of positive integers. Let γ_r be the resulting original edge chain and $R(r)$ its endpoint. An unlisted root remains itself, with zero added path.

Put

$$K(v_n) = \gamma_{r(n)}, \quad h' = h + K, \quad Q'(v_n) = v_{R(r(n))}, \quad F' = I - h'\partial. \quad (13)$$

The implementation gives an already compiled first-descent cell priority over the identical word certified by a tail. The original rules remain recorded. Distinct occupied first-descent words cannot compete at a source. Tail words also end at the first coefficient crossing, so distinct noncontracting prefixes have disjoint tail supports.

4.3.1 Theorem 3. Exact root-level extension

These are maps on all original finite-support relative chain modules. They satisfy

$$\begin{aligned} \partial K &= Q - Q', \quad KQ = K, \quad KQ' = 0, \quad hQ' = 0, \\ \partial h' &= I - Q', \quad h'Q' = 0, \quad Q'Q = QQ' = Q', \\ F'^2 &= F', \quad \partial F' = Q'\partial, \quad F'F = FF' = F', \quad FK = K. \end{aligned} \quad (14)$$

Every closed chain is fixed literally: $F'c = c$ for $\partial c = 0$. The new image complex is a chain-homotopy retract of the old image complex; its homotopy is K on the old vertex image. Original homology and its algebraic-dual cohomology are retained by these specific maps.

Proof. Telescoping (12) proves $\partial\gamma_r = v_r - v_{R(r)}$. Final roots have no new applicable block and were old roots, so both homotopies vanish there. This gives the first two lines. In particular, $h\partial K = h(Q - Q') = 0$, hence $FK = K$: the correction really lies in the old projected edge module. Also $h'\partial h' = h'(I - Q') = h'$, proving idempotence of F' . The chain equation follows from $\partial h' = I - Q'$. Direct multiplication using $h'Q = K$ and $hQ' = 0$ gives both compositions with F . The difference of the old and new projections on the old image is $(K\partial, \partial K)$, the stated chain homotopy. Substitution of $\partial c = 0$ proves literal cycle preservation. Applying the algebraic dual to these identities supplies the actual cochain maps and homotopy. ■

A nonperiodic component is not erased by an absence of cycle periods: its degree-zero component obstruction is retained by the same homotopy equivalence. No finite cutoff root has been promoted to such a component.

4.3.2 The 27 cylinder

The full 37-letter word is recorded in `verification.json`. Its exact data are

$$A = 59, \quad L = 450283905890997363, \quad C = 1100931843921811423, \quad D = 126176846412426125.$$

They give the entire family

$$27 + 2^{60}t \mapsto 23 + 900567811781994726t, \quad t \geq 0. \quad (15)$$

Every proper prefix has $D_j \leq 0$, and $27D - C > 0$, so (15) is a first descent on its whole cylinder. In contrast, its original formal reference value is

$$F_p(3) = 306472945199350439/72057594037927936 > 3.$$

Thus this exact source family supplies an extension that the reference-3 stop does not supply at this word. At cutoff 64 the old root of 27 is 27. The new block ends at 23, whose old root is 1; (12) adds that already checked path. The resulting 41-edge path has boundary v_{27} .

4.4 4. Multiplicative period morphism and finite arithmetic exclusion

For an actual orbit define the positive rational edge label

$$g(n) = \frac{3n+1}{4n} = 2^{a(n)-2} \frac{T(n)}{n}. \quad (16)$$

The factor $T(n)/n$ is the displayed multiplicative coboundary of the actual vertex coordinate. On a finite path it telescopes to terminal/source. On a closed cycle its product is 1. All factors are retained; the following inequalities are bounds on this same map, not a replacement of the affine equations.

For a positive cycle with m odd returns, exponent sum A , and every vertex at least s ,

$$3^m < 2^A = \prod_{i=1}^m \left(3 + \frac{1}{x_i} \right) \leq \frac{(3s+1)^m}{s^m}. \quad (17)$$

Let $k = \#\{i : a_i = 1\}$. Since

$$A = 2m - k + E, \quad E = \sum_{a_i \geq 3} (a_i - 2) \geq 0,$$

(17) implies the exact integer inequality

$$(4s)^m \leq 2^k (3s+1)^m. \quad (18)$$

The projection to (m, A, k, s) retains its fiber: all original labelled words with these counts whose unique primitive $x_i = C(p_i)/D$ is integral, satisfies $x_i \geq s$, and satisfies the declared minimum bound. The matrix forcing and word order in this fiber are not declared solvable merely because a window (17) is allowed. `integral_blocks.md` gives the integral comparison maps for this remaining fiber.

4.4.1 Period-driven feedback actually executed

The atlas initially verifies an actual first descent for every odd source $3 \leq n \leq 4095$. Strong induction on the source, beginning at 1, proves that all these sources reach 1. Consequently every nontrivial cycle has minimum at least 4097.

For each current lower bound s , select the least period for which (17) has any integer exponent A . The least possible exponent is the bit length of 3^m . For that pair, compute the largest integer u satisfying

$$2^A u^m \leq (3u+1)^m. \quad (19)$$

The ratio $(3+1/u)^m$ strictly decreases with positive u , so a successful comparison at u and failed comparison at $u+1$ prove the exact ceiling. The original minimum of this entire period window lies in a finite interval. The executed continuation proves first descent throughout that interval, then recomputes the next window.

Current minimum lower bound	First window (m, A)	Exact minimum ceiling	New verified odd-source bound
4097	(147,233)	6724	6723
6725	(200,317)	12824	12823
12825	(253,401)	27113	27113
27115	(306,485)	99780	99779

Every new seed is iterated only to its actual first lower odd value. Independent repeated-division replay verifies the entire final interval, and strong induction supplies its boundary paths to 1. These seeds produce **5,175 occupied infinite first-**

descent cells; their symbolic lift statements follow from Proposition 1, not from sampling. The construction additionally compiles **4,011 unbounded last-exponent tails** from the admitted prefix words.

4.4.2 Theorem 4. Certificate excluding all cycles with at most 402 exponent-1 steps

There is no nontrivial positive integer Collatz cycle whose primitive odd-return word contains at most 402 occurrences of exponent 1.

Proof. The actual seed certificates above establish $s = 99781$. The following finite integer comparisons are executed, with an independent least-power-of-two loop in the verifier:

$$2^{\text{bitlength}(3^m)} s^m > (3s + 1)^m \quad (1 \leq m \leq 970). \quad (20)$$

They exclude every integer exponent in (17) at those lengths. Separately,

$$(4s)^{969} > 2^{402} (3s + 1)^{969}. \quad (21)$$

Since $4s > 3s + 1 > 0$, the same strict comparison holds at every larger length with any $k \leq 402$. Thus (18) forces $m \leq 968$, whereas (20) excludes all those periods. The finite premises are part of the executable certificate; neither a sampled frequency nor an unproved analytic estimate occurs. ■

This is a self-contained improvement of the present workbench's previous zero-or-one-exponent-1 exclusion, **not a claimed advance over published cycle bounds**. The executed feedback used a seed budget of 100,000; that budget restricts the computation, not the all-length scope of Theorem 4.

The first unexcluded window at the new minimum is

$$(m, A, k_{\min}) = (971, 1539, 403), \quad 99781 \leq \min x_i \leq 330749. \quad (22)$$

For $k = 403$, (18) forces $m \leq 971$, and (20) forces $m = 971$. The only exponent in (17) at this length is 1539, as the next power of 2 fails the upper bound. Then $E = 0$: the original word consists of 403 ones and 568 twos. This window is retained as an unresolved arithmetic fiber, not as a found cycle. Its integral compression is in `integral_blocks.md`.

The precise clock comparison with conventional shortened-map local-minimum counts is also explicit. An odd-return exponent a expands to one odd shortened step followed by $a - 1$ even shortened steps. Expansion sends total shortened length to A and odd-step count to m ; grouping between successive odd states is its inverse. Each $a \geq 2$ contributes exactly one even-run termination and hence one local minimum in a nontrivial shortened cycle. Thus that count is $\mu = m - k$. The original ordered word remains available. This supplies the map between these counts rather than interchanging them with the m -cycle terminology in [R2].

4.5 5. First-crossing exception boxes and the retained infinite source

At a first **completed odd-return coefficient crossing**, $D > 0$ and every proper prefix has $D_j \leq 0$. Its exact positive sources with no descent even at the terminal step are

$$n = r + 2Ut, \quad \max\left(0, \left\lceil \frac{s-r}{2U} \right\rceil\right) \leq t \leq \left\lfloor \frac{C-Dr}{2UD} \right\rfloor. \quad (23)$$

This is the complement of the descent interval within that original cylinder above s . It is a finite **source** box, not an absent label. The original affine offset is essential to its upper endpoint.

There is a general bound on its occupancy:

$$3C \leq mL, \quad \#\text{box} \leq \left\lceil \frac{m}{6D} \right\rceil. \quad (24)$$

Indeed each summand of C is at most 3^{m-1} , using $2^{A_{i-1}} \leq 3^{i-1}$. The possible sources lie in $(0, C/D]$, are spaced by $2U$, and $C/(2UD) \leq mL/(6UD) < m/(6D)$. This proves (24). It does not assert the box to be empty.

Every first-crossing word with total exponent sum at most 18 was checked: 4,082 words, 4,081 empty boxes at minimum 1, and one occupied box containing the actual fixed source 1. This is a bounded scan. The decomposition in Section 2 applies at arbitrary finite prefixes, without asserting that their infinite-depth residual source is empty.

An actual non-base component minimum has no strict descent. It must avoid every certified cell and tail. A finite coefficient crossing places it in exactly (23); absence of any coefficient crossing retains all the original noncontracting prefix cylinders. In both cases ordinary positive-integer support still requires the predecessor's canonical residue tower to stabilize at that same integer. No reference-measure zero is used to remove a vertex. The root splicing in Section 3 preserves the original component obstruction in either case.

The distinction between coefficient contraction and actual descent is connected to the explicit remainder in [R1] by the clock expansion above: at a completed return, its coefficient is L/U and its remainder is C/U . Their expression $(L/U)n + C/U$ is exactly (1). No coefficient-stopping-time conjecture is assumed here.

4.6 6. Reproduction and inspection scope

`verify.py` regenerates the atlas and feedback, checks source lifts by independent integer repeated division, checks the exact root-level homotopies, exercises the unbounded tail strata, checks integral block comparisons, and verifies the finite all-length exclusion certificate. It also retains zero masks and rejects malformed data. `replay.py` checks source hashes, replays the entire unchanged import, and requires the new ordinary/optimized outputs to be byte-identical to `verification.json`.

The current-source reads include the workbench method chapter, issue #3, PR #2 and its discussion, and the cycle note at the pinned commit. The uploaded publication archive supplied executable local bytes; its note Git blob was compared with the authenticated source, and the unchanged 21-file manifest and six replay outputs passed before continuation. No unpushed local Codex state was accessed.

[P] `../cycle_relative_cohomology_20260914/note.md`, Sections 1–7, especially the actual relative graph and cutoff retractions; `../stopped_affine_transport_20260914/` is retained through that original dependency.

[R1] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948v2 (13 February 2025), Introduction, Definitions 1.1–1.2 and their displayed affine expression. Readable primary text: <https://arxiv.org/html/2502.0094>

8v2. Cited for the existing coefficient/remainder question, not as a proof premise or an adoption of a conjecture.

[R2] C. Hercher, *There are no Collatz m -Cycles with $m \leq 91$* , Journal of Integer Sequences 26 (2023), Article 23.3.5. Publisher page: <https://cs.uwaterloo.ca/journals/JIS/VOL26/Hercher/hercher5.html>, which also links a corrigendum dated 14 June 2026. The page and terminology were inspected; no detailed numerical theorem or external verification bound from it is a premise. Its TeX retrieval was not readable through the web tool. No priority or cycle-record claim is made.

Chapter 5

Integral block maps, retained forcing, and the next cycle sector

Source: repository/collatz_reconstruction/research_program/residual_splice_20260914/integral_blocks.md.

14 September 2026. Continuation of the original Collatz cycle complex in ../cycle_relative_cohomology_20260914/note.md at repository commit 32b8f3b9cc6dcfc4387ff783c4f4a6f6a365f717.

This note supplies the chain equivalence used by residual_descent.block_comparison. The original ordered word, the chosen block boundaries, all original coefficients, and the forcing vector remain part of the input. The compressed matrix alone is not substituted for that input.

5.1 1. Source complex and literal block telescoping

Let $p = (a_1, \dots, a_m)$, with $a_j \geq 1$, and put $A_j = \sum_{v \leq j} a_v$, $A_0 = 0$. The source complex, in degrees zero and one, is

$$K_p = [\mathbb{Z}^m \xrightarrow{B} \mathbb{Z}^m], \quad (Bx)_j = 2^{a_j} x_{j+1} - 3x_j, \quad x_{m+1} = x_1. \quad (\text{B1})$$

Set $L = 3^m$, $U = 2^{A_m}$, $D = U - L$. The integer D is nonzero and coprime to 6. Retain the functional

$$f_j = 3^{m-j} 2^{A_{j-1}}, \quad fB = De_1^T. \quad (\text{B2})$$

The determinant has absolute value $|D|$. The first and last coefficients of f are powers of 3 and 2 respectively, so their greatest common divisor is one. Thus f induces an isomorphism

$$\text{coker } B \longrightarrow \mathbb{Z}/D\mathbb{Z}, \quad [b] \mapsto [fb]. \quad (\text{B3})$$

For completeness, (B2) proves well-definedness. Primitivity of f proves surjectivity, and the determinant proves that the two finite groups have the same order. In particular, an integral vector b lies in $B\mathbb{Z}^m$ exactly when $D \mid fb$. The matrix itself is injective over $\mathbb{Z}$.

Partition the ordered word into nonempty consecutive blocks $q_1, \dots, q_k$, starting at original positions $s_1 = 1, s_2, \dots, s_k$. For each block retain

$$L_i = 3^{|q_i|}, \quad U_i = 2^{\sum q_i}, \quad C_i = \sum_{j=1}^{|q_i|} 3^{|q_i|-j} 2^{\sum_{v < j} (q_i)_v}.$$

Define $P_0 : \mathbb{Z}^m \rightarrow \mathbb{Z}^k$ by $(P_0 x)_i = x_{s_i}$. Define $R : \mathbb{Z}^m \rightarrow \mathbb{Z}^k$ by placing the local row (B2) of block q_i in the columns of that block, with zeros elsewhere. Finally define

$$(Gz)_i = U_i z_{i+1} - L_i z_i, \quad z_{k+1} = z_1. \quad (\text{B4})$$

For one block, the two contributions are added in the same entry, giving $G = (D)$, not a two-entry matrix.

The local telescoping identity is precisely

$$RB = GP_0. \quad (\text{B5})$$

No coefficient is divided out. Its affine instance is $U_i x_{s_{i+1}} - L_i x_{s_i} = C_i$ for the original forcing $Bx = \mathbf{1}$.

Write

$$\bar{f}_i = \left(\prod_{v>i} L_v \right) \left(\prod_{v<i} U_v \right).$$

Then direct multiplication gives

$$\bar{f}G = De_1^T, \quad f = \bar{f}R. \quad (\text{B6})$$

The endpoint coefficients of $\bar{f}$ are again coprime powers of 3 and 2. The determinant of G has absolute value $|\prod U_i - \prod L_i| = |D|$. The same argument as (B3) gives the marked cokernel isomorphism induced by $\bar{f}$. Hence the map induced by R is the identity on the original $\mathbb{Z}/D\mathbb{Z}$ coordinate:

$$[b] \mapsto [Rb], \quad [fb] = [\bar{f}Rb]. \quad (\text{B7})$$

5.2 2. Integral inverse and both homotopies

Each local row of R is primitive. Construct an integral right inverse $S_1 : \mathbb{Z}^k \rightarrow \mathbb{Z}^m$ as follows. For a block of one letter use its single coefficient 1. For a longer block, use the extended Euclidean algorithm to choose integers u_i, v_i with

$$u_i 3^{|q_i|-1} + v_i 2^{\sum_{j<|q_i|} (q_i)_j} = 1.$$

Place u_i, v_i in the first and last rows of column i of S_1 , and zero elsewhere. Blocks have disjoint column supports, so

$$RS_1 = I_k. \quad (\text{B8})$$

Use the rational inverse of the original nonsingular matrix to specify

$$S_0 = B^{-1}S_1G, \quad H = B^{-1}(I_m - S_1R). \quad (\text{B9})$$

These formulas define **integral matrices**. Indeed, by (B6)–(B8),

$$fS_1G = \bar{f}G = De_1^T, \quad f(I_m - S_1R) = 0.$$

Criterion (B3), applied to each column, places both numerators in $B\mathbb{Z}^m$. Thus the unique rational preimages in (B9) have integer coordinates. This proves integrality rather than discarding denominators after a rational calculation.

The resulting identities are

$$BS_0 = S_1G, \quad P_0S_0 = I_k, \quad RS_1 = I_k, \quad (\text{B10})$$

$$BH = I_m - S_1R, \quad HB = I_m - S_0P_0, \quad P_0H = 0. \quad (\text{B11})$$

The first identity of (B10) and the first of (B11) follow from (B9). For the second identity of (B10), multiply by G :

$$GP_0S_0 = RBS_0 = RS_1G = G.$$

Injectivity of G implies the claim. Also

$$HB = B^{-1}(B - S_1RB) = I_m - S_0P_0.$$

Finally $GP_0H = RBH = R - RS_1R = 0$, giving $P_0H = 0$.

Therefore $(P_0, R) : K_p \rightarrow [\mathbb{Z}^k \xrightarrow{G} \mathbb{Z}^k]$ and (S_0, S_1) are integral chain maps, their composition on the block complex is the identity, and their composition on the original complex is chain homotopic to the identity by the explicitly specified H .

5.2.1 Forced equations over every coefficient ring

Let b be a specified forcing vector. For every commutative coefficient ring, including $\mathbb{Z}$ and $\mathbb{Z}/N\mathbb{Z}$, the maps

$$x \mapsto z = P_0x, \quad z \mapsto x = S_0z + Hb \quad (\text{B12})$$

are inverse bijections between the solution sets

$$Bx = b \quad \text{and} \quad Gz = Rb.$$

In fact,

$$B(S_0z + Hb) = S_1Gz + (I_m - S_1R)b = b,$$

$P_0(S_0z + Hb) = z$, and for a source solution, $S_0P_0x + HBx = x$, by (B10)-(B11). These polynomial matrix identities remain true after reduction modulo N ; no inversion of D in the coefficient ring occurs.

The offset Hb is indispensable. Retaining only the homogeneous chain map S_0 would not give the inverse of the original affine problem. Positivity and actual valuations are tested on the reconstructed full coordinates. For a positive integral solution with $b = \mathbf{1}$, reduction of each original equation modulo 2 gives oddness of every x_j ; the original equation then gives its exact valuation a_j . This recovers an actual positive Collatz cycle, not merely a block congruence.

5.3 3. Anchor at the actual fixed orbit, without erasing the word

Use the coordinate map $y = x - \mathbf{1}$ with inverse $x = y + \mathbf{1}$. For the original +1 forcing the new right-hand side is

$$By = \mathbf{1} - B\mathbf{1} = (4 - 2^{a_j})_{j=1}^m.$$

By (B5), its exact image is

$$G(P_0 y) = (C_i + L_i - U_i)_{i=1}^k. \quad (\text{B13})$$

Both the original forcing and this specified affine coordinate map remain recorded.

Consider a block consisting of a 1 followed by $r \geq 0$ twos. Its unreduced data are

$$\begin{aligned} U_r &= 2 \cdot 4^r, & L_r &= 3^{r+1}, & C_r &= 2 \cdot 4^r - 3^r, \\ C_r + L_r - U_r &= 2 \cdot 3^r. \end{aligned} \quad (\text{B14})$$

To prove the numerator formula, apply the recurrence $C_{r+1} = 3C_r + 2 \cdot 4^r$ with $C_0 = 1$; substituting (B14) satisfies the recurrence exactly. Thus a cyclic word with one 1 at the beginning of each retained block has anchored equations

$$2 \cdot 4^{r_i} y_{i+1} - 3^{r_i+1} y_i = 2 \cdot 3^{r_i}. \quad (\text{B15})$$

Here a row represents the original block $(1, 2, \dots, 2)$ of length $r_i + 1$. The gap dictionary (r_i) and the rotation offset recover the original word. A cyclic rotation by offset h is inverted by rotation by $m - h$; on vertex coordinates the index correspondence is $i \mapsto i + h \pmod{m}$. No passage to unmarked rotation classes is performed.

5.4 4. Actual edge chains and a prime-power control

The weighted affine complex above and the actual orbit graph are connected through the solved original coordinates, not by identifying their differentials. For a positive integral solution, map original position j to actual vertex x_j and its edge to e_{x_j} . The original equation proves that this is an actual orbit edge.

The ordinary chain map from the selected block circle to the full position circle sends block vertex i to position s_i , and block edge i to the sum of original edges in q_i . Its boundary telescopes from s_i to s_{i+1} . Composing with the actual orbit map gives the original path chain for that block. The block circle's fundamental cycle maps to the same full cycle chain $\sum_j e_{x_j}$. Passing to the predecessor's relative graph kills only the vertex and loop at the actual fixed point 1. Repeated traversals remain their explicit integer multiplicities.

For the word $p = (1, 1, 2, 4, 3)$, split into blocks of lengths 1 and 4. The result is

$$G = \begin{pmatrix} -3 & 2 \\ 1024 & -81 \end{pmatrix}, \quad R\mathbf{1} = \begin{pmatrix} 1 \\ 197 \end{pmatrix}, \quad D = 1805 = 5 \cdot 19^2.$$

The block solution is $(5/19, 17/19)$. The marked original cokernel coordinate is

$$\bar{f} = (81, 2), \quad \bar{f}R\mathbf{1} = 81 + 2 \cdot 197 = 475.$$

Its order in $\mathbb{Z}/1805\mathbb{Z}$ is 19. The first equation modulo any odd N gives $z_2 = (1 + 3z_1)/2$.

Substitution into the second gives

$$1805z_1 = 475 \pmod{N}.$$

Consequently there are exactly 19 solutions at $N = 19$ and none at $N = 361$. The bijection (B12) transfers these counts to the full five-equation system. This is the predecessor's original prime-power obstruction, retained under the new block comparison. The checker also exhausts the possible first coordinate at both moduli.

5.5 5. The next arithmetic sector after the executed exclusion

The accompanying `note.md`, Theorem 4, excludes all nontrivial positive cycles with at most 402 exponent-1 positions in their primitive word. Its next unexcluded 403-position sector is forced by the exact period comparisons to have

$$m = 971, \quad A = 1539, \quad k = 403, \quad 99781 \leq \min x_j \leq 330749.$$

Since $A = 2m - k + \sum_{a_j \geq 3} (a_j - 2)$, every letter in this sector is 1 or 2. Cut at the 403 occurrences of 1, retaining the original rotation offset. Then

$$r_i \geq 0, \quad \sum_{i=1}^{403} r_i = 568.$$

Equation (B15) gives a 403-equation **integral** system with total

$$D = 2^{1539} - 3^{971},$$

and the original right-hand side retained as (B13). Formula (B12) reconstructs every original coordinate over $\mathbb{Z}$ or modulo any selected prime power. Integrality, positivity, the chosen minimum placement, and the original valuation word remain required. No solution or exclusion of this entire sector is asserted here.

This identifies an exact next computation: either discharge its full possible minimum interval by additional original-source descent certificates, or prove a marked forcing obstruction for every retained gap tuple, with any finite-coefficient test transported by (B12). An unmarked diagonal spectrum, a prime-only test, or the deletion of a zero label cannot replace that calculation.

5.6 6. Executed scope

`verify.py` checks 360 block/partition fixtures of lengths at most four, the displayed five-letter prime-power example, arbitrary-gap formulas for $0 \leq r \leq 15$, both affine inverse compositions, marked forcing transport, and finite-coefficient chain identities. The proofs above cover all nonempty words and nonempty consecutive block partitions. Finite checks are regression evidence for their implementation; they are not independent external review or formal proof-assistant verification.

Chapter 6

Two-clock cohomology, original integral forcing, and a source-complete finite certificate

Source: repository/collatz_reconstruction/research_program/clocked_support_20260914/note.md.

14 September 2026. Collatz-only continuation of KokunoYumeto/collatz-workbench, authenticated source head 6a30f1dc23110fae0abbe3b3bd5e60072a842385.

The contribution retains the workbench's original two-clock weight, not a new normalization. It gives its marked cyclic complex, exact integral-arithmetic specialization, component cokernel, completion maps, and a clock-preserving lift of the previously constructed descent retractions. A reversible source database closes the previously retained 403-rise sector and gives an all-length 678-rise exclusion. The actual infinite residual source remains.

6.1 0. Source relationships and notation

[W] research_companion/chapters/01_weighted_path_and_finite_residue_actions.tex, Git blob f0142c0d4c237ed068a0762a08c6622df5dbcfce, in the same authenticated commit. The first-return category, odd-block expansion, two-clock weights, endpoint series, finite-support transfer operator, and completed inverse are already present there. This continuation does not claim those constructions as new. The inspected source passages are lines 1–155 and 260–555.

[C] research_program/cycle_relative_cohomology_20260914/note.md, the relative actual-orbit complex and exact descent retractions.

[S] research_program/residual_splice_20260914/note.md and integral_blocks.md, the complete first-descent fibers, root splicing, original integral comparison and retained 403-rise sector. The source note has Git blob b31acf3b485b9565b1bbf878f1e1a98d5e704ad8.

All paths above are relative to collatz_reconstruction/. Original forcing is +1 unless a control is expressly labelled otherwise. Nothing here supplies a morphism from the entire Zeta theta quotient to the entire Collatz orbit complex, or imports an RH conclusion. The finite support/Hurwitz relationship already recorded in the repository remains available. The new arrows below begin with the actual Collatz path operator of [W].

Put

$$X = \{1, 3, 5, \dots\}, \quad T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1), \quad R = \mathbb{Z}[u, v].$$

The weight of the original edge at n is

$$w_n = uv^{a(n)-1}. \quad (C1)$$

For a path of m odd returns and exponent sum A it is exactly $u^m v^{A-m}$. This is [W]'s convention. The explicit expansion of exponent a into one shortened odd step and $a-1$ shortened even steps recovers both clocks and has the inverse obtained by grouping at odd returns.

The variable v in R is a clock variable; the symbol V_n below is an original vertex basis vector. No original integer is rescaled in these definitions.

6.2 1. The relative complex comes from the original transfer operator

Let E and V be free R -modules on the original nonbase edges E_n and vertices V_n , respectively, for n in X excluding 1. Define

$$d_{u,v}E_n = V_n - w_n V_{T(n)}, \quad V_1 = 0. \quad (C2)$$

Give E degree zero and V degree one, and write

$$\mathcal{K}_{u,v}^\bullet = [E \xrightarrow{d_{u,v}} V].$$

At $u=v=1$, this is precisely [C]'s relative chain complex with degree reindexing $\mathcal{K}^i = C_{1-i}$: the map on every original edge and vertex is the identity. Thus H^1 of this reindexed complex is the old H_0 , and H^0 is the old H_1 . This states the degree correspondence explicitly.

The transfer formula of [W], on finite-support vectors, is

$$(Lf)(y) = u \sum_{T(x)=y} v^{a(x)-1} f(x).$$

Each input basis vector has exactly one image, so $LV_n = w_n V_{T(n)}$. Identifying E_n with V_n gives $d = I - L$, literally. On the original module including 1, RV_1 is invariant because $LV_1 = uvV_1$. Passing to the specified quotient by this submodule gives (C2). On complexes, only the subcomplex

$$[RE_1 \xrightarrow{1-uv} RV_1]$$

is removed. No other source or zero coefficient is used as a reason to remove a basis label.

6.2.1 Proposition 1. Polynomial injectivity and original path boundaries

The map d in (C2) is injective. For a finite actual path $n_0 \rightarrow \dots \rightarrow n_m$, stopping on its first hit of 1, put

$$W_0 = 1, \quad W_j = \prod_{i < j} w_{n_i}, \quad P_p = \sum_{j=0}^{m-1} W_j E_{n_j}.$$

Then

$$dP_p = V_{n_0} - W_m V_{n_m}. \quad (\text{C3})$$

Proof. The two terms belonging to each intermediate vertex cancel with their identical original prefix monomial, giving (C3). For injectivity, identify E and V by labels and write $L = uS$, where $SV_n = v^{a(n)-1}V_{T(n)}$. In $(I - uS)f = 0$, comparison of the constant coefficient in u gives $f_0 = 0$; the subsequent coefficients give $f_j = Sf_{j-1} = 0$. A polynomial vector has finitely many coefficients, all zero. This works on the declared finite-support modules, without any convergence assumption. ■

The original graph's closed-cycle kernel at $u=v=1$ is therefore not obtained by declaring the generic polynomial kernel to be nonzero. The cycle information is retained in the polynomial cokernel and in its explicitly computed specialization below.

6.3 2. A faithful marked cyclic complex

Keep a nonempty original labelled exponent word $p = (a_1, \dots, a_m)$. Define

$$A_j = \sum_{i=1}^j a_i, \quad A_0 = 0, \quad w_i = uv^{a_i-1}, \quad W_j = \prod_{i=1}^j w_i, \\ W = W_m = u^m v^{A_m-m}, \quad N_p(u, v) = \sum_{i=1}^m W_{i-1}. \quad (\text{C4})$$

The position-circle boundary sends edge i to vertex i minus w_i times vertex $i+1$, cyclically. Its algebraic dual on the finite free modules is

$$(J_p h)_i = h_i - w_i h_{i+1}, \quad h_{m+1} = h_1. \quad (\text{C5})$$

The duality is the original evaluation pairing: $\langle de, h \rangle = \langle e, J_p h \rangle$. Neither (C5) nor (C2) is identified with the integer matrix B before the specialization in Section 3.

6.3.1 Theorem 2. Explicit polynomial comparison, both homotopies, and forcing

The maps

$$\phi_0(h) = h_1, \quad \phi_1(b) = \sum_{i=1}^m W_{i-1} b_i$$

give a chain map

$$[R^m \xrightarrow{J_p} R^m] \longrightarrow [R \xrightarrow{1-W} R]. \quad (\text{C6})$$

A section is

$$(\psi_0 t)_1 = t, \quad (\psi_0 t)_i = \left(\prod_{k=i}^m w_k \right) t \quad (i \geq 2), \quad \psi_1 t = (t, 0, \dots, 0).$$

Define

$$(Hb)_1 = 0, \quad (Hb)_i = \sum_{j=i}^m \left(\prod_{k=i}^{j-1} w_k \right) b_j \quad (i \geq 2). \quad (C7)$$

They satisfy

$$\begin{aligned} \phi_1 J_p &= (1 - W)\phi_0, & J_p \psi_0 &= \psi_1(1 - W), & \phi_0 \psi_0 &= \phi_1 \psi_1 = I, \\ J_p H &= I - \psi_1 \phi_1, & H J_p &= I - \psi_0 \phi_0. \end{aligned} \quad (C8)$$

Consequently the marked cohomology is

$$H^0 = 0, \quad H^1 \cong R/(1 - W), \quad [b] \mapsto [\phi_1(b)]. \quad (C9)$$

For the unit forcing vector its mark is $[N_p]$. The exact inverse of the forced equation is

$$(1 - W)t = \phi_1(b), \quad h = \psi_0 t + Hb. \quad (C10)$$

All maps and identities are over R , and remain valid after every specified coefficient homomorphism.

Proof. Multiplication of (C5) by $W_{\{i-1\}}$ telescopes to $(1 - W)h_1$, proving the first chain equation. In $J_p \psi_0$, all rows except the first cancel; the first is $(1 - W)t$. The section identities follow from the first coordinate. For $i \geq 2$, the tail sum (C7) obeys $(Hb)_i - w_i(Hb)_{i+1} = b_i$, including the final row with $(Hb)_1 = 0$. The first row is $-\sum_{i \geq 2} W_{i-1} b_i = b_1 - \phi_1(b)$. This proves the first homotopy. Applying the same telescope to $J_p h$ in each tail gives $h_i - (\prod_{k=i}^m w_k) h_1$, proving the second. The scalar $1 - W$ is nonzero in the integral domain R , giving (C9). Formula (C10) follows by applying J_p and ϕ_0 and then using both homotopies. For $m=1$ the formulas give $H=0$ and $J_p=(1-w_1)$; the two cyclic contributions are in the same matrix entry. ■

For completeness, the position-chain complex, rather than its dual, has an equally explicit comparison with $[R \rightarrow R]$. Send vertex 1 to 1, vertex $i \geq 2$ to $Q_i = \prod_{k=i}^m w_k$, edge 1 to 1, and other edges to zero. The scalar sections send 1 to vertex 1 and to $\sum_i W_{i-1} E_i$. The homotopy sends vertex 1 to zero and vertex $i \geq 2$ to its weighted forward tail to vertex 1. Telescoping proves both homotopy identities. These maps connect the two cyclic cokernel descriptions through the same scalar complex; an unexplained equality of a matrix and its transpose is not used.

6.3.2 Proposition 3. The retained determinant and forcing polynomial recover the word

The pair (W, N_p) determines the entire original labelled word. Its inverse is explicit. Write $W = u^m v^B$. In N_p the coefficient of u^j is a single monomial v^{b_j} , for $0 \leq j < m$. Put $b_m = B$. The exact image conditions are

$$b_0 = 0, \quad 0 = b_0 \leq b_1 \leq \dots \leq b_m,$$

with coefficient 1 at each of these m original positions. The inverse is

$$a_i = 1 + b_i - b_{i-1} \quad (1 \leq i \leq m). \quad (\text{C11})$$

Proof. By definition $b_j = A_j - j$, giving every stated condition and (C11). Conversely, (C11) yields positive integers with cumulative exponents $j+b_j$, whose total weight and forcing polynomial are exactly the supplied pair. Both compositions are identities on the specified image. ■

The raw polynomial N_p is retained, not only its quotient class. Merely retaining W forgets precisely the original words with the same m, A . For example,

$$W_{(1,3)} = W_{(2,2)} = u^2v^2, \quad N_{(1,3)} = 1 + u, \quad N_{(2,2)} = 1 + uv. \quad (\text{C12})$$

Thus the determinant alone has a nontrivial word fiber, and (C11) supplies the actual repair. This faithful marker uses the original weights of $[W]$; no priority claim is made for polynomial encoding of finite words.

6.4 3. The original integral cycle obstruction is an exact specialization

For the same word retain

$$L = 3^m, \quad U = 2^{A_m}, \quad D = U - L, \quad C = \sum_{i=1}^m 3^{m-i} 2^{A_{i-1}},$$

$$(B_p x)_i = 2^{a_i} x_{i+1} - 3x_i. \quad (\text{C13})$$

Let $\Lambda = \mathbb{Z}[1/3]$. The specified coefficient homomorphism is

$$\alpha : R \longrightarrow \Lambda, \quad u \longmapsto 2/3, \quad v \longmapsto 2. \quad (\text{C14})$$

It sends w_i to $2^{a_i}/3$. Directly, including all original signs and factors,

$$\alpha(J_p) = -B_p/3, \quad \alpha(1 - W) = -D/3^m, \quad \alpha(N_p) = C/3^{m-1}. \quad (\text{C15})$$

Thus the original forced system $B_{px}=b$ becomes $\alpha(J_p)x = -b/3$; its scalar forcing mark in (C6) becomes

$$\alpha(\phi_1)(-b/3) = -F_p(b)/3^m, \quad F_p(b) = \sum_{i=1}^m 3^{m-i} 2^{A_{i-1}} b_i. \quad (\text{C16})$$

Multiplication by the specified unit -3^m identifies the scalar equation with

$$Dt = F_p(b). \quad (\text{C17})$$

The degree-zero map from the localized original complex to the evaluated position complex is the identity, and its degree-one map is multiplication by $-1/3$. Its inverse is the identity in degree zero and multiplication by -3 in degree one. This is the chain isomorphism responsible for (C15)-(C17).

6.4.1 Proposition 4. Localization at 3 retains the original integral forcing class

For original integral b , solvability of $B_p x = b$ in Λ^m is equivalent to solvability in $\mathbb{Z}^m$. The map on original cokernels is the explicit isomorphism

$$\mathbb{Z}/D\mathbb{Z} \rightarrow \Lambda/D\Lambda. \quad (C18)$$

It preserves the distinguished original class $[F_p(b)]$, in particular $[C]$ for $b=1$. It also retains the reduced denominator of the unique rational primitive.

Proof. The integer D is odd and coprime to 3, since U is a power of 2 and L a power of 3. The telescoping row F_p satisfies $F_p B_p = D e_1^T$, is primitive (its endpoint coefficients are coprime powers of 3 and 2), and both cokernels have determinant order $|D|$ before localization. Its explicit integral identification is $[C]$'s calculation and $[S]$'s integral block comparison. For (C18), the inverse sends $a/3^r$ to $a(3^r)^{-1} \bmod D$. Independence of the chosen fraction and both inverse identities follow by multiplication by powers of 3, invertible modulo D .

An integral b with a localized preimage has $3^r b$ in $B_p \mathbb{Z}^m$ for some r . Its original cokernel class is killed by 3^r , hence is zero in $\mathbb{Z}/D$ by coprimality. An integral preimage therefore exists and equals the localized one, since B_p is injective. Equivalently its unique rational coordinates are integer adjugate numerators divided by D , and localization at 3 cannot remove a denominator coprime to 3. ■

The coefficient embedding $\mathbb{Z} \rightarrow \Lambda$ is not replaced by an identity on lattices. Its extra degree-wise quotient is the 3-primary group $(\Lambda/\mathbb{Z})^m$. On this group B_p is an automorphism: its adjugate multiplied by the inverse of D is an explicit inverse, because D acts invertibly on 3-primary torsion. Hence that quotient complex is acyclic. This accounts for the extra localized elements through their actual quotient and maps.

The original positive cycle criterion remains $D > 0$ and $D|C$; the full original coordinates and actual valuations are reconstructed as in $[C, S]$. The ring change alone does not assert positivity or the existence of an integer cycle. The word $(1, 3)$ has $(D, C) = (7, 5)$, whereas $(2, 2)$ has $(7, 7)$. Their identical clock determinants in (C12) give different original forcing classes through (C16). The second word maps to the actual fixed vertex 1 and is removed only by the declared relative graph quotient.

The prime-power control $(1, 1, 2, 4, 3)$ retains $D = 1805$ and $C = 475$. Its scalar equation has 19 solutions modulo 19 and none modulo 361, because $\gcd(1805, 19) = 19$ divides 475 while $\gcd(1805, 361) = 361$ does not. Formula (C18) and $[S]$'s integral inverse preserve that obstruction; no passage to rational coefficients substitutes for this test.

6.5 4. What the clocked cokernel records for actual components

Let $\mathcal{C}$ be the set of actual nonbase periodic components, each equipped with one chosen point on its primitive cycle. Let $\mathcal{A}$ be the set of actual nonperiodic components, each equipped with a chosen original forward ray $x_0, x_1, \dots$. These are sets defined by the actual map, not assertions that either kind of nonbase component exists.

For a primitive cycle c let m_c, A_c be its actual clocks and $W_c = u^{m_c} v^{A_c - m_c}$. For a chosen nonperiodic ray let

$$B_j = \sum_{i < j} (a(x_i) - 1), \quad W_j = u^j v^{B_j}, \quad M_x = \bigcup_{j \geq 0} W_j^{-1} R \subset \mathbb{Z}[u^{\pm 1}, v^{\pm 1}]. \quad (C19)$$

The union is increasing because W_j divides W_{j+1} . Each element remains a finite Laurent polynomial; the union is not silently replaced by a completion.

6.5.1 Theorem 5. Component classification with explicit maps

The cohomology of the actual relative complex is

$$H^0(\mathcal{K}) = 0, \quad H^1(\mathcal{K}) \cong \bigoplus_{c \in \mathcal{C}} R/(1 - W_c) \oplus \bigoplus_{x \in \mathcal{A}} M_x. \quad (\text{C20})$$

The original base component contributes zero. Every cycle summand has its displayed annihilator ideal $(1 - W_c)$; every nonperiodic summand is a nonzero torsion-free R -module inside the stated Laurent domain.

Proof. Finite-support modules decompose by actual components, and d respects this decomposition. Proposition 1 gives $H^0 = 0$.

On the component reaching 1, the actual finite path from n to 1 has boundary V_n by (C3), so every generator of the cokernel vanishes. Equivalently, before taking the relative quotient, this component has exactly $R/(1 - uv)$, and the subcomplex at 1 maps onto that summand by its original generator. Taking the relative quotient removes precisely it.

For a periodic component, use the weighted finite path from every vertex to the chosen cycle point to express its class as a monomial times that point's class. One traversal of the primitive cycle gives exactly the relation $(1 - W_c)V = 0$. Send each vertex to the weight of its first path to the chosen point, reduced modulo $(1 - W_c)$. Every edge relation maps to zero; at the chosen point it is precisely $1 - W_c$. The reverse map sends 1 to the chosen vertex. The path relations and the primitive return prove both inverse identities. This gives the displayed quotient, whose annihilator is its defining principal ideal.

For a nonperiodic component, every vertex eventually meets the chosen ray. To see this, adjacent vertices have forward orbits that meet, and this property is transitive along each finite undirected path. For $T^d(n) = x_j$, send

$$[V_n] \mapsto \frac{\prod_{i < d} w_{T^i(n)}}{W_j} \in M_x. \quad (\text{C21})$$

Changing the meeting point to a later point multiplies numerator and denominator by the same actual path weight. Nonperiodicity ensures that two meeting descriptions align in this way, so the map is well defined. It respects the edge relation by the first weight factor. Conversely send f/W_j to $f[V_{x_j}]$. Equality of two fractions can be tested at a common later denominator W_k , and the original ray path relations make their images equal. These are inverse maps on generators, and therefore on all finite sums. The target is a nonzero submodule of a domain, proving torsion-freeness. ■

This classification does not decide the actual component of a source whose behavior has not been determined. It specifies exactly what a component would contribute through the displayed maps, with no deletion from a reference-measure statement.

6.5.2 Repetitions and the actual orbit map

For a solved positive integral word, the position vertex i maps to V_{x_i} and edge i maps to E_{x_i} . The original equations prove that the exponent weight is the actual

$w_{\{x_i\}}$, so this is a chain map for (C2). Its degreewise kernel consists exactly of coefficient vectors whose sums over each repeated original vertex/edge label vanish. The full coordinate dictionary is retained.

For r repetitions of a primitive cycle of weight W_c , the source cokernel maps by

$$R/(1 - W_c^r) \longrightarrow R/(1 - W_c), \quad [f] \longmapsto [f]. \quad (\text{C22})$$

Its kernel is $(1 - W_c)/(1 - W_c^r)$. The weighted fundamental path maps to

$$(1 + W_c + \cdots + W_c^{r-1})P_c.$$

At $u=v=1$ this gives exactly r times the original cycle chain, not one traversal. Repetitions of (2) map entirely to the removed base vertex and loop.

6.5.3 Recovery of the old cycle kernel

The coefficient homomorphism $u \rightarrow t, v \rightarrow 1$ has kernel $(v-1)$ and sends (C20)'s cycle summand to $\mathbb{Z}[t]/(1-t^{m_c})$, and its ray summand to $\mathbb{Z}[t, t^{-1}]$. On a ray this follows directly by specializing the explicit increasing union: the denominator W_j becomes t^j . The kernel of the map on that union is $(v-1)M_x$, as follows by writing an element at one denominator and dividing its polynomial numerator by $v-1$.

Now use the exact sequence of complexes given by multiplication by $t-1$ and evaluation $t=1$. Its connecting map is explicit. For an ordinary primitive cycle chain $\sum_i E_i$, lift it to $\sum_i t^{i-1} E_i$; its differential is

$$(1 - t^m)V_1 = -(t-1)(1 + t + \cdots + t^{m-1})V_1.$$

Thus the connecting class is $-[1 + t + \cdots + t^{m-1}]$, the generator of the $(t-1)$ -torsion subgroup in $\mathbb{Z}[t]/(1-t^m)$. The Laurent ray module has no such torsion. Quotienting either component module by $t-1$ gives $\mathbb{Z}$. This recovers exactly [C]'s ordinary cycle kernel and the degree-zero component obstruction, with their signs and degree correspondence accounted for.

6.6 5. Two completion maps and the support they retain

Let $E_0 = \mathbb{Z}[v]^{(X \setminus \{1\})}$ and $\widehat{R} = \mathbb{Z}[v][[u]]$. There is an explicit injective map

$$\iota : \widehat{R}^{(X \setminus \{1\})} \longrightarrow E_0[[u]], \quad (f_n(u, v))_n \longmapsto \sum_{j \geq 0} \left(\sum_n [u^j] f_n V_n \right) u^j. \quad (\text{C23})$$

Its image is exactly the series with one common finite vertex-support set for all u coefficients. The target permits the finite vertex support to depend on the coefficient index. The quotient is precisely the latter series modulo the common-finite-support series. These formulas specify the relation between the two constructions, rather than merely giving them different names.

On $E_0[[u]]$,

$$(I - uS)^{-1} = \sum_{j \geq 0} u^j S^j \quad (\text{C24})$$

is defined coefficientwise. Each coefficient is a finite original vertex sum. This is the

forward-column form of [W]'s completed transfer inverse, specialized to the relative source; [W] already contains the existence of the completed inverse.

6.6.1 Theorem 6. Exact source-support test for the inverse column

For an original source n , the unique completed inverse column is

$$G_n = \sum_{j \geq 0} u^j v^{A_j(n)-j} V_{T^j(n)}, \quad V_1 = 0. \quad (C25)$$

The formula stops contributing after the first hit of 1. Its membership has the following complete description:

- On an orbit reaching 1, G_n is a polynomial vector, with u -degree exactly its positive number of odd returns to 1 minus one (and the zero column for $n=1$).
- On an orbit eventually entering a nonbase cycle, G_n lies in the image of (C23) and is not a polynomial in u . Its actual tail is a finite weighted cycle numerator divided by $1-W_c$, with its original entry prefix retained.
- On a nonperiodic orbit, G_n is outside the image of (C23): its nonzero coefficients have infinitely many distinct original vertex labels.

Consequently

$$[V_n] = 0 \text{ in } H^1(\mathcal{K}_{u,v}) \iff G_n \text{ is a polynomial vector} \iff n \text{ reaches 1.} \quad (C26)$$

Proof. Coefficient comparison in $(I-uS)G_n=V_n$ uniquely gives the displayed successive terms, retaining both clocks and the actual label. Arrival at 1 makes all subsequent relative terms zero. A repeated nonbase vertex makes the deterministic orbit periodic from its first repetition; the finite prefix and geometric cycle tail give the stated rational expression. Without any repeated vertex, the support is unbounded in vertex labels. Conversely, finitely many encountered vertex labels force a repetition or a base hit. These statements prove every support characterization and (C26), since Proposition 1 makes a polynomial preimage unique and its inclusion in the completion must equal G_n . ■

The scalar extension from R to $\widehat{R}$ does not by itself produce (C24) on the common-finite-support module: on a ray its cokernel is the nonzero increasing union $\bigcup_j W_j^{-1}\widehat{R}$, with the same fraction maps as (C21). For a finite cycle, $1-W_c$ is a unit in $\widehat{R}$, so that cycle summand becomes zero. Passing further through (C23) makes the ray column available and the entire completed complex acyclic. This computes what each map removes.

Also, evaluation $u=1$ on polynomials does not extend to a unital homomorphism $\mathbb{Z}[v][[u]] \rightarrow \mathbb{Z}[v]$ fixing v : $1-u$ is a unit in the source and would map to zero. The exact inclusion of polynomials and this unit obstruction explain why an available formal inverse is not automatically a finite-support contraction at the original parameter value.

The global Collatz target is therefore the source-dependent polynomial membership in (C26) for every n . No uniform degree bound over all positive integers is required or asserted. A completed inverse already exists without that membership; its existence is not the missing proof.

6.7 6. Retraction and root splicing retain the original clocks

Let h, Q, F be one of $[C, S]$'s actual path retractions. Retain the original path from n to its chosen root $r(n)$, its length ℓ_n , and its full exponent word. Let P_n be its chain (C3) and M_n its terminal monomial. Define

$$h_w(V_n) = P_n, \quad Q_w(V_n) = M_n V_{r(n)}, \quad F_w = I - h_w d. \quad (C27)$$

Root paths have length zero, and the relative base vertex is zero. These are maps on the original finite-support polynomial modules. The predecessor's positive edge multiplicities and the original starting point reconstruct the path's chronological order: iterate its deterministic successor for the sum of the multiplicities, then verify both the endpoint and the complete edge-count dictionary. The code does exactly that rather than inventing an ordering of a sparse vector.

6.7.1 Theorem 7. Clocked homotopy and extension identities

The maps satisfy

$$\begin{aligned} dh_w &= I - Q_w, & h_w Q_w &= 0, & Q_w^2 &= Q_w, \\ F_w^2 &= F_w, & dF_w &= Q_w d. \end{aligned} \quad (C28)$$

For an extension attached at old roots as in $[S]$, let Γ_r be its full weighted path from an old root to a final old root, including each old endpoint path. Define

$$K_w(V_n) = M_n \Gamma_{r(n)}, \quad h'_w = h_w + K_w. \quad (C29)$$

Then

$$dK_w = Q_w - Q'_w, \quad F_w K_w = K_w, \quad Q'_w Q_w = Q_w Q'_w = Q'_w, \quad F'_w F_w = F_w F'_w = F'_w. \quad (C30)$$

At $u=v=1$ these are exactly the old retraction and root-splice maps.

Proof. Equation (C3) proves the first equation of (C28). The maps h vanish at their roots, and their root monomials are 1, proving the other vertex identities. Then $h_w dh_w = h_w(I - Q_w) = h_w$, yielding edge idempotence and the chain equation. In a concatenation, the second path chain is multiplied by the first path's terminal monomial. Thus (C29) is the actual added chain and telescopes to $dK_w = Q_w - Q'_w$. Final roots are old roots with no added path, so $K_w Q_w = K_w$, $K_w Q'_w = 0$ and $h_w Q'_w = 0$. In particular $h_w dK_w = 0$ and hence $F_w K_w = K_w$. The remaining equations follow by multiplying $I - h_w d$ and $I - (h_w + K_w)d$ and applying those identities. The evaluation homomorphism $u=v=1$ sends every prefix monomial to 1, recovering the original chains coefficient for coefficient. ■

For a genuine weighted cycle path P_c based at n ,

$$F_w P_c = P_c - (1 - W_c) h_w(V_n). \quad (C31)$$

This is the exact relation before specialization. At $u=v=1$, it becomes literal equality with the old closed cycle chain. One must not assert that a generic weighted cycle path has zero boundary: its boundary is exactly $(1 - W_c)V_n$.

6.7.2 An actual omitted-clock defect and its repair

Use the old one-edge descent $5 \rightarrow 1$, with exponent 4. At the old root 7 the new original block is

$$7 \xrightarrow{1} 11 \xrightarrow{1} 17 \xrightarrow{2} 13 \xrightarrow{3} 5.$$

It has terminal monomial u^4v^3 . The correct new column is

$$P_{7 \rightarrow 5} + u^4v^3E_5.$$

Its boundary is V_7 . Appending E_5 without that monomial produces instead

$$V_7 + (1 - u^4v^3)V_5. \quad (C32)$$

The verifier computes this residual literally, then checks the corrected path and both compositions in (C30). This is a concrete example in which retaining a clock is required for the equation to hold.

6.8 7. A reversible source database closes the 403-rise sector

The previous retained sector was $m=971, A=1539$ with 403 exponent-1 positions and possible minimum in $[99781, 330749]$. The new certificate retains every positive odd source $3 \leq n \leq 330749$, its exact first-descent word, the full affine cylinder and target progression, and its source parameter.

The database uses a finite ordered word dictionary and one dictionary ID for each source row. The source for row i is $n=3+2i$. For its exact word p , the original cell is

$$n = r + 2Ut, \quad T^m(n) = q + 2Lt,$$

with all of r, U, q, L, C, D and the first-descent interval retained. Its decoder returns

$$n = 3 + 2i, \quad p = \text{dictionary}[\text{id}_i], \quad t = (n - r)/(2U), \quad y = q + 2Lt. \quad (C33)$$

The inverse encoder looks up that exact word and checks the same original n, t, y . This is an inverse on the specified valid source records; it is not an assertion that a count or hash recovers a word. Gzip is only a byte container: decompression is verified to recover the exact canonical JSON.

The declared labels (1) and (2,2) remain present with empty first-descent fibers and zero source counts. They are not removed from the dictionary. Counts may be placed in $G(\mathbb{Z}) = \mathbb{Z} \sqcup \{\tau\}$ by assigning an integer to each declared label and τ to an undeclared label. The reflection fixes each integer and maps τ to 0. Its zero fiber is exactly $\{\tau, 0\}$; the separate declaration mask distinguishes them. The full word, cell and original source maps are retained in addition to this mask, so neither an integral forcing class nor a vanished coefficient is inferred from the support label alone.

6.8.1 Theorem 8. Executed source-complete finite polynomial contraction

The delivered database and independent integer replay establish that every positive odd $n \leq 330749$ reaches 1. They also construct a closed actual witnessed graph with 262615 nonbase vertices, on which the original unit edge cochain has an explicitly

reconstructed nonnegative integral primitive of maximum 164. On that witnessed graph, the clocked operator satisfies

$$S^{164} = 0, \quad (I - uS)^{-1} = \sum_{j=0}^{163} u^j S^j. \quad (C34)$$

Here $SV_n = v^{a(n)-1}V_{T(n)}$, with $V_1=0$, so every coefficient and original edge remains in (C34).

Proof and executable certificate. `sector_certificate.py` generates the actual first-descent word for each of the 165374 source rows, retaining any failure rather than accepting it. The saved database has 13989 occupied word cells and the two present zero cells. `independently_verify` replays the original +1 operation using repeated integer division, separately from the generator's valuation implementation. It checks every exponent, absence of an earlier descent, the original affine source/target parameter inverses, every complete cell's count on the interval, and the coverage of every original source row. There are 575110 replayed original return equations. Every target is a strictly smaller positive odd integer, so strong induction beginning at 1 supplies convergence for each seed.

More explicitly, assign height 0 to 1 and work through the source rows in increasing order. For a first-descent path ending at $y < n$, $\text{height}(y)$ is already defined. Assign the intermediate vertices the remaining path length plus $\text{height}(y)$; all overlaps must agree and are checked. The resulting original edge table is closed under the actual map except for the explicitly included basepoint, and $\text{height}(n) - \text{height}(T(n)) = 1$ is verified on all 262615 nonbase vertices. Its maximum is 164; its largest original vertex is 8216025965. Hence every sequence of 164 original edges reaches the relative zero vertex. This proves the first identity in (C34) on every basis column. Multiplication of the finite geometric sum proves both inverse identities. The bound concerns this finite witnessed graph, not all original positive integers. ■

The new source interval includes 115485 additional odd seeds above the prior bound 99779. It closes the entire old 403-rise sector without enumerating its gap tuples: the possible minimum of each tuple is an actual source in the certified interval, and therefore cannot lie on a nonbase cycle.

6.8.2 Theorem 9. All-length exclusion through 678 exponent-1 positions

No nontrivial positive integer cycle has at most 678 exponent-1 positions in its primitive odd-return word.

Proof. Theorem 8 gives the minimum lower bound $s=330751$. For an actual positive cycle,

$$3^m < 2^A = \prod_i (3 + 1/x_i) \leq (3s + 1)^m / s^m.$$

With k exponent-1 positions, the unchanged identity $A = 2m - k + \sum_{a_i \geq 3} (a_i - 2)$ implies

$$(4s)^m \leq 2^k (3s + 1)^m. \quad (C35)$$

The certificate computes the least power of 2 strictly above 3^m by repeated doubling and verifies

$$2^{\lceil \log_2(3^m) \rceil} s^m > (3s + 1)^m \quad (1 \leq m \leq 1635). \quad (C36)$$

The logarithm is notation for that exact least-power integer; no floating-point logarithm is used. It also verifies

$$(4s)^{1634} > 2^{678}(3s+1)^{1634}. \quad (\text{C37})$$

Because $4s > 3s+1$, (C37) persists at every larger length and for all $k \leq 678$. Thus (C35) forces $m \leq 1633$, while (C36) excludes every such length. All finite comparisons and every source premise are regenerated by the delivered checker. ■

This is an internal, self-contained workbench bound, not a claim to exceed published Collatz cycle bounds. In particular k counts exponent-1 positions, not the local-minimum count used in other cycle nomenclature. The original clock map in [S] still gives local-minimum count $m-k$ in a nontrivial shortened cycle.

The next retained sector with $k=679$ is forced to

$$m = 1636, \quad A = 2593, \quad 330751 \leq \min x_i \leq 583287. \quad (\text{C38})$$

The checker verifies the two adjacent minimum-endpoint inequalities, excludes the next power of 2 at that period, and excludes larger lengths for $k=679$ by (C35). Then $\sum_{a_i \geq 3} (a_i - 2) = 0$: the word contains 679 ones and 957 twos. Cutting at the ones retains gap lengths $r_i \geq 0$, $\sum r_i = 957$, the original rotation offset, and [S]’s anchored equations

$$2 \cdot 4^{r_i} y_{i+1} - 3^{r_i+1} y_i = 2 \cdot 3^{r_i}, \quad y_i = x_i - 1. \quad (\text{C39})$$

The complete integral inverse in [S] and the marked clock specialization in Section 3 both remain attached. Neither a solution nor an exclusion of this entire next sector is asserted.

6.9 8. What is new here, what is retained, and the outstanding task

The existing weighted sums, path category, ordinary graph homology, affine packet identities and completed inverse are sourced, not rediscovered under new names. The continuation adds the explicit marked cyclic R-comparison and its original integral specialization; the component cokernel and scalar-versus-support completion maps; the clocked residual retraction and omitted-clock residual (C32); and the reversible finite certificate establishing (C34)–(C37).

The original word is recoverable from (W,N_p), the original integral forcing is recoverable through (C14)–(C18), and actual component behavior is attached through original integer vertices. The available formal inverse is kept in its actual completion. This identifies the productive role of retained data without a priority claim about an undefined acronym or a claim that no other method can prove these results.

The global remaining task is polynomial membership of G_n for each original positive source, or an actual nonzero original source class with a certified orbit obstruction. Merely constructing G_n in the completion has already been done and does not settle that membership. The next finite sector is (C38)–(C39); the old infinite nonperiodic and first-crossing exception sources are unchanged. Zero reference mass never removes one of those vertices.

This session used authenticated GitHub reads and the uploaded source archive. The accessible GitHub tool surface exposed read operations only; no write operation or CLI connection was available. Consequently this continuation is supplied as an ad-

ditive patch against the authenticated head, not described as a pushed commit, PR update, or successful new GitHub CI run. The original eight replays and the new ordinary/optimized pair were executed locally; their receipts distinguish that from the earlier remote CI.

6.10 9. Verification controls

The symbolic suite checks 364 full-word fixtures: every word of lengths 1 through 5 with letters 1 through 3, and the original five-letter prime-power fixture. Each homotopy is checked as a polynomial matrix identity on its basis columns; the arithmetic evaluation is checked against the unchanged original B matrix. The source database is independently replayed and every saved cell is compared with the untouched predecessor constructor. The suite has 33503 named algebraic/source checks in addition to its separately counted 575110 original return equations; the counts are not advertised as independent mathematical reviews.

The periodic positive control is $T_-(n) = (3n-1)/2^{\{v_2(3n-1)\}}$, with the genuine orbit $5 \rightarrow 7 \rightarrow 5$ and weight u^2v . Its exact relation to the signed original map is $T_-(n) = -T_+(-n)$, using $v_2(-q) = v_2(q)$. It is not a positive +1 Collatz counterexample. Its r -fold weighted path is checked against $(1 + u^2v + \dots + (u^2v)^{r-1})$ times the primitive path, for $r \leq 16$.

The ray control is expressly the synthetic map $n \rightarrow n+2$, assigned the test weight uv . Its length- j boundary is $V_3 - u^{jv} v^j V_{\{3+2j\}}$, with j distinct original test labels. This checks the support-completion mechanism; it is never assigned to an unverified +1 orbit. Twelve malformed inputs are rejected, including an erased forcing coefficient, a removed original source row, a changed forcing sign, and erased supported-zero labels.

Chapter 7

Recovering supported Collatz classes from a vanishing completion

Source: repository/collatz_reconstruction/research_program/completion_defect_20260914/note.md.

14 September 2026. Continuation of KokunoYumeto/collatz-workbench, read at ce8f784da8fb48aa19065c036cd815c05628cec4 on PR #2. This contribution concerns infinite formal paths and the spaces carried to supported zero. It makes no new numerical cycle-bound claim.

7.1 1. Source objects, support, and comparisons

The coefficient and homology source is the Split-Zero programme's formal/splitzero/DERIVED_MATHEMATICS.md, Sections 1–4, at Zeta commit 91ed3b7c358a1f444d0d48c292e4a400ca509eff, blob fe05d32ccd314daac7c59d225c9cac3d4c1bf3d3. It constructs support-indexed modules and their actual transition maps. For a chain map i , its module of representatives mapped to target boundaries, modulo the *original* source boundaries, is explicitly isomorphic to $\ker H(i)$. This note instantiates that construction on the original Collatz operator and gives a second realization of the kernel by infinite primitives modulo finite primitives.

The actual operator comes from the Collatz research companion, research_companion/chapters/01_weighted_path_and_finite_residue_actions.tex, blob f0142c0d4c237ed068a0762a08c6622df5dbcfee. It already defines the two clocks, the finite-support transfer, and its completed inverse. The current anchored continuation, anchored_defect_20260914/note.md, Sections 1–3, retains the supported-zero scalar, original integral forcing, and the fixed-point relative convention. The preceding local clocked-support note also treats polynomial membership and completion support. Those are predecessors, not discoveries attributed to this note.

All Collatz paths in this note are relative to collatz_reconstruction/. The general connecting-sequence and filtered-colimit principles are standard; see the references in Section 11. The formulas, finite source-kernel basis, original residual certificates, and Split-Zero diagram below are proved explicitly. Neither a theta-to-Collatz isomorphism nor an estimate imported from the Riemann-hypothesis programme is a premise.

Retain the original map and exponents:

$$X = \{1, 3, 5, \dots\}, \quad T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1).$$

Put $\Lambda = \mathbb{Z}[v]$, $R = \Lambda[u]$, and

$$A = \bigoplus_{n \in X \setminus \{1\}} \Lambda V_n, \quad M = A[u].$$

The label V_n is the original positive integer, not its residue or a rescaled coordinate. A finite vector has finite original vertex support. Define

$$SV_n = v^{a(n)-1}V_{T(n)}, \quad V_1 = 0, \quad d = I - uS. \quad (1)$$

Identifying the original edge E_n with its starting-vertex basis V_n gives the relative two-clock boundary

$$dE_n = V_n - uv^{a(n)-1}V_{T(n)}.$$

The identification on bases is invertible; the complex is in cochain degrees 0 and 1. At $u = v = 1$ the map sends each original edge to source minus target, exactly the earlier relative graph boundary, with degree relation $K^i = C_{1-i}$. Before relativity, the fixed point is the actual subcomplex $[RE_1 \xrightarrow{1-uv} RV_1]$. The quotient removes precisely that subcomplex. An edge entering 1 and its source remain in the relative complex. Each input vector has finite support and one image per source, so S is well-defined even though a vertex can have infinitely many predecessors. All constructions below also apply with $\mathbb{Q}$ in place of $\mathbb{Z}$; their source modules are then vector spaces over $\mathbb{Q}$. The integer version remains available and no rationalization is used to dispose of an integral obstruction.

7.2 2. Completion and its exact defect module

Write

$$K = [M \xrightarrow{d} M], \quad \widehat{M} = A[[u]], \quad \widehat{K} = [\widehat{M} \xrightarrow{\hat{d}} \widehat{M}]. \quad (2)$$

Here $A[[u]]$ means that each coefficient is a finite-support original vertex vector. There need not be a common finite vertex-support set for all coefficient degrees. It is precisely $\lim_N M/u^N M$: a compatible family gives each coefficient once, and conversely its successive truncations give that family. These operations are inverse. S acts coefficientwise. No analytic convergence or evaluation of an infinite series at $u = 1$ is used. Define

$$Gq = \sum_{j \geq 0} u^j S^j q. \quad (3)$$

Each coefficient uses only finitely many summands, including for $q \in \widehat{M}$. Telescoping gives $\hat{d}G = G\hat{d} = I$. Equivalently, for $q = \sum u^j q_j$, the coefficients of $g = Gq$ satisfy

$$g_{-1} = 0, \quad g_j = q_j + Sg_{j-1}. \quad (4)$$

Thus $H^0(\widehat{K}) = H^1(\widehat{K}) = 0$. On M the same lowest-coefficient argument proves $\ker d = 0$, but it does not prove surjectivity: the inverse (3) has been constructed in $\widehat{M}$.

Retain the actual inclusion $i : M \hookrightarrow \widehat{M}$ and its quotient

$$D_\infty = \widehat{M}/M, \quad \mathcal{D} = [D_\infty \xrightarrow{\bar{d}} D_\infty]. \quad (5)$$

The quotient is by *finite polynomial vectors*, not by a closure of them. It is an R-module. No action of all formal scalars is imposed: V_3 represents zero modulo M, but multiplying it by the formal scalar $(1-u)^{-1}$ gives $\sum_{j \geq 0} u^j V_3 \notin M$, a nonzero quotient representative. Thus that proposed action would depend on the representative. The original polynomial scalar action is retained. Since $dM \subset M$, $\bar{d}$ is well-defined. This gives a degreewise exact sequence $0 \rightarrow K \rightarrow \widehat{K} \rightarrow \mathcal{D} \rightarrow 0$ with all maps specified.

7.2.1 Theorem 1. The entire original cohomology is recovered from the completion defect

The maps

$$\begin{aligned} \eta : M/dM &\longrightarrow \ker \bar{d}, & [q] &\longmapsto [Gq] \pmod{M}, \\ \delta : \ker \bar{d} &\longrightarrow M/dM, & [g] &\longmapsto [\hat{d}g] \end{aligned} \quad (6)$$

are inverse R-linear isomorphisms. Also $\text{coker } \bar{d} = 0$.

Proof. For polynomial q , $\hat{d}Gq = q \in M$, so its quotient class lies in $\ker \bar{d}$. Replacing q by $q+df$ replaces Gq by $Gq+f$; hence η is well-defined modulo M. A class $[g]$ in $\ker \bar{d}$ has $\hat{d}g \in M$. Changing g by a polynomial f changes its differential by df , so δ is well-defined. Applying $\hat{d}G = G\hat{d} = I$ proves both inverse compositions. The inverse G is R-linear. Finally every $[h] \in D_\infty$ is $\bar{d}[Gh]$, proving surjectivity. QED.

The map δ is the ordinary connecting homomorphism, with sign fixed by applying the original differential to a lifted degree-zero quotient representative. The direct proof above also specifies its inverse, every representative change, and the degree shift:

$$H^1(K) \cong H^0(\mathcal{D}), \quad H^1(\mathcal{D}) = 0. \quad (7)$$

This is a presentation of the exact space sent to zero by completion. It does **not** assert that this space is nonzero for the original positive Collatz map. Its vanishing or nonvanishing is still a substantive source question.

7.3 3. The literal Split-Zero support diagram

Use the join-semilattice $\mathcal{L} = \mathcal{P}(X \setminus \{1\})$, bottom $\emptyset$, join union. For a seed label F define its original forward carrier

$$\Omega_F = \{1\} \cup \{T^j(n) : n \in F, j \geq 0\}.$$

Use (1)–(5) on $A_F = \Lambda^{(\Omega_F \setminus \{1\})}$. These are well-defined modules for finite or infinite F . The label F remains the declared seed set; two labels whose forward carriers happen to coincide are not silently identified. For $F \subset F'$ the transition is inclusion on every original basis vector, coefficient, and formal series. It commutes with S , d , G , and the quotient map. In particular (6) commutes with every support transition, as follows by applying the identical coefficient formula (3).

For any of these module diagrams V , reconstruct

$$\text{Tot}(V) = \coprod_{F \in \mathcal{L}} V_F,$$

with $(F, x) + (F', y) = (F \cup F', \iota x + \iota y)$. A supported scalar r acts by $(F, x) \mapsto (F, rx)$, and the external scalar τ acts by $(F, x) \mapsto (\emptyset, 0)$. Thus the supported scalar $e = 0$ sends a vector to $(F, 0)$ and retains its original carrier. This is the reconstruction used by the cited Split-Zero source, with the actual inclusions just specified. Ring laws and composition of inclusions prove the scalar and addition identities.

At degree one the completed homology diagram has zero fibres. Its map is

$$(F, [q]) \mapsto (F, 0). \quad (8)$$

For nonempty F this supported zero is different from $(\emptyset, 0)$. More than the label is retained: the source complex, its full representative module M_F , original boundary submodule dM_F , completion inclusion, and primitive Gq are specified. In the source's notation,

$$\begin{aligned} K_{\text{rep}, F} &= M_F, & B_{\text{old}, F} &= dM_F, \\ K_{\text{rep}, F}/B_{\text{old}, F} &= \ker H^1(i_F) \xrightarrow[\eta_F]{\sim} \ker \bar{d}_F. \end{aligned} \quad (9)$$

Every vector is a completed boundary by (3), which proves the first equality. Quotienting only by original boundaries gives the second line. Equation (6) supplies the final comparison. The natural family η reconstructs a split-linear equivalence on totals: addition is preserved after both vectors are included in $F \cup F'$, supported scalars are preserved by R -linearity, and τ is preserved at the bottom label.

Even when $H^1(K_F) = 0$, M_F can remain nonzero. Its original relations and source vectors are part of the diagram. A supported zero of homology does not identify that source module with the zero module.

7.4 4. Why arbitrarily precise finite clock calculations lose the same obstruction

S commutes with d and induces $\bar{S}$ on $Q = M/dM$. The defining relation gives

$$u\bar{S} = \bar{S}u = I_Q. \quad (10)$$

Thus multiplication by u is invertible on Q , with the **specified inverse** $\bar{S}$. For every N ,

$$Q/u^N Q = 0, \quad \lim_N Q/u^N Q = 0. \quad (11)$$

No conclusion $Q = 0$ follows from (11). For example the control in Section 8 has $Q = R/(1 - u^2 v) \neq 0$, with these same zero quotients.

At the level of original representatives, for every polynomial q ,

$$q = d \left(\sum_{j=0}^{N-1} u^j S^j q \right) + u^N S^N q. \quad (12)$$

All summands in this equation are original finite vectors. Consequently dM is u -adically dense in M : (12) gives $M = dM + u^N M$ for every N . The map to the closure quotient is the explicit projection

$$M/dM \longrightarrow M/\overline{dM} = 0,$$

whose kernel is the whole original Q and is recovered by (6). This calculates the information lost by imposing the closure, rather than assuming the original and closed relation submodules are equal.

Each truncated complex $K/u^{N+1}K$ is acyclic, with inverse $\sum_{j=0}^N (uS)^j$. A retained inverse jet for a constant source V_n has the **original untruncated boundary**

$$d \left(\sum_{j=0}^N u^j S^j V_n \right) = V_n - u^{N+1} S^{N+1} V_n. \quad (13)$$

The last term is precisely what reduction modulo u^{N+1} discards. For a singleton source before its first arrival at 1 it is

$$u^{N+1} v^{A_{N+1}(n)-(N+1)} V_{T^{N+1}(n)}. \quad (14)$$

Its exponent, coefficient, and original endpoint are retained. `completion_certificate` returns the complete jet and this term, not just the zero class in the truncated target. The status is unresolved at the cutoff whenever that exact residual is nonzero.

7.5 5. Recovering the original source from a single finite terminal vector

There is another concrete description of Q . Let

$$L = \operatorname{colim} \left(A \xrightarrow{S} A \xrightarrow{S} A \xrightarrow{S} \dots \right).$$

Its elements $[a, j]$ obey $[a, j] = [Sa, j+1]$; equality means equality at one common finite later stage. Define $u[a, j] = [a, j+1]$ and let v have its original scalar action. The inverse u -action is $[a, j]$ mapped to $[Sa, j]$.

7.5.1 Theorem 2. Exact direct-limit and finite-terminal comparison

The maps

$$\begin{aligned} \Phi : Q &\longrightarrow L, & \left[\sum_{j=0}^r u^j q_j \right] &\longmapsto \sum_{j=0}^r [q_j, j], \\ \Psi : L &\longrightarrow Q, & [a, j] &\longmapsto [u^j a] \end{aligned} \quad (15)$$

are inverse R -linear isomorphisms. For q of degree r as above retain

$$b_j = \sum_{i=0}^j S^{j-i} q_i, \quad P = \sum_{j=0}^{r-1} u^j b_j.$$

Then

$$q = dP + u^r b_r, \quad \Phi[q] = [b_r, r]. \quad (16)$$

The exact membership criterion is

$$q \in dM \iff S^N b_r = 0 \text{ for some finite } N. \quad (17)$$

Its finite primitive is $P + u^r \sum_{j=0}^{N-1} u^j S^j b_r$, with the sum empty when $N=0$.

Proof. The image of $u^j(a - uSa)$ under Φ is $[a, j] - [Sa, j+1] = 0$. The defining direct-limit relation is respected by Ψ for exactly the same reason. Both compositions fix the stated generators. To prove (16), compare coefficients using $b_j = q_j + Sb_{j-1}$: all lower coefficients of $q - dP$ cancel and its last coefficient is b_r . Alternatively bring the finitely many direct-limit representatives to the common stage r . A direct-limit representative $[b, r]$ is zero exactly when its image at a finite later stage is zero; this is also immediate from the defining equality relation. Substitution into (12) proves the stated primitive and both directions of (17). QED.

Thus the map of the original constant source module $A \rightarrow Q$ has kernel

$$\bigcup_{N \geq 0} \ker S^N. \quad (18)$$

The operator acts on this entire increasing family of source submodules. A linear combination can enter the kernel because distinct original paths merge with precisely canceling clock coefficients, even before the individual sources have reached 1. It is important to retain those relations rather than to confuse them with singleton convergence certificates.

7.6 6. Complete, constructive spaces of supported-zero vectors at each finite stage

Let F be a finite declared set of nonbase original sources and $A_F^0 = \bigoplus_{n \in F} \Lambda V_n$. This is its finite source span inside A_{Ω_F} ; its target under S^N retains all actual endpoints, including those outside F .

For every n in F , follow N original returns or stop at the first arrival at 1. An arrived source has $S^N V_n = 0$. For every other source retain

$$y_n = T^N(n) > 1, \quad e_n = A_N(n) - N, \quad S^N V_n = v^{e_n} V_{y_n}. \quad (19)$$

For each nonbase endpoint y , choose n_y in its actual source fibre with least e_n , breaking ties by least original n . Put $b_y = e_{n_y}$. These choices are retained as a dictionary, not substitutions for the original sources.

7.6.1 Theorem 3. Source-kernel basis and inverse decomposition

A free Λ -basis of $\ker(S^N|_{A_F^0})$ consists of

$$\begin{aligned} &V_n \quad (y_n = 1), \\ &V_n - v^{e_n - b_{y_n}} V_{n_y} \quad (y_n = y > 1, n \neq n_y). \end{aligned} \quad (20)$$

The original image has basis $v^{b_y} V_y$, one for each occupied nonbase endpoint. The complementary source submodule has basis V_{n_y} ; its map onto that original image

sends V_{n_y} to $v^{b_y}V_y$, with inverse **on that image** sending the displayed generator back. No inverse of v is imposed on the full target module.

For $f = \sum c_n(v)V_n$ the inverse source decomposition is

$$f = \sum_{y_n=1} c_n V_n + \sum_{y>1} \sum_{\substack{y_n=y \\ n \neq n_y}} c_n (V_n - v^{e_n-b_y} V_{n_y}) + \sum_{y>1} \left(\sum_{y_n=y} c_n v^{e_n-b_y} \right) V_{n_y}. \quad (21)$$

Proof. Every vector in (20) has zero original image by (19). All exponents are non-negative by the retained minimum choice. Each vector in (20) has a distinct non-selected pivot coefficient $+1$, so they are linearly independent and independent of the selected-source complement. Expanding (21) cancels the introduced selected-source terms and returns f exactly. Its last summand maps to the original image at each distinct y with factor v^{b_y} . Multiplication by a monomial is injective in Λ , so f is in the kernel precisely when all the parenthesized coefficients vanish. This proves spanning and both inverse decompositions. QED.

The kernel rank is $|F|$ minus the number of occupied nonbase endpoint fibres. It records the whole source-zero space, with its inclusions and original coefficient relations. These kernels form an increasing sequence in N , because S maps zero to zero. Over field coefficients the same formulas describe actual vector subspaces; no dimension comparison replaces (20)–(21).

7.6.2 Original Collatz example

Take $F = \{3, 5, 13\}$ and $N=1$. The original edges are

$$3 \xrightarrow{1} 5, \quad 13 \xrightarrow{3} 5, \quad 5 \xrightarrow{4} 1.$$

Therefore $SV_3 = V_5$, $SV_{13} = v^2V_5$, $SV_5 = 0$. The kernel is the free rank-two module

$$\Lambda V_5 \oplus \Lambda(V_{13} - v^2V_5), \quad (22)$$

and the complement is ΛV_3 , mapping to the unchanged image ΛV_5 . The vector $v^2V_3 - V_{13}$ is nonzero in its original source space, but maps to its supported zero at the next step. Its finite primitive under d is itself. The complete source labels, both paths to 5, and coefficient v^2 remain in the certificate. At $N=2$ all three sources have reached 1 and the kernel is the full original rank-three source span. Neither zero space is replaced by the absent label.

7.7 7. What the exact defect detects on singletons and cycles

For a singleton source n ,

$$\eta[V_n] = \left[\sum_{j \geq 0} u^j v^{A_j(n)-j} V_{T^j(n)} \right] \pmod{M}. \quad (23)$$

Before arrival at 1, every coefficient is a nonzero monomial at one actual vertex. Consequently

$$\eta[V_n] = 0 \iff GV_n \in M \iff S^N V_n = 0 \text{ for some } N \iff n \text{ reaches } 1. \quad (24)$$

All implications have been proved: (6), (17), and the original singleton formula (19) give them. For finitely many polynomial source coefficients, each finite arrival certificate combines to a finite common bound; thus the whole conjecture is also equivalent to local nilpotence of S on A . This is a source-membership criterion, not an assertion that local nilpotence has been established globally.

For an actual finite path p from n to y , let $W_p = u^m v^{A_m - m}$ and retain its original weighted path P . Splitting the series after those m edges gives

$$GV_n = P + W_p GV_y, \quad \eta[V_n] = W_p \eta[V_y]. \quad (25)$$

These are the comparison maps for advancing a residual along its actual orbit. An actual cycle with primitive clocks m, A gives

$$(1 - u^m v^{A-m}) \eta[V_n] = 0. \quad (26)$$

It is nonzero for a nonbase cycle: the infinitely many monomial coefficients in (23) cannot be a polynomial vector. The original relative quotient makes (23) zero for the base orbit, through its actual finite arrival path. This distinguishes those cases using specified primitives and the original quotient.

For completeness the cyclic component coordinate is the explicit map to $R/(1 - W_p)$ that sends each vertex to its first path weight to a chosen primitive-cycle vertex. Its inverse sends 1 to that original vertex. Original path relations give both compositions, and the cycle relation is exactly $1 - W_p$. Repetitions are retained by $(1 - W_p^r) = (1 - W_p)(1 + W_p + \dots + W_p^{r-1})$ and the actual r -fold edge expansion.

On a nonperiodic component choose an original forward ray x_j with prefix monomials W_j . A vertex meeting x_j after a path of weight w is mapped to w/W_j in $\bigcup_j W_j^{-1} R$. A later meeting multiplies numerator and denominator by the same original path weight; this proves independence. The inverse sends f/W_j to $f[V_{x_j}]$. The original edge relations give both compositions. This realizes the component in a Laurent polynomial module and gives a nonzero source class for every original vertex in such a component. No existence of such a component in positive Collatz is assumed or claimed.

7.7.1 7.1 Returning the retained infinite primitive to the integral cycle obstruction

For a nonempty original word $p = (a_1, \dots, a_m)$, retain $A_j = \sum_{i \leq j} a_i$, $A_0 = 0$, and

$$W = u^m v^{A_m - m}, \quad a_p = 1 - W, \quad N_p = \sum_{j=0}^{m-1} u^j v^{A_j - j},$$

$$U = 2^{A_m}, \quad L = 3^m, \quad D = U - L, \quad C = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}.$$

The finite position complex has $(J_p h)_i = h_i - uv^{a_i-1} h_{i+1}$. The retained functional $b \mapsto \sum_{j=0}^{m-1} u^j v^{A_j - j} b_{j+1}$ telescopes J_p to $a_p h_1$. Its scalar forcing for $J_p h = -\mathbf{1}$ is $-N_p$. Thus its completed scalar primitive is $-N_p/a_p$, retaining the original word, polynomial, and sign. The full comparison has $\phi_0(h) = h_1$ and $\phi_1(b) = \sum_{i=1}^m W_{i-1} b_i$, where $w_i = uv^{a_i-1}$ and $W_j = \prod_{i \leq j} w_i$. Its sections are $\psi_1(t) = (t, 0, \dots, 0)$, $(\psi_0 t)_1 = t$ and $(\psi_0 t)_i = (\prod_{j=i}^m w_j) t$ for $i \geq 2$.

Set $(Hb)_1 = 0$ and $(Hb)_i = \sum_{j=i}^m (\prod_{l=i}^{j-1} w_l) b_j$ for $i \geq 2$. Each tail telescopes, giving $J_p H = I - \psi_1 \phi_1$, $H J_p = I - \psi_0 \phi_0$ and $\phi_0 \psi_0 = \phi_1 \psi_1 = I$. Thus the scalar and full position complexes are related by inverse chain maps up to these specified polynomial homotopies. The full forced primitive is $\psi_0(-N_p/a_p) - H\mathbf{1}$, so its correction and original coordinates remain available. This is the same comparison in the preceding local clocked note, now transported through its quotient of primitives.

Let $B'_p = R[a_p^{-1}]$. It embeds in $\widehat{R} = \mathbb{Z}[v][[u]]$ by the original geometric expansion of a_p^{-1} . Injectivity follows by multiplying a zero image by a finite power of the nonzero polynomial a_p . The map

$$R/(a_p) \longrightarrow \ker(a_p : B'_p/R \rightarrow B'_p/R), \quad [q] \mapsto [q/a_p] \quad (30)$$

has inverse $[z] \mapsto [a_p z]$. Indeed being in that kernel means $a_p z$ is represented by a polynomial; both inverse identities then hold before passing to the indicated quotients. The inclusion $B'_p/R \hookrightarrow \widehat{R}/R$ is injective: a rational element whose formal expansion is a polynomial equals that polynomial after multiplication by its denominator. The entire kernel of multiplication by a_p in the latter quotient is already (30), since $a_p g = q \in R$ forces $g = q/a_p$. Thus no relevant scalar cyclic defect is lost by using this specified rational subcomplex.

Now put $\Lambda_3 = \mathbb{Z}[1/3]$, $F_D = \Lambda_3[1/D]$. The coefficient map

$$\alpha : R \rightarrow \Lambda_3, \quad u \mapsto 2/3, \quad v \mapsto 2$$

satisfies

$$\alpha(a_p) = -D/3^m, \quad \alpha(N_p) = C/3^{m-1}. \quad (31)$$

It is surjective, with kernel $(v - 2, 3u - 2)$: in that quotient $3(1 - u) = 1$, so $1/3$ has the explicit inverse-coordinate $1 - u$, while u and v are recovered as $2/3$ and 2 . The coefficient map extends to $\beta : B'_p \rightarrow F_D$ by evaluating the numerator and the specified finite denominator. The extended inverse image of a_p is $-3^m/D$; it has not been declared a polynomial scalar.

The two complexes and their quotients are related by the actual chain maps

$$\begin{aligned} [R \xrightarrow{a_p} R] &\rightarrow [\Lambda_3 \xrightarrow{D} \Lambda_3], \quad f^0 = \alpha/3, \quad f^1 = -3^{m-1}\alpha, \\ [B'_p \xrightarrow{a_p} B'_p] &\rightarrow [F_D \xrightarrow{D} F_D], \quad \hat{f}^0 = \beta/3, \quad \hat{f}^1 = -3^{m-1}\beta. \end{aligned} \quad (32)$$

Their chain equation is $D\beta(z)/3 = -3^{m-1}\beta(a_p z)$, exactly (31). They commute with the original inclusions, hence induce the quotient maps with the same formulas. On the retained degree-zero defect, the map is

$$[z] \mapsto [\beta(z)/3] \in \ker(D : F_D/\Lambda_3 \rightarrow F_D/\Lambda_3). \quad (33)$$

The target connecting isomorphism is $[t] \mapsto [Dt] \in \Lambda_3/D\Lambda_3$, with inverse $[b] \mapsto [b/D]$. Both are well-defined by their literal representative differences. Equations (30)–(33) therefore give the commutative identity

$$\left[\frac{\beta(q/a_p)}{3} \right] = \left[\frac{-3^{m-1}\alpha(q)}{D} \right]. \quad (34)$$

For the original forcing $q = -N_p$, this is exactly

$$\boxed{[-N_p/a_p] \mapsto [C/D] \mapsto [C] \pmod{D}.} \quad (35)$$

The integer D is nonzero and coprime to 6. The final map to the original integral cokernel is the inverse of $\mathbb{Z}/D\mathbb{Z} \rightarrow \Lambda_3/D\Lambda_3$, explicitly $[b/3^j] \mapsto [b(3^j)^{-1}] \pmod{D}$. Its inverse laws follow because 3 is invertible modulo D . In the full integer position matrix $(B_p x)_i = 2^{a_i} x_{i+1} - 3x_i$, the original telescoping functional is $f_i = 3^{m-i} 2^{A_{i-1}}$; it satisfies $f B_p = D e_1^T$. Its endpoint coefficients are coprime, and $|\det B_p| = |D|$, so it identifies the original forcing class with $[C]$. This completes the comparison to the unchanged integral equations. Positivity and the original valuations still have to be established on the reconstructed full coordinates to obtain an actual positive cycle.

The map on scalar cokernels in (32) is surjective; its kernel is the image of the ideal $(v-2, 3u-2)$ in $R/(a_p)$. This follows from the surjectivity and kernel of α and the fact that $\alpha(a_p)$ is D times a unit. Thus its coefficient loss is identified, not suppressed.

There is also an exact obstruction to evaluating **every** completed scalar in this arithmetic target. Choose a prime ℓ dividing $6|D|+1$; then ℓ is odd, different from 3, and does not divide D . For $k = (3+\ell)/2$, the series ring makes $1 - ku$ invertible, but α sends it to $-\ell/3$, which is not a unit of F_D : an inverse $b/(3^r D^s)$ would force $\ell b = -3^{r+1} D^s$, contradicting coprimality. A unital ring homomorphism must send units to units. Therefore α has no extension $\widehat{R} \rightarrow F_D$. The rational subcomplex (30) is the explicitly successful replacement: it contains the full cyclic defect, admits (32), and returns the exact original integer class (35).

For $(1, 3)$, the returned residue is $5/7$ modulo Λ_3 , corresponding to the nonzero integral class [5]. For $(2, 2)$ it is $7/7$, zero modulo Λ_3 , with the source word and original fixed-point primitive retained. For $(1, 1, 2, 4, 3)$ it is $475/1805 = 5/19$, of order 19, preserving the original failure to lift from modulus 19 to modulus 361. The code checks (31), (34), the original mark, and the denominator order on finite word fixtures. The formulas prove the statement for every nonempty original word.

7.8 8. Controls that test the infinite information

7.8.1 A genuine cycle for an explicitly different forcing

Use $T_-(n) = (3n-1)/2^{\nu_2(3n-1)}$. On positive odd n , its comparison with the signed original map is $T_-(n) = -T_+(-n)$, because $3(-n)+1 = -(3n-1)$ and the valuations agree. The control cycle is

$$5 \xrightarrow{1} 7 \xrightarrow{2} 5.$$

Retain $S_- V_5 = V_7$ and $S_- V_7 = v V_5$. Its primitive and defect are

$$G_- V_5 = \frac{V_5 + u V_7}{1 - u^2 v}, \quad \eta_-[V_5] = [G_- V_5] \pmod{M} \neq 0. \quad (27)$$

Multiplying the denominator proves the first equality coefficientwise. Infinitely many nonzero coefficients prove the second. The component is $R/(1 - u^2 v)$ under $V_5 \mapsto 1$, $V_7 \mapsto uv$; the inverse is generated by V_5 . This is an explicit nonzero module whose every u -adic quotient vanishes by (10). It is not a positive $3n+1$ counterexample.

7.8.2 An explicitly declared ray control

Use vertices $x_j = j + 2$, transition $x_j \mapsto x_{j+1}$, and exponent label 1, as a separate test operator. Its exact primitive is $\sum_{j \geq 0} u^j V_{x_j}$. Its source cohomology is $\Lambda[u, u^{-1}]$, by $V_{x_j} \mapsto u^{-j}$; the inverse sends u^{-j} to $[V_{x_j}]$. Positive u powers follow from the R-module structure, and the original relations $V_{x_j} = u V_{x_{j+1}}$ verify both compositions.

The quotient class of its primitive is nonzero and has no nonzero polynomial annihilator, through this explicit Laurent module map. Its vertex support grows at every coefficient degree. This control is not described as an observed positive Collatz ray.

7.9 9. No uniform finite clock is available, on an actual infinite family

For each $N \geq 1$, take the original positive integer

$$n_N = 2^{N+1} - 1.$$

For every $0 \leq j \leq N$,

$$T^j(n_N) = 3^j 2^{N+1-j} - 1. \quad (28)$$

Indeed, at $j < N$, its $3x+1$ value is $2(3^{j+1} 2^{N-j} - 1)$, with the parenthesis odd. The actual next exponent is exactly 1. Induction proves (28), and each successive value increases. In particular

$$S^N V_{n_N} = V_{2 \cdot 3^{N-1}} \neq 0. \quad (29)$$

Thus S is not nilpotent: no single exponent kills all original source vectors. The requirement in (24) is local, with a finite exponent for each fixed source, not one common bound for the whole space.

The sources n_N in (28) differ. The infinite all-one word has canonical residues $2^{N+1} - 1$, whose dyadic limit is the signed integer -1 . A single positive n realizing every one of these prefixes would make $n+1$ divisible by all powers of 2, forcing $n=-1$. The infinite prefix intersection therefore has no positive integer source. This is the original residue-tower obstruction already retained in the cycle note, now also furnishing the explicit witnesses (29). Nonempty positive fibres at each finite depth are not assigned a common positive source without their intersection map.

7.10 10. Executable results and next source calculation

`completion.py` implements (4), (13), (16), and the entire basis/decomposition (20)-(21) with integer polynomial coordinates. Every result retains the declared sources, original transition triples, terminal monomial, and mode. Sources outside the original plus map are accepted only through the separately named controls. Certificates are validated by exact reconstruction; a nonzero residual is never called a divergence certificate.

`verify.py` replays 1,677 original path fixtures independently by repeated division, 130 finite kernel presentations, 180 polynomial-forcing comparisons, and 512 original source jets. It tests both controls, actual all-one prefixes through length 512,

carrier-preserving zero cancellation, and false or malformed input rejection. It produces 7,928 named exact checks in the recorded execution. Ordinary and optimized outputs must be byte-identical. These finite tests do not stand in for the infinite proofs (6), (15), or (20).

The next source calculation is now specified on the retained module, rather than on an already acyclic completion: extend the proved original-source coverage of the increasing kernels (18), using complete arithmetic families and their exact S-action; or identify a nonzero original singleton class with a valid infinite certificate. The defect presentation (6) and the untruncated residual (13) must be retained throughout. The present work supplies neither global coverage nor a positive counterexample.

The earlier local `clocked_support_20260914` package is a separately delivered predecessor. Its large database and 678-rise certificate are not silently imported into the newer live anchored branch by this contribution. The definitions and proofs used here are self-contained over the original companion operator. The current remote anchored contribution is preserved unchanged. This separates actual publication from the existence of a local package without deleting either mathematical history.

7.11 11. References and attribution

SZ. Split-Zero research programme, `formal/splitzero/DERIVED_MATHEMATICS.md`, Sections 1–4, pinned above. The support reconstruction and killed-class kernel construction are source-derived. This note contains no new Lean execution or assertion that the infinite Collatz application was already formalized there.

W. Kokuno Yumeto, Collatz research companion, `research_companion/chapters/01_weighted_path_and_finite_residue_actions.tex`, same repository, pinned blob above. The original clocks, transfer and completed inverse are taken with their existing attribution. Odd-return exponent a expands to one shortened odd step and $a-1$ shortened even steps; parsing at odd returns is the inverse.

C. `research_program/cycle_relative_cohomology_20260914/note.md` and `residual_splice_20260914/`, preserved in the current source. Original relative graph, source towers, integral comparison maps, and descent splicing.

A. `research_program/anchored_defect_20260914/note.md`, blob `b9b5d07514676eb2b33989e301029910602c2e99`, current source `ce8f784da8fb48aa19065c036cd815c05628cec4`. The retrieved Sections 1–3 supply the live support/forcing conventions. Its numerical results were not used as premises in this note.

H. The Stacks Project, Lemma 12.13.12, long exact cohomology sequence: <https://stacks.math.columbia.edu/tag/0117>. Theorem 1 is an explicit instance with a constructed inverse.

L. The Stacks Project, Section 10.8 and Lemma 10.8.8, filtered colimits of modules: <https://stacks.math.columbia.edu/tag/00DB>. Theorem 2 proves its required representatives and inverse directly.

P. The Stacks Project, Section 10.96, completion and its comparison maps: <https://stacks.math.columbia.edu/tag/00M9>. The actual source completion and nonseparated cohomology in Sections 2 and 4 are computed directly, with no finite-generation hypothesis imposed on M .

The contribution is a proof-bearing application and implementation of the source's support-preserving method and standard exact-sequence algebra. No global priority claim, independent external review, full Collatz proof, or automatic continuation is

asserted.

Chapter 8

A structural Split-Zero restatement of Collatz

Source: repository/collatz_reconstruction/research_program/supported_structure_20260915/note.md.

15 September 2026. Source checkpoint: KokunoYumeto/collatz-workbench, PR #2, commit 6b3ded8880946b2c3140a9d79f1aa1d658d36b8c.

This note answers a structural question: what are the complete modules behind the supported zeros, and which arithmetic assertions would remove every nonbase obstruction? It retains the original source, the integral lattice, both clocks, original word order, maps, and representative fibres. It does not claim a proof or disproof of Collatz, a new numerical cycle bound, an independently reviewed theorem, or a Lean execution.

The completion-defect source already constructs the weighted complex, the formal inverse, its defect isomorphism, the finite endpoint-kernel basis, and component coordinates. The additions here are the maximal-one-run comparison; a proof that both clocks are invertible on the original cohomology; the resulting full Laurent-module classification; its torsion/generic-fibre decomposition; the positive-cone interpretation of the death filtration; an absolute/base-relative exact sequence; and a proof-obligation and certificate design. Component index sets below mean the actual components of the original map. They may be empty. They are not hypothetical replacement graphs.

8.1 1. The actual source and the history presentation

Let

$$X = \{1, 3, 5, \dots\}, \quad T(n) = (3n + 1)/2^{a(n)}, \quad a(n) = \nu_2(3n + 1),$$

$$\Lambda = \mathbb{Z}[v], \quad R = \Lambda[u], \quad A = \Lambda^{(X \setminus \{1\})}, \quad M = A[u].$$

The superscript $()$ denotes finite support. Its basis vector V_n retains the original positive integer. Define

$$SV_n = v^{a(n)-1}V_{T(n)}, \quad V_1 = 0, \quad d = I - uS, \quad K = [M \xrightarrow{d} M], \quad Q = \text{coker } d. \quad (1)$$

The complex is in cochain degrees 0 and 1. In its source degree the basis is also

written E_n , and the identification $E_n \leftrightarrow V_n$ is explicit and invertible. Thus $dE_n = V_n - uv^{a(n)-1}V_{T(n)}$. An exponent a expands to one shortened odd step and $a-1$ shortened even steps. Parsing at odd endpoints is its inverse. Hence u and v retain the companion's original clocks.

A useful retained object is the diagram consisting of M, dM, Q , the original labelled basis, all finite path words, $\ker S^N$, the completion inclusion, and its inverse primitives. Q alone is not the entire history. For example equal source classes have representatives differing by d of an original finite vector. Keeping that vector and d is how the relation is retained; the quotient element alone does not determine it uniquely.

For each seed label $F \subseteq X \setminus \{1\}$, form the original forward carrier $\Omega_F = \{1\} \cup \{T^j(n) : n \in F, j \geq 0\}$. Use (1) on this carrier. Inclusions of seed sets act by the original basis inclusions, not by renaming vertices. The Split-Zero total is the disjoint union over F . A supported scalar zero sends (F, z) to $(F, 0)$, while the external absent scalar sends it to $(\emptyset, 0)$. Addition transports both terms into the union label. These are precisely the source's reconstruction formulas [SZ]. This note uses the full modules and maps in each fibre, not only their labels.

8.2 2. What the zero in the completed complex means

Let $\widehat{M} = A[[u]]$, coefficientwise finite in the original vertex basis. The predecessor's inverse is

$$Gq = \sum_{j \geq 0} u^j S^j q, \quad \hat{d}G = G\hat{d} = I. \quad (2)$$

Every coefficient involves a finite sum. The original kernel of d is zero as well: comparing the lowest u coefficient of $df=0$ successively gives every coefficient of f equal to zero. Surjectivity on M is not inferred from (2).

With $D_\infty = \widehat{M}/M$ and the induced $\bar{d}$, the original cohomology is recovered by

$$Q \xrightarrow{\eta} \ker \bar{d}, \quad [q] \mapsto [Gq] \bmod M, \quad [g] \xrightarrow{\delta} [\hat{d}g] \bmod dM. \quad (3)$$

Changing q by df changes Gq by f ; changing g by f changes its differential by df . Equations (2) prove both inverse identities. These maps commute with every seed inclusion. Thus the fibre of the completed homology map over its supported zero is the full original Q_F , explicitly realized by (3). The class is not reconstructed from the zero symbol alone: the specified source, relation space and comparison kernel are retained.

For a singleton, before arrival at 1,

$$S^j V_n = v^{E_j(n)} V_{T^j(n)}, \quad E_j(n) = \sum_{i=0}^{j-1} (a(T^i(n)) - 1).$$

Consequently the following are exact equivalent statements:

$$[V_n] = 0 \text{ in } Q \iff GV_n \in M \iff S^N V_n = 0 \text{ for some } N \iff T^N(n) = 1 \text{ for some } N. \quad (4)$$

To prove the middle step directly, the series coefficients are distinct u degrees, with one nonzero original singleton monomial at every surviving degree. They vanish

eventually exactly on arrival at 1. Equation (3) proves the first step. Its polynomial primitive is the uniquely determined original weighted path. No signed alternate primitive can shorten a genuinely infinite singleton series, by injectivity in (2).

8.3 3. An arithmetic lemma: maximal strings of exponent 1

8.3.1 Proposition 1 (exact block, inverse parameters, and retained clocks)

For every positive odd integer n , define

$$r = \nu_2(n+1) - 1, \quad t = (n+1)/2^{r+1}.$$

Then $r \geq 0$, t is positive odd, and $n = 2^{r+1}t - 1$. These are mutually inverse parameter maps between positive odd n and pairs (r, t) with r nonnegative and t positive odd. The first r exponents are exactly 1, with

$$T^j(n) = 3^j 2^{r+1-j} t - 1 \quad (0 \leq j \leq r).$$

The next exponent and endpoint are

$$b = 1 + \nu_2(3^{r+1}t - 1) \geq 2, \quad y = \frac{3^{r+1}t - 1}{2^{b-1}}. \quad (5)$$

Thus the complete block is $(1)^r(b)$; it has $r+1$ odd returns, total exponent $r+b$, and weight

$$W = u^{r+1}v^{b-1}.$$

Proof. For $j < r$, the displayed value plus 1 is divisible by four. Hence it is 3 modulo four, and its next valuation is exactly 1. Applying $(3x+1)/2$ gives the next displayed value. At $j=r$ its value plus 1 is twice an odd integer, so the value is 1 modulo four. Its $3x+1$ is $2(3^{r+1}t - 1)$, whose inner factor is positive even. This proves (5), including the exact valuation. The parameter inverse follows from the unique 2-adic valuation of $n+1$. For $n=1$, $(r,t,b,y)=(0,1,2,1)$; its original fixed edge remains explicit before the relative quotient. QED.

For $n > 1$ let P be the original weighted edge chain of this block. Its telescoping boundary is

$$dP = V_n - u^{r+1}v^{b-1}V_y. \quad (6)$$

Its exact strict-descent test is

$$y < n \iff (2^{b+r} - 3^{r+1})t > 2^{b-1} - 1. \quad (7)$$

This is obtained by multiplying the original endpoint inequality by the positive integer 2^{b-1} and substituting the inverse source formula. No affine term is omitted. Grouping an orbit into these maximal blocks retains (r,b) , t , every intermediate integer in (5), and the edge expansion. Parsing the original exponent list into maximal strings of ones followed by one exponent at least two is its inverse.

8.3.2 Corollary 1 (both clocks are cofinal on every infinite nonbase orbit)

Along every infinite actual orbit remaining above 1, the odd-return count j and the even-step count E_j both tend to infinity. Indeed bounded nondecreasing integer E_j would be eventually constant, so all later exponents would be 1. Proposition 1 proves a finite end to that string at its actual positive starting integer. Equivalently, an infinite all-one string would force $n+1$ divisible by every power of two. This argument proves cofinality and does not supply an upper bound for the total stopping time.

8.4 4. Inverting both clocks loses no original cohomology

Put $L = \mathbb{Z}[u, u^{-1}, v, v^{-1}]$. The assertion here concerns Q , not M : M and all original primitives remain unlocalized.

8.4.1 Theorem 1 (explicit Laurent structure on Q)

Multiplication by u and by v are bijections on Q . The canonical map

$$Q \longrightarrow L \otimes_R Q$$

is therefore an isomorphism.

Proof and inverse. The relation $[q] = u[Sq]$ makes the induced S the inverse of multiplication by u . For v -surjectivity, (6) gives

$$[V_n] = v[u^{r+1}v^{b-2}V_y], \quad (8)$$

where $b \geq 2$. This defines a v -preimage for every original generator and hence for every polynomial class.

For injectivity, write $vq = df$. Reduce modulo v . In the polynomial module over $\mathbb{Z}[u]$, the reduced map is $I - uS_0$; its lowest-coefficient recursion is injective. Thus $\bar{f} = 0$, so $f = vf_1$. Cancelling v in the original free module gives $q = df_1$. Hence multiplication by v has zero kernel on Q . Equation (8) specifies its unique inverse on every original singleton, independent of representative after passage to Q . This inverse commutes with the original R -action by uniqueness.

The inverse of the localization map sends $q/(u^a v^b)$ to $\bar{S}^a J_v^b(q)$, where J_v is (8). These commuting inverses prove the fraction relations and both compositions. QED.

The corresponding statement holds before relativity, with $n=1$ giving $v^{-1}[V_1] = u[V_1]$. This is a proved comparison, not a silent replacement of the finite source by Laurent coefficients.

8.5 5. Complete component modules, not only a vanishing predicate

Partition the actual functional graph into weak components. Vertices in one component have forward paths that meet: adjacency gives that property and finite chains of adjacencies preserve it. A component either meets 1, contains one other primitive directed cycle, or has an injective forward ray and no cycle. A repetition of a forward value gives the cycle; distinct cycles cannot have meeting forward paths. These observations prove the exhaustive partition.

Write $\mathcal{C}_*$ for actual primitive nonbase cycle components and $\mathcal{B}_\infty$ for actual components with no periodic vertex. These symbols do not assert that either set has members.

8.5.1 Theorem 2 (classification with explicit maps)

As an L-module, and hence through Theorem 1 as its original R-module,

$$Q \cong \bigoplus_{C \in \mathcal{C}_*} L/(1 - u^{m_C} v^{e_C}) \oplus \bigoplus_{B \in \mathcal{B}_\infty} L, \quad e_C = A_C - m_C. \quad (9)$$

The component meeting 1 has relative cohomology zero, while its original source vectors and finite contraction columns remain retained. On every actual cycle, $m_C \geq 1$ and $e_C \geq 1$.

Proof with component maps.

Base component. For each of its vertices take the actual first-arrival path to 1. Its finite weighted chain P_n obeys $dP_n = V_n$ and $P_n = E_n + w_n P_{\{T(n)\}}$. These two identities give both contraction equations on the original basis. Thus its relative complex is contractible with specified finite columns, not an absent source module.

Cycle component. Choose one original cycle vertex c , retaining that choice. For n other than c let P_n be its first path to c and w_n its weight; take $P_c = 0, w_c = 1$. Send $[V_n]$ to $[w_n]$ in $R/(1 - W_C)$; the inverse sends 1 to $[V_c]$. Every original edge relation is respected. At c it becomes exactly $1 - W_C$; all other edges telescope along their first path. These calculations prove both inverse compositions and the exact relation ideal.

There is also a full chain comparison. The scalar complex is $[R \xrightarrow{1 - W_C} R]$. The target-degree projection sends V_n to w_n . The source-degree projection sends E_c to 1 and all other original E_n to zero. Its section sends the scalar target generator to V_c and the source generator to the full weighted primitive cycle chain. The homotopy sends V_n to P_n . The path recurrence gives $dH = I - \psi_1 \phi_1$ and $Hd = I - \psi_0 \phi_0$, including $HdE_c = E_c - P_C$. All columns are finite despite arbitrarily many incoming vertices.

Corollary 1, or the same impossible all-one argument on a cycle, gives $e_C > 0$. In $R/(1 - u^{m_C} v^{e_C})$, u and v already have inverses $u^{m_C-1} v^{e_C}$ and $u^{m_C} v^{e_C-1}$. Hence the map to $L/(1 - u^{m_C} v^{e_C})$ is an isomorphism with these inverse actions. This proves the displayed summand without losing an integral class.

Nonperiodic component. Choose an original forward ray $x_0, x_1, \dots$, with prefix weights $W_j = u^j v^{E_j}$. A vertex n meeting x_j along a path of weight w maps to w/W_j . A later meeting multiplies numerator and denominator by the identical intervening path monomial, proving independence. The inverse sends f/W_j to $f[V_{x_j}]$. Original edge relations give both compositions, realizing the module as $\bigcup_j W_j^{-1} R$.

By Corollary 1, both j and E_j are unbounded and nondecreasing. For every Laurent monomial $u^a v^b$, select a finite j with $j + a \geq 0$ and $E_j + b \geq 0$. Then

$$u^a v^b = \frac{u^{a+j} v^{b+E_j}}{W_j}. \quad (10)$$

This proves that the union is the whole L and supplies the inverse with an actual finite meeting index. Equal choices of j are compared by the original path monomials.

Every original vertex maps to a unit Laurent monomial and is nonzero.

Finally d never mixes different actual components; every finite vector meets finitely many of them. Splitting its coefficients by component gives the direct-sum map in (9), and adding their representatives gives its inverse. QED.

An injective positive-integer ray eventually leaves every bounded interval, since that interval has only finitely many original vertices. The theorem does not claim that such a component exists; it identifies precisely the module carried by one in the actual component decomposition.

8.6 6. Two complementary retained obstructions

Let $\mathbb{K} = \mathbb{Q}(u, v)$, the fraction field of L , and retain the canonical map

$$\ell : Q \longrightarrow \mathbb{K} \otimes_L Q, \quad q \mapsto 1 \otimes q.$$

Define $\mathcal{T} = \ker \ell$ and $\mathcal{F} = \mathbb{K} \otimes_L Q$. An element maps to zero exactly when some nonzero element of L annihilates it: this is the fraction equivalence relation. This standard torsion-kernel principle is recorded in [LOC]; no finite-generation assumption is needed.

Applying (9) gives the complete descriptions

$$\mathcal{T} \cong \bigoplus_{C \in \mathcal{C}_*} L/(1 - W_C), \quad Q/\mathcal{T} \cong \bigoplus_{B \in \mathcal{B}_\infty} L, \quad \mathcal{F} \cong \bigoplus_{B \in \mathcal{B}_\infty} \mathbb{K}. \quad (11)$$

On a cycle, $1 - W_C$ is nonzero in L and becomes a unit in the field, killing exactly that component. On a ray, L includes injectively in its fraction field. The kernel and image statements follow component by component, with finite support across components. The sequence $0 \rightarrow \mathcal{T} \rightarrow Q \rightarrow Q/\mathcal{T} \rightarrow 0$ has the displayed component splitting. The kernel is canonical; the coordinates for each rank-one ray retain the chosen original ray.

For an original singleton the annihilator ideal in L is precisely

$$\text{Ann}_L[V_n] = \begin{cases} L, & n \text{ reaches } 1, \\ (1 - W_C), & n \text{ belongs to the nonbase cycle component } C, \\ (0), & n \text{ belongs to a component in } \mathcal{B}_\infty. \end{cases} \quad (12)$$

The first statement uses (4). In the second, its image is a unit monomial times 1 in $L/(1 - W_C)$, so the annihilator is exactly that ideal. In the third, its image is a unit in the domain L . This is an ideal-valued classification rather than a binary flag.

Polynomial torsion here means annihilation by a nonzero polynomial or Laurent polynomial; it is not an assertion of additive integer torsion. In fact $L/(1 - u^m v^e) \cong \mathbb{Z}[\mathbb{Z}^2 / \langle (m, e) \rangle]$: send each Laurent monomial to its exponent coset, and send the coset basis back to its residue monomial. The relation identifies exactly those cosets, proving inverse ring maps. The group ring is free as an abelian group on its cosets. The ray module L is likewise free abelian. Thus Q has no additive integer torsion, and Q embeds in $\mathbb{Q} \otimes_{\mathbb{Z}} Q$. The integer lattice is nevertheless retained, especially for the different, explicitly related integral forcing complex in Section 9.

Every summand in (11) also maps to zero under the original formal completion (2). The retained localization kernel distinguishes periodic torsion from nonperiodic persistence; the completion-defect isomorphism (3) recovers both. Neither projection

is used without its actual kernel and source diagram.

8.7 7. The distinguished trivial cycle before taking the relative quotient

Use the absolute module $M_{\text{abs}} = R^{(X)}$ and the unchanged absolute edge differential. Its original fixed-loop subcomplex is

$$K_1 = [RE_1 \xrightarrow{1-uv} RV_1].$$

The quotient complex is exactly K in (1). Every I-uS complex is injective in source degree by coefficient recursion. The degreewise exact sequence of these original complexes consequently gives

$$\boxed{0 \longrightarrow R/(1-uv) \xrightarrow{1 \mapsto [V_1]} Q_{\text{abs}} \longrightarrow Q \longrightarrow 0.} \quad (13)$$

One may also verify injectivity and exactness directly from the component maps in Theorem 2. The actual component meeting 1 is $R/(1-uv)$, and the other components are unchanged. Its map sends an original source to its first-arrival path weight times $[V_1]$, with its entire finite path retained.

The same abstract binomial type occurs for the trivial cycle and other cycles. What singles out the trivial one is the specified actual fixed vertex, the original equation $4x-3x=1$ forcing $x=1$, its actual subcomplex, and all finite incoming histories. No abstract zero or unmarked spectrum makes that selection.

8.8 8. What distinguishes a supported zero caused by merging from one caused by arrival

For a finite original source set F , set $K_N(F) = \ker(S^N|_{\Lambda(F)})$. Retain every endpoint $y_n = T^N(n)$ and exponent $E_N(n)$, stopping at first arrival at 1. At each nonbase endpoint y select an original source n_y minimizing its exponent, with that choice recorded. The predecessor proves the exact basis

$$V_n \ (y_n = 1), \quad V_n - v^{E_N(n)-E_N(n_y)} V_{n_y} \ (y_n = y > 1, n \neq n_y). \quad (14)$$

The inverse decomposition subtracts each nonselected pivot and collects the selected coefficient at the original y . This is a direct sum of the absorbed coordinate submodule and the specified coalescence-relation submodule; the complement maps to the unchanged original image generators $v^{E_N(n_y)} V_{n_y}$. Thus two different mechanisms of supported zero coexist in a single actual kernel.

Let $A_+ = \mathbb{N}[v]^{(X \setminus \{1\})}$, the cone of finite vectors with nonnegative polynomial coefficients. Then

$$\boxed{\ker S^N \cap A_+ = \left\{ \sum c_n(v) V_n : c_n \in \mathbb{N}[v], c_n \neq 0 \Rightarrow T^N(n) = 1 \right\}.} \quad (15)$$

Proof. At every surviving endpoint the image coefficient is a sum of nonnegative polynomials times nonzero monomials. It is zero exactly when no nonzero input coefficient contributes. Distinct endpoint basis vectors do not cancel. The reverse implication is immediate from $V_1=0$. QED.

For $F=\{3,5,13\}$, $S V_3=V_5$, $S V_{13}=v^2 V_5$ and $S V_5=0$. Hence the first kernel is $\Lambda V_5 \oplus \Lambda(V_{13} - v^2 V_5)$. The second generator retains a genuine merger relation, but neither V_3 nor V_{13} has vanished after one return. At two returns all three source generators vanish. Formula (15) prevents a large signed kernel from being counted as singleton absorption.

8.9 9. The cycle word must pass its original arithmetic realization map

The binomial W_C in (9) retains both total clocks, not their complete order. The original ordered word and its forcing polynomial must remain marked. Put

$$W_p = u^m v^{A-m}, \quad N_p = \sum_{j=0}^{m-1} u^j v^{A_j-j}.$$

Write $b_j = A_j - j$ for $j < m$ and read b_m from the v -degree of W_p . Then $a_i = 1 + b_i - b_{i-1}$ recovers the word. Nonnegative nondecreasing b_j with $b_0=0$ give its exact image conditions. These differences and prefix sums are inverse operations. The raw N_p , not an unspecified quotient representative of it, is retained.

On the finite position circle the weighted graph differential has columns $E_i \mapsto V_i - w_i V_{i+1}$. Its algebraic dual has

$$(J_p h)_i = h_i - u v^{a_i-1} h_{i+1}.$$

Thus the transpose is the explicit finite free duality map relating graph positions to this equation; its forcing is specified separately. Under $\alpha : R \rightarrow \mathbb{Z}[1/3]$, $u \mapsto 2/3, v \mapsto 2$,

$$\alpha(J_p) = -B_p/3, \quad (B_p x)_i = 2^{a_i} x_{i+1} - 3x_i,$$

$$D = 2^A - 3^m, \quad C = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}, \quad \alpha(1 - W_p) = -D/3^m, \quad \alpha(N_p) = C/3^{m-1}. \quad (16)$$

Consequently the original equation $B_{px}=1$ is the forced equation $\alpha(J_p)x = -1/3$, and its scalar primitive is $x_1=C/D$. The integral cokernel is $\mathbb{Z}/D\mathbb{Z}$ with marked forcing $[C]$, via the original telescoping functional. The predecessor's exact localization inverse and the rational-subcomplex maps retain that integral class; they are not replaced by evaluation of arbitrary formal series.

Every positive integral cycle has $D>0$ and $[C]=0$. Both the trivial cycle and any other integral cycle obey that same marked vanishing. The unique primitive distinguishes them: $x=1$ for the trivial word, and every $x_i \geq 3$ on a nonbase cycle. Under the explicitly invertible affine coordinate $y=x-1$,

$$B_p y = (4 - 2^{a_i})_i,$$

with unchanged integral forcing class because its difference is $-B_{p1}$. This is a bijection of forced solution sets, not a claim that translation is linear.

For $p=(1,3)$ and $p'=(2,2)$, both $W=u^2 v^2$ and $D=7$, but $N_p=1+u$, $N_{p'}=1+uv$, $C=5$ versus 7. The former has a nonintegral rational cycle; the latter is the repeated actual fixed orbit. A proof programme must reject unrealized word cycles before assigning them an actual component in (9). Repeated words retain the covering

factor $1 + W + \dots + W^{r-1}$ and their actual r -fold edge expansion.

8.10 10. The refined restatement

The following equivalences are proved by (3), (4), (9), (11), and (13):

$$\begin{aligned}
 &\text{Every original positive odd integer reaches 1;} \\
 &\text{the canonical fixed-loop map } R/(1 - uv) \rightarrow Q_{\text{abs}} \text{ is an isomorphism;} \\
 &\mathcal{T} = 0 \text{ and } \mathcal{F} = 0; \\
 &\ker(\bar{d} : A[[u]]/M \rightarrow A[[u]]/M) = 0; \\
 &A = \bigcup_{N \geq 0} \ker S^N.
 \end{aligned} \tag{17}$$

Each line is read together with its original labelled source and maps. In particular a zero relative module does not erase the base basin's vectors, their merger kernels, clocks, or finite primitives. The isomorphism assertion in (17), applied to the exact sequence (13), says that the only recurrent module in the absolute object is the actual fixed-loop module. The exact sequence alone does not assert that its final quotient Q vanishes. Lines (11) separate periodic polynomial torsion from generic nonperiodic persistence, without discarding either.

There is no common finite N with $S^N = 0$. For $N \geq 1$, the original source $n_N = 2^{N+1} - 1$ has $T^j(n_N) = 3^j 2^{N+1-j} - 1$ for $0 \leq j \leq N$, so $S^N V_{n_N} = V_{2 \cdot 3^{N-1}} \neq 0$. The sources vary with N . This proves why the correct exhaustion in (17) is local rather than one finite global clock. The proof is Proposition 1 specialized to $t=1$.

8.11 11. A proof programme: established lemmas and outstanding arithmetic targets

This section is a dependency design, not a collection of assumed conditional theorems. The algebraic statements in Sections 1-10 have proofs. The following arithmetic targets remain unproved here.

8.11.1 P0. Preserve the presented source throughout

Already available: exact original source cylinders, arbitrary-exponent strata, original block expansions, integral forcing inverses, clocked finite primitives, completion-defect recovery, the positive cone (15), and the component classification (9). The kernel checker must verify these interfaces and retain source hashes, failed candidates, and original maps. A support label by itself is not an adequate source certificate.

8.11.2 P1. Eliminate the periodic torsion on the actual positive source

Target: determine the full positive integral solutions of $B_{px}=1$ for every primitive original word. The required conclusion is that only $p=(2)$ with $x=1$ is primitive. Retain repetitions explicitly afterwards.

Subtasks: prove symbolic exclusions for complete families of original gap words, transport every congruence test through the integral block inverse, and combine

the marked class [C], positivity, minimum placement, the original affine inequalities, and all prime powers. A prime-only test does not establish divisibility by D; a zero marked class does not identify x with 1. A surviving candidate must yield original positive integer coordinates and an independently replayed actual cycle chain. Finite windows closed by descent remain valid, but they are not an all-word theorem until their coverage is proved.

8.11.3 P2. Eliminate the generic nonperiodic component module

Target: establish $\mathcal{F} = 0$ on the actual source. By (11) this is exactly the absence of a component whose module is L and whose every original singleton remains nonzero over the fraction field.

An analytic completion that kills all Q has already failed to distinguish these classes; (3) gives its full successful recovery. A useful new comparison must therefore carry this retained defect, not just show another completed inverse. Any norm, trace or asymptotic estimate must be evaluated on the original labelled residuals, with its kernel, topology, scalar action, and return map specified. No such global estimate is supplied here.

The direct arithmetic route is the survivor-source tree in P3. A no-descent minimum of a nonbase component has every iterate at least its original minimum; all original affine inequalities therefore remain attached. A nonperiodic component has such a minimum by well ordering of positive integers. This makes minimum-source analysis address the generic obstruction as well as cycles.

8.11.4 P3. Prove arithmetic exhaustion, not only finite-jet exhaustion

A finite word p has original data L,U,C, legal residue r_p modulo 2U, and the actual affine map $(Ln+C)/U$. Its no-descent source is exactly

$$\mathcal{S}_p = \{n = r_p + 2Ut \geq 3 : t \in \mathbb{Z}, D_j n \leq C_j \ (1 \leq j \leq |p|)\}. \quad (18)$$

For $D_j \leq 0$ there is no upper restriction; for $D_j > 0$ it is the explicit upper bound $n \leq C_j/D_j$. Appending a gives $L'=3L, U'=2^a U, C'=3C+U$, the original congruence lift, and set inclusion $\mathcal{S}_{pa} \subseteq \mathcal{S}_p$. Empty sets remain declared supported nodes with their equations, not deleted records. The prior unbounded-tail construction represents all large a by one exact progression plus its valuation map; Proposition 1 gives a second full block dictionary with (5)-(7).

Target: prove that every compatible infinite sequence of these no-descent source sets has empty intersection in the ordinary positive integers. The finitely branching reformulation produced by the prior whole-tail decomposition retains all integer parameters and exceptional boxes. An infinite path of nonempty finite-prefix cylinders alone is not a positive integer witness.

A concrete subtarget is a height-escape statement for the canonical residues of all surviving paths. Those residues obey $r_{j+1} = r_j + 2^{A_j+1} b_j$, $b_j \geq 0$. A bounded sequence is eventually constant; then its value realizes every prefix by the original cylinder theorem. Conversely the residues of a fixed positive realizing integer become that integer once the modulus exceeds it. Thus ordinary-integer support is exactly the eventual-constant case, including the source inequalities (18). An all-one branch is already excluded by $r_j = 2^{\{j+1\}-1}$. The remaining nonconstant-word and first-coefficient-crossing families require new arithmetic control; the finite boxes cannot

be discarded merely because each individual box is finite.

This target addresses every nonbase component: its least integer belongs to all of the source sets (18) for its itinerary. Its first exponent is 1, since $T(n)1$ whenever $a(n) \geq 2$. This is a proved initial restriction, not an assumed distribution of exponents.

8.11.5 P4. Assemble a finite-column, positive contraction

Target: construct for each original n a finite polynomial column $h(V_n)$, with nonnegative monomial path coefficients, satisfying $dh=I$ and $hd=I$ on original finite-support modules. The exact predecessor splice identities permit an accepted actual descent to attach at an old retained root, with the old terminal clock multiplied into the new path. The extra homotopy lies in the old projected edge module. At the explicit coefficient specialization $u=v=1$, the original unweighted closed cycle chains are fixed literally. Before that specialization the weighted cohomology and its cycle relation are retained by the actual chain-homotopy equivalence; a weighted period has boundary $(1-W_C)V_c$ and is not being called a zero-boundary chain. Thus new descent coverage cannot accidentally delete a genuine counterexample.

The original source number itself is a well-founded ranking for strict-descent splices. One must prove applicability on every residual source, or supply another explicitly defined well-founded ranking with a proved transition decrease. Calling the stopping time a ranking does not prove that it is finite. This is the still-missing global applicability assertion, not an unrecorded premise.

8.12 12. The program to build

Separate a small exact certificate checker from a potentially elaborate search engine. The search engine may propose words, parameter families, period obstructions, or survivor invariants; no search timeout is a mathematical certificate.

Required records:

- `source_state`: original seed label, exact integer or parameter domain, ordered exponent word, every affine prefix, integral lattice, original and weighted boundary, terminal clock, and next-step partition with all empty and exceptional cases.
- `finite_primitive`: a finite weighted edge column; the checker proves its full untruncated boundary. A symbolic family also needs its exact parameter-domain proof and word/valuation inverse. A point sample never validates its universal quantifier.
- `cycle_realization`: original primitive word, D, C , every rational primitive coordinate, verified positive integer valuations, repetition map, and actual relative period chain. Failure modes are marked forcing, positivity, valuation, and period failures separately.
- `survivor_certificate`: a fully proved original-source invariant set or exact infinite-support obstruction, excluding 1 under all its actual transitions. Finite residual jets are not admitted as this type.
- `projection_record`: exact map and coefficient domain, killed-representative module, old relation submodule, inverse/comparison maps, and original source support. A reported supported zero cites this record.

Useful computation priorities are complete symbolic arithmetic families, increasing positive-coordinate absorption, and retained affine exception boxes. Kernel dimension alone is insufficient because of (14)-(15). Generic rank and periodic torsion are different outputs of the same explicitly computed module, not substitute success criteria.

Completion-defect CI already checks the prior retained maps. This increment adds finite tests for (5)-(8), the full cycle chain comparison in Theorem 2, Laurent coset and ray-coordinate inverses, ordered forcing recovery, the positive-cone test, and false-certificate rejection. These tests are executable regression evidence; they do not establish P1-P4 globally. The notes supply the general algebraic proofs, not the number of test cases.

8.13 13. Sources and attribution

[SZ] KokunoYumeto/zeta-function-research-reader, commit 91ed3b7c358a1f444d0d48c292e4a400ca509eff, formal/splitzero/DERIVED_MATHEMATICS.md, blob fe05d32ccd314daac7c59d225c9cac3d4c1bf3d3, Sections 1-4. This is the source of support reconstruction and the killed-representative quotient. No new Lean execution is claimed.

[CD] KokunoYumeto/collatz-workbench, commit 6b3ded8880946b2c3140a9d79f1aa1d658d36b8c, collatz_reconstruction/research_program/completion_defect_20260914/note.md, blob 9f00643cb135acd77defec9e05ad63b632443a50. This supplies (1)-(4), the direct-limit comparison, finite endpoint-kernel basis, and preceding component maps. The uploaded full proof was checked against this Git blob before continuation.

[W] The same repository's research companion 01_weighted_path_and_finite_residue_actions.tex, blob f0142c0d4c237ed068a0762a08c6622df5dbcfef, supplies the original path category, clocks and completed transfer inverse. The older residual-splice and integral-block notes remain attached through [CD].

[LOC] The Stacks Project, Lemma 15.22.2, <https://stacks.math.columbia.edu/tag/0537> : torsion as the kernel of the fraction-field map. Section 15.22, <https://stacks.math.columbia.edu/tag/0549> . Definition 10.9.6, <https://stacks.math.columbia.edu/tag/07JZ> . The relevant fractions and kernels are also proved directly above.

Existing operator formulations are not assigned priority to this work: M. Neklyudov, *Functional analysis approach to the Collatz conjecture*, arXiv:2106.11859; T. Mori, *Application of Operator Theory for the Collatz Conjecture*, arXiv:2411.08084. Their abstracts were inspected for this attribution boundary, not used as proof premises. No isomorphism with their analytic operator spaces is asserted or needed for the maps proved here.

Chapter 9

Intrinsic Split-Zero support and the integral first-jet defect of Collatz

Source: repository/collatz_reconstruction/research_program/intrinsic_zero_firstjet_20260915/notes.md.

15 September 2026. Source checkpoint: KokunoYumeto/collatz-workbench, PR #2, 2a77e3bc9435f181bc4c9c3957224a7d93b4e887. This is an additive continuation. It corrects a terminology overlap, not the previously proved equations: the user’s informal account of retained information at a supported zero is not a definition of an exponent-word history in the weighted-automata companion.

9.1 1. The actual foundation and what is retained

The foundational scalar object is the supplied split-zero construction

$$G(R) = R \sqcup \{\tau\}, \quad e = \text{ofR}(0_R) \neq \tau.$$

The supported copy has its original ring addition and multiplication; tau is the semiring additive identity and multiplicative absorber. The archive’s original `SplitZero.lean` states exactly these operations. Its path-specific commit history reaches the archived July 4, 2026 commit; the June 26 date in its directory name is not being asserted as the invention date. The current formalization explicitly cites that archived file and the chapter-14 linear-join-diagram theorem. This is the documented source chain recovered here, not a claim to have recovered all earlier private development.

The exact scalar comparison is

$$G(R) \longrightarrow R \times \mathbb{B}, \quad \tau \mapsto (0, 0), \quad \text{ofR}(r) \mapsto (r, 1),$$

with image $\{(0, 0)\} \cup (R \times \{1\})$ and inverse given by those same cases. Addition in the second coordinate is OR, multiplication is AND. Case substitution proves both operations and inverse identities. Reflection to R alone has zero fibre $\{\tau, e\}$. The separate support coordinate prevents that collapse. Every map to an additively cancellative ring sends e to zero, since $e+e=e$ forces $h(e)+h(e)=h(e)$, and therefore $h(e)=0$.

For a $G(R)$ -semimodule M , intrinsic support is $s(m) = em$. The source theorem identifies the support semilattice with the additive idempotents and the fibre over l with

the R-module

$$M_l = \{m : em = l\},$$

whose zero is l . For $l \leq k$ its original transition is $m \mapsto m + k$. Conversely a coherent diagram V_l, ρ_{lk} reconstructs the disjoint union with

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad e(l, x) = (l, 0), \quad \tau(l, x) = (\perp, 0).$$

These are the source's actual quasi-inverse constructions; they do not require trajectory words. The general theorem permits a nonzero bottom fibre. The application below has zero bottom fibre, stated explicitly.

There is an exact qualification to the phrase "the zero remembers." On any nonzero fibre V_l the map $x \mapsto (l, 0)$ has the whole V_l as its fibre. In particular it sends both 0 and a nonzero x to the same element. That element cannot select a unique prior vector. Keeping the fibre, its relation submodule and a specified comparison retains the space of possible representatives and the operation, not an automatically encoded chronological log. Conversely external τ also has a computable inverse image once its original map is retained; its output alone supplies no non-bottom support. These statements follow from the displayed maps, rather than an assertion that there can be no relation to path history.

An infinite-dimensional fibre or a ring of formal series can be used in this construction, but its scalar/module operations and comparison must be specified. The carrier R disjoint union $\{\tau\}$ itself adds no formal summation rule or numerical infinity to R.

For a two-term cochain map $f: C \rightarrow D$, the source's killed-class module is

$$\{z \in C^1 : f^1 z \in d_D D^0\} / d_C C^0 \cong \ker H^1(f). \quad (1)$$

The forward map takes a representative to its original source class; its inverse takes any representative of that source class. Well-definedness and inverse identities are exactly the original boundary relations. This is the retained structure used below.

The early Rees continuation computes a comparison defect and retains a first variation when specialization makes a boundary-character difference zero. Its particular trace identity and arithmetic estimates are not transported to Collatz by vocabulary. The common operation here is explicit: retain a first-order coefficient and the connecting map of its exact coefficient sequence. The Collatz calculation below is proved from its own differential.

9.2 2. A support diagram of equations, not a sequence of trajectory words

Retain the original odd-return map

$$X = \{1, 3, 5, \dots\}, \quad T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1).$$

For any declared subset E of X, interpret n in E as the actual original equation/edge $e_n : n \rightarrow T(n)$. Put

$$V(E) = E \cup T(E), \quad C_E^0 = \mathbb{Z}^{(E)}, \quad C_E^1 = \mathbb{Z}^{(V(E))}.$$

Parentheses mean finite sums, even when E is infinite. Define

$$J_E e_n = V_n, \quad P_E e_n = V_{T(n)}, \quad d_E = J_E - P_E. \quad (2)$$

This is an absolute complex: the actual fixed vertex and loop at 1 are NOT removed. A declared edge with zero differential, notably e_1 , remains a source basis vector. In finite calculations every target outside E remains a separate frontier vertex.

The support semilattice is $\mathcal{P}(X)$, with union as join and empty set as bottom. For E subset F all maps are literal inclusions on original labelled edges and vertices. They commute with J, P, d . Its finite-subset subdiagram provides executable approximations; its full label X defines the infinite object. Reconstruction by the preceding source formula gives $G(Z)$ -semimodules and internal supported-zero differentials. The bottom fibres are zero. The order of entering the equations into a file has no role in E or these maps.

The earlier seed-carrier diagram assigned to E its forward carrier $\Omega_E = \bigcup_{j \geq 0} T^j(E)$, including 1 where used by that construction. The natural comparison from this equation diagram to that carrier diagram is inclusion on each displayed original edge and vertex. The carrier contains all those labels and their targets; hence this is a cochain map and commutes with support inclusion. It is not an assertion that the finite set of known equations already equals its whole forward carrier.

9.2.1 Exact comparison with the elementary Collatz map

The odd-return source is the existing source of this workbench, not an unrecorded change to the conjecture. Every positive integer has unique form $2^q \cdot n$ with n odd; q initial divisions reach n . An odd-return edge $n \rightarrow T(n)$ expands to the original odd step $n \rightarrow 3n+1$ followed by exactly $a(n)$ divisions. Grouping at successive odd vertices is its inverse on those elementary blocks. In particular the original elementary cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ is the one odd-return loop at 1.

To compare first jets, assign the factor $1+\epsilon$ to each elementary odd step and factor 1 to each division. Let $h(V_y)$ be the finite division chain from y to its odd part, with zero chain for odd y . Send an odd-return edge to its odd elementary edge plus $(1+\epsilon)$ times its following division chain. Send its vertex to the same odd vertex. The reverse map sends any elementary vertex to its odd part, every odd elementary edge to its odd-return edge, and every division edge to zero. Telescoping proves both are chain maps, their composition on the odd-return complex is the identity, and on the elementary complex

$$d_\epsilon h = I - \iota\pi, \quad h d_\epsilon = I - \iota\pi$$

in the respective vertex and edge degrees. For an odd elementary edge the second equation is the negative of $(1+\epsilon)$ times its division tail; for a division edge it is that original edge. These check both cases. All columns are finite. Thus the first-jet invariant is preserved by this specified expansion and retraction. Assigning a new $1+\epsilon$ factor to EVERY elementary division would be a different coefficient assignment, counting $m+A$ instead of m ; it is not used here.

9.3 3. The first-order coefficient map

Use the ring of dual integers

$$\mathbb{D} = \mathbb{Z}[\epsilon]/(\epsilon^2).$$

It has unique coordinates $a+\epsilon b$ and multiplication $(a,b)(c,d) = (ac, ad + bc)$. Its split extension distinguishes $\text{ofR}(\epsilon)$, e , and τ ; $\text{ofR}(\epsilon)^2 = e$, not τ .

Constant reduction $r : \mathbb{D} \rightarrow \mathbb{Z}$, $a+\epsilon b \rightarrow a$, induces $G(r)$. The complete supported-zero fibre is

$$G(r)^{-1}(e) = \{\text{ofR}(\epsilon b) : b \in \mathbb{Z}\}, \quad G(r)^{-1}(\tau) = \{\tau\}. \quad (3)$$

Thus already at the scalar level the projected zero has a specified nontrivial kernel. No trajectory has been stipulated.

On the SAME original graph take

$$K_E = [C_E^0 \xrightarrow{d_E} C_E^1], \quad K_{E,\epsilon} = [C_E^0 \otimes \mathbb{D} \xrightarrow{d_{E,\epsilon}} C_E^1 \otimes \mathbb{D}],$$

$$d_{E,\epsilon} = J_E - (1 + \epsilon)P_E = d_E - \epsilon P_E. \quad (4)$$

For integral original edge vectors b, c ,

$$d_{E,\epsilon}(b + \epsilon c) = d_E b + \epsilon(d_E c - P_E b). \quad (5)$$

Multiplication by epsilon and constant reduction give the degreewise exact sequence

$$0 \longrightarrow K_E \xrightarrow{\epsilon} K_{E,\epsilon} \xrightarrow{r} K_E \longrightarrow 0. \quad (6)$$

Here K_E is a D-module through r . Equation (5) proves both chain equations. These maps commute with every support transition.

There is also an exact comparison with the existing weighted companion, without identifying its term “history” with Split-Zero support. On its ABSOLUTE coefficient ring $R = \mathbb{Z}[u, v]$, take

$$\alpha : R \rightarrow \mathbb{D}, \quad u \mapsto 1 + \epsilon, \quad v \mapsto 1. \quad (7)$$

It sends its edge $V_n - uv^{a(n)-1}V_{T(n)}$ to (4). Its kernel is $(v - 1, (u - 1)^2)$; indeed $\alpha(f) = f(1, 1) + \epsilon \partial_u f(1, 1)$, and successive polynomial remainders in $v-1$ and $u-1$ prove both inclusions. It is surjective, with section of abelian groups $a+\epsilon b \rightarrow a+(u-1)b$. $G(\alpha)$ preserves the two zero labels; its supported-zero fibre is the supported copy of that ideal. The raw original endpoints and exponent $a(n)$ remain in the graph record. Formula (7) is a particular comparison out of the weighted presentation, not the definition of intrinsic support.

The map (7) does not extend to a unital map from all of $\mathbb{Z}[v][[u]]$ to $\mathbb{D}$: $1-u$ is a unit in that source, with its geometric-series inverse, while its prescribed image $-\epsilon$ is not a unit. The same obstruction holds after G because a supported inverse would have to multiply $-\epsilon$ to 1. The successful replacement used here is the finite first-jet ring (7) and the original finite-support complex, not evaluation of that completed inverse.

9.4 4. The actual connecting map and its entire zero fibre

The connecting map of (6) is

$$\beta_E : H^0(K_E) \rightarrow H^1(K_E), \quad c \mapsto [-P_E c], \quad d_E c = 0. \quad (8)$$

Indeed lift c as a constant vector in K_ϵ . Its differential is $-\epsilon P_E c$; removing the ϵ factor gives the displayed class. All signs are fixed by (4).

Define the integral first-jet defect to be the ACTUAL comparison kernel

$$\mathfrak{B}_E = \ker(H^1(r) : H^1(K_{E,\epsilon}) \rightarrow H^1(K_E)). \quad (9)$$

9.4.1 Theorem 1. Explicit presentation of the retained first-jet zero fibre

There are natural isomorphisms

$$\mathfrak{B}_E \cong \text{coker } \beta_E \cong C_E^1 / (d_E C_E^0 + P_E \ker d_E) \cong \epsilon H^1(K_{E,\epsilon}). \quad (10)$$

Proof with inverse maps. Send the class of y in the middle quotient to $[\epsilon y]$. An original relation $d_E c$ maps to $d_\epsilon(\epsilon c)$. For a cycle b , $\epsilon P_E b = -d_\epsilon b$. Hence the map is well-defined.

Every class in (9) has representative $z_0 + \epsilon z_1$ with $z_0 = d_E b$. Subtracting $d_\epsilon b$ leaves $\epsilon(z_1 + P_E b)$. Send the class to $[z_1 + P_E b]$. Another choice of b differs by an original cycle, giving exactly a $P_E \ker d_E$ relation. Changing the representative by $d_\epsilon(a + \epsilon c)$ replaces b by $b + a$ and changes $z_1 + P_E b$ by $d_E c$. Thus this inverse is well-defined. Both compositions are identity by the subtraction just performed. Finally every ϵ multiple is represented by ϵy , giving the last equality. QED.

In the Split-Zero source's notation the killed representatives and old relations are exactly

$$K_{\text{rep},E} = d_E C_E^0 + \epsilon C_E^1, \quad B_{\text{old},E} = d_{E,\epsilon}(C_E^0 \otimes \mathbb{D}). \quad (11)$$

Their quotient is (9); equation (10) computes it. The supported map is $(E, [z]) \mapsto (E, 0)$ on this kernel. At a nonempty label that is not the external absent point. The maps (10) commute with the original equation inclusions, so reconstruct a split-linear equivalence of the full diagrams. The additive and scalar identities are checked fibrewise after transport to E union F , not by treating the total as an abelian group with a single zero.

9.5 5. Component calculation, including infinite source sets

For the directed graph with declared edges E , every vertex has at most one outgoing edge. The free abelian degree-one quotient has one generator η_A for each weak component A . The map is the component augmentation

$$[\sum a_n V_n] \mapsto (\sum_{n \in A} a_n) \eta_A. \quad (12)$$

For its inverse choose the least original integer s_A in each nonempty component and map η_A to $[V_{s_A}]$. A finite undirected path from n to s_A , with a plus or minus sign for each directed original edge, has boundary $V_n - V_{s_A}$. This proves both compositions and that its kernel is exactly the ORIGINAL edge-relation image. Finite sums meet only finitely many components.

The kernel of d_E has one free integral generator

$$c_C = \sum_{n \text{ on } C} e_n \quad (13)$$

per actual primitive directed cycle C in E . To prove spanning, restrict any kernel vector to its finite edge support. A tree leaf forces the coefficient on its incident edge to vanish. Removing such edges leaves cycles. Outdegree at most one makes every remaining undirected cycle directed, and its boundary equation forces all its coefficients to be equal. Independence follows from disjoint cycle edges. Each weak component has at most one directed cycle: any two adjacent vertices have forward paths that meet (or the same finite sink); this property is preserved across a finite connecting path, and two different cycles cannot have meeting forward paths.

A component with a cycle of m distinct vertices has

$$\beta_E(c_C) = -m\eta_C. \quad (14)$$

Every component without a cycle receives no generator under beta. This proves beta is injective over $\mathbb{Z}$, including for infinite E , and proves the full defect classification

$$\mathfrak{B}_E \cong \bigoplus_{C \text{ cycles in } E} \mathbb{Z}/m_C\mathbb{Z} \oplus \bigoplus_{A \text{ components without a cycle}} \mathbb{Z}. \quad (15)$$

Every coordinate map is (12), with reduction modulo m in a cyclic component; the inverse sends the coordinate generator to $[\epsilon V_{sA}]$. Its exact order is m by (12)-(14). A period-one component contributes a zero GROUP with its original nonempty module diagram and all equations retained. It is not relabelled absent.

For finite E , a component without a cycle has one frontier vertex with no declared outgoing equation. Its $\mathbb{Z}$ summand is unresolved under that finite set of equations, NOT evidence of an infinite Collatz orbit. For $E=X$, all outgoing equations are present. A component without a cycle has an injective forward ray and supplies the infinite-component summand in (15).

9.6 6. The complete first-jet cohomology modules

The preceding defect is not merely a count. On a cycle component of primitive length m ,

$$H^1(K_\epsilon|_C) \cong \mathbb{D}/(m\epsilon). \quad (16)$$

Choose an original cycle point c . For every vertex n in its component let $r(n)$ be the first hitting length to c , with $r(c)=0$. The map sends V_n to $(1+\epsilon)^{r(n)}$ in the quotient. The relation at c is $1-(1+\epsilon)^m = -m\epsilon$; every other original edge respects its first hitting length. The inverse sends 1 to $[V_c]$. The actual finite path to c gives $[V_n] = (1+\epsilon)^{r(n)}[V_c]$; the weighted finite cycle gives the single relation $m\epsilon V_c = 0$. These identities prove both inverse compositions. All paths used for individual vertices are finite, even with infinitely many incoming vertices.

For a full nonperiodic component choose its actual forward ray x_j from its least vertex. A vertex n meeting x_j after r steps maps to $(1+\epsilon)^{r-j}$. Negative powers are explicitly $1+k\epsilon$ for integer k , because $(1+\epsilon)^{-1} = 1-\epsilon$. A later

meeting adds the same number to r and j . The inverse sends 1 to $[V_{x0}]$; original edges give both inverse identities. Thus

$$\boxed{H^1(K_\epsilon|_A) \cong \mathbb{D}, \quad \mathfrak{B}_A = \epsilon\mathbb{D} \cong \mathbb{Z}.} \quad (17)$$

The same D presentation holds for a finite open component by choosing its frontier as reference and using the finite path to it. That statement has a different declared support and does not turn it into a nonperiodic component of the full graph.

As abelian groups the cycle module in (16) is $\mathbb{Z}$ plus $(\mathbb{Z}/m\mathbb{Z})^*\epsilon$. Constant reduction forgets exactly the second term. The base loop $m=1$ therefore has $H^1=\mathbb{Z}$ and zero first-jet kernel; a longer cycle has the retained torsion kernel $\mathbb{Z}/m\mathbb{Z}$; a full nonperiodic component has the free kernel $\mathbb{Z}$. The original constant source spaces remain present in all three cases.

9.7 7. The integral first-jet restatement of Collatz

A positive fixed point obeys

$$(2^a - 3)n = 1.$$

For $a=1$ its solution is negative, for $a=2$ it is $n=1$, and for $a \geq 3$ the positive factor 2^{a-3} is at least 5, so no positive integer solves the equation. The actual positive fixed point is therefore exactly 1.

On the FULL original absolute graph, (15) now gives

$$\boxed{\mathfrak{B}_X \cong \bigoplus_{C \text{ nontrivial positive cycles}} \mathbb{Z}/m_C\mathbb{Z} \oplus \bigoplus_{A \text{ nonperiodic components}} \mathbb{Z}.} \quad (18)$$

All periods in the first sum are at least 2. The actual fixed loop contributes $\mathbb{Z}/1\mathbb{Z}=0$ through the unit map -1 , without deleting its edge, vertex, or source fibre.

Consequently the following are equivalent, with the proofs provided by (10)-(18):

$$\boxed{\begin{aligned} &\text{Every positive odd integer reaches 1;} \\ &\ker H^1(r : K_{X,\epsilon} \rightarrow K_X) = 0; \\ &\epsilon H^1(K_{X,\epsilon}) = 0; \\ &\beta_X : H^0(K_X) \rightarrow H^1(K_X) \text{ is an integral isomorphism.} \end{aligned}} \quad (19)$$

Surjectivity of β_X suffices because injectivity was proved in (14). This is a refinement by a specific comparison kernel, not a claim that its vanishing is established. It asks that the particular infinitesimal sources ϵV_n already be original finite boundaries. Their images under constant reduction are always supported zero, whether or not those boundaries exist. This is precisely why retaining the original kernel and relation map matters.

No global endpoint follows from the algebraic classification: determining that the nonbase component indexing sets are empty remains the arithmetic content. No new numerical cycle bound is asserted here.

9.8 8. Finite witnesses are extractable, not just formal zeros

The exact certificate for ϵV_n to be an original boundary is a pair of finite INTEGRAL original edge vectors b, c with

$$d_X b = 0, \quad d_X c - P_X b = V_n. \quad (20)$$

It is equivalent to $d_{X,\epsilon}(b + \epsilon c) = \epsilon V_n$, by (5).

9.8.1 Theorem 2. Finite integral witness and actual path extraction

Every pair (20) yields an actual finite Collatz path from n to 1 using only its finite edge support. Conversely, an actual first-arrival path p from n to 1 yields (20) with

$$b = -e_1, \quad c = \sum_{e \text{ in } p} e. \quad (21)$$

Proof. For the converse, $d_X c = V_n - V_1$ and $-P_X b = V_1$; also $d_X e_1 = 0$. This includes $n=1$ with $c=0$.

For extraction form the finite graph on the support of b and c , retaining all its actual targets. In the component containing n , b is an integer multiple k of its unique cycle sum, or zero when it has no cycle. Component augmentation of (20) says $-k \cdot m = 1$. Thus a cycle exists and $m=1$. The fixed-point calculation proves its vertex is 1. In a finite functional component with a cycle, every vertex has a forward route to that cycle: a different terminal frontier or cycle would contradict the meeting-path property proved in Section 5. Following the declared original edges therefore reaches 1 within the number of vertices of this finite graph. The checker executes this extraction after verifying both integral equations; it does not infer it from a tolerance or a scalar zero. QED.

This theorem also computes what rationalization would discard. On a period- m component, $m \cdot \epsilon V_n$ is a boundary via minus the finite weighted cycle. Dividing that certificate by m over Q would erase its order- m class. Over Z , (20) rejects such a division for a singleton unless $m=1$.

For the explicitly changed map $T_-(n) = (3n-1)/2^{\nu_2(3n-1)}$, the actual cycle is $5 \rightarrow 7 \rightarrow 5$. Its first-jet defect is $Z/2Z$ and

$$d_\epsilon(-(e_5 + e_7) - \epsilon e_7) = 2\epsilon V_5. \quad (22)$$

The checker rejects its half as a nonintegral singleton witness. T_- is a control, related to the original SIGNED map by $T_-(n) = -T_+(n)$; it is not a positive $3n+1$ counterexample.

On the original map, the edges $3 \rightarrow 5 \rightarrow 1$ yield

$$d_\epsilon(-e_1 + \epsilon(e_3 + e_5)) = \epsilon V_3. \quad (23)$$

The relation involving the supported zero of the fixed loop is an actual part of this certificate, not a deleted row.

9.9 9. Support transitions, repeated words, and original arithmetic

For E subset F , the map on (15) sends the generator for an old component to the generator of its containing new component, with coefficient 1, followed by the target's modulus. It respects every old relation: a retained actual cycle remains the same cycle, so its period does not change; two different cycles cannot later join in a functional graph. Open components may merge or enter a known cycle. The matrices compose exactly, including modulus-one labels. This is a diagram indexed by equation support, not by chronological time.

Every finite vector and every relation in the full construction uses finitely many equations. Thus

$$\mathfrak{B}_X \cong \operatorname{colim}_{E \subset X, E \text{ finite}} \mathfrak{B}_E. \quad (24)$$

Direct proof: represent a class by ϵ times finitely many original vertex vectors, include edges from those vertices in E , and use (10). Any equality in the full quotient has an original finite d -vector plus a finite cycle vector; putting their supports in a larger F proves the equality at that finite stage. This gives both inverse descriptions of the colimit. It agrees with the standard exactness of filtered module colimits, without assuming compactness of a space of infinite integer orbits.

A declared position word is not automatically an actual primitive cycle. An actual solution x_i of $2^{\{a_i\}}x_{(i+1)} - 3x_i = 1$ gives the position-to-orbit maps $e_i \rightarrow e_{(x_i)}$, $V_i \rightarrow V_{(x_i)}$. They commute with both J and P , hence with d_ϵ and β . A word traversing an actual primitive cycle r times maps its position cycle to r times the actual cycle. In particular the position circle for $(2,2)$ has a $\mathbb{Z}/2\mathbb{Z}$ first-jet defect, but its original realization $x=(1,1)$ maps both edges to e_1 , and maps that defect to $\mathbb{Z}/1\mathbb{Z}=0$. The kernel is accounted for by this explicit map. Using position length in place of actual primitive length would give a false counterexample.

The previous integral forcing class $[C]$ in $\mathbb{Z}/(2^A - 3^m)\mathbb{Z}$ remains a separate retained invariant connected through this actual realization map. Integrality, positivity and exact valuations are verified before a position word is assigned an actual cycle. No equality of this forcing modulus with the first-jet index m is asserted. The numerical endpoints themselves recover the original exponent by $2^a = (3n+1)/T(n)$; its labelled edge is not replaced by an unmarked abstract cycle.

9.10 10. Coefficient and infinity comparisons

From (18), coefficient extension to $\mathbb{Q}$ kills exactly the finite cyclic defect summands and maps each free nonperiodic summand injectively to $\mathbb{Q}$. Thus rational vector-space dimensions alone would miss all finite cycle-index obstructions in this particular comparison. Split-Zero support is preserved by $G(\mathbb{Z}) \rightarrow G(\mathbb{Q})$, but the original integral kernel is still required. This is a computed loss, not a claim of an absence of relations between the coefficient settings.

For a prime p and integer $r \geq 1$, the exact quotient presentation gives

$$\mathfrak{B}_X \otimes \mathbb{Z}/p^r \cong \bigoplus_C \mathbb{Z}/\gcd(m_C, p^r)\mathbb{Z} \oplus \bigoplus_A \mathbb{Z}/p^r\mathbb{Z}. \quad (25)$$

Every nontrivial finite m has a prime divisor, so vanishing of these reductions for all primes is equivalent to vanishing of (18). That detection claim uses the PROVED

direct-sum form (18); it is not claimed for arbitrary abelian groups.

The infinite graph in (18) is handled without infinite sums of incident-edge coefficients. All chain columns remain finite. Along a specified injective original ray $x_0 \rightarrow x_1 \rightarrow \dots$, an enlarged module allowing locally finite edge chains contains $\epsilon \sum_j e(x_j)$, whose componentwise boundary is $\epsilon V(x_0)$. Each vertex of that ray has at most two contributing edges, so this calculation is well-defined. It displays a free source class killed by that enlarged comparison. It does not compute the entire kernel of every locally finite completion. Retaining the original finite-support relation submodule prevents this completed primitive from being accepted as (20).

9.11 11. What the source rereading changes in the research programme

The scalar core is not a device that automatically stores words. The useful research object is a specified support diagram and a specified map, with its full fibre of representatives carried to supported zero. The original Rees work demonstrates that a specialization may erase a zeroth-order observation while a first-order comparison retains a defect. Sections 3-7 implement that operation directly on the original Collatz equations.

The newly exposed arithmetic target is the vanishing of the actual first-jet kernel (9) at full support, or equivalently the integral surjectivity in (19). It separates, within the SAME comparison, finite recurrence index from nonperiodic persistence and retains the fixed-loop source rather than starting by deleting it.

There are two concrete proof-development tasks, neither assumed here:

- Construct complete arithmetic families of finite pairs (20), retaining their source parameters and original valuations, and prove their coverage of all original positive sources. The small checker can verify individual instances and extracted paths; universal family coverage requires a written proof, not a sample or timeout. Equation (24) identifies the precise finite-support exhaustion that such a proof must establish.
- Control the original nonzero defect classes with arithmetic obstructions. The torsion part requires excluding actual primitive periods $m \geq 2$, not merely generic matrix cycles. The free part requires excluding an original nonperiodic component. Prime tests, tensor amplification, or geometric estimates are useful only with maps returning to this same original integral kernel and a proof of their coverage or bound. No estimate from the Riemann-hypothesis programme is imported.

A finite support's open free generator is a question to extend, not a divergence verdict. Example: $E = \{3, 13\}$ has one open component at 5; adding the edge at 5 reaches an open vertex 1; adding the ACTUAL fixed-loop equation makes its first-jet defect zero. The source modules and all previously entered equations remain in the same diagram throughout.

This is a new exact presentation and executable witness interface, not a claimed proof of the global arithmetic vanishing. It gives no new numerical cycle bound, no new Lean execution, and no independent external mathematical audit.

9.12 12. Reproduction and scope

`firstjet.py` constructs the declared equation diagram, both original matrices, its full first-jet cohomology defect presentation, support-transition matrices and integral singleton certificates. `verify.py` checks 128 finite original equation sets; 2,187 support inclusions; 1,024 support-composition diagrams; abstract position-circle controls of lengths 1 through 32; the actual -1-forcing cycle; and 1,024 original positive odd singleton certificates, with 25,437 separately replayed original return equations. Twelve malformed or false inputs are rejected. All tests remain active under optimized Python.

The separate integer matrix calculation constructs unimodular row and column maps U, V and checks $U \cdot A \cdot V = D$, $\det U = \det V = \pm 1$, and the actual diagonal divisibility. Its row additions, column additions, swaps and sign changes have the explicit inverse elementary operations. It does not substitute the diagonal matrix for the retained original source. Prime reductions of the original matrices provide additional independent rank checks.

The general statements, including full infinite source (18), are proved above. Finite matrices and path replays are implementation regression evidence. Ordinary and optimized runs each passed 39,211 exact checks and produced the same recorded bytes. `replay.py` verifies the source manifest and both executions.

9.13 13. Source locators and attribution

- [S0] Archived scalar core: KokunoYumeto/modern-latex-manuscripts, revision f7ff59b176c7dc3941babd4cb9272dffc653070d, formalization/lean/classical_candidates_20260626/split_support_sidecar/SplitZero.lean, blob ff991f7383922e71cdf0e4a3bc85e89e18f808ef. Read in full. Its retrieved path history has the archived July 4 commit d7def24933004584b687842fe75e92ed5ee655a1.
- [S1] Chapter 14, satellites/14_globalization_cue_split_zero_and_shifted_sheets.tex, Zeta revision 42d00e359b16d52ca71568ce5e3db5341949d929, blob 49318a578b96462384000afa16794cfd5fe492c1: scalar coordinate formulas, intrinsic support and linear-join-diagram equivalence. Read the algebraic source through its ideal/zero-divisor sections. Its historical presentation and empty-join qualifications are corrected in [S2]; those historical errors are not adopted here.
- [S2] formal/splitzero/README.md and MATHEMATICAL_NOTE.md, Zeta revision 652d3c7e269cef11e0f667be0401abb6014314c2, blobs 4279feaeb3a80f203684d7f6b00a4e208b71d92d and 7531d36aa44d8213e0e80d15f79668b6c8839ce5. Read in full, including the required $[\tau]=0$ presentation relation and the non-bottom ideal-poset correction.
- [S3] formal/splitzero/DERIVED_MATHEMATICS.md, same Zeta revision, blob fe05d32ccd314daac7c59d225c9cac3d4c1bf3d3, Sections 1-6: reconstruction, internal quotient, killed-representative kernel, and actual homology maps. The earlier source's own Lean scope is not a Lean verification of this new note.
- [S4] workbenches/split-support-rees-trace/RESEARCH_NOTE.md, same Zeta revision, blob 168775d2be296d74b8243cebd3e9fc8942f2eb31, Sections 1-5. The retained first-variation calculation motivated the distinct comparison proved here; its tensor-growth estimates are not reassigned to Collatz.
- [C] Preserved Collatz predecessor: supported_structure_20260915/note.md, revision 2a77e3bc9435f181bc4c9c3957224a7d93b4e887, blob 115af68c1684147e840cdd6cf690269490058f95; the uploaded copy matches this blob. Its automata/word use of "history" remains that named presentation, not an intrinsic definition of Split-

Zero cohomology.

- Standard homological constructions: Stacks Project, cones and termwise split sequences, <https://stacks.math.columbia.edu/tag/014D> ; Tor exact sequences, <https://stacks.math.columbia.edu/tag/00M0> ; exact filtered module colimits, <https://stacks.math.columbia.edu/tag/00DB> . Equations (8)-(11) and (24) include direct proofs rather than relying on an unspecified connecting-map convention.

Research direction and Split-Zero framework: KokunoYumeto. This note's Collatz first-jet application, proof exposition and executable checks were developed in this continuation. No global priority claim is made for the standard scalar, homological or graph-theoretic operations. No Zeta file, inherited contribution, frozen edition, or remote main branch is changed by this addition.

Chapter 10

Supported anchored defects, exact cycle periods, and an alphabet-exit certificate

Source: repository/collatz_reconstruction/research_program/anchored_defect_20260914/note.md.

14 September 2026. Collatz workbench continuation.

Source: KokunoYumeto/collatz-workbench, commit 6a30f1dc23110fae0abbe3b3bd5e60072a842385, PR #2. All inherited files remain unchanged. The proofs below use the actual original odd-return map and the preserved integral cycle and residual-source constructions. The finite certificate is executed by `verify.py`; source preservation and all ten ordinary/optimized executions are checked by `replay.py`. No independent external mathematical review, Lean verification, or global convergence result is asserted.

10.1 1. What the Split-Zero technique contributes, with an exact comparison

For a ring R , use the supplied support extension $G(R)=R$ disjoint union $\{\tau\}$. The external τ is the additive identity and multiplicative absorber. Operations on supported elements are their original ring operations. Thus $e=0_R$ is a supported zero, with e distinct from τ . There are explicit semiring maps

$$\pi : G(R) \longrightarrow R, \quad \chi : G(R) \longrightarrow \mathbb{B},$$

where π fixes R and sends τ to 0 , while χ sends τ to 0 and every element of R , including e , to 1 . The Boolean operations are OR and AND. Direct case checking shows that both maps preserve addition and multiplication. The map

$$(\pi, \chi) : G(R) \longrightarrow R \times \mathbb{B} \tag{S1}$$

is an isomorphism onto the subsemiring

$$\{(0,0)\} \cup \{(r,1) : r \in R\}.$$

Its inverse sends $(0,0)$ to τ and $(r,1)$ to the supported r . Both compositions are identities. The zero fiber of π alone is $\{\tau, e\}$; every nonzero fiber is a singleton. This also specifies exactly how ordinary coefficient calculations with a retained Boolean support coordinate implement the supplied technique. No exclusivity or

priority claim for these elementary operations is made.

Here $R=Z$ or Q . The exponent word, source lattice, forcing vector and graph edges are additional retained data, not information reconstructed from χ alone. Supported implements (S1) over Q and tests both zero values, cancellation, multiplication and distributivity.

10.1.1 A zero-forcing row is still an actual relation

For the one-letter word (2), the anchored cycle equation is $(4-3)y=0$. Its complex is $[Z \rightarrow -1 \rightarrow Z]$, with both cohomology groups zero. Deleting the zero-forcing row produces $[Z \rightarrow 0]$, whose H^0 is Z . The exact comparison is $q^0=\text{id}$ and $q^1=0$. Its kernel complex is $[0 \rightarrow Z]$. In the associated long exact sequence the connecting map from $H^0([Z \rightarrow 0])$ to $H^1([0 \rightarrow Z])$ is multiplication by 1, computed by lifting the source generator and applying the original differential. The extra kernel created by row deletion is therefore explicitly accounted for. The supported zero never licensed deletion of that differential.

10.1.2 The coefficient map to the complex analytic setting

The inclusion $Z \rightarrow C$ induces $G(Z) \rightarrow G(C)$, taking τ to τ and the supported integer zero to the supported complex zero. Through (S1), this is the restriction of inclusion times id_B , so it preserves the two zero labels injectively. For the original return complex $[Z \rightarrow D \rightarrow Z]$, coefficient extension gives $[C \rightarrow D \rightarrow C]$. At D nonzero, its contracting homotopy is multiplication by $1/D$ in degree one. Thus the induced H^1 map is the explicitly zero map $Z/DZ \rightarrow 0$, even though the support coordinates remain present. For (1,3), $D=7$ and the forcing class [5] is nonzero integrally; its complex primitive is $5/7$. This proves the precise loss under this coefficient change. We retain the original integral complex rather than substitute its complex-coefficient contraction.

The peer Zeta guide currently studies spectral cohomology and original period/Gram calculations. That guide provides context, not an arithmetic estimate used here. No identification of its theta complex with the Collatz cycle complex is asserted. The coefficient maps just specified are the comparisons used in this note.

10.2 2. Original forcing and anchored primitive

Let

$$T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1), \quad X = \{1, 3, 5, \dots\}.$$

For an original nonempty exponent word $p=(a_1, \dots, a_m)$, retain

$$A_j = \sum_{i=1}^j a_i, \quad A_0 = 0, \quad U = 2^{A_m}, \quad L = 3^m, \quad D = U - L,$$

$$C = \sum_{i=1}^m 3^{m-i} 2^{A_{i-1}}, \quad F_p(n) = (Ln + C)/U.$$

The preserved cycle differential and functional are

$$\begin{aligned}(B_p x)_i &= 2^{a_i} x_{i+1} - 3x_i, \quad x_{m+1} = x_1, \\ f_i &= 3^{m-i} 2^{A_{i-1}}, \quad f B_p = D e_1^T.\end{aligned}\tag{A1}$$

The predecessor proves $H^0=0$ and $H^1=\mathbb{Z}/D\mathbb{Z}$ with $[b]$ sent to $[fb]$. The actual cyclic forcing is $b=1$, with marked coordinate $[C]$. Its rational primitive is $x_i=C(p_i)/D$, for the retained rotations p_i . Integrality, positivity, and the actual valuation equations remain required to obtain a positive integer orbit.

The affine coordinate map $y=x-1$ has inverse $x=y+1$. Substitution gives

$$B_p y = b_p, \quad (b_p)_i = 4 - 2^{a_i}, \quad f b_p = C - D.\tag{A2}$$

This is a bijection of the two specified solution sets, not a claim that translation is a linear chain map. Its forcing difference is the original boundary $-B_p 1$. Consequently

$$[b_p] = [1], \quad [C - D] = [C] \text{ in } \mathbb{Z}/D\mathbb{Z}.\tag{A3}$$

The same integral class is retained. What changes is its specified primitive and the exact observable below. The trivial word repetitions $(2)^m$ have $y=0$. Conversely $y=0$ forces every right-hand side $4-2^{a_i}$ to vanish, so every letter is 2 and $x=1$. An actual nontrivial positive cycle never meets 1, since $T(1)=1$; therefore all its x_i are at least 3 and y_i are positive even integers.

The block maps from `../residual_splce_20260914/integral_blocks.md` still apply. In that notation the anchored right-hand side maps to Rb_p , and the inverse is $y=S_0 z + Hb_p$, followed by $x=y+1$. Dropping Hb_p does not give this affine inverse. Equations (A2)-(A3) make the comparison with the existing marked obstruction literal.

10.3 3. An anchored period character with its complete zero domain

For $n>1$ define

$$\eta(n) = \frac{2^{a(n)}(T(n) - 1)}{3(n - 1)} = 1 + \frac{4 - 2^{a(n)}}{3(n - 1)}.\tag{P1}$$

This is a nonnegative rational value. A step into 1 gives $\eta=0$, retained as a supported zero. At $n=1$ the ratio is not defined: its retained equation is $4*0=3*0+0$, with exponent 2 and zero forcing. No $0/0$ is assigned a value.

For an actual finite path starting above 1 and stopped no later than its first arrival at 1, multiplication of the exact edge equations gives

$$\prod_{j=0}^{m-1} \eta(T^j(n)) = \frac{2^{A_m} T^m(n) - 1}{3^m (n - 1)}.\tag{P2}$$

The identity includes terminal value 1: both sides are then supported zero. Its proof cancels only the positive intermediate factors. No canceled intermediate factor is zero.

On the subgraph with both endpoints above 1, all η values are positive. Let C_1 be the free abelian group on its actual edges, C_0 the free abelian group on its vertices, and $\text{partial}(e_n)=v_n-v_{T(n)}$. Define multiplicative homomorphisms

$$\rho(v_n) = n - 1, \quad \lambda(e_n) = 2^{a(n)}/3.$$

They extend by products with the original integer chain coefficients. Their exact relation is

$$\eta(c) = \lambda(c)\rho(\partial c)^{-1}. \quad (\text{P3})$$

Thus eta and lambda differ by the specified vertex coboundary, and define the same class in multiplicative degree-one graph cohomology. In particular every actual closed cycle c satisfies

$$\eta(c) = \lambda(c) = 2^A/3^m. \quad (\text{P4})$$

The kernel of lambda consists exactly of finite chains with both $\sum c_n a(n)=0$ and $\sum c_n=0$, by unique prime factorization. The kernels of rho and eta are likewise the simultaneous integer prime-valuation equations of their displayed rational products. These describe the fibers as cosets of the stated kernels; no unspecified spectral quotient is used.

The predecessor's retractions preserve closed chains literally, $F(c)=c$. Formula (P4) is therefore unchanged on every actual nontrivial cycle under those retractions. We do not extend the positive-group character to a general retraction path that enters 1: that path has the supported zero endpoint in (P2), not an invertible character value.

10.3.1 Theorem P. A cycle inequality controlled by the original exponent-1 positions

For a nontrivial positive integer cycle with every vertex at least $s \geq 3$, and k occurrences of exponent 1,

$$1 < \frac{2^A}{3^m} \leq \left(\frac{3s-1}{3s-3} \right)^k. \quad (\text{P5})$$

Proof. The ordinary cycle equations give $2^A = \text{product}(3+1/x_i) > 3^m$. By (P1), an exponent-1 edge has factor $1+2/(3(x_i-1))$, at most $(3s-1)/(3s-3)$. An exponent-2 edge has factor exactly 1: its zero anchored forcing remains a supported relation and its original exponent still contributes to A . Every exponent at least 3 has a factor strictly between 0 and 1, since the next cycle vertex exceeds 1. Bound those latter factors above by 1, retain the original k factors, and use (P4). This proves (P5). In particular k cannot be zero. No independence, asymptotic estimate, or probabilistic coefficient model occurs. QED.

For a fixed contracting pair (m,A) , (P5) is an exact integer test

$$2^A(3s-3)^k \leq 3^m(3s-1)^k. \quad (\text{P6})$$

Its right-to-left ratio strictly decreases with integer $s \geq 2$. An accepted integer and rejection of the next one therefore certify its exact ceiling. For $(m,A,k)=(971,1539,403)$, the ceiling is 274546. The prior unanchored ceiling was 330749. Both endpoint comparisons are checked on integers.

10.4 4. The complete next exponent partition and its exact parameter maps

Keep a legal prefix q , including the empty prefix, with the original chart

$$n = r + 2Ut, \quad x = F_q(n) = u + 2Lt, \quad Uu = Lr + C. \quad (C1)$$

For the empty word use $L=U=1$, $C=0$, $r=u=1$. Every prefix source under consideration is a positive odd integer. The next three cases are

$$a = 1, \quad a = 2, \quad a \geq 3. \quad (C2)$$

The first two cases are precisely the original legal cylinders of q_1 and q_2 . The last case is exactly

$$\begin{aligned} t &= t_0 + 4v, & t_0 &\equiv -\frac{3u+1}{2}(3L)^{-1} \pmod{4}, & 0 \leq t_0 < 4, \\ n &= r + 2Ut_0 + 8Uv. \end{aligned} \quad (C3)$$

The full exponent in this last case remains recoverable:

$$a = \nu_2(3u + 1 + 6Lt). \quad (C4)$$

Indeed, $3x+1$ is positive and even. Its valuation is at least 3 precisely when $(3u+1)/2+3Lt$ is divisible by 4. Since $3L$ is odd, (C3) is its complete solution. Its inverse is $v=(n-r-2Ut_0)/(8U)$, and $t=(n-r)/(2U)$. This proves both the partition and its full label recovery. The lower bound $a \geq 3$ is not a substitution $a=3$.

For each of the two exact-exponent children, the original descent test remains

$$F_p(n) < n \iff D_p > 0 \text{ and } D_p n > C_p. \quad (C5)$$

At a node that has not previously descended, (C5) splits the current source interval into a descent part above $\text{floor}(C_p/D_p)$ and a retained part at or below it. For $D_p \leq 0$ there is no descent part. Empty parts keep their labels and original equations.

10.4.1 A synchronized congruence has its own retained map

The cover also imposes $n=c \pmod{M}$. To intersect this with $n=r \pmod{Q}$, put $g=\text{gcd}(Q,M)$. Compatibility is g divides $c-r$. In the compatible case choose

$$\begin{aligned} t_0 &\equiv ((c-r)/g)(Q/g)^{-1} \pmod{M/g}, & 0 \leq t_0 < M/g, \\ \alpha &= r + Qt_0, & \Delta = Q(M/g), & n = \alpha + \Delta z. \end{aligned} \quad (C6)$$

Both original residue equations are recovered by this substitution. Conversely every joint solution has this form, because its original $t=(n-r)/Q$ solves the displayed linear congruence. The inverses are $z=(n-\alpha)/\Delta$ and $t=(n-r)/Q$. Intersecting with the retained interval $[l,h]$ gives exactly

$$\left\lceil \frac{l-\alpha}{\Delta} \right\rceil \leq z \leq \left\lfloor \frac{h-\alpha}{\Delta} \right\rfloor. \quad (C7)$$

The code retains incompatible residue defects, compatible empty intervals, the original r, Q, c, M , and both parameter maps. It does not replace the source progression by a uniformly sampled interval.

10.5 5. An actual alphabet-exit map, not a convergence certificate

Define the graph G_{12} on the actual positive odd vertices by retaining exactly the original edges e_n for which $a(n)$ is 1 or 2. Let E be the actual vertex set with $a(n) \geq 3$. The inclusion of G_{12} into the original orbit graph is identity on vertices and on the retained edges and commutes with partial. Every E -vertex is present but has no outgoing edge in G_{12} .

An original positive integral solution for a word in $\{1,2\}$ maps its position circle into G_{12} : position i goes to x_i and its edge goes to e_{x_i} . The original equations establish those edge memberships and the boundary identity. The block-circle expansion of the predecessor sends each block edge to the sum of its original edges, so its fundamental cycle maps to the same full cycle chain. Repetitions keep their integer multiplicities.

There is also the explicit relative chain map from the complex relative to (v_1, e_1) to the one additionally relative to E : q_1 is the identity, and q_0 kills the E -vertices and fixes the others. Its degree-zero kernel is the free group on E ; its degree-one kernel is zero. Because both complexes have no degree-two chains, q induces an injection on H_1 . In particular, a nontrivial $\{1,2\}$ cycle stays a nonzero cycle on its original edges.

A prefix in (C3) gives a path of allowed original edges from n to z in E , with boundary $v_n - v_z$. This becomes a boundary certificate for v_n in the exit-relative complex. It remains $v_n - v_z$ in the original graph. There is no assertion that z reaches 1.

For a proposed cycle minimum n , either of two original facts excludes membership: a path descends below n , or a path of allowed edges reaches E . The former contradicts minimality; the latter contradicts the proposed $\{1,2\}$ word. This is the exact relationship used by the finite certificate. It does not erase the possibility of a cycle having an exponent at least 3.

10.6 6. Discharge of the complete 403-rise sector

The preserved residual-splice certificate proves an actual first descent for every odd source from 3 through 99779. Strong induction proves that all those sources reach 1. Its existing all-length theorem excludes at most 402 exponent-1 positions. The new calculation keeps that common source bound unchanged.

10.6.1 Theorem Q. No nontrivial positive cycle has at most 403 exponent-1 positions in its primitive odd-return word

Proof. Its minimum is at least $s=99781$ by the replayed predecessor certificate. The original product and exponent identities give

$$(4s)^m \leq 2^k(3s+1)^m, \quad A = 2m - k + E, \quad E = \sum_{a_i \geq 3} (a_i - 2) \geq 0. \quad (Q1)$$

For $k \leq 403$, exact integer comparison at $m=972$ and monotonicity of $(4s)/(3s+1) > 1$

force $m \leq 971$. For every length 1 through 970, the exact comparison is $2^{\text{bitlength}(3^m)} * s^m > (3s+1)^m$; thus the least power of 2 greater than 3^m already exceeds the upper cycle bound. These 970 comparisons are checked independently by a least-power-of-two loop. At $m=971$ the only admissible exponent sum is $A=1539$; the next power fails the upper bound. Then $E=k-403 \geq 0$, giving $k=403$ and $E=0$. Thus the word consists of 403 ones and 568 twos.

Theorem P bounds its minimum by 274546. Its outgoing exponent must be 1, since an exponent 2 would descend at a source greater than 1. Its incoming exponent must be 2, since exponent 1 strictly increases a positive source. The first fact gives $n \equiv 3 \pmod{4}$. The second gives $4n=3x+1$ for its original predecessor x , hence $n \equiv 1 \pmod{3}$. The CRT inverse (C6) identifies these two conditions with $n \equiv 7 \pmod{12}$. The possible minima are exactly

$$n = 99787, 99799, \dots, 274543, \quad (\text{Q2})$$

with 14564 elements. No word-order choice has been discarded: every original cycle in the sector maps its minimum to a member of (Q2).

Starting from this precise source, the complete partitions (C2)-(C7) give a finite proof tree. At every nonempty node all previous actual returns have been at least the original source. The tree retains the original word, congruences, all coefficients, and all empty child masks. This instance uses the code's excess budget 0; the positive-budget comparison is proved in the companion note. It closes after 2957 branching nodes; no nonempty prefix exceeds 32 allowed returns and no OPEN leaf remains. Of its 8871 terminal labels, 7721 have empty integer fibers and keep their present labels. Exactly 1150 labels are occupied: 1133 actual alphabet-exit tails and 17 first-descent cells. Their cardinalities sum to

$$13399 \text{ exit sources} + 1165 \text{ descent sources} = 14564. \quad (\text{Q3})$$

The certificate checks disjointness and conservation at every node and checks every terminal fiber by independently enumerating the ORIGINAL dyadic progression and imposing $n \equiv 7 \pmod{12}$, without using the generator's CRT inverse. A separate repeated-division interpreter reads every n in (Q2), verifies its full word and endpoint classification, and accumulates the original path boundary. Every one descends or exits the allowed alphabet. Section 5 excludes both cases for the proposed cycle minimum. This proves the theorem. QED.

This is a computer-assisted finite certificate combined with the displayed all-length inequalities. It advances the self-contained workbench bound, not a claimed published cycle record. It does not depend on independent random exponents or on the distributional estimates in the earlier notes.

The proof-tree generator accepts arbitrary finite depth caps. A cap reached before closure creates an explicit OPEN leaf; the verifier rejects its use as a proof of this exclusion. Other rejected controls include an erased empty label, a forged original word, a deleted occupied fiber, and an alphabet exit incorrectly promoted to descent.

10.7 7. What remains and how to reproduce the work

The companion `turns_and_unit_excess.md` proves the ordered adjacent-turn period identity, retains the exponent-3 negative forcing, and completely discharges the 404-rise sector too. Its two source classes have different exact incoming-edge constraints.

The next 405-rise sector remains at $m=971$, $A=1539$ with total excess 2: either two threes and 564 twos, or one four and 565 twos, together with the 405 ones. These original letters and their placement remain part of the task; a zero-budget alphabet exit is not a certificate for that larger fiber.

For the nonperiodic direction, (P2) retains the endpoint displacement as well as both clocks. No uniform bound on that endpoint or on the residual integer-supported source has been proved here. The old residual roots and their degree-zero obstruction remain. The alphabet-exit operation is confined to its specified word fiber and does not supply a source-complete contraction of the original graph.

Run from the repository root:

```
python -B collatz_reconstruction/research_program/anchored_defect_20260914/replay.py
```

To materialize every tree node and every retained zero label:

```
python -B collatz_reconstruction/research_program/anchored_defect_20260914/anchored_cycles.py --full-cover --output
↪ cover.json
```

The full-tree hash, independent witness hash, example actual paths, all period endpoints, and rejected controls are retained in `verification.json`. The generator, independently enumerated fiber checks, and repeated-division interpreter are supplied in full.

10.8 Sources and attribution

[S] The supplied Split-Zero support extension, used in the pinned Collatz `split_zero_history_20260913/note.tex`, subsection “Support, specialization, and what each forgets.” Equation (S1) proves the representation used here. User direction is retained as Split-Zero cohomology and the related exact-map technique; an earlier speech-to-text acronym is not used as a mathematical attribution.

[P] Pinned Collatz `cycle_relative_cohomology_20260914/note.md`, Sections 1, 5 and 6: original marked cycle complex, actual relative graph and chain retractions. `residual_splice_20260914/note.md`, Theorem 4 and the executed 99779 source bound; `integral_blocks.md`, equations B5-B13 and its integral affine inverse. All are retained unchanged and replayed.

[Z] The connected peer Zeta `CURRENT_RESEARCH.md`, blob `9eefdcb14ca6fa22efb29fd60a01429512bfbadc`, read 14 September 2026, opens with original Gamma volumes, intrinsic sums, period Gram matrices and section-corrected residuals. This current programme context is not a premise of Theorems P or Q. No Zeta file is changed.

[R] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948v5, 17 May 2026, Introduction and equation (2), <https://arxiv.org/html/2502.00948v5>. The shortened-clock affine expression maps to ours by expanding each odd-return a into one odd shortened step followed by $a-1$ even steps: shortened length is A , odd count is m , and the remainder is C/U . This explicit comparison places the coefficient/offset calculation in the existing literature. Their conjectural global assertions and numerical bounds are not premises here. No priority claim for affine coding, cycle products, or elementary cohomological algebra is made.

Chapter 11

Ordered turn periods and the complete unit-excess cycle sector

Source: repository/collatz_reconstruction/research_program/anchored_defect_20260914/turns_and_unit_excess.md.

14 September 2026. Continuation of note.md, with the same original coefficients and source checkpoint. The definitions of T, a, eta, m, A, k and the source bound s=99781 are unchanged. No inferred convergence from a budget exit is used.

11.1 1. Two consecutive original edges and their exact product

Take an actual positive orbit segment

$$x \xrightarrow{b} n \xrightarrow{1} z, \quad b \geq 2, \quad n > 1.$$

The labels are exact valuations; in particular

$$x = \frac{2^b n - 1}{3}, \quad z = \frac{3n + 1}{2}. \quad (\text{T1})$$

These formulas and the intermediate n are retained, with the original two edges and their boundary v_x-v_z. The anchored character from (P1) gives

$$\eta(x)\eta(n) = \frac{2^b(n-1)}{2^b n - 4} \frac{3n-1}{3(n-1)} = 1 + \frac{12 - 2^b}{3 \cdot 2^b n - 12}. \quad (\text{T2})$$

Both equalities follow by cross-multiplication of the displayed positive denominators. Formula (T2) is the value of the original character on e_x+e_n, not the replacement of the two-edge path by an unmarked edge. Its fibers retain the original b and n; (T1) recovers both endpoints. For b=2 the factor is the outgoing exponent-1 factor, since the incoming exponent-2 factor is 1. For b=3 it is

$$1 + \frac{1}{6n - 3}. \quad (\text{T3})$$

For b>=4 it is positive and strictly less than 1. The signs follow from 12-2^b; positivity also follows from the two positive original factors. No affine coefficient, edge or valuation has been removed.

11.1.1 Theorem T. An ordered-turn refinement for every positive nontrivial cycle

Let h_3 count the original cyclic adjacent patterns (3,1), and let h_4 count patterns (b,1) with $b \geq 4$. Then

$$\frac{2^A}{3^m} \leq \left(\frac{3s-1}{3s-3} \right)^{k-h_3-h_4} \left(1 + \frac{1}{6s-3} \right)^{h_3}, \quad s \leq \min_i x_i, \quad (\text{T4})$$

for $s \geq 3$.

Proof. These pairs are disjoint: the first edge of a pair has exponent at least 3, and its second has exponent 1, so an edge cannot occupy both roles. Group exactly those original edge factors. Their values are given by (T2). For $b=3$, (T3) decreases with the intermediate vertex and is at most its value at s ; for $b \geq 4$ the product is at most 1. The remaining $k-h_3-h_4$ exponent-1 factors obey (P5). Every unpaired exponent at least 3 contributes a positive factor less than 1, and every exponent 2 contributes 1. Apply the original cycle period identity (P4). QED.

The projection to (m, A, k, h_3, h_4) keeps its fiber: all ordered original words with these counts and their integral, positive marked primitives. Nothing here declares every such aggregate tuple realizable. In particular, the adjacency counts cannot be inferred from the marginal exponent counts alone.

11.2 2. The incoming exponent at a minimum gives an exact additional bound

For a minimum n of a positive nontrivial cycle, its outgoing exponent is 1. When its incoming exponent is 3, those two edges give (T3), and hence

$$\frac{2^A}{3^m} \leq \left(\frac{3n-1}{3n-3} \right)^{k-1} \left(1 + \frac{1}{6n-3} \right). \quad (\text{T5})$$

Equivalently, retaining the original incoming factor explicitly,

$$\frac{2^A}{3^m} \leq \left(\frac{3n-1}{3n-3} \right)^k \frac{2n-2}{2n-1}. \quad (\text{T6})$$

Equality of the two right sides is the exact multiplication identity in (T2) for $b=3$. Both factors in (T5) decrease for $n > 1$, so the expression strictly decreases. For $(m, A, k) = (971, 1539, 404)$, its exact integer ceiling is 274717. The verifier checks (T6) at 274717 and failure at 274718 on integers. With the incoming exponent unrestricted, the original anchored ceiling for this triple is 275227.

11.3 3. Prefix excess as a retained observation

For an original finite word q define

$$E(q) = \sum_{a \text{ in } q} \max(0, a-2). \quad (\text{T7})$$

This is an additive map from word concatenation to nonnegative integers: $E(uv) = E(u) + E(v)$. The original ordered word remains attached; the fiber over e

is exactly the set of those words whose displayed sum is e . Both exponent 1 and exponent 2 have zero contribution, but are not identified. Their original affine maps and full labels remain.

On a prefix q with $E(q) \leq E_0$, put $R = E_0 - E(q)$. The next allowable exponents are the finite list $1 \leq a \leq R+2$. Every other next exponent satisfies $a \geq R+3$ and exceeds the total budget. The exact source progression for this last tail is the original (C3) with $B = R+3$ and 4 replaced by $2^{(B-1)}$:

$$\begin{aligned} t &= t_0 + 2^{B-1}v, & t_0 &\equiv -\frac{3u+1}{2}(3L)^{-1} \pmod{2^{B-1}}, \\ n &= r + 2Ut_0 + 2U2^{B-1}v. \end{aligned} \tag{T8}$$

Divisibility of $3u+1+6Lt$ by 2^B proves this identity and its inverse $v = (n - r - 2Ut_0) / (2U2^{B-1})$. The exact next exponent is still (C4). The finite exact-exponent children plus (T8) partition the whole original prefix source. Each permissible child has the unchanged affine descent split (C5). Every branch, including empty ones, remains recorded.

A prefix excess budget is not a cyclic-state quotient. An actual cycle can be read for one full traversal with cumulative excess increasing from 0 to E_0 . It cannot be required to return that counter to 0. We use the budget only on explicitly bounded prefixes within the first traversal. The underlying original source and endpoint are connected by the same path chain as before; no acyclicity of an auxiliary counter is used to rule out a cycle.

11.4 4. Discharge of the full 404-rise sector

11.4.1 Theorem U. No nontrivial positive cycle has at most 404 exponent-1 positions in its primitive odd-return word

Proof. Theorem Q has discharged at most 403. At rise budget 404, the integer inequality (Q1) bounds m by 973: the strict integer comparison at $m=974$ persists for every larger m . The complete period comparisons leave only $(m,A)=(971,1539)$. The implementation recomputes these inequalities without floating-point logarithms. The original exponent equation gives $E=k-403$. Thus the still-unexcluded case has $k=404$, $E=1$. It contains exactly 404 ones, 566 twos, and one three, in their original order.

Let n be its actual minimum. The outgoing exponent is 1, so $n \equiv 3 \pmod{4}$. The incoming exponent is not 1 and is either 2 or 3. Its actual equation $2^b n = 3x+1$ determines $n \pmod{3}$: it is 1 for $b=2$ and 2 for $b=3$. Therefore the two cases have exact residue classes 7 and 11 modulo 12.

Incoming exponent 2. Its minimum lies in $[99781, 275227]$ by (P5), and $n \equiv 7 \pmod{12}$. The finite cover uses excess budget 1, allowing the original exponent-3 step once. An excess exit is either an exponent at least 4 or a second exponent 3. The full source has 14621 points. Its tree has 7267 nonempty branching nodes, and the independent replay gives

$$5798 \text{ descents} + 8823 \text{ excess exits} = 14621. \tag{T9}$$

No allowed prefix exceeds 39 returns. Thus every exit is witnessed within the first 40 returns, strictly inside one traversal of the prescribed 971-return word. No con-

tradiction is inferred from a second traversal.

Incoming exponent 3. The sharper (T5) bounds the minimum by 274717, and $n=11 \bmod 12$. Starting at the retained minimum, the unique exponent 3 is the last, 971st edge of that traversal. Hence its first 970 edges must lie in $\{1,2\}$. The zero-excess cover is applied only inside that justified prefix. The full source has 14578 points, its tree has 2857 nonempty branching nodes, and independent replay gives

$$1168 \text{ descents} + 13410 \text{ forbidden-prefix exits} = 14578. \quad (\text{T10})$$

No allowed prefix exceeds 32 returns; every exit is therefore witnessed by return 33, before the last permissible exponent-3 position. Those exits contradict this original marked incoming-edge case. They are not declared convergence certificates.

For both cases, every node is the exact partition (C5)-(C7), (T8). Independent enumeration along the original dyadic progression verifies the terminal fibers and their disjoint complete coverage. A repeated-division interpreter verifies every original exponent and every source-to-endpoint chain. Descent contradicts a cycle minimum; an excess exit within the justified prefix contradicts the retained word counts or the specified last-edge position. All 29199 possible minima in the two cases are covered. This excludes the remaining 404 sector and proves the theorem. QED.

The common convergence certificate remains only the inherited odd-source range through 99779. The 22233 excess-exit counts in (T9)-(T10) are not additions to that range. The result is an exact exclusion of the specified cycle fiber and a workbench improvement, not a claim to exceed established published cycle bounds.

11.5 5. The next retained fiber and scope of the calculation

The next budget 405 still leaves $m=971$ and $A=1539$ in the executed period range. After the exclusions above, $k=405$ gives $E=2$. Its full original alphabet data have exactly two possibilities: two threes and 564 twos, or one four and 565 twos, together with the 405 ones. Their placement and the actual incoming minimum edge remain retained. Equation (T2) supplies the original pair factor for incoming four; (T8) supplies the full tail partition at every remaining budget. No exhaustion of those two fibers is claimed in this contribution.

The ordered-turn bound (T4) applies at arbitrary lengths, with arbitrary positive exponents, to every actual nontrivial positive cycle. The finite exclusions Q and U have the stated count scope. None of these assertions removes a nonperiodic component of the original graph. The source-complete nonnegative primitive required by the predecessor remains unconstructed here.

Chapter 12

Arithmetic descent from retained section defects

Source: repository/collatz_reconstruction/research_program/defect_rank_descent_20260915/note.md.

15 September 2026. Continuation of the original positive-integer Collatz workbench at d0c987b12c20be5978f496ab1ef2a60018caa5e5.

12.1 0. Scope and source order

This contribution advances two arithmetic operations rather than proposing another vanishing reformulation. The original integral affine-section defect gives an exact, source-dependent limit on repeated packets. Its change under a packet switch gives a precise resonance equation. A separate argument proves strict descent for an unbounded family of rise/fall words, with arbitrarily large fall exponents. The first-jet maps then turn these original paths into supported boundary relations. A proof-carrying symbolic cover provides the finite-source computation.

The predecessor is `intrinsic_zero_firstjet_20260915/note.md`, with original code blob 97f9ef5ad535f7ec18259d41fab6a3557be17d98 and note blob d336b734a090b72bffffc70cdd9a4543504d3ded3. Its support is a declared set of original equations, not an automata trajectory log. Its differential is $d_{\epsilon} = J \cdot P \cdot \epsilon$ over the dual integers. The current source was read through the connected repository; the uploaded original bytes match those Git blob identities. The structural predecessor is also replayed. All inherited files remain unchanged.

The companion `logarithmic_run_exclusion.md` completes the positive-gap domain using the explicitly cited Matveev logarithm bound and exact rational Farey certificates. It also proves the all-length restriction $m > 226a$ for the initial exponent-one run a at an actual cycle minimum, and excludes the complete family with one cyclic run of exponent-one letters. Nine previously certified original sources in the supported-zero equation fibre sharpen that run bound from 213 to 226 without adding discovery starts. Those are established-theorem-dependent proofs, not consequences inferred from a sampled frequency.

The original cylinder, affine numerator and descent-cell formulas occur in the earlier cycle and residual-splice contributions. They are rederived where needed below. The dyadic coding/periodicity mechanism has a substantial prior literature; [BL] identifies the classical conjugacy programme. No priority claim is made for periodic

coding, monotone-run formulas, or cylinder sieves. [RT] explicitly distinguishes coefficient contraction from actual descent. Its current v5 text was consulted for that distinction; no unproved coefficient-stopping-time assertion is a premise here. The new derivations, finite certificates and their exact scope are the subject of this note.

The code is standard-library Python. General arguments are written below; finite checks are not being counted as general proofs. The rational run cone and the numerical cycle bound additionally use the explicitly listed finite integer certificates. No global Collatz resolution, independent external mathematical review, or proof-assistant execution is asserted.

12.2 1. The original section defect, not only its quotient class

For a nonempty positive-exponent word $p=(a_1, \dots, a_m)$, retain

$$A_j = \sum_{i=1}^j a_i, \quad A_0 = 0, \quad L = 3^m, \quad U = 2^{A_m}, \quad D = U - L,$$

$$C = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}, \quad g_p(n) = \frac{Ln + C}{U}. \quad (1)$$

The underlying branch is $g_a(x)=(3x+1)/2^a$, in chronological order. Appending a letter gives $(L, U, C) \rightarrow (3L, 2^a U, 3C+U)$. Thus (1) is proved by induction, including its original additive numerator. Both C and D are odd, D is nonzero, and $\gcd(D,6)=1$.

The integral $Z[t]$ -module on Ze_1+Ze_2 with t -matrix

$$\begin{pmatrix} L & C \\ 0 & U \end{pmatrix}$$

has specified submodule Z_{Le_1} and quotient Z_U . A Z -linear quotient section at an integer n is $s_n(1)=e_2+ne_1$. Its failure to intertwine t is precisely

$$ts_n(1) - s_n(U) = h_p(n)e_1, \quad \boxed{h_p(n) = C - Dn}. \quad (2)$$

Changing the section by k changes h by $-Dk$. The resolution $0 \rightarrow Z[t] \xrightarrow{(t-U)} Z[t] \rightarrow Z_U \rightarrow 0$, after applying $\text{Hom}(\cdot, Z_L)$, gives the marked extension group Z/DZ and class $[C]$. Equation (2), rather than that class alone, is retained in this continuation. Every $h_p(n)$ has the same class $[C]$, but can have a different exact dyadic valuation.

The section coordinate map $n \rightarrow s_n$ is a bijection from Z to the Z -linear quotient sections, with inverse the e_1 coefficient of $s(1)-e_2$. The positive-odd source is its specified subset. No affine translation of section coordinates is being called a linear chain equivalence.

For $p=(1)$, $D=-1$ and the quotient extension group is zero, but $h_p(7)=8$ and $h_p(31)=32$. The dyadic depths 3 and 5 will give different repeated-packet lengths. The supported source, section and defect distinguish these data despite the same zero quotient.

12.3 2. Exact cylinder membership through that same defect

12.3.1 Theorem A. Original source-cylinder identity

For an original positive odd integer n ,

$$\boxed{\text{the next exact exponent word is } p \iff Ln + C \equiv U \pmod{2U} \iff h_p(n) \equiv 0 \pmod{2U}.} \quad (3)$$

Consequently its complete positive source cylinder is

$$n = r_p + 2Ut, \quad t \geq 0, \quad r_p \equiv CD^{-1} \pmod{2U}, \quad 1 \leq r_p < 2U. \quad (4)$$

Proof. An actual word gives (1) with an odd integer endpoint, hence the first congruence. Conversely, expand C by its first letter. Divisibility of $Ln+C$ by 2^{a_1} forces $3n+1$ divisible by that power. The new integer $n_1=(3n+1)/2^{a_1}$ satisfies the suffix congruence. Repeating this argument gives all intermediate integers. Divisibility at the next letter makes each preceding intermediate integer odd; the last is odd by the final congruence. Thus each prescribed exponent is exact. Positive sources have positive intermediate values.

Since n is odd, $Un=U \pmod{2U}$. Therefore $Dn=C \pmod{2U}$ is equivalent to $Ln+C=U \pmod{2U}$. D is odd, so its modular inverse exists; the residue in (4) is odd. All positive elements in that residue class have exactly $t \geq 0$. This proves both equivalences and the inverse parameter $(n-r_p)/(2U)$. QED.

This is the same cylinder as $(U-C)2^{-1} \pmod{2U}$ in the earlier contribution: the preceding congruence calculation proves equality of the selected representatives. The new expression connects it to the original section defect.

12.4 3. Repetition has a finite arithmetic rank, except at a supported return

12.4.1 Theorem B. Exact repetition rank

Let $A=A_m$. For each positive integer r ,

$$\boxed{n \text{ realizes } p^r \iff 2U^r \mid h_p(n).} \quad (5)$$

For $h_p(n) \neq 0$, the number of complete initial copies of p is exactly

$$\boxed{R_p(n) = \left\lfloor \frac{\nu_2(h_p(n)) - 1}{A} \right\rfloor.} \quad (6)$$

At every admitted copy,

$$h_p(g_p(n)) = \frac{L}{U} h_p(n), \quad \nu_2(h_p(g_p(n))) = \nu_2(h_p(n)) - A. \quad (7)$$

At $h_p(n)=0$, n is an actual positive return of p , and every copy is admitted. The original word, its possible repetitions, and the actual orbit are retained; word length is not substituted for primitive orbit length.

Proof. Composition gives

$$S_r = \sum_{j=0}^{r-1} L^{r-1-j} U^j, \quad C(p^r) = C S_r, \quad D(p^r) = D S_r.$$

The integer S_r is odd: its first term is odd and every other term is even. Apply (3) to p^r . Its modulus is $2U^r$ and its section defect is $S_{rh_p(n)}$, so (5) follows without dividing out an unrecorded even factor. This proves (6). Direct substitution proves (7). If h vanishes, (3) admits p and (1) gives $g_p(n)=n$; all intermediate valuations are actual by Theorem A. QED.

The endpoint after r admitted copies is specified without expanding the path:

$$g_p^r(n) = \frac{C - L^r h_p(n)/U^r}{D}. \quad (8)$$

Every division is exact on the stated original source. Expanding the original word p exactly r times is the inverse path dictionary, including all intermediate integers and odd-return clocks. Formula (8) alone is not substituted for that dictionary.

The full source of exactly r copies followed by failure of another complete copy has the parameterization

$$n = \frac{C - 2^{Ar+1}z}{D} > 0, \quad z \in \mathbb{Z} \setminus \{0\},$$

$$2^{Ar+1}z \equiv C \pmod{D}, \quad 0 \leq \nu_2(z) < A. \quad (9)$$

Its inverse is $z=h_p(n)/2^{Ar+1}$. Oddness follows from the odd numerator and denominator. Signs of z are not dropped. At $r=0$ this still means an original odd source with no complete initial copy.

For a fixed ordinary integer, arbitrarily long initial p -powers force $h_p(n)=0$, since an integer divisible by all powers of two is zero. Thus the intersection of all original p -power cylinders is either empty or the single positive integer C/D ; it is present precisely when $D>0$ and D divides C . In the 2-adic source those compatible cylinders have the single point C/D , since D is odd. Its positive-integer support is tested separately by the displayed conditions. For $D<0$ that point is negative and is not an original positive source.

This proves in particular that an eventually periodic exponent itinerary on a positive integer gives an eventually periodic actual orbit. It does not eliminate an aperiodic positive orbit. A bound for a fixed nonzero defect is

$$R_p(n) \leq \left\lfloor \frac{\text{bitlength}(|h_p(n)|) - 2}{A} \right\rfloor.$$

No bound independent of n is claimed.

12.5 4. Exact packet switching, and the resonance that prevents a false global rank

For two words p, q define their original integer resultant

$$\Delta_{p,q} = D_p C_q - D_q C_p.$$

Then for every integer x ,

$$D_p h_q(x) - D_q h_p(x) = \Delta_{p,q}. \quad (10)$$

For an actual p-step $y=g_p(n)$, this becomes

$$D_p h_q(y) = D_q L_p \frac{h_p(n)}{U_p} + \Delta_{p,q}. \quad (11)$$

The quotient on the right is an original integer by (3). No comparison category has changed. Equations (10)-(11) are obtained by substituting (2); they retain both off-sets and all coefficients.

Let $t=\text{nu}_2(h_p(n))-A_p$ and $r=\text{nu}_2(\Delta_{pq})$, including infinity for a zero. When t and r differ, (11) gives the exact new depth $\min(t, r)$, because its other multiplicative factors are odd. When they coincide, their leading dyadic terms can cancel and the new depth can be arbitrarily large. The code retains the full integer right-hand side in that resonant layer. It does not promote its lower valuation bound to an upper bound. At $\Delta=0$ the common fixed-point relation remains specified, including any supported zero.

12.5.1 A failed uniform rank bound leads to an actual descent family

Take $p=(1)$, $q=(2)$, so $h_p(n)=n+1$, $h_q(n)=1-n$, and $\Delta=-2$. For

$$r \geq 2, \quad v \geq 0, \quad s = 6v + 1, \quad n = \frac{2^{2r+3}s + 1}{3}, \quad (12)$$

one has $\text{nu}_2(h_p(n))=2$, but after the first exponent-1 edge,

$$T(n) = 4^{r+1}s + 1, \quad \nu_2(h_q(T(n))) = 2r + 2.$$

Thus a source of fixed first-packet depth produces arbitrarily large second-packet depth. Formula (11) places it exactly in the resonant layer; a proposed common decreasing depth would fail here.

The same complete family has the actual word $(1, 2, \dots, 2)$ with r twos, and

$$T^{r+1}(n) = 4 \cdot 3^r s + 1 < n. \quad (13)$$

Proof. The source is integral because $s \equiv 1 \pmod{3}$. Its first step is the displayed integer. Each following exponent is exactly 2, with intermediate values $1+4^{\{r+1-j\}}3^j s$ for $0 \leq j \leq r$. Their original minus-one depths give exactly r consecutive twos because s is odd. The final difference is

$$n - T^{r+1}(n) = \frac{(8 \cdot 4^r - 12 \cdot 3^r)s - 2}{3}.$$

At $r=2$ the coefficient is 20; its recurrence is $A_{(r+1)}=3A_r+8 \cdot 4^r$, so it remains greater than 2. Since $s \geq 1$, the difference is positive. QED.

The source progression at fixed r is

$$\frac{2^{2r+3} + 1}{3} + 2^{2r+4}v,$$

and its target is $4 \cdot 3^{r+1} + 24 \cdot 3^r v$. Both inverses to v are retained in the code. This proves the full unbounded family rather than testing large r as a substitute for its

proof.

12.6 5. A complete run calculation with the affine correction retained

Let $a \geq 1$, $b \geq 1$, and consider an original word consisting of a ones followed by b exponents at least two. Put

$$D_2(a, b) = 2^{a+2b} - 3^{a+b}, \quad \lambda_2 = (3/2)^a (3/4)^b.$$

These refer to the explicit all-two comparison tail. The actual tail word and all its coefficients remain separately recorded.

12.6.1 Lemma C1. Contracting pure-two runs cannot increase

On the complete original source of $(1)^a(2)^b$, with $D_2 > 0$, the terminal value is at most the original source. Equality is an actual return and is not deleted.

Proof with the complete parameter domain. The initial ones and the next even valuation give

$$n = 2^{a+1}t - 1, \quad t > 0 \text{ odd}, \quad x = 2 \cdot 3^at - 1.$$

The b actual exponent-two steps give a unique integer $z \geq 1$ with

$$3^at - 1 = 4^bz, \quad y = 2 \cdot 3^bz + 1. \quad (14)$$

To see integrality, $(3/4)^b(x-1)=y-1$ and the latter is even; since 3 is odd, $x-1$ is divisible by $2 \cdot 4^b$. Conversely every $z \geq 1$ with $4^bz \equiv -1 \pmod{3^a}$ gives $t = (4^bz + 1)/3^a$, which is positive odd, and the displayed original source and intermediates realize exactly that word. These operations are inverse. Increasing z by 3^a increases n by 2^{a+2b+1} and y by $2 \cdot 3^{a+b}$, exactly the original cylinder steps.

The endpoint difference is

$$\boxed{n - y = 2(k - 1), \quad k = \frac{D_2z + 2^a}{3^a} \in \mathbb{Z}_{>0}.} \quad (15)$$

The numerator is a positive multiple of 3^a because $4^bz \equiv -1 \pmod{3^a}$. Hence $k \geq 1$ and $n - y \geq 0$. The zero difference $k=1$ is retained as an actual periodic source. QED.

12.6.2 Lemma C2. A higher tail exponent makes this descent strict

On the original source of $(1)^a(c_1, \dots, c_b)$, where each $c_i \geq 2$ and at least one $c_i \geq 3$, the inequality $D_2(a, b) > 0$ gives strict descent.

Proof. All branch maps are increasing on positive rational inputs. Replacing one chosen higher exponent by 3 and the other tail exponents by 2 gives an upper bound on the SAME original terminal value. Moving that 3 to the start of the comparison tail increases the bound, since

$$g_2g_3(x) - g_3g_2(x) = 1/8 > 0.$$

This is an inequality between two displayed affine composites, not a substitution of their words for the actual word. If $x=g_{-1}^a(n)=(3/2)^{a(n+1)-1}$, the resulting upper bound is

$$g_2^{b-1}g_3(x) = 1 + \frac{\lambda_2}{2}(n+1) - \frac{5}{3}(3/4)^b. \quad (16)$$

Since $\lambda_2 < 1$ and $n \geq 3$, this is strictly less than $1+(n+1)/2 \leq n$. All original exponents, source congruences and target integers remain in the source record. QED.

12.6.3 Theorem C. An all-length, unbounded-alphabet descent cone

Every actual positive source whose word begins with $a \geq 1$ ones and then $b \geq \text{ceil}(3a/2)$ consecutive exponents at least two strictly descends by the end of those $a+b$ returns.

Proof. Higher-exponent tails are handled by Lemma C2. For pure-two tails, it suffices in (15) to prove $D_2 > 3^a - 2^a$, because $z \geq 1$ then forces $k \geq 2$. At the minimum b , even $a=2h$ gives

$$D_2 = 256^h - 243^h \geq 13 \cdot 243^{h-1} > 9^h - 4^h.$$

Odd $a=2h+1$ gives

$$D_2 = 32 \cdot 256^h - 27 \cdot 243^h \geq 5 \cdot 243^h > 3 \cdot 9^h - 2 \cdot 4^h.$$

Both are the required comparison. Increasing b changes D_2 to $3 D_2 + 2^{a+2b}$, so the strict comparison persists. No bound on the individual c_i is imposed. QED.

12.6.4 Theorem C-rational. Certified rational cone

The same strict-descent conclusion holds for

$$b \geq \left\lceil \frac{2124a}{1507} \right\rceil. \quad (17)$$

This is a computer-assisted all-length theorem with **1,507 finite base inequalities**, not a test of only 1,507 word lengths. The finite certificate is `slope_certificate(2124, 1507)` and is independently replayed in `verify.py`.

Proof. Set $q=1507, p=2124$. The exact integer block comparison is

$$U_* = 2^{q+2p} = 2^{5755} > 3^{q+p} = 3^{3631} = L_*.$$

For every $1 \leq r \leq q$, the certificate verifies

$$D_2(r, \lceil pr/q \rceil) > 3^r - 2^r. \quad (18)$$

Now retain the unique decomposition $a=hq+r$, with $h \geq 0$ and $1 \leq r \leq q$. The least b in (17) is $hp + \text{ceil}(pr/q)$. The exact extension step gives

$$D_2(a+q, b+p) = L_* D_2(a, b) + (U_* - L_*) 2^{a+2b}. \quad (19)$$

Moreover $3^a - 2^a \geq 3^{a-1}$, so multiplication by $L_* = 3^{q+p}$ sends a strict bound above $3^a - 2^a$ to a strict bound above $3^{a+q} - 2^{a+q}$. Induction from (18) proves the

required inequality for every a . Increasing b again uses the preceding recurrence. Lemmas C1-C2 complete the proof. QED.

The record includes the full recipe and checksum of (18), and also checks the elementary slopes $3/2$ and $17/12$ and the refinements $179/127$ and $568/403$. The smallest ratio $D_2/(3^r-2^r)$ in the 1,507 base cases is $13/5$, at $(r,b)=(2,3)$. This finite minimum is auxiliary; the proof uses the individual strict comparisons.

A smaller slope cannot simply be assumed. The original word $(1)^{22}(2)^{31}$ has

$$D_2 = -40432553845953101497907.$$

Its actual source 31455430079290429512089599 has terminal 31521181806453574636140115, which is larger. Both are independently replayed. More generally $(1)^{22h}(2)^{31h}$ has negative D_2 for every $h \geq 1$, and every positive source in its original nonempty cylinder grows. Thus the proposed universal slope $31/22$ is concretely refuted by original sources, not merely rejected numerically.

12.6.5 Complete positive-gap domain

The companion `logarithmic_run_exclusion.md`, Theorem E, proves $D_2(a,b) > 3^a - 2^a$ for **every** positive pair with $D_2(a,b) > 0$, using the stated published transcendence theorem and a full finite rational certificate. Its Corollary E1 therefore strengthens the preceding rational cone to the exact condition $2^{a+2b} > 3^{a+b}$, with the same arbitrary tail exponents and actual source equations. The earlier elementary and rational-cone arguments remain shorter, independently replayable routes on their stated domains.

12.7 6. First-jet transport of the new arithmetic paths

Keep the original absolute complex, with $J_{e_n} = V_n$, $P_{e_n} = V_{T(n)}$, $d = J - P$, and

$$\mathbb{D} = \mathbb{Z}[\epsilon]/(\epsilon^2), \quad d_\epsilon = d - \epsilon P.$$

For an actual finite path p from n to y , let $B_p(n) = \text{sume}_-(T^j(n))$. Then

$$dB_p(n) = V_n - V_y, \quad \boxed{d_\epsilon(\epsilon B_p(n)) = \epsilon V_n - \epsilon V_y.} \quad (20)$$

The original path has finitely many edges for every finite parameter value, although the family permits unbounded lengths and source labels. The domain-to-chain map is the linear map on the free module on the original parameter fibre that sends its n -labelled generator to $B_p(n)$; its boundary is the difference of the specified source and terminal maps. This is the actual connection between the section/rank calculation and the first-jet complex, not an asserted equivalence between their differentials.

For a specified rule choosing one such strict-descent path for each admitted original source, this map on finite free source modules is injective. In a nonzero finite combination choose the largest original source n with nonzero coefficient. No endpoint from a smaller source can equal n , so the boundary coefficient at V_n is exactly that source coefficient. This proves injectivity. It also gives the inverse on the image by successively recovering the largest source coefficient and subtracting its original path column. The rule and its domain remain part of that inverse.

Adding (20) to an existing integral witness at y yields one at n . For the finite cover in

Section 8, every endpoint is smaller, so these additions terminate by strong induction on original positive integers. The final witness has the exact form

$$d_\epsilon(-e_1 + \epsilon c_n) = \epsilon V_n, \quad (21)$$

with c_n an original finite integral path chain. The predecessor checker verifies (21) and independently extracts the actual route to 1. No rational division by a cycle period is permitted. The original equation support is the union of supports in (20)-(21), not a record of the order in which they were discovered.

12.8 7. A whole-source first-return reduction whose first-jet clock is not shortened

The subset $Y = \{n > 0 \text{ odd} : n \equiv 1 \pmod{4}\}$ contains the actual fixed vertex. Every original vertex reaches Y through its maximal initial string of exponent 1. In exact coordinates,

$$\begin{aligned} \tau(n) &= \nu_2(n+1) - 1, \quad t = (n+1)/2^{\tau(n)+1}, \\ T^j(n) &= 3^j 2^{\tau(n)+1-j} t - 1 \quad (0 \leq j \leq \tau(n)), \quad \theta(n) = 2 \cdot 3^{\tau(n)} t - 1 \in Y. \end{aligned} \quad (22)$$

The integer t is positive odd. These identities prove exact valuations and finiteness without assuming global convergence.

For y in Y define $F(y) = \theta(T(y))$ and $\ell(y) = 1 + \tau(T(y))$. Use a block complex on the SAME original Y -vertices with differential

$$d_Y E_y = V_y - (1 + \ell(y)\epsilon) V_{F(y)}. \quad (23)$$

Let $w = 1 + \epsilon$. The projection on vertices is $V_n \rightarrow w^{\tau(n)} V_{\theta(n)}$. On edges it sends e_y to E_y for y in Y , and every other original edge to zero. The inclusion fixes Y -vertices and sends E_y to its entire original path with coefficients $w^{j-1+\tau(y)}$. The homotopy sends V_n to the similarly weighted original path from n to $\theta(n)$.

Proof of both comparison identities. Outside Y , the first exponent is 1, $\tau(n) = 1 + \tau(T(n))$, and the two projected endpoint terms cancel. At y in Y they give exactly (23). Telescoping proves that the inclusion is a chain map. A block has only its first source in Y , so projection after inclusion is the identity in both degrees. The homotopy equations are

$$d_\epsilon H = I - \iota\pi, \quad H d_\epsilon = I - \iota\pi \quad (24)$$

on vertices and edges respectively: outside Y , subtracting w times the next rising tail leaves precisely e_n ; at y in Y it leaves e_y minus the full block expansion. All columns are finite. QED.

The kernel of H is exactly the span of the Y -vertices. It vanishes on them. For a nonzero finite combination of other vertices, take the least original source with nonzero coefficient. Its first edge occurs with coefficient one in its own rising tail and cannot occur in a tail starting at a larger original vertex, since exponent-one edges strictly increase positive vertices. The coefficient survives. This also proves the stated kernel over the dual-integer ring without cancellation by a nonunit.

For a finite declared equation set E , adjoin the rising tails of its vertices and targets. This enlargement is finite, is idempotent, preserves unions, and supplies the support

of (24). Its Y-equations are exactly E intersect Y. These explicit support maps make the comparison compatible with the Split-Zero reconstruction; no unknown forward orbit is used to define the finite enlargement.

A block cycle retains original total length $\text{sumell}(y)$, so its connecting-map index is that sum. A compressed self-loop with $\text{ell} > 1$ is not silently called the trivial cycle. For example the actual block $9 \rightarrow 7 \rightarrow 11 \rightarrow 17$ has length 3. Assigning it the coefficient $1 + \epsilon$ rather than $1 + 3\epsilon$ leaves the explicit nonzero residual $-2\epsilon V_{17}$. The checker verifies this failed shortening and the correct map.

12.9 8. The symbolic source-cover certificate

For an arbitrary actual first-descent word p , let r, U, L, C, D be as in (1),(4). The complete source set is empty when $D \leq 0$. For $D > 0$ it consists of

$$n = r + 2Ut, \quad \ell \leq t \leq h, \\ \ell = \max \left(0, \left\lfloor \frac{C - Dr}{2UD} \right\rfloor + 1 \right), \quad h = \min_{\substack{j < m \\ D_j > 0}} \left\lfloor \frac{C_j - D_j r}{2UD_j} \right\rfloor, \quad (25)$$

where the empty minimum is infinity. These are exactly $D_n > C$ at the terminal return and $D_{j_n} \leq C_j$ at every earlier return. Positivity of C_j makes earlier $D_j \leq 0$ impose no extra bound. Thus all constraints and the affine offset remain. The target progression is $u + 2Lt$, with $Uu = Lr + C$; both parameter inverses are explicit.

Two occupied first-descent cells cannot intersect: an original source has one actual first-descent time and one original exponent word at that time. This proves disjointness even for nested legal cylinders whose first-descent intervals separate them.

For an odd upper source bound B , the number of sources of one cell in $[3, B]$ is the exact integer interval count obtained from (25) and $3 \leq r + 2Ut \leq B$. If the sum of these disjoint counts equals $(B - 1)/2$, the cells cover every odd source from 3 to B . Each target is a lower positive odd integer. Strong induction, beginning at 1, then proves finite first-jet witnesses (21) for the entire interval.

The search takes only the least uncovered original source, obtains its first descent, and adds its COMPLETE cell. It never treats a search cap as a proof. The acceptance checker does not assume that discovery is correct: it reconstructs every cell with the unreduced final parity congruence, verifies the constant and slope of every original prefix equation, checks the exact floor interval, and verifies the complete disjoint counting identity. A second implementation enumerates every interval slot and checks the lower positive terminal. Source labels (1) and (2,2) remain declared zero-count first-descent cases.

The delivered certificate has:

Scope	Exact result
Original odd interval	3 through 1,166,399
Sources covered	583,199
Distinct complete first-descent cells	41,122
Cells with unbounded positive parameter	41,122
Declared empty first-descent labels	2
Original sources actually iterated in discovery	41,122
Original return equations in those discovery words	754,655
Independently verified constant and slope equalities	1,509,310
Maximum retained word length	141

Scope	Exact result
Maximum retained exponent sum	227

These symbolic statements extend to every parameter in each cell. Convergence to 1 is asserted for the completely covered finite interval, not for every larger member of the cell without its lower-endpoint certificate.

12.10 9. The actual finite first-jet support behind that coverage

For every source occurrence in (25) within the certified interval, expand its original finite word and retain all its original edge labels. Include e_1 . The resulting equation set is E_B . Every internal edge target is the next retained edge source; every terminal is a lower source in the covered interval or 1. Thus E_B is forward closed. Every vertex lies on a path to a lower covered source and hence to 1 by the proved induction.

The code separately materializes this support and recomputes each edge by repeated integer division, rather than the valuation-bit operation in discovery. It constructs a nonnegative integer height H with

$$H(1) = 0, \quad H(n) = 1 + H(T(n)) \quad (n \neq 1). \quad (26)$$

Its recorded source scope is:

- 926,548 distinct original equations and vertices;
- 2,029,720 prefix-equation occurrences before deduplication;
- largest original vertex 30,079,718,549;
- maximum height 196;
- one actual component, with actual cycle (1).

All original edges and heights enter the transcript hash. The connecting map for this finite supported complex has matrix [-1]; the first-jet reduction kernel is zero, while its original nonempty source module and relations remain. This is an actual supported-zero calculation on E_B . It does not set the kernel at full infinite support to zero.

There are additionally 399 original singleton witnesses assembled from the symbolic cover and replayed by the unchanged predecessor's integral first-jet witness extractor. They contain 11,451 original returns. These samples test the interface; the complete interval claim comes from the covering proof, not from sampling.

12.11 10. The executed all-length cycle exclusion

For an actual nontrivial positive cycle, write m for its primitive number of odd returns, A for its exponent sum, k for its number of exponent-1 positions, and s for a lower bound on every vertex. Multiplying the ORIGINAL equations yields

$$3^m < 2^A = \prod_{i=1}^m (3 + 1/x_i) \leq (3s + 1)^m / s^m. \quad (27)$$

Also $A = 2m - k + E$, where $E = \sum_{i=3} (a_i - 2) \geq 0$. Hence

$$(4s)^m \leq 2^k(3s+1)^m. \quad (28)$$

The observation (m,A,k,s) does not replace the word. Its fibre still consists of the ordered words with those counts whose original forced cycle equations have a positive integral primitive with that minimum. The comparisons below exclude the entire fibre without declaring a permitted numerical window realizable.

The source certificate gives $s=1,166,401$. The exact integer comparisons in `independent_period_check` establish

$$2^{\text{bitlength}(3^m)} s^m > (3s+1)^m \quad (1 \leq m \leq 3630), \quad (29)$$

and

$$\boxed{(4s)^{3631} > 2^{1506}(3s+1)^{3631}.} \quad (30)$$

The independent implementation obtains the least power of two strictly above 3^m by an integer doubling loop, not by calling `bit_length`. Inequality (29) excludes every possible A at those m. Since $4s > 3s+1$, (30) persists at every larger m and every $k \leq 1506$. Therefore:

12.11.1 Theorem D. Finite arithmetic certificate, all primitive lengths

There is no nontrivial positive integer Collatz cycle whose primitive odd-return word has at most **1,506 exponent-1 positions**.

The proof is (25)-(30) and the recorded complete integer certificate. This improves this workbench's self-contained certificate, not a published cycle record. It does not use an external large-source verification bound.

The next retained budget $k=1507$ is forced to

$$m = 3631, \quad A = 5755, \quad 1166401 \leq \min x_i \leq 1505447 \quad (\text{odd minimum}). \quad (31)$$

The exact ceiling over ALL integers is 1,505,448, so the odd ceiling is one smaller. The next power of two fails (27), and (28) at $m=3632$ excludes that and every larger length for $k=1507$. Thus $E=0$ and this sector contains exactly 1,507 ones and 2,124 twos. The all-rise-then-all-fall ordering is already excluded by Theorem C-rational; the companion also excludes every cycle with one cyclic exponent-one run, including arbitrary larger exponents. Its all-length minimum-run bound $m > 226a$ remains attached to the actual minimum placement. The other ordered placements have not been exhausted here. The original gap dictionary, integral forcing and actual primitive-period requirement remain attached to (31).

12.12 11. What is resolved here, and the exact surviving arithmetic task

The section-defect calculation handles arbitrarily long repetitions of a fixed packet: its nonzero dyadic rank strictly decreases, and its zero is an actual return with its original primitive retained. Packet switching cannot be treated as a common monotone rank; (11)-(13) give an unbounded reset and then a descent proof for that same family. The run cone provides another unbounded family of original descent relations, without bounding the individual exponents. Equations (20)-(24) preserve the

original first-jet object when using those arithmetic reductions.

The remaining source is not identified with all locally compatible residue paths. An actual least vertex of a nonbase component must avoid every proved descent family. Its initial rise/fall block is outside every applicable descent cone. The companion removes the positive-gap equality case in the weak pure-two lemma by an explicit established-theorem-dependent proof; arbitrary packet-return zeros outside that family remain retained. Every nontrivial cycle has at least two cyclic exponent-one runs and satisfies the companion's original minimum-run bound. Its packet changes obey (11), including their full resonant residuals. Its compatible dyadic residues must still stabilize at one original positive integer. No proof is supplied that every such residual source is covered, and no original nontrivial cycle or divergent source is produced.

The concrete next work is to control the remaining resonant packet-switch fibres with their exact offsets, or add further complete descent families on those fibres. The atlas and first-jet witness interfaces accept such families without weakening the integer or support requirements. Increasing the finite source bound remains a separate finite calculation; it is not a substitute for that all-source task.

12.13 12. Reproduction and publication status

Run `python-Breplay.py` in this directory. The replay checks source hashes, independently verifies the full atlas and first-jet support, and compares both ordinary/optimized outputs byte-for-byte. Add `--predecessors` to run the unchanged structural and first-jet predecessors as well. Add `--rebuild-atlas` to regenerate the symbolic cover from the least-uncovered-source search and compare its canonical uncompressed bytes. The gzip container's incidental platform header is not used as a mathematical identity for regenerated data.

`query.py` exposes repetition certificates, resonant-family examples, peak blocks and finite integral first-jet witnesses from the atlas. Source paths and clocks remain explicit. `morphisms.jsonl` records domains, formulas, inverses, support and information-loss scope for the new maps.

The current connector successfully read the repository and pinned source but exposes no write operation in this session. The local Git HTTPS read also failed to resolve the host. Plugin discovery found the installed GitHub plugin, not an additional available write action. This increment is therefore packaged as a tested additive patch against the read head, not claimed as a remote commit or CI run. The remote status is recorded in the separate delivery receipt. No inherited file or Zeta source is modified by the patch.

12.14 References and attribution

[P1] The original workbench `intrinsic_zero_firstjet_20260915/note.md`, Sections 3-9, and its unchanged `firstjet.py`. Current read commit `d0c987b12c20be5978f496ab1ef2a60018caa5e5`.

[P2] The original `residual_ssplice_20260914/note.md` and `integral_blocks.md`: affine source cylinders, exact first-descent masks and integral comparison maps. The new checker reconstructs its data independently rather than silently replacing the formulas.

[P3] supported_structure_20260915/note.md: the original maximal exponent-1 run and supported component calculation. Its corrected source blob is 115af68c1684147e840cdd6cf690269490058f95.

[BL] D. J. Bernstein and J. C. Lagarias, *The $3x+1$ Conjugacy Map*, Canadian Journal of Mathematics 48 (1996), 1154-1169, DOI 10.4153/CJM-1996-060-x. The publisher abstract and authors' publication index were consulted for established dyadic-coding context; no uninspected theorem from that paper is a premise.

[RT] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv 2502.00948v5, current readable text at <https://arxiv.org/html/2502.00948v5>, especially Definitions 1.1-1.2 and the coefficient/remainder distinction. The elementary shortened-map clock expands an odd-return exponent a into one odd shortened step and $a-1$ divisions; grouping between successive odd states is its inverse. Thus its affine coefficient/remainder at a completed return are exactly L/U and C/U in (1). No coefficient-stopping-time conjecture is assumed.

Research direction and Split-Zero framework: KokunoYumeto. This continuation's derivations, exposition and executable finite checks were developed in the present tool-assisted session. General source provenance and standard mathematical operations are not relabelled as a priority claim.

Chapter 13

Completing the run argument with an explicit logarithm bound

Source: repository/collatz_reconstruction/research_program/defect_rank_descent_20260915/logarithmic_run_exclusion.md.

15 September 2026. This is part of the defect_rank_descent_20260915 continuation. The original map, word order, integral sections and first-jet differentials are those of note.md. This companion proves two additional all-length results. It does not assert an all-source Collatz conclusion.

13.1 1. Established transcendence input and the exact specialization

We use the rational-number specialization of Matveev's lower bound, as printed in Theorem 2.1 of Languasco, Luca, Moree and Togbe, *Sequences of integers generated by two fixed primes*, Abhandlungen aus dem Mathematischen Seminar der Universitat Hamburg 95 (2025), 123-148, DOI 10.1007/s12188-025-00293-9. The complete theorem statement and its logarithmic-height convention were read in the publisher's HTML. That paper cites Matveev's original theorem and the Bugeaud-Mignotte-Siksek formulation. This external transcendence theorem is an explicit proof dependency, not a theorem claimed to have been proved by our finite checker. In that statement, for positive rational numbers γ_i , nonzero integers b_i , and nonzero $\Lambda = \prod \gamma_i^{b_i - 1}$,

$$\log |\Lambda| > -1.4 \cdot 30^{k+3} k^{4.5} (1 + \log B) \prod_i h(\gamma_i), \quad B \geq \max_i |b_i|. \quad (\text{L1})$$

Here $h(r/s) = \max(\log |r|, \log s)$ for reduced positive rationals. We use only $\gamma_1 = 2, \gamma_2 = 3$, with exponents $A, -m$, so their heights are $\log 2$ and $\log 3$. The original coefficient expression is

$$\Lambda = 2^A 3^{-m} - 1 > 0, \quad A > m \geq 1, \quad B = A.$$

Unique prime factorization proves it nonzero. No exponent in this application is zero. The coefficient in (L1) has the rigorous rational upper bound

$$1.4 \cdot 30^5 \cdot 2^{4.5} \log 2 \log 3 < \frac{14}{10} 30^5 \cdot 24 \frac{7}{10} \frac{11}{10} = 628689600 < 10^9.$$

We used $\sqrt{2} < 3/2$, $1/2 < \log 2 < 7/10$, and $1 < \log 3 < 11/10$. The latter inequalities follow

from the explicit series bounds in Section 2. Put $c_*=10^9$. Thus the only transcendence estimate used below is

$$\boxed{\log \Lambda > -C_*(1 + \log A)}. \quad (\text{L2})$$

We retain the original A, m in every substitution. This does not identify a coefficient contraction with an actual descent. The additive correction and integer source congruences enter separately through the run argument in `note.md`.

13.2 2. A fully rational interval excluding an infinite collection of approximations

For an integer $x \geq 2$, set $z = (x-1)/(x+1)$. Integration of the geometric series, or its finite identity with remainder, gives

$$2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1} < \log x < 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1} + \frac{2z^{2N+1}}{(2N+1)(1-z^2)}. \quad (\text{L3})$$

All omitted terms are positive, and replacing their denominators by $2N+1$ bounds their sum by the last geometric tail. The implementation uses $N=128$ for $x=2$ and $x=3$. Write the resulting rational intervals as (l_2, u_2) and (l_3, u_3) . Then

$$\gamma_- := l_3/u_2 < \gamma := \log 3 / \log 2 < \gamma_+ := u_3/l_2.$$

The following certificate consists entirely of integer and rational comparisons:

$$\frac{83130157078217}{52449289519716} < \gamma_- < \gamma_+ + 2^{-128} < \frac{350861368503572}{221368876767703}. \quad (\text{L4})$$

Its two endpoint numerators and denominators satisfy

$$\begin{aligned} 350861368503572 \cdot 52449289519716 - 83130157078217 \cdot 221368876767703 &= 1, \\ 52449289519716 + 221368876767703 &= 273818166287419 > 226 \cdot 10^{12}. \end{aligned} \quad (\text{L5})$$

The final endpoint pair is obtained by a mediant search whose steps and hash are recorded; the proof needs only (L4)–(L5), not confidence in that search.

13.2.1 Elementary denominator lemma

For reduced fractions $l=p/q < h=r/s$ with $rq-ps=1$, every rational x/y strictly between them, with x, y positive integers, satisfies $y \geq q+s$.

Proof. The integers $k=xq-py$ and $j=ry-sx$ are both positive. Direct expansion gives $y=sk+qj$, hence $y \geq q+s$. This holds even when x/y has not been reduced. QED.

Applying this lemma proves

$$\boxed{0 < A/m - \gamma < 2^{-128}, \quad 1 \leq m \leq 226 \cdot 10^{12} \quad \text{has no integral solution } (A, m)}. \quad (\text{L6})$$

The map from the original period coordinates to the rational number is $(A, m) \rightarrow A/m$. Its fibres consist of integer multiples of a reduced numerator/denominator pair. The denominator lemma applies directly to every unreduced member; no original

multiplicity is lost.

13.3 3. The contracting pure-two comparison is always strictly separated

Recall

$$D_2(a, b) = 2^{a+2b} - 3^{a+b}, \quad a, b \geq 1.$$

13.3.1 Theorem E. Complete positive-gap inequality

On the complete integer domain $D_2(a, b) > 0$,

$$\boxed{D_2(a, b) > 3^a - 2^a.} \quad (\text{L7})$$

Proof. The elementary run-cone proof in `note.md` establishes (L7) for all $b \geq \lceil 3a/2 \rceil$. Consider an alleged remaining pair with $0 < D_2 \leq 3^a - 2^a$. It has

$$a < b < \lceil 3a/2 \rceil, \quad m = a + b < 5a/2, \quad A = a + 2b < 4a. \quad (\text{L8})$$

The lower inequality $b > a$ follows from $(2/3)^a (4/3)^b \leq (8/9)^a < 1$ when $b \leq a$. The integer endpoint for b gives the strict $A < 4a$ inequality in both parity cases.

The original relative gap obeys

$$0 < \Lambda = D_2/3^{a+b} < 3^{-b} < 3^{-a}.$$

Together with (L2) and $\log 3 > 1$, this yields

$$a < C_*(1 + \log(4a)). \quad (\text{L9})$$

It forces $a < 10^{12}$. For an explicit check, the positive finite sum $\sum_{j=0}^{10036} j/j!$ exceeds $452 \cdot 10^{12}$, so $\log(4 \cdot 10^{12}) < 36$. At $a = 10^{12}$, the left minus right of (L9) is positive; its derivative $1 - C_*/a$ is positive thereafter. These inequalities are checked by rational arithmetic.

For $a \geq 128$, therefore $m < 226 \cdot 10^{12}$ and

$$0 < A/m - \gamma = \frac{\log(1 + \Lambda)}{m \log 2} < 2 \cdot 3^{-b}/m < 2 \cdot 3^{-128} < 2^{-128}.$$

This contradicts (L6). For the remaining $1 \leq a \leq 127$, the finite certificate computes the least b with $D_2 > 0$ and verifies (L7) there. Increasing b sends D_2 to $3D_2 + 2^{a+2b}$, so the inequality persists. These 127 exact rows cover all remaining b , rather than truncating b at a search bound. QED.

13.3.2 Corollary E1. Strict original descent at the exact comparison threshold

For every actual positive odd source whose word is $1^a, c_1, \dots, c_b$, with $a, b \geq 1$ and every $c_i \geq 2$, the original endpoint is strictly below the original source throughout the domain

$$\boxed{2^{a+2b} > 3^{a+b}.} \quad (\text{L10})$$

Proof. For a pure-two tail, the exact source parameter and displacement from `note.md` are

$$3^a t - 1 = 4^b z, \quad z \geq 1, \quad n - y = 2 \left(\frac{D_2 z + 2^a}{3^a} - 1 \right).$$

Equation (L7) makes the inner integer at least two, so $n - y \geq 2$. A tail containing a higher exponent is handled by the proved monotone affine comparison in Lemma C2 of `note.md`, on precisely the same original source. QED.

This is stronger than the elementary 3/2 cone and the certified 2124/1507 cone. Those earlier arguments remain useful as shorter self-contained certificates that do not use (L1). The new condition refers to the pure-two COMPARISON multiplier; it does not assert that every arbitrary word with contracting actual multiplier descends. The actual higher exponents, offsets, source congruences, and intermediate integers remain attached. A zero displacement left possible by the earlier weak pure-two lemma is now excluded on its actual positive domain by (L7), not deleted from its source declaration.

13.4 4. A global restriction on the rise beginning at an actual cycle minimum

Let an actual nontrivial positive cycle have primitive odd-return length m , original exponent sum A , and least original vertex n . It cannot have exponent ≥ 2 leaving n , because that would give $(3n+1)/4 < n$. Its first $a \geq 1$ exponents are therefore a maximal run of ones. The exact original source formula gives

$$n = 2^{a+1}t - 1, \quad t > 0 \text{ odd}. \quad (\text{L11})$$

The common source cover in `note.md` gives $n \geq s_0 = 1166401$. In the complete original source of this run, the least allowed vertex is more precise than the maximum of two inequalities. It is

$$t_a = \text{least positive odd integer at least } (s_0 + 1)/2^{a+1}, \quad s_a = 2^{a+1}t_a - 1. \quad (\text{L12})$$

This is an exact progression intersection, with inverse $t = (n+1)/2^{a+1}$. For example at $a=19$, (L12) gives 3145727, not the weaker bound 1166401. At $a=17$ it gives 1310719. The certificate retains every t_a and s_a .

There is further source information in the already certified supported-zero fibre. The complete equation support E_B in `note.md` has its actual finite integral contraction to 1. A nontrivial cycle cannot contain a vertex of E_B . Define $\tilde{t}_a$ to be the least odd $t \geq t_a$ for which $2^{(a+1)*t} - 1$ is not in E_B , and put

$$\tilde{s}_a = 2^{a+1}\tilde{t}_a - 1. \quad (\text{L12a})$$

This exists because E_B is finite. For $a \leq 127$ the delivered `retained_zero_minima.json` lists every skipped original parameter and its finite integral first-jet witness. There are only nine skipped sources, with 639 original returns in their witness expansions. Every one is independently checked to belong to the previously constructed E_B with its recorded height. Thus this reuses the retained source behind a supported zero; it is not a new sweep of starting integers or an assertion that a zero value determines

its preimage without the map.

For example, at $a=17$ the original value 1310719 is in E_B and has height 105, although it lies above the contiguous interval bound. Its exclusion makes the next allowed parameter 7 and the next minimum bound 1835007. The unweighted equations and the finite first-jet witness behind that exclusion remain in the record. For larger a the inequality $\widetilde{m} \geq 2^{a+1} - 1$ suffices; no unknown infinite orbit is queried.

13.4.1 Theorem F. All-length minimum-run exclusion

Every actual nontrivial positive cycle satisfies

$$\boxed{m > 226a}, \quad (\text{L13})$$

where a is the maximal initial exponent-one run at its actual minimum. The coefficient 226 is a proved workbench bound, not a claimed optimal constant or published record.

Proof. Assume instead $m \leq 226a$. Multiplying the original cycle equations gives

$$0 < \delta := A \log 2 - m \log 3 = \sum_{i=1}^m \log \left(1 + \frac{1}{3x_i} \right) \leq \frac{m}{3\tilde{s}_a}. \quad (\text{L14})$$

All $x_i \geq 3$, so $2^A \leq (10/3)^m < 4^m$, hence $A < 2m \leq 452a$. Also $\Lambda = e^{\delta} - 1$ is precisely the positive expression in (L2).

First let $a \geq 128$. Equation (L11) gives $\delta \leq 226a / [3(2^{a+1} - 1)] < 1/2$. The last bound is checked at 128 and persists: $a / (2^{a+1} - 1)$ decreases, as direct cross-multiplication shows. For $0 < \delta < 1/2$, integration of e^x and $e^{1/2} < 2$ give $e^{\delta} - 1 < 2\delta$. Consequently

$$\Lambda < 226a 2^{-a}.$$

Combining with (L2) gives

$$a \log 2 < C_*(1 + \log(452a)) + \log(226a). \quad (\text{L15})$$

Since $\log 2 > 1/2$, the rational certificate at $a_0 = 10^{12}$,

$$\log(452a_0) < 36, \quad a_0/2 > 37C_* + 36, \quad a_0 > 2(C_* + 1),$$

and the derivative $1/2 - (C_* + 1)/a$ prove $a < a_0$. Thus $m < 226 \cdot 10^{12}$. On the other hand (L14) gives

$$0 < A/m - \gamma \leq \frac{1}{3\tilde{s}_a \log 2} < 2^{-a} \leq 2^{-128},$$

contradicting (L6).

It remains to cover $1 \leq a \leq 127$. For each such a , the certificate constructs a rational bracket $p_a/q_a < \gamma_- < \gamma_+ < r_a/s'_a$ with determinant one, denominator sum greater than $226a$, and

$$\gamma_+ + \frac{1}{3\tilde{s}_a l_2} < \frac{r_a}{s'_a}. \quad (\text{L16})$$

The exact table contains all 127 source parameters, bounds and endpoint pairs. Equation (L14) places A/m strictly within this bracket. The denominator lemma gives $m > 226a$, the desired contradiction. All rows are checked with exact fractions and the same certified logarithm intervals. QED.

Retaining the original odd parameter first improves the weaker interval-only ratio 191 to 213. The nine original source exclusions in (L12a), already supported by E_B , then improve 213 to 226. The checker verifies both the plain-progression baseline and the support-refined table. No additional discovery source was iterated for this improvement. The previously retained zero fibre is used with its actual original source and boundary witness, not as a bare zero symbol.

13.4.2 Corollary F1. A complete ordering family of cycles is excluded

There is no nontrivial positive cycle whose primitive original exponent word has all its exponent-one letters in a single cyclic run. The other exponents may be arbitrarily large and their number is unbounded.

Proof. Such a cycle has, after its actual minimum, a consecutive ones and then $b \geq 1$ exponents ≥ 2 . The ones strictly increase each vertex, while every later edge strictly decreases its vertex above 1, so the beginning of the run is indeed the actual minimum and $m = a + b$. The exponent sum satisfies $A \geq a + 2b = 2m - a$. The same original cycle product gives

$$(6/5)^m \leq 2^a.$$

Since $(6/5)^4 > 2$, this forces $m < 4a$, contradicting (L13). A word with no ones decreases at every nonbase vertex and cannot be such a cycle; a word with only ones increases at every vertex. Thus no excluded edge case is assigned a positive periodic source. QED.

This result excludes an all-length family of actual period-torsion modules through their original arithmetic realizations. It does not assert that arbitrary abstract circles with this word pattern are absent. Their rational affine primitives, integral forcing obstructions and supported zero values remain available; what has been excluded is a positive integral cycle realizing that pattern.

13.5 5. Exact relationship with the first-jet object

No new cohomological quotient is introduced. Every path certified by Corollary E1 uses the same original equations and satisfies

$$d_\epsilon(\epsilon B_p(n)) = \epsilon V_n - \epsilon V_{g_p(n)}.$$

Since the endpoint is lower, an existing integral endpoint witness splices to the source witness. Original cycle modules have first-jet index equal to their ORIGINAL primitive period. Theorem F and Corollary F1 restrict the original words and original minimum placements that can produce such a module. They do not infer disappearance merely from a zero scalar or from passing to rational coefficients.

The proven source-cover bound s_0 in Theorem F is a finite arithmetic dependency verified in this same package. The transcendence estimate (L1) is a sourced established theorem. Equations (L3)–(L16) and the full run proofs specify every inter-

vening inequality. No probability law, unproved stopping-time estimate, or assumed common nilpotent exponent is used.

13.6 6. Execution and remaining source

`logarithm_certificate.py` uses only exact rational arithmetic. It recomputes the positive series bounds, the large Farey bracket and its determinant, the 127 small pure-run rows, all 127 exact minimum-run source brackets, and the nine reused original supported-zero witnesses. `logarithm_verification.json` records the complete output. `verify.py` includes this certificate and rejects changed bracket endpoints and an invalid coefficient claim. No numerical logarithm or unchecked floating-point continued fraction is a proof premise.

The finite rational certificate does not independently verify Matveev's theorem. The theorem is explicitly sourced; the subsequent finite steps and arithmetic instantiation are checkable here. These all-length results remain additional to, not a rebranding of, the finite-source cover and its 1506-exponent-one-count exclusion.

The actual remaining cycle source has at least two cyclic exponent-one runs and obeys the original minimum-run restriction. The aperiodic source still requires control of genuine packet switches and their resonant source fibres. The nonzero-defect rank in `note.md` handles repeated packets, not arbitrary switching. Neither new argument proves all such sources converge.

13.7 Sources

[M] E. M. Matveev, *An explicit lower bound for a homogeneous rational linear form in the logarithms of algebraic numbers. II*, *Izvestiya: Mathematics* 64 (2000), 1217–1269. DOI 10.1070/IM2000v064n06ABEH000314. The original article's bibliographic record was checked; its full proof was not re-audited in this session.

[LLMT] A. Languasco, F. Luca, P. Moree and A. Togbe, *Sequences of integers generated by two fixed primes*, *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 95 (2025), 123–148, Theorem 2.1 and the immediately preceding height definition. DOI 10.1007/s12188-025-00293-9. Full readable source: <https://link.springer.com/article/10.1007/s12188-025-00293-9>. This is the inspected formulation of [M] used in (L1).

Chapter 14

Positive shadows of retained integral cycle sections

Source: repository/collatz_reconstruction/research_program/negative_shadow_descent_20260916/note.md.

16 September 2026. Collatz workbench continuation.

The published Collatz source was read at d0c987b12c20be5978f496ab1ef2a60018caa5e5. Its first-jet comparison remains unchanged. The separately delivered defect_rank_descent_20260915 package is still an additive local predecessor; it is not silently assigned a remote commit. This contribution gives new all-length arithmetic families and a small independent gap certificate. The large preceding source atlas is **not** a premise of the new all-length theorem.

The elementary descriptions of the signed cycles and the general logarithm theorem are not claimed as discoveries. The arithmetic deductions below are independently proved. Neither a published cycle record nor a resolution of all positive sources is claimed.

14.1 1. Original map and the nine retained integral sections

Use the same formula on nonzero odd integers of either sign:

$$T(x) = \frac{3x + 1}{2^{\nu_2(3x+1)}}.$$

The positive source used for the conclusion is always $X = \{1, 3, 5, \dots\}$. No replacement of the forcing by $3x - 1$ is made. For an ordered word $p = (a_1, \dots, a_m)$, retain

$$A_j = \sum_{i=1}^j a_i, \quad A = A_m, \quad U = 2^A, \quad L = 3^m, \quad C = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}.$$

Composition in chronological order is

$$f_p(x) = \frac{Lx + C}{U}. \tag{1}$$

The following two signed cycles are verified directly under this original formula:

$$-5 \xrightarrow{1} -7 \xrightarrow{2} -5,$$

$$-17 \xrightarrow{1} -25 \xrightarrow{1} -37 \xrightarrow{1} -55 \xrightarrow{2} -41 \xrightarrow{1} -61 \xrightarrow{1} -91 \xrightarrow{4} -17. \quad (2)$$

For each of their nine distinct vertices $-h$, let p_h be the original exponent word beginning at that vertex. All rotations and their original starting vertices are retained:

h	p_h	(m, A)
5	(1,2)	(2,3)
7	(2,1)	(2,3)
17	(1,1,1,2,1,1,4)	(7,11)
25	(1,1,2,1,1,4,1)	(7,11)
37	(1,2,1,1,4,1,1)	(7,11)
55	(2,1,1,4,1,1,1)	(7,11)
41	(1,1,4,1,1,1,2)	(7,11)
61	(1,4,1,1,1,2,1)	(7,11)
91	(4,1,1,1,2,1,1)	(7,11)

For each row,

$$L > U, \quad C = h(L - U), \quad f_{p_h}(-h) = -h. \quad (3)$$

Thus the original marked class $[C]$ in $\mathbb{Z}/(U-L)\mathbb{Z}$ vanishes, with its actual integral primitive $-h$. This is a **negative** primitive, not a positive Collatz cycle. For the two-letter rows the ambient group has modulus one; for the seven-letter rows it has modulus 139. Their zero forcing classes do not identify those different presented objects or their different sections.

The retained extension matrix is

$$M_p = \begin{pmatrix} L & C \\ 0 & U \end{pmatrix}.$$

For the integral section $s_n(1) = e_2 + ne_1$, the actual intertwining defect is

$$M_p s_n(1) - s_n(U) = (C - (U - L)n)e_1 = (L - U)(n + h)e_1. \quad (4)$$

Changing the section changes (4), although its quotient class remains $[C] = 0$. This is the information from the supported section which will be used on the positive source. The exact coordinate change $z = n + h$, inverse $n = z - h$, gives $z \mapsto (L/U)z$ for this packet. Neither the original positive domain nor the inverse translation is removed.

14.2 2. Complete positive shadow lattice, with inverse maps

14.2.1 Proposition 1. Positive realizations of a repeated retained packet

For every h in (2) and integer $r \geq 1$, the complete positive source realizing p_h^r is

$$\boxed{n = 2U^r t - h, \quad t \in \mathbb{Z}_{\geq 1}, \quad T^{mr}(n) = 2L^r t - h.} \quad (5)$$

The inverse source and target parameters are $(n + h)/(2U^r)$ and $(y + h)/(2L^r)$, on these stated images. The source step $2U^r$ and target step $2L^r$ remain different, original integers.

Proof. Formula (3) gives by composition

$$f_{p_h^r}(x) = \frac{L^r(x+h)}{U^r} - h. \quad (6)$$

For an actual positive source, both its initial and terminal values are odd. The section defect for the repeated word is $(L^r - U^r)(n+h)$; its equality to $U^r(f(n) - n)$ makes it divisible by $2U^r$. Since $L^r - U^r$ is odd, it is coprime to $2U^r$, proving the source congruence in (5).

Conversely write $-h_j$ for the successive vertices of the signed orbit repeated r times, and A_j for the original prefix sums of that repeated word. At every position,

$$x_j = -h_j + 2^{Ar+1-A_j}3^jt. \quad (7)$$

These are integers and are odd, because each correction is even. The original affine equations give $3x_{j-1} + 1 = 2^{a_j}x_j$; hence their valuations are exactly a_j . The starting value in (5) is positive: $U \geq h+1$ for the listed anchors. Positivity propagates along the displayed equations. Formula (7) proves every intermediate original value and gives (5) at the endpoint. Positivity of n forces $t \geq 1$; every such integer is allowed. The inverse identities are substitutions. ■

There is an exact common path object, not an assumed map from a negative cycle to a positive cycle. Take the finite position **interval** with vertices $0, \dots, mr$ and its consecutive edges. Evaluation at $-h_j$ and at x_j gives two maps into the original signed and positive graphs. They preserve both source and target maps, and hence their incidence differentials and the first-jet differentials. The negative evaluation identifies its endpoints. The positive evaluation has endpoints (5); identifying them would require the additional, generally false, equality $2U^rt - h = 2L^rt - h$. That endpoint defect is retained.

The source module, its original relation submodule, the supported label p_h , and its selected integral primitive are therefore all used. No chronological meaning is assigned to intrinsic Split-Zero support.

14.3 3. The second section: arbitrary falling tails around the fixed point 1

Let $q = (c_1, \dots, c_s)$, where $s \geq 1$ and every $c_i \geq 2$. Put

$$B_j = \sum_{i=1}^j c_i, \quad B = B_s, \quad U_q = 2^B, \quad L_q = 3^s, \\ C_q = \sum_{j=0}^{s-1} 3^{s-1-j} 2^{B_j}, \quad E_q = C_q + 3^s - 2^B.$$

The change of coordinate $w = x - 1$, inverse $x = w + 1$, gives the **exact** formula

$$f_q(x) - 1 = \frac{3^s(x-1) + E_q}{2^B}, \quad E_q = \sum_{i=1}^s 3^{s-i} 2^{B_{i-1}} (4 - 2^{c_i}). \quad (8)$$

It follows that $E_q \leq 0$, with equality exactly for the pure-two tail. Every coefficient in the sum is positive and each final factor is nonpositive, proving both assertions. Equivalently, appending c changes E to $3E + (4 - 2^c)2^{B_{\text{old}}}$.

A zero E_q remains a supported forcing value. The pure-two tail still has all its original

edges, original differential, and multiplier $3^s/4^s$. It is not removed from the word or its module.

The two section coordinates are related by the exact affine map

$$w = z - (h + 1), \quad z = w + (h + 1). \quad (9)$$

The translation $h + 1$ is the critical retained term. Dropping it would falsely replace coefficient contraction by actual descent.

14.4 4. The arithmetic gap that controls that translation

14.4.1 Theorem 2. Two all-length integer gap inequalities

For either row

$$(U, L, K) = (8, 9, 4) \quad \text{or} \quad (2048, 2187, 46),$$

and integers $r, s \geq 1$, $B \geq 2s$, define

$$D = U^r 2^B - L^r 3^s.$$

On the complete positive-gap domain $D > 0$,

$$\boxed{D > K(2^B - 3^s)}. \quad (10)$$

The next section proves the theorem by an explicit logarithm bound and a complete finite certificate. The coarser cones at the end of that section have a separate elementary proof.

The result is a statement about the indicated original integers, not a numerical approximation to the ratio of the two sides. All r, s, B are unrestricted within the displayed domain.

14.5 5. Proof of the gap theorem

Write the required positive margin as

$$H(r, s, B) = (U^r - K)2^B - (L^r - K)3^s. \quad (11)$$

The listed constants satisfy $U \geq 2K$, $U < L$, and

$$16U - 9L > 7K. \quad (12)$$

The first row has difference 47 versus 28; the second has difference 13085 versus 322.

14.5.1 5.1 An elementary whole half-space

For $s \geq 2r$, $B \geq 2s$, it suffices to take $B = 2s$, because $U^r - K > 0$. After division by 3^s , the margin is increasing in s , so it suffices to take $s = 2r$. At that value the desired comparison is

$$(16U)^r - (9L)^r > K(16^r - 9^r).$$

Use $a^r - b^r = (a - b) \sum_{j=0}^{r-1} a^{r-1-j} b^j$. By (12), and the termwise bounds $16U \geq 16$, $9L \geq 9$, the left side is strictly larger than $7K \sum 16^{r-1-j} 9^j = K(16^r - 9^r)$. This proves the entire half-space.

14.5.2 5.2 Reduce every other case to its first crossing

It remains to take $1 \leq s < 2r$. For fixed r, s , let B_0 be the least integer $B \geq 2s$ with $D > 0$. It exists and

$$2s \leq B_0 \leq 4r, \quad (13)$$

because $U^r 2^{4r} > L^r 3^{2r} \geq L^r 3^s$ by (12). The margin (11) increases with B . A failure anywhere therefore gives a failure at B_0 .

Define the **same original** logarithmic gap

$$\Lambda = \frac{U^r 2^{B_0}}{L^r 3^s} - 1 = 2^{A_*} 3^{-m_*} - 1 > 0,$$

$$A_* = Ar + B_0 \leq 15r, \quad m_* = mr + s \leq 9r. \quad (14)$$

The letters (m, A) here are (2,3) or (7,11). A supposed failure of (10) gives

$$D \leq K(2^{B_0} - 3^s) < K2^{B_0},$$

and hence

$$\Lambda < \frac{K}{U^r - K} \leq \frac{2K}{U^r} \leq \frac{92}{8^r}. \quad (15)$$

The retained correction K has not been omitted in this estimate.

14.5.3 5.3 The established transcendence input

Use the rational-number instance of Matveev's theorem printed as Theorem 2.1 in [M]. The positive rational numbers are exactly 2 and 3, and their nonzero integer exponents are A_* and $-m_*$. Unique prime factorization proves that the expression is nonzero. The heights are $\log 2$ and $\log 3$. Its coefficient is safely bounded by

$$1.4 \cdot 30^5 \cdot 2^{4.5} \log 2 \log 3 < \frac{14}{10} 30^5 \cdot 24 \cdot \frac{7}{10} \cdot \frac{11}{10} = 628689600 < 10^9.$$

Since $2^{A_*} > 3^{m_*}$, we have $A_* > m_*$, so the exponent maximum may be taken as A_* . The resulting bound is

$$\log \Lambda > -10^9(1 + \log A_*). \quad (16)$$

This external transcendence theorem is an explicit dependency. Neither the finite Python tests nor this note are offered as a new proof of Matveev's theorem.

Combining (15) and (16), and using $\log 8 > 1$, forces

$$r < 10^9(1 + \log(15r)) + \log 92. \quad (17)$$

This gives $r < 10^{12}$. Indeed the finite rational sums $\sum_{j=0}^{100} 36^j/j! > 15 \cdot 10^{12}$ and $\sum_{j=0}^6 5^j/j! > 92$ prove the needed upper logarithm bounds at 10^{12} . At that endpoint the right side of (17) is less than $37 \cdot 10^9 + 5 < 10^{12}$. The derivative of left minus right is $1 - 10^9/r > 0$ thereafter.

14.5.4 5.4 A completely rational denominator certificate

For $x = 2, 3$, put $z = (x - 1)/(x + 1)$. The positive series and its geometric remainder give

$$2 \sum_{j=0}^{127} \frac{z^{2j+1}}{2j+1} < \log x < 2 \sum_{j=0}^{127} \frac{z^{2j+1}}{2j+1} + \frac{2z^{257}}{257(1-z^2)}. \quad (18)$$

Let the corresponding rational bounds be l_2, u_2, l_3, u_3 . The certificate checks, entirely by integer cross-multiplication,

$$\begin{aligned} \frac{83130157078217}{52449289519716} &< \frac{l_3}{u_2} < \frac{\log 3}{\log 2} < \frac{u_3}{l_2}, \\ \frac{u_3}{l_2} + 2^{-128} &< \frac{350861368503572}{221368876767703}. \end{aligned} \quad (19)$$

The two outer fractions have determinant one, and their denominators sum to

$$273818166287419 > 9 \cdot 10^{12}. \quad (20)$$

Here is the precise denominator fact used. Between $p/q < t/u$ with $tq - pu = 1$, an original fraction a/b has positive integers $k = aq - pb$, $j = tb - au$. Direct expansion gives $b = uk + qj \geq u + q$. The fraction a/b need not be reduced; original multiplicities are retained.

For $r \geq 128$, the alleged failure satisfies

$$0 < \frac{A_*}{m_*} - \frac{\log 3}{\log 2} = \frac{\log(1+\Lambda)}{m_* \log 2} < \frac{184}{8^r} < 2^{-128}. \quad (21)$$

The last inequality is checked at 128 by $184 \cdot 2^{128} < 8^{128}$ and persists thereafter. We used $\log(1+\Lambda) < \Lambda$ and $\log 2 > 1/2$. Equations (17), (20) give $m_* < 9 \cdot 10^{12}$, contradicting the denominator fact and (19). This excludes every $r \geq 128$ at once.

14.5.5 5.5 The complete remaining finite base

For each of the two triples, the verifier checks every

$$1 \leq r \leq 127, \quad 1 \leq s \leq 2r,$$

computes the least $B_0 \geq 2s$ by integer doubling, and verifies

$$(U^r - K)2^{B_0} > (L^r - K)3^s.$$

There are exactly 16256 rows for each triple, 32512 total. The full triples (r, s, B_0) are delivered in `gap_bases.json.gz`. The checker regenerates them, verifies first-crossing minimality and the strict margin, and compares every byte of the canonical table. It

also records a hash of every exact integer margin. The smallest margins are 1 and $1631-46=1585$ respectively. Increasing B preserves the comparison, while Section 5.1 treats every longer tail. This completes the proof of Theorem 2. ■

14.5.6 5.6 Stronger elementary cones requiring no logarithm theorem

For the first triple, the gap margin is positive throughout $s \geq \lceil r/2 \rceil$, $B \geq 2s$. For the second it is positive throughout $s \geq \lceil r/4 \rceil$, $B \geq 2s$.

Here is an elementary induction. Let $d = 2$ or 4 respectively, and $\mu = 3L^d$. The listed constants satisfy $4U^d > 3L^d$ and $\mu > 7$. For the pure-two margins,

$$H(r+d, s+1, 2s+2) - \mu H(r, s, 2s) = (4U^d - 3L^d)U^r 4^s + K((\mu-4)4^s - (\mu-3)3^s) > 0. \quad (22)$$

The last bracket is positive for $s \geq 1$, since its value after division by 3^s is at least $(\mu-7)/3$. The finitely many seeds $r = 1, \dots, d$, $s = 1$ have positive margins and are explicitly checked. This gives the boundary $s = \lceil r/d \rceil$; monotonicity gives all larger s, B . This result is independent of [M]. The full positive-gap domain in (10) is wider and uses the proof above.

14.6 6. The actual positive descent theorem

14.6.1 Theorem 3. Repeated-shadow/falling-tail descent, at the actual coefficient crossing

For any of the nine p_h , let an original positive source realize

$$p_h^r(c_1, \dots, c_s), \quad r, s \geq 1, \quad c_i \geq 2.$$

On the entire domain

$$\boxed{U^r 2^B > L^r 3^s, \quad B = \sum_i c_i,} \quad (23)$$

its endpoint is strictly below its source. The domain uses the **actual total tail exponent**, not the all-two comparison total. All repetitions, tail lengths and individual exponents are unbounded.

Proof. By Proposition 1, $n = 2U^r t - h$, $t \geq 1$. The original combined numerator is

$$C_{\text{total}} = 3^s h(L^r - U^r) + U^r C_q.$$

Use the original endpoint equation and (8) to obtain the exact displacement

$$\boxed{n - y = \frac{2Dt - (h+1)(2^B - 3^s) - E_q}{2^B}.} \quad (24)$$

Equivalently its numerator is $2[D - K_h(2^B - 3^s)] + 2D(t-1) - E_q$, with $K_h = (h+1)/2$. For the two-letter anchors $K_h \leq 4$, and for the seven-letter anchors $K_h \leq 46$. Since $2^B > 3^s$, Theorem 2 proves the first bracket positive. The remaining terms are nonnegative. Thus $n - y > 0$. Both endpoints are actual odd integers, so their difference is at least two. Every term in (24), including a supported zero $E_q = 0$, remains present. ■

This also locates the formal positive fixed point of the whole word below the least possible positive source of its repeated prefix:

$$0 < \frac{C_{\text{total}}}{D} < 2U^r - h. \quad (25)$$

Indeed substituting $t = 1$ in the original section displacement proves $D(2U^r - h) - C_{\text{total}} > 0$; this is an integer comparison, not a claim that this lower bound already realizes the tail.

14.6.2 Corollary 3.1. Complete cycle-word exclusion

There is no positive integer cycle with a traversal word which is a cyclic rotation of $p_h^r(c_1, \dots, c_s)$, with the parameters above. Repetitions of such a traversal are not new cases.

Proof. An actual positive cycle obeys $2^{A+B} = \prod (3 + 1/x_i) > 3^{mr+s}$. Thus (23) holds on that original traversal. Theorem 3 makes its endpoint strictly smaller than its starting vertex, contradicting closure. A traversal consisting only of p_h^r has the sole affine fixed point $-h$ by (6), outside the positive source. Rotation is the invertible reindexing of the same original vertices and edges. ■

This excludes arbitrary numbers of separated rises, not just one consecutive run of exponent one. For example every word $(1, 2)^r$ followed by any nonempty all-falling tail belongs to this exclusion family. The seven-letter motifs give another family with several rise and fall transitions in each repeated packet.

The result is not an all-word cycle exclusion. In the earlier count sector $(m, A, k) = (3631, 5755, 1507)$, it eliminates the original ordering $(1, 2)^{1507}(2)^{617}$, and its retained cyclic rotations. The other gap orderings remain present. No stronger global numerical cycle bound is inferred.

14.7 7. Full source/target cylinders for every admitted falling word

The universal statement can be used without replacing its source by sampled starts. Let r_q be the least positive odd residue for the original tail word:

$$r_q \equiv (2^B - C_q)(3^s)^{-1} \pmod{2^{B+1}}.$$

At the repeated-packet endpoint $x = 2L^r t - h$, the legal tail condition is exactly

$$t \equiv t_0 := \frac{r_q + h}{2}(L^r)^{-1} \pmod{2^B}. \quad (26)$$

Take $0 \leq t_0 < 2^B$, and let $t_{\min} = t_0$ for $t_0 > 0$, otherwise $t_{\min} = 2^B$. The complete positive source and target are

$$\boxed{\begin{aligned} n(v) &= 2U^r t_{\min} - h + 2U^r 2^B v, \\ y(v) &= y_0 + 2L^r 3^s v, \quad v \in \mathbb{Z}_{\geq 0}, \end{aligned}} \quad (27)$$

where $y_0 = (L^r 3^s n(0) + C_{\text{total}})/(U^r 2^B)$. The inverse parameters are $(n - n(0))/(2U^r 2^B)$ and $(y - y_0)/(2L^r 3^s)$. Formula (26) follows by dividing the even congruence $2L^r t - h \equiv r_q \pmod{2^{B+1}}$ by two, whose inverse step is multiplication by two. The odd number L^r is invertible modulo 2^B . These calculations prove compatibility, completeness, and

both inverse identities. The zero residue case is retained and has its correct positive lower bound.

In this chart

$$n(v) - y(v) = n(0) - y_0 + 2Dv.$$

The whole-cylinder descent comes from (24), not from point sampling.

14.8 8. Return to the unchanged integral first-jet comparison

The absolute original graph, its maps J, P , and the ring $\mathbb{Z}[\epsilon]/\epsilon^2$ are the predecessor's. In particular

$$d_\epsilon = d - \epsilon P, \quad d = J - P.$$

For the actual finite edge chain $c_{p,q}(n)$ of (27),

$$d_\epsilon(\epsilon c_{p,q}(n)) = \epsilon V_n - \epsilon V_y. \quad (28)$$

The constant boundary is zero because the constant chain coefficient here is zero; the linear boundary is the original telescoping edge boundary. Thus in the retained reduction kernel $\mathfrak{B}_X$, $[\epsilon V_n] = [\epsilon V_y]$ with $y < n$ on (23).

An existing finite integral witness $d_\epsilon(b_y + \epsilon c_y) = \epsilon V_y$ splices to

$$d_\epsilon(b_y + \epsilon(c_{p,q}(n) + c_y)) = \epsilon V_n. \quad (29)$$

No division by a cycle length or change to rational coefficients occurs. The unchanged predecessor validator checks both coefficients and extracts an actual path to 1. The new verifier executes (29) on the delivered examples.

Equation (28) alone is a descent relation, not a claim that every variable endpoint $y(v)$ is already known to reach 1. It excludes every admitted source from being the least vertex of a nonbase component, whether that component is periodic or nonperiodic. Full source coverage still requires more arithmetic.

14.9 9. The exact remaining falling-tail language

For a fixed p_h^r , a falling tail not yet certified by (23) must have

$$U^r 2^B < L^r 3^s, \quad B \leq H_s := \lfloor \log_2(L^r 3^s) \rfloor - Ar. \quad (30)$$

There is no equality between the original powers of 2 and 3. The logarithm floor is computed exactly as an integer bit length. A tail of s exponents at least two has $B \geq 2s$. Only $s < 2r$ can remain by Section 5.1, and the stronger elementary cones give the smaller cutoffs as well.

For each remaining s , the complete number of such words is

$$\binom{H_s - s}{s}. \quad (31)$$

Subtracting one from each original exponent gives a composition with positive parts and sum at most $H_s - s$; stars and bars gives (31), with inverse adding one. The

ordered parts themselves and their source cylinders remain available. The empty tail is retained separately.

Thus at fixed repetition count, the next exponent-one switch can occur only after a finite, exactly specified list of uncontracted falling tails. Crossing tails have no positive exception box: Theorem 3 proves descent on their complete positive source. This does not make the whole orbit finite; it identifies the original mixed-switch frontier left by these new rules.

14.10 10. An exact reset law, followed by an actual repair

For the $h = 5$ packet, take the entire stratum $t \equiv 1 \pmod{8}$ in (5). Its next exponent is exactly three, because

$$3(2 \cdot 9^r t - 5) + 1 = 2(3 \cdot 9^r t - 7), \quad \nu_2(3 \cdot 9^r t - 7) = 2.$$

After that step the original value is

$$x = \frac{3 \cdot 9^r t - 7}{4}, \quad \nu_2(x + 1) = \nu_2(9^r t - 1) - 2.$$

Consequently its next maximal exponent-one run has exact length

$$\boxed{k = \nu_2(9^r t - 1) - 3}, \quad (32)$$

with complete parameter stratum

$$t \equiv 9^{-r}(1 + 2^{k+3}) \pmod{2^{k+4}}. \quad (33)$$

The endpoint after those k ones is $3^{k+1}(9^r t - 1)/2^{k+2} - 1$. The next exponent is at least two. Formulas (32)-(33) are inverse descriptions of the exact valuation, and their residues automatically lie in $1 \pmod{8}$. They retain the dependence on t , which permits arbitrary reset depths.

On the particular anchor ladder $t = 1$, this becomes

$$k = \nu_2(r). \quad (34)$$

An elementary proof suffices: an odd exponent in $9^d - 1$ leaves its valuation at 3, since its geometric quotient has an odd number of odd summands. Doubling multiplies by $9^d + 1$, of valuation one. Hence $\nu_2(9^r - 1) = 3 + \nu_2(r)$. This is an exact reset law, not a global decreasing-rank assertion for arbitrary t .

At $r = 12, t = 1$, the source 137438953467 follows $(1, 2)^{12}$, then exponent three, but the latter endpoint is still 211822152359. The next exponent is one, so the falling-tail theorem correctly retains this switch. Formula (34) determines the next run to have length two.

Continuing on that same original source gives the complete first-descent word

$$w = (1, 2)^{12}(3, 1, 1, 2, 1, 2, 1, 2, 1, 7), \quad |w| = 34, \quad A(w) = 57.$$

Its original coefficients are

$$L = 16677181699666569, \quad U = 144115188075855872, \quad C = 81607860757265581.$$

Every proper prefix has $D_j \leq 0$, and the final source satisfies $Dn_0 > C$. Therefore the entire original cylinder has its first strict descent at that last return:

$$137438953467 + 2^{58}v \mapsto 15904599857 + 33354363399333138v, \quad v \geq 0. \quad (35)$$

The proof is the original prefix inequality $T^j(n) < n \iff D_j n > C_j$, the literal cylinder congruence, and the positive final constant and slope of $Dn - C$. The checker recomputes every prefix and independently replays three lifts. Thus the failed applicability of one rule produces another complete arithmetic family. Formula (35) is not generalized to every r without proof.

14.11 11. Evidence, reproduction, and remaining scope

`verify.py` checks the signed original cycles, complete positive shadow congruences, 32512 integer gap rows, the rational logarithm intervals, original falling-word cylinders, supported zero tail forcings, first-jet boundaries, finite splices and reset strata. It rejects malformed anchor, exponent, lattice and status records. `gap_bases.json.gz` contains the complete base tables; `examples.json` contains the actual original paths and integral witnesses. `verification.json` separates the executed scopes. Ordinary and optimized outputs must agree byte for byte.

`crosscheck_atlas.py` is an optional crosswalk into the earlier local source atlas. It does not participate in the all-length proof. It identifies already-proved first-descent cells that are instances of this theorem; it neither rediscovers their points nor treats that finite collection as proof for the unrestricted repetition parameter.

No positive infinite survivor is certified by a finite remaining word label. No new common interval bound or numerical cycle record is claimed. The next arithmetic calculation is on the exact switch residues (33), the residual word families (30), and their original source parameters. The full conjecture still requires coverage of every ordinary positive source, or a rigorously surviving one. All statements above retain the ordinary integer lattice and the exact maps into the original first-jet kernel.

14.12 Sources

[P] `intrinsic_zero_firstjet_20260915/note.md` and `firstjet.py`, Collatz repository commit `d0c987b12c20be5978f496ab1ef2a60018caa5e5`. The executed dependency is the unchanged `firstjet.py`, Git blob `97f9ef5ad535f7ec18259d41fab6a3557be17d98`. This supplies the original integral kernel and finite-witness validator, not Theorem 2.

[A] The separately delivered `defect_rank_descent_20260915` local contribution: section defects and their exact repetition congruences; original source atlas; all-length rise/fall and logarithm certificate. Its files remain unchanged. The gap proof above re-establishes the needed integer and rational comparisons independently and has no dependency on that atlas or its convergence interval.

[M] A. Languasco, F. Luca, P. Moree and A. Togbe, *Sequences of integers generated by two fixed primes*, *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 95 (2025), 123-148, DOI [10.1007/s12188-025-00293-9](https://doi.org/10.1007/s12188-025-00293-9), Theorem 2.1. Its rational-number formulation of Matveev's established theorem was read in the primary publisher text during this continuation. The paper attributes its formulation to Bugeaud, Mignotte and Siksek and the underlying theorem to Matveev. No primary-literature file is copied into this package.

The original Split-Zero construction and research direction remain attributed to the owner's programme. Standard affine composition, finite graph incidence, geometric sums, and rational denominator lemmas are not assigned new priority here. This is a written mathematical contribution with exact computational certificates, not an independent external review or a new Lean proof.

Chapter 15

Mixed-support section control and arbitrary mixed-core Collatz descent

Source: repository/collatz_reconstruction/research_program/mixed_switch_control_20260916/note.md.

16 September 2026. Additive continuation in KokunoYumeto/collatz-workbench. Authenticated Collatz base: d0c987b12c20be5978f496ab1ef2a60018caa5e5. Read-only Zeta source pin: 1efd53337561d8f67cf0ac1d119acdaa222644c3.

This note proves a new arithmetic descent result for every ordering of two specified packet types, every exponent-level phase, every repetition count at least two, and every contracting falling tail. It also computes the exact mixed coordinate compatibility and order-dependent affine correction. The proof does not identify Split-Zero support with the weighted-automata term “history.” It retains the original integral source, masks, maps, affine numerators and first-jet witnesses. The full quantitative Zeta estimates are not premises.

15.1 1. Source intake and the mathematical arrow from mixed extension control

The current Zeta guide describes original source/target quotient metrics, simultaneous mixed rows, and a retained boundary quotient. Its remaining native long-return allocation is not imported as a Collatz estimate. The specific source read and used structurally is the earlier complete mixed-control fragment MRE.tex, equations MRE7-12 and MRE30-33. It retains a rank-two extension, its original action, and the actual section defect even when an auxiliary diagonal action admits a section. MCF.tex, MCF6-10, retains all independent coefficient masks and their zero-insertion maps. The current guide and these exact source identities are recorded in SOURCE_INTAKE.json.

Here is a common typed construction, rather than a claim based on the word “mixed.” Set

$$R_0 = \mathbb{Z}[\lambda, \mu, \kappa], \quad E_0 = R_0 e_1 \oplus R_0 e_2, \quad t \mid E_0 = \begin{pmatrix} \lambda & \kappa \\ 0 & \mu \end{pmatrix}.$$

The maps $a \mapsto ae_1$ and $ae_1 + be_2 \mapsto b$ give the exact sequence of $R_0[t]$ -modules

$$0 \longrightarrow R_{0,\lambda} \longrightarrow E_0 \longrightarrow R_{0,\mu} \longrightarrow 0. \quad (1)$$

Each term is free over R_0 . Every coefficient evaluation below therefore preserves the displayed underlying split exact sequence, and the matrices prove the intertwining equations. For the section $s_z(1) = e_2 + ze_1$,

$$(ts_z - s_z\mu)(1) = (\kappa + (\lambda - \mu)z)e_1. \quad (2)$$

In MRE, interchange its fixed basis order from $(1, u)$ to $(u, 1)$, with inverse the same permutation. The arithmetic operator becomes upper triangular. Evaluation

$$(\lambda, \mu, \kappa) \mapsto (a^\rho, a^\rho, a^\rho \log a)$$

gives exactly MRE10-11 and MRE32, including the original parameters and coefficient of the nilpotent map. For an original Collatz packet, evaluation

$$(\lambda, \mu, \kappa) \mapsto (L, U, C)$$

gives its original integral extension, and (2) gives $C - (U - L)z$. These are specified base changes of the same free extension presentation. They are not an asserted isomorphism between the two evaluated representations, or a transport of a Zeta numerical bound.

For two present packet slots, chronological composition is the actual polynomial matrix map

$$M_q M_p = \begin{pmatrix} L_q L_p & L_q C_p + U_p C_q \\ 0 & U_p U_q \end{pmatrix}. \quad (3)$$

The mixed coefficient in (3) remains. Taking a face of the two-slot diagram keeps its own label and coefficients; inserting a slot with supported zero is the original zero-insertion arrow, not an assertion that a missing packet has been supplied. All source records below keep the full ordered word in addition to its composite matrix.

15.2 2. The two original packets and their interaction

Use the original odd-return map on positive odd integers

$$T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1).$$

For a word $w = (a_1, \dots, a_m)$, set

$$A_j = \sum_{i \leq j} a_i, \quad A_0 = 0, \quad L = 3^m, \quad U = 2^{A_m}, \quad C = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}. \quad (4)$$

Its affine map is $(Lx + C)/U$. Its complete positive source is

$$n = \rho + 2Uz, \quad z \geq 0, \quad \rho \equiv (U - C)L^{-1} \pmod{2U}, \quad 0 < \rho < 2U. \quad (5)$$

The unchanged target is $(L\rho + C)/U + 2Lz$. The inverse parameters are the original differences divided by $2U$ or $2L$, on those images only. Formula (5) is the workbench's original exact valuation cylinder: composing the original equations gives the terminal odd congruence, and reversing the congruences successively recovers every

intermediate odd state and prescribed valuation.

Retain

$$P = (1, 2), \quad (L_P, U_P, C_P) = (9, 8, 5),$$

$$Q = (1, 1, 1, 2, 1, 1, 4), \quad (L_Q, U_Q, C_Q) = (2187, 2048, 2363).$$

Their signed integer fixed points are -5 and -17 . Direct substitution verifies the complete signed cycles

$$-5 \rightarrow -7 \rightarrow -5, \quad -17 \rightarrow -25 \rightarrow -37 \rightarrow -55 \rightarrow -41 \rightarrow -61 \rightarrow -91 \rightarrow -17.$$

They are used as coefficient identities for the original $+1$ maps; they are not positive-integer counterexamples. The earlier local positive-shadow contribution used each anchored packet separately. This continuation allows arbitrary mixtures of the two original packet types.

Order already changes the result at the first mixed pair:

$$C_{PQ} = 29839, \quad C_{QP} = 31507, \quad L_{PQ} = L_{QP} = 19683, \quad U_{PQ} = U_{QP} = 16384, \\ C_{PQ} - C_{QP} = -1668. \quad (6)$$

Inside a larger original word, replacing an adjacent PQ by QP changes the complete numerator by

$$C_{\text{prefix } PQ \text{ suffix}} - C_{\text{prefix } QP \text{ suffix}} = -1668 U_{\text{prefix}} L_{\text{suffix}}. \quad (7)$$

Apply (3) before and after the change; the common outer multipliers give (7). Thus diagonal data or packet counts alone do not determine this mixed coefficient. In this proof counts are used for an upper bound; the exact original numerator is still used in every source and endpoint map.

The full affine record even has a word inverse. For $m > 1$,

$$a_1 = \nu_2(C - 3^{m-1}), \quad C_{\text{tail}} = (C - 3^{m-1})/2^{a_1}. \quad (8)$$

The parenthesized tail sum is odd, which proves the valuation. Iteration recovers all but the last exponent, and the retained total A_m recovers the last. The last numerator is one. These tests give the exact image conditions; recomposition proves both inverse identities. The P/Q symbol parsing is also unique, since P starts with $(1,2)$ and Q with $(1,1)$.

15.3 3. A complete mixed overlap calculation and exact switch fibres

The two integral anchor coordinates at one original integer are $x + 5$ and $x + 17$. Their full compatibility is not two independent smallness statements. At the level of polynomial evaluations there is the exact sequence

$$0 \rightarrow \mathbb{Z}[X]/((X+5)(X+17)) \xrightarrow{\text{ev}} \mathbb{Z} \oplus \mathbb{Z} \xrightarrow{(a,b) \mapsto b-a \bmod 12} \mathbb{Z}/12\mathbb{Z} \rightarrow 0, \quad (9)$$

where $\text{ev}(f) = (f(-5), f(-17))$. The product equals the intersection of the two principal ideals: dividing a polynomial in the intersection by $X+5$, then evaluating at -17 , shows its quotient is divisible by $X+17$, because multiplication by 12 is injective in $\mathbb{Z}$. The

image consists precisely of pairs with $12 \mid b - a$. Its inverse representative of degree below two is

$$f(X) = a + \frac{a-b}{12}(X+5). \quad (10)$$

Both evaluations and both inverse compositions are direct. The supported-zero fibre of the last map is this complete rank-two image lattice. The actual source diagonal is the stronger affine slice $(a, b) = (x+5, x+17)$, corresponding to $f(X) = x - X$; that additional original condition is retained. Equation (9) is proved here over the integers, not identified with an unread particular Zeta conductor complex.

15.3.1 The induced filtration, including the non-strictness defect

Write $\Lambda = \text{im}(\text{ev}) \subset \mathbb{Z}^2$. The image of the original coefficient filtration is $2^k \Lambda$; the induced target filtration is $I_k = \Lambda \cap 2^k \mathbb{Z}^2$. Put $g_k = \gcd(12, 2^k) = 2^{\min(k, 2)}$. There is the exact isomorphism

$$I_k/2^k \Lambda \longrightarrow \mathbb{Z}/g_k \mathbb{Z}, \quad [(a, b)] \mapsto \frac{b/2^k - a/2^k}{12/g_k} \pmod{g_k}. \quad (10a)$$

Its inverse sends j to the class of $2^k(0, (12/g_k)j)$. Indeed 12 divides $b-a$ precisely when $12/g_k$ divides $b/2^k - a/2^k$. The kernel of the displayed residue is exactly $2^k \Lambda$. These facts prove both inverse identities. The original image basis is $2^k(1, 1), 2^k(0, 12)$; the induced basis is $2^k(1, 1), 2^k(0, 12/g_k)$. Thus the quotient has exact order $1, 2, 4, 4, \dots$ as $k=0, 1, 2, 3, \dots$. This computes the failure of strictness; assigning the two independent target depths to the source without this quotient would lose actual integral classes. The repair is to retain the induced lattice I_k and its displayed original-image sublattice together. No inverse of 2 or 3 was used in (10a).

All four coefficient masks (empty, P only, Q only, both) are retained in the implementation. Zero insertion means an actual labelled coordinate with value zero. Applying the compatibility map to a single-slot face has kernel $12\mathbb{Z}$; on the joint face its kernel is Λ . The computed kernels, rather than every arbitrary single-slot amplitude, pass into the compatible joint face by zero insertion. This identifies the exact restriction introduced by synchronization.

In particular

$$\min\{\nu_2(x+5), \nu_2(x+17)\} \leq 2. \quad (11)$$

Otherwise their difference 12 would have valuation at least three. This is an induced compatibility constraint; it does not bound a later coordinate after divisions.

For a switch from P^r to Q^s , retain all three original stages:

$$n = 2 \cdot 8^r t - 5, \quad y = 2 \cdot 9^r t - 5, \quad 9^r t + 6 = 2048^s z, \quad y' = 2 \cdot 2187^s z - 17. \quad (12)$$

Its entire parameter fibre is

$$\begin{aligned} t &= t_0 + 2048^s v, & t_0 &\equiv -6(9^r)^{-1} \pmod{2048^s}, & 0 < t_0 < 2048^s, \\ z &= z_0 + 9^r v, & z_0 &= (9^r t_0 + 6)/2048^s, & v &\geq 0. \end{aligned} \quad (13)$$

Both inverse parameters follow by subtraction and division on the displayed images. The full source step is $2 \cdot 8^r 2048^s$; the full target step is $2 \cdot 9^r 2187^s$. They agree with (5).

For the reverse switch replace the cross-term +6 by -6 and exchange the original packet data. All quantities remain integral.

Equation (12) forces

$$\nu_2(t) = 1, \quad \nu_2(n+5) = 3r+2, \quad \nu_2(y+5) = 2. \quad (14)$$

The reverse switch gives $\nu_2(n+17) = 11r+2$. Thus the old dyadic depth must be transported through the actual packet, not recentered independently at its endpoint. For fixed r , every s has nonempty original source fibre (13); the new depth is unbounded across those sources.

More generally, at the endpoint of the $h=5$ packet, the exact stratum

$$\nu_2(9^r t + 6) = k$$

is the single residue class

$$t \equiv (2^k - 6)(9^r)^{-1} \pmod{2^{k+1}}. \quad (15)$$

This retains every reset stratum without a finite cutoff on k . It also shows why a bare claim of uniform switch-depth decay is unavailable: arbitrarily large k have positive original representatives. The construction that works on the repeated mixed source is given next.

15.4 4. A uniform bound on every ordered mixed core, with its primitive retained

Let s be ANY finite symbol string in P and Q , containing $\alpha \geq 1$ P 's and $\beta \geq 1$ Q 's. Expand it to its original exponent word, then allow ANY cyclic rotation at an original exponent position. Call the retained resulting word p . Set

$$m = 2\alpha + 7\beta, \quad A = 3\alpha + 11\beta, \quad L = 3^m, \quad U = 2^A, \quad \delta = L - U > 0.$$

Its exact numerator C depends on the complete ordering and rotation. Put

$$g = \gcd(C, \delta), \quad c = C/g, \quad d = \delta/g, \quad H = C/\delta = c/d. \quad (16)$$

15.4.1 Lemma 1. Every such original phase has $0 < H \leq 91$.

At a packet boundary write each packet map as $f_j(x) = a_j x + h_j(a_j - 1)$, with $a_j = L_j/U_j > 1$ and $h_j \in \{5, 17\}$. The composite translation is

$$\sum_j h_j(a_j - 1) \prod_{k>j} a_k,$$

and the same sum with every $h_j=1$ is $\prod_j a_j - 1$. Hence H at that boundary is the explicitly weighted average of 5 and 17 with positive weights

$$\omega_j = \frac{(a_j - 1) \prod_{k>j} a_k}{\prod_j a_j - 1}, \quad \sum_j \omega_j = 1. \quad (17)$$

This gives $5 \leq H \leq 17$ at every packet boundary, by rotating the symbol string. In-

side a P packet, the intermediate magnitude is $(3H-1)/2 \leq 25$. Inside a Q packet, each affine prefix is increasing as a function of H, so its value at $H \leq 17$ is at most the corresponding magnitude in the explicitly verified -17 cycle, hence at most 91. Positivity at every phase also follows from its rotated $C > 0$ and the unchanged $\delta > 0$. This proves the bound at every original exponent position. The coefficients in (17) and the unaveraged C are retained; no substitution of a common anchor is made. QED.

Consequently c,d are positive odd coprime integers, $\gcd(d,6)=1$, and

$$\frac{c+d}{2} \leq 46d \leq 46\delta < 46L. \quad (18)$$

The original integral return class has order d: it is [C] in $\mathbb{Z}/(U-L)\mathbb{Z}$. For the basic mixed core PQ, $(c,d)=(29839,3299)$; this class is nonzero and has order 3299. Its specified rational primitive is -29839/3299, not the integer anchor of either constituent. Repetitions keep that same primitive and its original covering multiplicity.

The coordinate $z = dx + c$ has inverse $x = (z - c)/d$ on the affine lattice $c + d\mathbb{Z}$. It is not treated as an invertible integral change on every ambient coordinate or modulo every prime. Its homogenized matrix has determinant d; that image lattice and its quotient are retained.

15.5 5. The complete positive source of a repeated mixed core

For $r \geq 2$, put $L_r = L^r$, $U_r = U^r$, and

$$C_r = c(L_r - U_r)/d \in \mathbb{Z}.$$

The integrality follows from d dividing L-U and from the original composition formula. The exact positive source and image of $p^{\wedge r}$ are

$$\boxed{n = \frac{2U_r t - c}{d}, \quad T^{mr}(n) = \frac{2L_r t - c}{d}, \quad t \geq 1, \quad 2U_r t \equiv c \pmod{d}. \quad (19)}$$

To prove completeness, multiply the original cylinder condition $L_r n + C_r \equiv U_r \pmod{2U_r}$ by the odd unit d. It becomes $L_r(dn + c) \equiv (c + d)U_r = 0 \pmod{2U_r}$. Since L_r is odd, this is exactly the divisibility in (19). Reversing the same congruences proves the converse. No original intermediate valuations are discarded.

Also $2U^r > c$. Indeed $c \leq 91L$, $r \geq 2$, and at $\alpha = \beta = 1$ the integer comparison $2 \cdot 8^2 \cdot 2048^2 > 91 \cdot 9 \cdot 2187$ holds. Increasing α or β multiplies U^2/L by 64/9 or 4194304/2187, both greater than one; increasing r multiplies U^r by U . Thus every admitted $t \geq 1$ is positive in (19).

The congruence modulo d has one residue because $\gcd(2U_r, d) = 1$. Let $t_{\min}$ be its least positive representative (one for $d=1$). Then $t = t_{\min} + dv$. Equation (19) yields the original steps $2U_r$ and $2L_r$, with $v \geq 0$, and the anchor is the unique positive cylinder representative below $2U_r$. This gives both parameter inverses and proves agreement with (5). The denominator constraint is not dropped.

15.6 6. The all-length arithmetic gap

The following finite-plus-logarithmic proof is independent of any large source atlas. Its external premise is the cited established two-logarithm estimate in Section 11.

Its finite part consists of a COMPLETE stated integer domain, not sampled word frequencies.

15.6.1 Theorem 1. Uniform mixed-count gap.

For all $\alpha, \beta \geq 1$, $r \geq 2$, $s \geq 1$ and $B \geq 2s$, retain m, A, L, U above and set

$$M = rm, \quad D = U^r 2^B - L^r 3^s.$$

On the positive-gap domain $D > 0$,

$$\boxed{D > 46L(2^B - 3^s).} \quad (20)$$

Proof. Put $K=46L$ and

$$F(B) = (U^r - K)2^B - (L^r - K)3^s.$$

Here $M \geq 18$, A/m is between $3/2$ and $11/7$, and $r \geq 2$. Therefore

$$K/U^r \leq 46(3/8)^{M/2} \leq 46(3/8)^9 < 1/2. \quad (21)$$

In particular $F(B)$ increases with B .

Long falling tails. For $s \geq M$, it suffices to take $B=2s$. After division by 3^s , the expression is increasing with s , so take $s=M$. The relative upper bounds for the two subtractions are

$$\frac{L^r 3^M}{U^r 4^M} \leq (81/128)^{M/2}, \quad \frac{K(4^M - 3^M)}{U^r 4^M} < 46(3/8)^{M/2}.$$

Their sum is at most $(81/128)^9 + 46(3/8)^9 < 1$, an exact rational comparison. Thus $F > 0$ throughout $s \geq M$. This part uses no logarithm theorem.

A bounded tail before the first crossing. Now $s < M$. Take B_0 as the least integer $B \geq 2s$ for which $D > 0$. It satisfies $B_0 \leq 2M$, since $U^{r \geq 2} (3M/2)$ and $2^{(7/2)M} > 3^{2M}$, the latter following from $2^7 > 3^4$. Define

$$A_* = rA + B_0 \leq 4M, \quad m_* = M + s < 2M, \quad \Lambda = 2^{A_*} 3^{-m_*} - 1 > 0.$$

A failure $F(B_0) \leq 0$ implies

$$0 < \Lambda < \frac{K}{U^r - K} < \frac{2K}{U^r}.$$

Using $\log 92 < 5$, $\log 2 > 2/3$, and $\log 3 < 11/10$ gives

$$\log \Lambda < 5 - \frac{9}{20}M. \quad (22)$$

Matveev's rational two-log bound, with the explicit safe constant computed in Section 11, gives

$$\log \Lambda > -10^9(1 + \log(4M)). \quad (23)$$

At $M=10^{12}$, $4M < 2^{42}$ and $\log 2 < 7/10$ imply $(9/20)M > 5 + 10^9(1 + 42 \cdot 7/10)$. The difference between the left linear term and the right logarithmic term is strictly increasing for $M \geq 10^{12}$, as its derivative is $9/20 - 10^9/M > 0$. Thus (22)–(23) force

$M < 10^{12}$.

Exact rational separation. The certificate proves

$$\frac{83130157078217}{52449289519716} < \frac{\log 3}{\log 2} < \frac{\log 3}{\log 2} + 2^{-128} < \frac{350861368503572}{221368876767703}. \quad (24)$$

For $M \geq 256$, (22) implies

$$0 < \frac{A_*}{m_*} - \frac{\log 3}{\log 2} = \frac{\log(1 + \Lambda)}{m_* \log 2} < \exp(6 - 9M/20) < 2^{-128}.$$

The fractions at the ends of (24) have determinant one. Their denominator sum is $273818166287419 > 2 \cdot 10^{12}$. Any fraction x/y strictly between p/q and r/s with $rq - ps = 1$ satisfies $y \geq q + s$: the positive integers $k = xq - py$ and $j = ry - sx$ obey $y = sk + qj$. This uses the original, possibly unreduced x, y . But $m_* < 2M < 2 \cdot 10^{12}$, a contradiction. Thus any failed margin has $M < 256$.

Complete remaining arithmetic. The executable certificate visits exactly

$$\alpha, \beta \geq 1, \quad r \geq 2, \quad 18 \leq M = r(2\alpha + 7\beta) < 256, \quad 1 \leq s < M,$$

computes B_0 by literal integer doubling from $2s$, and checks $F(B_0) > 0$. There are 1,197 triples (α, β, r) and 210,527 complete rows. The least margin is 801844969, at $(\alpha, \beta, r, s, B_0) = (1, 1, 2, 2, 4)$. The row stream hash is in `verification.json`; `-ledger` materializes all rows. The committed receipt also pins the complete regenerated result by SHA-256 and byte length. This integer enumeration discharges precisely the residual domain. Monotonicity then proves the claim for every $B \geq B_0$. These cases exhaust the theorem. QED.

15.7 7. Descent after arbitrary repeated mixed cores

15.7.1 Theorem 2. Original positive-source descent.

Let p be any ordered mixed core from Section 4, at any original phase, and $r \geq 2$. Let $q = (c_1, \dots, c_s)$ be any nonempty falling word, every $c_i \geq 2$. Put $B = \sum c_i$. Every original positive source realizing $p^{\wedge r} q$ in the domain

$$U^r 2^B > L^r 3^s \quad (25)$$

has strict descent below its starting value at the end of that word.

Proof. Retain the exact falling numerator C_q and its anchored correction

$$E_q = C_q + 3^s - 2^B = \sum_{i=1}^s 3^{s-i} 2^{B_{i-1}} (4 - 2^{c_i}) \leq 0. \quad (26)$$

All powers and original exponents remain. The supported-zero case $E_q = 0$ is exactly the all-two tail. From (19), an original source and the endpoint y of $p^{\wedge r}$ obey $dn + c = 2U^r t$ and $dy + c = 2L^r t$, with $t \geq 1$ and its retained congruence. Let z be the final original endpoint after q . Direct substitution gives

$$d2^B(n - z) = 2Dt - (c + d)(2^B - 3^s) - dE_q. \quad (27)$$

This is the exact affine displacement, not a multiplier-only test. By (18) and Theorem

1,

$$D > 46L(2^B - 3^s) > \frac{c+d}{2}(2^B - 3^s).$$

Rewrite the right side of (27) as $2[D - (c+d)(2^B - 3^s)/2] + 2D(t-1) - dE_q$. The first term is strictly positive and the others nonnegative. Since $d2^B > 0$, $n > z$. Both original endpoints are positive odd integers, so the descent is at least two. QED.

The theorem has no bound on the number of packet switches in p , the counts α, β , the number r of repetitions, or any tail exponent. It does require repetition $r \geq 2$ and a falling tail reaching its actual coefficient crossing. The original first-descent index is not asserted to equal the endpoint of this word; earlier descent remains possible and is not erased.

15.7.2 Consequence for actual cycles and component minima

No positive cycle can have a full traversal equal to $p^r q$ from Theorem 2, or a cyclic rotation of such a traversal. Multiplication of its original equations gives $2^{(rA+B)} > 3^{(rm+s)}$, so (25) would hold, whereas Theorem 2 gives an endpoint smaller than its source. Repetitions and nonprimitive presentations remain explicit; the contradiction is on the actual original endpoint equation. A traversal p^r alone has its unique fixed point $-c/d < 0$, so has no positive return either.

Every nonbase actual component has a least positive integer. Its entire orbit cannot contain a strict descent below that initial minimum. The theorem therefore also excludes its domain as possible itineraries starting at such a minimum, whether the component would be periodic or nonperiodic. It does not by itself settle mixed itineraries outside that language.

15.8 8. Full cylinders and original first-jet witnesses

The implementation constructs $p^r q$ using

$$L_{\text{tot}} = 3^s L^r, \quad U_{\text{tot}} = 2^B U^r, \quad C_{\text{tot}} = 3^s C_r + U^r C_q. \quad (28)$$

Equation (5) applied to this exact triple supplies the complete source and target progressions, not a sample. For a lift parameter $\zeta \geq 0$, the anchor parameter in (19) changes by $d2^B \zeta$. Equation (27) is tested as an identity in the original anchor and its slope. The original word and phase recover every intermediate equation.

Examples recorded with independent +1 division replay include

$$T^{19}(2823006875) = 1527868265 \quad \text{with word } (PQ)^2(3),$$

$$T^{28}(8300162951835) = 5396745598079 \quad \text{with word } (PQ)^3(3).$$

Another retained mixed order is $p = \text{PPQQPQ}$, repeated twice with tail $(2,2,2,2)$; its source and endpoint are

$$9535679069403344498661387515 \mapsto 9070365730002260119033546999.$$

Each displayed anchor belongs to a complete infinite cylinder returned by family. The result file includes the unchanged steps, all coefficients, denominator constraint,

and raw margin. Separate point witnesses to 1 have respectively 93, 128 and 256 original odd returns. Those point certificates do not assert convergence of every variable endpoint in their cylinders.

The cohomological receiver is the published integral first-jet complex

$$d_\epsilon = d - \epsilon P, \quad \epsilon^2 = 0.$$

For the finite ORIGINAL edge chain B from n to z,

$$d_\epsilon(\epsilon B) = \epsilon V_n - \epsilon V_z. \quad (29)$$

The identity follows from the original boundary and epsilon squared equal to zero. An integral original witness at z extends by adding B to its epsilon coefficient. The checker uses the unchanged authenticated predecessor `firstjet.py`, verifies both coefficient equations, and extracts the actual finite path. No negative rational primitive is substituted for the positive original graph. No division by a period or by d is admitted in its integer witness module.

15.9 9. What the calculation does and does not control

The interaction in (6)–(7), the compatible source fibres (12)–(15), and the rational lattice in (19) are all retained. The order-independent bound in (18) applies to those exact data rather than replacing them. This is the successful use of mixed extension control: preserve the coupling, compute its complete effect on the original source, and then prove a bound uniform over the retained order fibre.

For a fixed repeated core, falling tails of length $s \geq M = rm$ necessarily cross (25), so they are covered without limiting their individual exponents. Before crossing, at any fixed s the sum is bounded by

$$2s \leq B \leq \text{bitlength}(3^{M+s}) - 1 - rA. \quad (30)$$

All ordered words with each exponent at least two and this sum bound can be enumerated exactly. An exponent-one return before crossing exits the falling-tail domain and remains an original mixed-switch case. There is no disappearance of that label.

The universal remaining issue is nonrepeating or more general switching, not the formal existence of infinite inverse series. Equations (12)–(15) show why a bounded conductor index cannot be claimed to bound all future reset depths. They supply exact residue input for a further source-cover argument. The theorem here controls a large repeatable class without making that false step. No original positive nontrivial cycle, certified divergent orbit, or global source coverage is asserted.

15.10 10. Executed verification and integration scope

`verify.py` reconstructs the full 210,527-row finite domain of Theorem 1 and its stream hash. It obtains logarithm bounds from 160 positive rational-series terms and explicit remainders. It separately checks all 4,914 family fixtures, with 449,723 original returns independently divided rather than trusting a valuation bit operation. The 3,149 accepted fixtures and 1,765 pre-crossing fixtures are distinct recorded outcomes. There are 351 retained core/phase fixtures, 321 order-swap comparisons, 128 two-packet switch charts with two source lifts each, and sixteen rejected malformed or

false inputs.

The ordinary and optimized outputs have 1,106,639 named exact checks and are byte-identical. The general ordering/anchor/source arguments are the written proofs, not inferred from these fixtures. The finite gap-domain enumeration is a stated computer-assisted proof component. Matveev's theorem is an external mathematical dependency, not a locally formalized result. No Lean execution or independent external review is claimed.

The new replay checks its source manifest and the original firstjet Git blob before running both modes. The new workflow also runs the unchanged firstjet predecessor pair. It has contents:read only, pinned action commits, and no retained checkout credentials.

This increment is standalone relative to the published firstjet dependency. It neither assumes nor silently publishes the two earlier local pending directories defect_rank_descent_20260915 and negative_shadow_descent_20260916. Their source formulas motivated this continuation; all numerical and logarithmic premises used here are reproduced in this contribution. Their separate pending package remains available with its original manifest.

15.11 11. Exact external input and source identities

[L] A. Languasco, F. Luca, P. Moree and A. Togbe, *Sequences of integers generated by two fixed primes*, Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg 95 (2025), 123–148, Theorem 2.1; DOI 10.1007/s12188-025-00293-9. The publisher's readable primary text was inspected in this continuation. For positive rationals γ_i and nonzero integer exponents b_i , the cited established theorem bounds a nonzero product minus one by

$$\log |\Lambda| > -1.4 \cdot 30^{k+3} k^{4.5} (1 + \log B) \prod_i h(\gamma_i).$$

Our actual substitution is $k=2$, $\gamma_1=2$, $\gamma_2=3$, $b_1=A^*$, $b_2=-m^*$, with $B=4M$. Unique prime factorization gives Λ nonzero, and the actual crossing gives Λ positive. Since $\sqrt{2} < 3/2$, $\log 2 < 7/10$, and $\log 3 < 11/10$, its coefficient is smaller than the checked rational number

$$(7/5)30^5 24 (7/10)(11/10) < 10^9.$$

This proves (23) with all hypotheses verified.

For $x=2,3$ and $z=(x-1)/(x+1)$, the interval calculation uses

$$L_N(x) = 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1}, \quad 0 < \log x - L_N(x) < \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

The identity follows by integrating the positive geometric series for $1/(1-z^2)$; the remainder bound follows by replacing its increasing denominators by the first. At $N=160$ the four rational endpoints prove (24), $\log 2$ and $\log 3$ bounds, and the listed determinant and denominator comparisons. No evaluation of a floating-point logarithm is a premise.

[Z1] Zeta mixed-control sources/MCF.tex, blob a8b1300ea8b842ab1340d67e44cd6da2f71e8e62, especially MCF6–10, at the read-only pin above. Its original independent slot

masks, zero insertion, and quotient fibres remain their source constructions. [Z2] sources/MRE.tex, blob e50623727112e925961ca9e7ad9accbe78758cf1, at the read-only pin above, especially MRE7–12 and MRE30–33. No unreturned file identity is used as a certificate.

[C1] Original Collatz firstjet firstjet.py, blob 97f9ef5ad535f7ec18259d41fab6a3557be17d98, at the authenticated base; the checker computes that Git blob from the local bytes. [C2] intrinsic_zero_firstjet_20260915/note.md, especially its original integral witness equations. The prior weighted automata presentation and preceding source-cylinder conventions remain their existing source contributions. No global priority claim is made for universal triangular extensions, exact overlap sequences, or the standard logarithmic method.

Chapter 16

Global cylinder control: arbitrary words, whole crossing tails, and retained singleton defects

Source: repository/collatz_reconstruction/research_program/global_cylinder_control_20260916/note.md.

16 September 2026. Additive Collatz workbench continuation, read at 9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b, PR #2. The construction uses the original +1 map and the published integral first-jet receiver. It does not import an unproved estimate from the Zeta programme. The preceding 33-file local arithmetic/shadow package is not an input. This note makes no priority claim and no assertion of global Collatz convergence.

The main arithmetic quantifiers run over **all finite exponent words**, not only the earlier P/Q languages, repeated cores or a bounded switch count. The price is explicit: one original source can remain at each contracting cylinder, and at most one original source remains across the entire unbounded next-exponent crossing tail of a pre-crossing prefix. Those sources are retained rather than declared zero. A separate finite proof certificate discharges the first-crossing exception through 18 odd returns.

16.1 1. Original objects and source maps

Let X be the positive odd integers and

$$T(n) = (3n + 1)/2^{a(n)}, \quad a(n) = \nu_2(3n + 1).$$

For an ordered nonempty word $p=(a_1, \dots, a_m)$, $a_i \geq 1$, retain

$$A_j = \sum_{i=1}^j a_i, \quad A_0 = 0, \quad U = 2^{A_m}, \quad L = 3^m, \quad D = U - L,$$

$$C = C(p) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}, \quad f_p(x) = (Lx + C)/U. \quad (1)$$

The branch order is $f_p = g_{(a_m)} \dots g_{(a_1)}$, $g_a(x) = (3x+1)/2^a$. The complete original positive source is

$$\mathcal{C}_p = \{\rho_p + 2Ut : t \in \mathbb{Z}_{\geq 0}\}, \quad \rho_p \equiv (U - C)L^{-1} \pmod{2U}, \quad 0 < \rho_p < 2U. \quad (2)$$

Its original image is $y_p + 2Lt$, $y_p = (L\rho_p + C)/U$, and both t inverses are given by the displayed source or target difference divided by its original step. In particular the target step is not changed to the source step. The positive representative is odd: U is even and C is odd.

For completeness, the terminal-odd congruence (2) gives all intermediate oddness conditions. Backwards from an odd last value, the recurrence $3x_{j+1} = 2^{a_{j+1}}x_{j+1}$, together with the prefix formula, shows each previous numerator divisible by its power of two and every previous x_j odd. Equivalently, modulo 2^{A_j+1} , the composite numerator has its first j terms plus a last odd contribution 2^{A_j} . Since every $a_i \geq 1$, terms past that contribution vanish. Dividing the resulting congruence by 2^{A_j} gives odd x_j . More explicitly, for $j < m$ the complete numerator reduces to $3^{(m-j)}(3^j n + C_j) + 3^{(m-j-1)}2^{A_j}$ modulo 2^{A_j+1} . The terminal congruence makes this zero; division by the odd coefficient and inversion of 3 give $3^j n + C_j$ congruent to 2^{A_j} modulo 2^{A_j+1} . The exact recurrence then forces the original valuation a_{j+1} . Positive forward values follow directly from g_a .

An explicit append map, also used by the checker, starts from $n = \rho + 2Ut$ and $x = y + 2Lt$. To append exponent a , put

$$k \equiv (2^{a-1} - (3y + 1)/2)(3L)^{-1} \pmod{2^a}, \quad 0 \leq k < 2^a. \quad (3)$$

Then the new anchor is $\rho + 2Uk$, the new target is $(3y + 1 + 6Lk)/2^a$, and the new coefficients are $L' = 3L$, $U' = 2^a U$, $C' = 3C + U$. Substituting $t = k + 2^a v$ proves the whole new progression and its inverse. Empty prefix data are $(\rho, y, L, U, C) = (1, 1, 1, 1, 0)$; its source is all positive odds.

16.2 2. A proved global separation of powers

16.2.1 Theorem 1. All-length separation

For every integer $m \geq 1$ and every integer A with $2^A > 3^m$,

$$2^A - 3^m > (3/2)^m - 1. \quad (4)$$

This is a computer-assisted theorem with one stated external input, Matveev's established rational-number logarithm bound [MAT]. The complete finite part is generated and checked in `power_gap_certificate.json`. It consists of rational interval inequalities and 127 integer rows, not a large Collatz-start computation. The general argument follows.

Take $A_0 = \text{floor}(m \log(3)/\log(2)) + 1$; then $A_0 \leq 2m$. Increasing A increases the left side of (4), so it suffices to use A_0 . A failure gives

$$0 < \Lambda = 2^{A_0} 3^{-m} - 1 < 2^{-m}. \quad (5)$$

Apply [MAT, Theorem 2.1] with its two positive rational numbers 2,3, integer exponents $A_0, -m$, and $B = 2m$. Their heights are $\log 2, \log 3$. Its coefficient is dominated by

$$(7/5)30^5 23 (7/10)(11/10) < 10^9, \quad (6)$$

because $2^{(9/2)} < 23$, $\log 2 < 7/10$ and $\log 3 < 11/10$. Thus

$$-10^9(1 + \log(2m)) < \log \Lambda < -m \log 2 < -(2/3)m. \quad (7)$$

At $m=10^{12}$, use $2 \cdot 10^{12} < 2^{41}$ and $\log 2 < 7/10$ to get

$$(2/3)10^{12} > 10^9(1 + 41 \cdot 7/10).$$

The difference $(2/3)m - 10^9(1 + \log(2m))$ has derivative $2/3 - 10^9/m > 0$ for $m \geq 10^{12}$. Hence a failure has $m < 10^{12}$.

All logarithm bounds are certified by rational series. For $x=2$ or 3 , $z=(x-1)/(x+1)$, use

$$\log x = 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1} + R_N, \quad 0 < R_N < \frac{2z^{2N+1}}{(2N+1)(1-z^2)}. \quad (8)$$

The remainder bound follows by replacing all later positive denominators by $2N+1$ and summing a geometric series. With $N=160$, exact rational arithmetic gives, for $\alpha = \log 3 / \log 2$,

$$\frac{83130157078217}{52449289519716} < \alpha < \alpha + 2^{-128} < \frac{350861368503572}{221368876767703}. \quad (9)$$

The outer determinant is one. A rational x/y strictly between p/q and r/s with $rq-ps=1$ has $y \geq q+s$: the positive integers $k=xq-py$ and $j=ry-sx$ satisfy $y=sk+qj$. The identity does not require reducing x/y . Here $q+s=273818166287419 > 10^{12}$.

For $m \geq 128$, (5) gives

$$0 < A_0/m - \alpha = \log(1 + \Lambda)/(m \log 2) < \frac{3}{2m} 2^{-m} < 2^{-128}. \quad (10)$$

This contradicts (9), since its original denominator m is below 10^{12} . It remains to check $1 \leq m \leq 127$. For each m the checker takes $A_0 = \text{bit_length}(3^m)$ and verifies the strict integer margin

$$((2^{A_0} - 3^m) + 1)2^m - 3^m > 0. \quad (11)$$

All 127 rows pass and are retained in full. These cases exhaust (4). No floating-point evaluation or guessed decimal bound occurs. QED.

16.3 3. Every contracting word has at most one non-descending source

The coefficient-only comparison below concerns the formal affine maps at $x=0$. It does not treat zero as an actual positive odd Collatz source. For $x \geq 0$,

$$g_a(x) = 2^{1-a} g_1(x) \leq g_1(x), \quad g_1(x) = (3x + 1)/2.$$

Monotonicity, applied at the same input at each step, gives

$$0 < C/U = f_p(0) \leq 2^{1-a_m} ((3/2)^m - 1). \quad (12)$$

The last factor $2^{(1-a_m)}$ is kept separately: bound the preceding $m-1$ maps by $g_1^{(m-1)}$, then use $g_{(a_m)} = 2^{(1-a_m)} g_1$. This proves (12) without identifying

different ordered affine maps or dropping C.

16.3.1 Theorem 2. All-word canonical-source isolation

For every original word p with $D > 0$,

$$C/D < 2^{A_m - a_m + 1} = 2^{A_{m-1} + 1}. \quad (13)$$

The set of its positive sources with $f_p(n) \geq n$ is exactly

$$\{\rho_p\} \quad \text{when } C \geq D\rho_p, \quad \emptyset \quad \text{when } C < D\rho_p. \quad (14)$$

All other sources in the infinite original cylinder strictly descend. The candidate in (14) is retained even when its resulting set is empty.

Proof. Combine (4) and (12) to get (13). The exact displacement $f_p(n) - n = (C - Dn)/U$ makes every non-descending positive n at most C/D . Every legal source except ρ_p is at least $\rho_p + 2U > 2U$, whereas the right side of (13) is at most $2U$. Thus only ρ_p is possible. Testing $C - D\rho_p$ gives exactly (14). On the whole source the unchanged identity is

$$n(t) - f_p(n(t)) = \rho_p - y_p + 2Dt. \quad (15)$$

Its positive slope proves the rest after its first accepted parameter. For $D < 0$, the same formula in n gives $f_p(n) > n$ on every positive source, since $C > 0$. $D = 0$ is impossible for a nonempty word. QED.

This theorem quantifies over every finite exponent word, with arbitrary switches, lengths, exponents and orderings. It does not say the exceptional canonical source always descends. For example the original word

(4, 1, 1, 1, 1, 2, 2, 1, 2, 1, 1, 2, 1, 1, 2, 3)

has $m = 17, A = 27$, canonical source 165 and endpoint 167, with $D > 0$. It already visited 31 on its first return, so this is not a first-crossing exception. The source and image of its whole cylinder are

$$165 + 268435456t \mapsto 167 + 258280326t. \quad (16)$$

Every $t \geq 1$ strictly descends; $t = 0$ is explicitly retained and has its own verified finite arrival witness. This is an example of the coefficient- contracting yet non-descending behavior investigated in [RT], not a new claim that such behavior is absent.

16.4 4. The entire unbounded next-exponent crossing family

A pre-crossing prefix q of length l has $U_j < 3^j$ at every nonempty prefix. Keep its data (ρ, y, U, L, C) as above. Define

$$h = \lfloor \log_2(3L) \rfloor - A_l + 1 \geq 2. \quad (17)$$

Thus $2^{(h-1)U} < 3L < 2^{hU}$. The next original exponent has not crossed exactly for $a = 1, \dots, h-1$. Every $a \geq h$ is a first coefficient crossing. Neither a frequency model nor independence of exponents is used.

Let

$$M = 2^{h-1}, \quad t_0 \equiv -((3y+1)/2)(3L)^{-1} \pmod{M}, \quad 0 \leq t_0 < M, \\ n_0 = \rho + 2Ut_0, \quad b_0 = (3y+1+6Lt_0)/2^h. \quad (18)$$

The full union of all crossing children is the one positive progression

$$n = n_0 + U2^h v, \quad T^{l+1}(n) = \frac{b_0 + 3Lv}{2^{\nu_2(b_0+3Lv)}}, \quad a_{l+1} = h + \nu_2(b_0 + 3Lv), \quad v \geq 0. \quad (19)$$

The inverse is $v=(n-n_0)/(U2^h)$ on the displayed source. Substitution into $3T^{l+1}(n)+1$ proves (19) and the full valuation, including arbitrarily large extra divisions. Congruence (18) is equivalent to its divisibility by 2^h , so the source is complete. The low children retain (3) and their own labels. Together, they exhaust every next exponent and every positive source of q .

16.4.1 Theorem 3. One original test controls the whole crossing alphabet

The only possible non-descending source in (19) is the OLD canonical prefix source ρ . It belongs to (19) precisely when $t_0=0$, and its actual next exponent is uniquely $a^*=\nu_2(3y+1)$. It is an exception precisely when

$$a^* \geq h \quad \text{and} \quad (3y+1)/2^{a^*} \geq \rho. \quad (20)$$

Every other source in the full unbounded crossing family has its first strict descent below its start at return $l+1$.

Proof. For any admitted child $p=qa$, (13) says a non-descending source n is less than $2^{(A_l+1)}=2U$. Its q -cylinder congruence then forces $n=\rho$. That source has just one actual next exponent, proving (20). Earlier prefixes have multiplier greater than or equal to one and positive forcing, so every proper nonempty endpoint exceeds n . Thus final descent is its first strict descent. The empty prefix case retains $n=1$ as the actual fixed-loop exception and all other crossing sources descend. QED.

Each individual family also has an elementary checkable proof once its original coefficients have been calculated. Its undivided target line is b_0+3Lv , its source line is $n_0+U2^h v$, and

$$U2^h > 3L, \quad n_0 + U2^h > b_0 + 3L. \quad (21)$$

The second inequality follows from (13) applied to the formal affine word qh and the positive source $n_0+U2^h>2U$; the inequality (13) itself is about all real affine inputs and does not need terminal oddness. The implementation checks both integer inequalities directly. They certify every $v \geq 1$ by their positive slope; $v=0$ is the single exact valuation check. Thus a delivered individual family can be verified without executing a transcendence theorem or enumerating its sources.

16.4.2 An original nonrepeating example

For the single mixed prefix

$$q=(1,2,1,1,1,2,1,1,4)=PQ,$$

retain $L=19683, U=16384, C=29839, \rho=10907, y=13105, h=2$. Formula (19) is

$$10907 + 65536v \mapsto \frac{9829 + 59049v}{2^{\nu_2(9829+59049v)}}, \quad v \geq 0. \quad (22)$$

This is first strict descent after ten odd returns. The actual last exponent is $2+\nu_2(9829+59049v)$. Indeed the endpoint is at most $9829+59049v < 10907+65536v$. Arbitrarily large last exponents have positive source representatives because 59049 is odd. No repetition of PQ and no unspecified weight for v occurs.

16.5 5. Whole-system finite audit with infinite source parameters

The exact symbolic audit starts at the empty prefix. At a prefix of length l it performs the single test (20), checks (21), and creates ALL low children $a=1, \dots, h-1$. The full high tail (19) is retained as a complete certificate and is not truncated by exponent size. The exact number of nodes at each depth is independently checked by the recurrence

$$N_0(0) = 1, \quad N_j(A) = \sum_{B < A} N_{j-1}(B), \quad j \leq A \leq \lfloor \log_2(3^j) \rfloor. \quad (23)$$

Every original pre-crossing word satisfies precisely these prefix bounds. Recurrence (23) and the explicit child ranges prove completeness of the finite tree. At EVERY node the generated ρ is checked against (2) using an independent modular inverse, the original affine equation $U*y=L*\rho+C$ is checked, and the next valuation is re-computed by repeated division. Both whole-tail margins (21) and the actual endpoint of the whole crossing family's first source n_0 are also checked. Thus this finite audit supplies its own elementary affine-tail proofs and does not need the external logarithm theorem as a premise: each $v=0$ is checked directly, and the positive affine margin at $v=1$ proves all $v \geq 1$.

The complete run for first crossing at return at most 18 visits 1,166,058 prefixes. It retains exactly one exceptional crossing: source 1, word (2), endpoint 1. Therefore every positive source $n \geq 3$ whose first odd-return coefficient crossing is at most 18 has first actual strict descent at that return. This is unbounded in the starting integer and the last exponent. Its source-time scope is not a bound on every orbit's crossing return, and it is not presented as a published stopping-time record.

The ordered stream can be materialized with `--ledger`. The independent `audit_ledger.py` imports neither the main module nor the predecessor; it reconstructs the entire depth-first domain and each original numerator, residue, endpoint, unbounded-tail margin and canonical-point test. Its successful full replay is included in the delivery evidence. The uncompressed SHA-256 and per-depth counts are in `verification.json`. The executable proof certificate regenerates the ENTIRE finite domain, not a sample. The independent count recurrence does not replace the original affine and valuation checks. The infinite-depth frontier remains unproved.

For example $n=27$ first crosses at return 37, reaches 23 there, and has an independently checked 41-return witness to 1. It is outside the 18-return finite theorem but its whole crossing cylinder is still covered by the all-length reduction and a single original root test.

16.6 6. Exact transfer into the retained Split-Zero comparison

The original absolute first-jet complex has original edge and vertex modules C^0, C^1 , source and target maps J, P , $d=J-P$, and

$$d_\epsilon = d - \epsilon P, \quad \epsilon^2 = 0.$$

The receiver is the already published kernel

$$\mathfrak{B}_X = \ker(H^1(K_\epsilon) \rightarrow H^1(K)).$$

Its integral presentation is $C^1/(dC^0 + P \ker d)$, with the inverse map to first-order classes $[z] \rightarrow [\epsilon z]$. The predecessor proves that a finite integral pair $db=0$, $dc-Pb=V_n$ yields an actual finite path to 1, and that the full obstruction has cyclic Z/m summands and nonperiodic Z summands. Those definitions and source maps remain unchanged [WB1].

Let $B_p(n)$ be the sum of the actual original edges in an accepted word from n to y . Its boundary and its new first-order use are

$$dB_p(n) = V_n - V_y, \quad d_\epsilon(\epsilon B_p(n)) = \epsilon V_n - \epsilon V_y. \quad (24)$$

Thus the all-word descent gives $[\epsilon V_n] = [\epsilon V_y]$ with $y < n$. It does NOT by itself assert either class is zero. An original witness at y is transported by the explicit addition

$$(b_y, c_y) \mapsto (b_y, c_y + B_p(n)). \quad (25)$$

Substitution in db and $dc-Pb$ proves the exact boundary at n . No original edge or cycle is divided by a period and no integral denominator is removed. The checker imports the unchanged, Git-blob-verified `firstjet.py` and verifies (24), including all intermediates and terminal vertices. For the examples with complete arrival, its unchanged witness checker extracts the actual path.

At an equation support label E , (24) maps a declared source module into the original equations in that support, or enlarges E by exactly the finitely many edges of the displayed path. Unions of such supports give the original inclusion diagram, and (24)-(25) commute with its inclusions. Their supported zero is at that retained E . An empty exceptional set in (14) or (20) is stored alongside its source label and integer equation, not substituted for the external absent scalar. The word index itself is not a definition of intrinsic Split-Zero support.

The family maps are finite-column maps on free modules even over the infinite progression: each original basis vector has its own finite path. There is no infinite sum in (24). Both the true singleton exception at 1 and the non-descending canonical point 165 remain available to the original operator and subsequent certificates.

16.7 7. Relation to the elementary maps, with the different source moduli

Let F be the shortened elementary map: $F(x)=(3x+1)/2$ at odd x and $F(x)=x/2$ at even x . The word p expands to the parity word

$$w(p) = 1 0^{a_1-1} 1 0^{a_2-1} \dots 1 0^{a_m-1}, \quad |w| = A_m. \quad (26)$$

The composition has the exact same L,U,C as (1). Grouping at the original odd vertices is its inverse for paths with odd terminal value. The parity cylinder alone has modulus 2^A ; additionally requiring terminal oddness selects the original odd-return cylinder modulo $2^{(A+1)}$. These moduli are not identified. On their common source, $T^m(n) = F^A(n)$.

For ANY shortened parity word with k letters and m odd letters, its original affine constant E satisfies $0 \leq E \leq (3/2)^{m-1}$: omitting a division only increases the non-negative formal constant. If $2^k > 3^m$, (4) gives $E < 2^k - 3^m$, so its non-descent threshold is strictly below 2^k . Thus every such parity cylinder also has at most its single canonical positive representative as an exception. The all-even word has zero constant and strictly decreases every positive source, including its least positive source 2^k ; the canonical residue zero is not a positive starting point.

The first shortened coefficient crossing can occur inside the final even divisions of an odd-return block. Consequently the finite theorem of Section 5 is stated with odd-return counts and is not silently asserted as the identical count in Terras' shortened coefficient-stopping problem. The exact expansion (26), its terminal condition and its clocks are the comparison used here. The standard elementary map without immediate odd-step division further expands each odd shortened step into its original odd step plus one division, with inverse grouping.

16.8 8. Global restrictions and the remaining source problem

This is an all-word source-isolation theorem, not a proof that its canonical candidates are always harmless. For any nonbase component, its least original positive integer s has every iterate at least s . A coefficient-crossing prefix at s is therefore a non-descending case of (14). More precisely, at its first crossing it must pass test (20) as the canonical source of the proper prefix. The all-word theorem eliminates every other point of that entire infinite crossing family. If there is no coefficient crossing, all original prefix multipliers remain at least one and the associated itinerary stays in the infinite pre-crossing tree. Both alternatives remain represented.

The same exact inequalities are useful for finite returns. Every actual positive cycle has $2^A > 3^m$, by multiplying the unchanged cycle equations. At every rotation its starting vertex must be the canonical representative in (2) and satisfy (13). Its integral forcing condition is still $D|C$; no canonical-root test replaces the actual cyclic solution and valuations.

At the first coefficient crossing m , all proper-prefix $2^A_j \leq 3^j$ give $C \leq m \cdot 3^{(m-1)}$. Hence a non-descending original endpoint $n+2k$ satisfies

$$0 \leq k < m/6. \quad (27)$$

Indeed $(f_p(n)-n) < C/U < m/3$ because $U > 3^m$ and $Dn > 0$. For $m=1$ the same strict upper bound applies to its equality at 1. This is an additional all-length exact small-overshoot restriction, not a claim that every allowed k occurs.

There is no fixed finite bound on the first crossing for all sources: $n_N = 2^{(N+1)-1}$ has its first N exponents equal to one, with $T^j(n_N) = 3^j 2^{(N+1-j)-1}$ for $0 \leq j \leq N$. All those prefix coefficients exceed one. The sources vary with N . This explicitly prevents promotion of the finite tree audit to one uniform clock bound.

The next arithmetic calculation is therefore sharply specified: extend source proofs on the actual canonical crossing points and control the ordinary-positive-integer sup-

port of the no-crossing infinite branches. The canonical residue towers have their original increasing representatives; boundedness gives eventual constancy, not an automatic contradiction. The current argument does not show their positive intersections empty. A finite failure to cross is not an infinite-orbit certificate. The workbench still retains both periodic and nonperiodic comparison-kernel summands.

16.9 9. Verification and source discipline

`power_gap_certificate.json` contains the exact finite arithmetic component of Theorem 1. `verify.py` re-creates it, checks all 262,143 nonempty exponent words of total sum at most 18, and checks every canonical shortened parity source through length 14. These word tests are regression evidence; the unrestricted quantifier comes from Sections 2-4.

The crossing audit is a DIFFERENT complete bounded proof domain with unbounded source and last-exponent parameters. Both ordinary and optimized runs reproduce `verification.json` byte for byte. All checks use explicit exceptions rather than assert statements, so optimization cannot remove them. Invalid-input controls include a false descent at 1, a false endpoint descent at 165, an altered source lattice, a false valuation and a divided nonintegral first-jet witness.

No Lean verification, independent external mathematical review or new remote GitHub execution is implied by these files. The delivery receipt records the actual publication state. Earlier valid source files and their original scope statements are not rewritten by this additive continuation.

16.10 Sources

[MAT] A. Languasco, F. Luca, P. Moree and A. Togbe, *Sequences of integers generated by two fixed primes*, Abh. Math. Semin. Univ. Hambg. 95 (2025), 123-148, Theorem 2.1. The inspected rational-number bound is attributed there to Matveev, in a version due to Bugeaud, Mignotte and Siksek. The proof here uses only that displayed bound and checks its constants explicitly. <https://doi.org/10.1007/s12188-025-00293-9>

[RT] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948v2 (13 February 2025), Definitions 1.1-1.2, Sections 2 and 4, and the explicit comparison of iteration conventions in Section 5. This supplies context and attribution, not a convergence premise or numerical bound imported into our proof. <https://arxiv.org/html/2502.00948v2>

[WB1] KokunoYumeto, Collatz workbench, `intrinsic_zero_firstjet_20260915/firstjet.py`, Git blob 97f9ef5ad535f7ec18259d41fab6a3557be17d98, and its complete `note.md` at 9afcb2dc9bace77c1ae705b1cd128da22fe8f8b. Original integral first-jet presentation and finite-witness extraction; used unchanged.

[WB2] Same repository and revision, `mixed_switch_control_20260916/note.md` and `TASK_STATE.md`. The source stages the nonrepeating/general-switch target. Its previous mixed-core gap certificate is not a premise of the new all-word gap.

[SZ] Owner's Split-Zero construction and intrinsic fibre/quotient programme, source chain recorded in [WB1]. The precise structure used is its comparison kernel with original representatives, relations and support inclusions, not a claim that an isolated scalar zero contains a unique trajectory.

Chapter 17

Global height reduction: sharp crossing windows and original coalescence maps

Source: repository/collatz_reconstruction/research_program/height_window_reduction_20260916/note.md.

16 September 2026. Additive continuation of KokunoYumeto/collatz-workbench, read at 9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b (PR #2). The source mathematical object is the original absolute integral first-jet complex in intrinsic_zero_firstjet_20260915. The immediately preceding local global_cylinder_control_20260916 is preserved; its all-word power-gap proof is not a dependency of the new results below. In particular, this increment uses no Matveev estimate, floating-point logarithm or unproved comparison imported from the Zeta programme.

The results are: a sharp envelope for **every first coefficient crossing**; a complete finite-window reduction valid at every chosen depth; a computer-assisted theorem through 10,280 odd steps, for unbounded starting integers; an exact affine calculus of **all two-path coalescence families**; and a global integral chain retraction using both directions of original equation relations. A newly proved infinite coalescence family removes previously retained arithmetic roots that have no incoming Collatz edge. The remaining roots and their modules are not declared zero.

17.1 1. Original maps, clocks, and attribution

Write

$$X = \{1, 3, 5, \dots\}, \quad T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1).$$

For a finite ordered exponent word $p = (a_1, \dots, a_m)$, retain

$$A_j = \sum_{i=1}^j a_i, \quad U_p = 2^{A_m}, \quad L_p = 3^m, \quad C_p = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{A_j}, \quad f_p(x) = \frac{L_p x + C_p}{U_p}. \quad (1)$$

The empty word has $(m, A, L, U, C) = (0, 0, 1, 1, 0)$. Its source and target are the same positive odd integer. Nonempty words use the original chronological order.

The source cylinder and image are

$$n = \rho_p + 2U_p t \mapsto \eta_p + 2L_p t, \quad t \in \mathbb{Z}_{\geq 0},$$

$$\rho_p \equiv (U_p - C_p)L_p^{-1} \pmod{2U_p}, \quad 0 < \rho_p < 2U_p, \quad \eta_p = (L_p \rho_p + C_p)/U_p. \quad (2)$$

For the empty word, $\rho = \eta = 1$. Original intermediate oddness follows from terminal oddness: modulo 2^{A_j+1} , the full numerator separates into the original prefix numerator times an odd power of three and its next term $3^{m-j-1}2^{A_j}$. All subsequent terms vanish. This gives $3^j n + C_j \equiv 2^{A_j} \pmod{2^{A_j+1}}$. The consecutive prefix equations now give each exact valuation a_j . Positivity propagates from the positive initial value. This proves the full source assertion in (2), not merely an end-value divisibility. The two inverse parameters are the source difference divided by $2U_p$ and the target difference divided by $2L_p$, on their displayed images.

For comparison with Terras's coefficient-stopping-time question, define the shortened map

$$S(n) = \begin{cases} (3n+1)/2 & n \text{ odd}, \\ n/2 & n \text{ even}. \end{cases}$$

An odd-return edge of exponent a expands to one shortened odd step and $a-1$ shortened even steps. Thus $T^j(n) = S^{A_j}(n)$; the parity block is 10^{a-1} . Parsing at successive odd terms is the inverse. A shortened first coefficient crossing can occur before all the last even divisions have finished; that intermediate endpoint must be checked separately. It is not automatically the last odd endpoint. We do check it.

The coefficient-stopping-time conjecture and the study of contracting but non-descending segments are established parts of the literature [RT]. The affine cylinder mechanism and the first-jet receiver are inherited workbench constructions. No priority claim for inverse iteration, common-future equivalence or that literature is made. The following formulas and proof certificates give the scope of this continuation.

17.2 2. The sharp first-crossing envelope at every depth

Set

$$b_j = \lfloor \log_2 3^j \rfloor = \text{bitlength}(3^j) - 1, \quad b_0 = 0,$$

$$C_m^* = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{b_j}, \quad D_m^* = 2^{b_m+1} - 3^m > 0, \quad \Theta_m = C_m^*/D_m^*. \quad (3)$$

The logarithm in this notation is never numerically evaluated. Integer powers and bit lengths give every value. The exact recurrence is

$$C_0^* = 0, \quad C_m^* = 3C_{m-1}^* + 2^{b_{m-1}}. \quad (4)$$

17.2.1 Theorem 1. Sharp, all-depth numerator and exception envelope

For every original word with its first odd-return coefficient crossing at return m , one has

$$A_j \leq b_j \ (j < m), \quad A_m \geq b_m + 1, \quad \frac{C_p}{2^{A_m} - 3^m} \leq \Theta_m. \quad (5)$$

The maximum in (5), over all such words, is exactly Θ_m , attained by the unique ordered word

$$p_m^* = (b_1 - b_0, \dots, b_{m-1} - b_{m-2}, b_m + 1 - b_{m-1}). \quad (6)$$

Every non-descending original source at that first crossing satisfies

$$n \leq \lfloor \Theta_m \rfloor. \quad (7)$$

The same bound applies at the **earlier shortened-map first crossing**, with its possibly even endpoint.

Proof. Before a crossing, $2^{A_j} < 3^j$, which is exactly the first set of integer inequalities. At the first crossing the denominator is at least D_m^* . The entire numerator difference is

$$C_m^* - C_p = \sum_{j=0}^{m-1} 3^{m-1-j} (2^{b_j} - 2^{A_j}) \geq 0. \quad (8)$$

All coefficients are retained in this identity. Positivity of numerator and denominator proves (5). Equality requires every summand of (8) to vanish and the final exponent sum to be $b_m + 1$. This is exactly (6). Its proper exponent increments are one or two; its last increment is two or three. The word has a nonempty actual cylinder by (2), and all its proper multipliers are above one. This proves attainment and uniqueness on the stated word domain. It does not identify different word cylinders or claim that the maximizing cylinder has an actual non-descending point.

The original endpoint difference is $(C_p - (2^{A_m} - 3^m)n)/2^{A_m}$, proving (7). For the shortened map, the first crossing with m odd steps is at exactly $b_m + 1$ total shortened steps. All earlier completed odd-return prefix sums satisfy the same inequalities. Truncating the last division run at $b_m + 1 - A_{m-1}$ retains the identical numerator C_p ; its denominator is exactly 2^{b_m+1} . Equation (8) therefore applies again, even when the intermediate endpoint is even. QED.

The comparison here is the identity on the formal input with the explicit coefficient order (8). It is an inequality, not an equivalence replacing the original branch by the critical branch. The maximizing word and its cylinder remain distinct declared data.

17.2.2 Theorem 2. Complete finite-window reduction

For every chosen integer $M \geq 1$, put

$$H_M = \max_{1 \leq m \leq M} \lfloor \Theta_m \rfloor. \quad (9)$$

All failures of first-coefficient-crossing descent with at most M odd steps are contained in the original odd source interval $1 \leq n \leq H_M$. For every $n > H_M$, a first crossing within that horizon is an actual first strict descent, both in the shortened map and at the subsequent odd endpoint.

Proof. Equation (7) excludes non-descent outside that finite interval. Every proper shortened prefix has coefficient at least one and a nonnegative additive remainder, so it has not strictly descended. For an odd source the remainder is positive after the first odd step. The later divisions down to the odd endpoint only decrease that already smaller value. This establishes both notions of first descent with their proper clocks. The domain of a source that has no crossing by this horizon remains unresolved; it

is not accepted as convergent or divergent. QED.

A finite verification for a horizon is therefore available by testing these finitely many original sources for that horizon. The high-throughput scan delivered here proves the stronger fact that every tested source actually crosses and descends by a much smaller measured number of returns. It does not infer that fact for untested sources.

The envelope itself cannot be capped at a universal constant: since

$$\frac{C_m^*}{3^m} = \frac{1}{3} \sum_{j=0}^{m-1} \frac{2^{b_j}}{3^j} > \frac{m}{6}, \quad 0 < D_m^*/3^m < 1,$$

one has $\Theta_m > m/6$. This concerns the sharp coefficient threshold, not the existence of actual exceptions at that threshold.

17.3 3. Complete certificate through 10,280 odd steps

The exact recurrence (4) gives

$$H_{10280} = 5624777, \quad (m, b_m + 1, \lfloor \Theta_m \rfloor) = (9616, 15241, 5624777)$$

at its largest record. The next record is

$$(10281, 16295, 6728242). \quad (10)$$

envelope.json retains every row and the full critical-prefix-sum dictionary. Its integer-state hash includes the complete numerator and denominator at every row, encoded as length-delimited positive big-endian integers. Those large coefficients are regenerated exactly; a floating-point approximation never chooses a record.

The source ledger contains **every one of the 2,812,388 positive odd sources** from 3 through 5,624,777 in strict order. Each row is `[n, endpoint, ordered_exponents]`. It records an actual first odd-return coefficient crossing and strict descent. The checker also examines the earlier shortened first crossing: at row end with exponent sum A and m odd steps, that value is

$$x_{\text{short}} = T^m(n) 2^{A-(b_m+1)}. \quad (11)$$

It is checked to be strictly below n , rather than inferred from the odd endpoint.

The full independent checker imports neither `height_control.py` nor a predecessor. It reconstructs (3)-(4) through the original critical floor-prefix affine path, reads every ledger row, checks the exact consecutive source labels, repeats every division, and checks all proper coefficient prefixes. It replays **9,808,846 original odd-return equations and 19,623,792 divisions**. The longest first crossing in this finite source interval has 141 odd steps (224 shortened steps), at source 1,126,015. These measured values do not replace the proved horizon 10,280 in Theorem 2.

It follows from Theorems 1-2 and the complete finite certificate that

$$\boxed{\text{Every positive source } n \geq 2 \text{ whose first shortened coefficient crossing uses at most 10280 odd steps descends at that crossing.}} \quad (12)$$

Even sources descend on their first division. The source 1 is retained separately: its odd-return loop is the actual fixed point and never strictly descends. The statement also implies the corresponding first odd-return crossing theorem. This is a finite-

horizon theorem on **unbounded starting integers and unbounded final valuations**, not the global coefficient-stopping-time conjecture.

Each source-row endpoint is a smaller positive odd integer in the same scanned interval. The recursive height $h(n) = m(n) + h(T^{m(n)}(n))$, $h(1) = 0$ is consequently defined by finite induction in increasing n . It constructs a finite arrival path for every scanned source. The maximum inductively obtained total is 222 odd returns, at 3,732,423. No guessed stopping time is used as a premise: each decrease and each height-table index is checked. The maximum odd vertex in the retained first-crossing paths is 439,600,764,977, on the row for 4,637,979; that maximum is not a contiguous verified source interval.

For comparison, the horizon 18 has integer ceiling 108, so only odd sources through 107 are needed to test its possible anomalies. The preceding local contribution's complete 1,166,058-prefix audit remains correct and retained, but that exponential tree is no longer needed for this particular bounded-horizon question.

This calculation intersects Terras's coefficient-stopping-time problem [RT]. No new record relative to published large-start verifications, no global proof, and no claim to originate that problem is made.

17.4 4. The original Split-Zero comparison and its arithmetic height filtration

Use the inherited **absolute** original graph, not a quotient deleting the fixed loop:

$$C^0 = \mathbb{Z}^{(X)}, \quad C^1 = \mathbb{Z}^{(X)}, \quad J e_n = V_n, \quad P e_n = V_{T(n)}, \quad d = J - P.$$

Parentheses denote finite sums. Over $\mathbb{D} = \mathbb{Z}[\epsilon]/(\epsilon^2)$, write $q = 1 + \epsilon$ and

$$d_\epsilon e_n = V_n - q V_{T(n)}, \quad q^k = 1 + k\epsilon \quad (k \in \mathbb{Z}). \quad (13)$$

The inverse $q^{-1} = 1 - \epsilon$ is integral in this stated ring. No nonunit integer or cycle period is inverted.

The source comparison kernel is

$$\mathfrak{B}_X = \ker(H^1(K_\epsilon) \rightarrow H^1(K)) \cong C^1/(dC^0 + P \ker d), \quad [V_n] \mapsto [\epsilon V_n]. \quad (14)$$

The predecessor proves both quotient inverse maps. Its component coordinates are sums of original coefficients, reduced modulo the actual primitive cycle length m on a cyclic component, and unreduced on a nonperiodic component. Thus each actual cycle contributes $\mathbb{Z}/m\mathbb{Z}$, and each actual nonperiodic component contributes $\mathbb{Z}$. The fixed loop $m=1$ contributes zero through its integral unit relation; its original support is still present. Neither nonbase indexing set is assumed nonempty.

Set $\xi_n = [\epsilon V_n]$, and retain the subgroup filtration

$$\mathcal{F}_N = \langle \xi_n : n \leq N, n \text{ odd} \rangle \subseteq \mathfrak{B}_X. \quad (15)$$

Under (14), this is the direct sum of component groups whose least original integer is at most N . Indeed every ξ_n is the coordinate generator of its own component, all vertices of one component have the same generator, and finitely supported sums meet only finitely many components. These coordinate maps and their inverse $1_C \mapsto \xi_{\min C}$ prove the assertion.

Consequently $\mathcal{F}_n/\mathcal{F}_{n-2}$ is zero unless n is the least integer of a nonbase component. At such a least integer it is exactly that component's $\mathbb{Z}/m\mathbb{Z}$ or $\mathbb{Z}$. This is a filtration of the actual integral comparison kernel, not a claim that the chain homotopies below preserve the maximum integer occurring in a raw chain.

A relation $\xi_n = \xi_m$ with $m < n$ gives zero in this graded quotient but need not give zero in $\mathfrak{B}_X$. Its original lower subgroup, source vectors, and boundary witness remain. This distinction is exactly a retained supported-zero comparison, and is not an assertion about chronological automata histories.

The rows in Section 3 give $d_\epsilon(\epsilon B_n) = \epsilon V_n - \epsilon V_y$ with $y < n$. Concatenation with the smaller-source witness produces the original integral certificate

$$d_\epsilon(-e_1 + \epsilon c_n) = \epsilon V_n. \quad (16)$$

The fixed loop supplies the constant term. The unchanged predecessor checker receives and extracts point examples from these certificates. The finite source scan proves $\mathcal{F}_{5624777} = 0$, with its original source ledger retained.

17.5 5. Every expanding word supplies a complete inverse height reduction

Take any original nonempty word p with $L_p > U_p$. Its inverse on its **actual positive target image** is

$$n = \eta_p + 2L_p t \rightsquigarrow x = \rho_p + 2U_p t = \frac{U_p n - C_p}{L_p}, \quad t \geq 0. \quad (17)$$

It satisfies $0 < x < n$ throughout. Equivalently, the target image consists exactly of odd positive n with $U_p n \equiv C_p \pmod{L_p}$ and $n > C_p/U_p$. Integrality and oddness recover the complete source word by (2).

This reduction follows an original path **from x to n** , not from n to x . The parameter inverses are $(n - \eta_p)/(2L_p)$ and $(x - \rho_p)/(2U_p)$ on the stated images. It is therefore legitimate for component-height induction even before an actual forward descent from n is known.

For the original weighted path $P_p(x) = \sum_{j=0}^{m-1} q^j e_{T^j(x)}$,

$$d_\epsilon P_p(x) = V_x - q^m V_n, \quad d_\epsilon(-q^{-m} P_p(x)) = V_n - q^{-m} V_x. \quad (18)$$

Multiplication by epsilon gives $\xi_n = \xi_x$. The reverse clock is retained as the integer coefficient $-m$, not reset to one.

For example, the original word (1,2) gives

$$13 + 18t \rightsquigarrow 11 + 16t,$$

with the original two-step path in the opposite direction. In particular $27 \rightarrow 41 \rightarrow 31$ gives the height reduction $31 \rightarrow 27$. Its exact chain is

$$-(1 - 2\epsilon)e_{27} - (1 - \epsilon)e_{41},$$

with boundary $V_{31} - (1 - 2\epsilon)V_{27}$.

17.6 6. All shared-future reductions have an exact integral parameterization

Take **any two finite original words**, including the empty word, called l and r . Their complete source/image pairs are (2). Put

$$g = \gcd(L_l, L_r) = 3^{\min(|l|, |r|)}.$$

Their target images intersect exactly when

$$\eta_l \equiv \eta_r \pmod{2g}. \quad (19)$$

No independence of the two sources is assumed. Both target steps are retained. A compatible intersection has period $P = 2L_l L_r / g$. Let y_0 be the least member of the common congruence class at least $\max(\eta_l, \eta_r)$. Define

$$\begin{aligned} u_0 &= (y_0 - \eta_l) / (2L_l), & v_0 &= (y_0 - \eta_r) / (2L_r), \\ n_0 &= \rho_l + 2U_l u_0, & m_0 &= \rho_r + 2U_r v_0. \end{aligned}$$

Then **every pair of positive original sources with this common-word endpoint** is

$$\boxed{\begin{aligned} n(t) &= n_0 + 2U_l(L_r/g)t, \\ m(t) &= m_0 + 2U_r(L_l/g)t, \\ y(t) &= y_0 + 2(L_l L_r / g)t, & t \in \mathbb{Z}_{\geq 0}. \end{aligned}} \quad (20)$$

The inverse is $t = (y - y_0) / P$; the source and original image-parameter inverses in (2) give the same t . To prove completeness, subtract the target equations to obtain $L_l u - L_r v = (\eta_r - \eta_l) / 2$. Its homogeneous integral solutions are $(L_r / g, L_l / g)t$. Positivity of both original parameters chooses precisely the least common target y_0 and $t \geq 0$. This also proves (19) and both inverse descriptions.

The strict height domain $n(t) > m(t)$ is an exact integer interval. With

$$a = n_0 - m_0, \quad b = 2(U_l L_r - U_r L_l) / g,$$

it is: $t \geq \max(0, \lfloor -a/b \rfloor + 1)$ for $b > 0$; all $t \geq 0$ or none according to $a > 0$ for $b = 0$; and $0 \leq t \leq \lfloor (a - 1) / (-b) \rfloor$ for $b < 0, a > 0$, otherwise none. Incompatible and empty-height faces remain declared records, rather than being identified with absent input data.

Let $P_l(n), P_r(m)$ be the two original weighted chains to $y(t)$, and $k = |l| - |r|$. The full chain comparison is

$$\boxed{d_\epsilon(P_l(n) - q^k P_r(m)) = V_n - q^k V_m.} \quad (21)$$

All common-target terms cancel with their original clock. The result is a relation between original sources, not a claim that either source vanishes. In the retained kernel, multiplying (21) by ϵ gives $\xi_n = \xi_m$.

This family calculus is complete for lower-component comparisons. Two vertices of a functional graph lie in one weak component exactly when their forward paths meet. Adjacent vertices have that property; it is transitive because two meetings on the same intermediate orbit have a later common meeting. Conversely a common endpoint gives an undirected path. Therefore every original n whose component

has a smaller integer occurs in one of the strict domains (20). A finite search need not find all such pairs within a predetermined word bound, and no global coverage assertion is made from that completeness characterization.

17.7 7. A newly eliminated infinite family of formerly retained roots

Choose

$$l = (1), \quad r = (1, 1, 1, 1, 1, 2, 3).$$

Equation (20) gives

$$\boxed{1731 + 2916t \rightsquigarrow 1215 + 2048t, \quad t \geq 0.} \quad (22)$$

The original paths, including all affine coefficients, are

$$1731 + 2916t \longrightarrow 2597 + 4374t$$

and

$$\begin{aligned} 1215 + 2048t &\longrightarrow 1823 + 3072t \longrightarrow 2735 + 4608t \\ &\longrightarrow 4103 + 6912t \longrightarrow 6155 + 10368t \\ &\longrightarrow 9233 + 15552t \longrightarrow 6925 + 11664t \longrightarrow 2597 + 4374t. \end{aligned}$$

Every displayed constant is odd and every slope is even. Direct substitution into $3x + 1 = 2^a y$ proves the exact valuations on the whole nonnegative integer parameter domain. The height difference is $516 + 868t > 0$. This is a family proof, not an extrapolation from its anchor.

The signed first-jet clock is -6. At $t=0$ the retained chain is

$$e_{1731} - (1 - 6\epsilon)e_{1215} - (1 - 5\epsilon)e_{1823} - (1 - 4\epsilon)e_{2735} - (1 - 3\epsilon)e_{4103} - (1 - 2\epsilon)e_{6155} - (1 - \epsilon)e_{9233} - e_{6925},$$

whose boundary is $V_{1731} - (1 - 6\epsilon)V_{1215}$. Formula (21) supplies the identical coefficient pattern at every parameter.

Each left source is divisible by three. It has **no incoming original Collatz edge**: $2^{a(x)}T(x) = 3x + 1$ is 1 modulo three, so $T(x)$ is never 0 modulo three. Thus this new lower-source relation cannot be supplied by an ancestor of the left source. It uses the two original legs meeting later. At parameters divisible by three the right source is also divisible by three, so neither distinct source is a forward iterate of the other. Their common endpoint and (21), rather than a fabricated reverse Collatz edge, prove the relation.

17.8 8. A genuine whole-source finite-column retraction

Fix the following **explicit** reduction rules, in order, for $n > 1$:

1. Take the original one-return edge when its endpoint is smaller (equivalently, $n \equiv 1$ modulo four).
2. Otherwise use the inverse of the original expanding words (1), (1,2), and (1,1,1,2,1,1,4), in that order, on their exact images (17).

3. Otherwise use the coalescence family (22).
4. Retain n as an arithmetic root.

Leave 1 as the specified base root with its original loop still present. Every rule has a smaller positive integer endpoint, so the recursion terminates for **every original input**, without assuming Collatz. It need not terminate at 1.

For a rule $n \rightarrow m$ write B_n for its exact chain (forward, inverse, or shared-future) and k_n for its signed clock; each satisfies

$$d_\epsilon B_n = V_n - q^{k_n} V_m. \quad (23)$$

Define recursively

$$H(V_n) = B_n + q^{k_n} H(V_m), \quad Q(V_n) = q^{k_n} Q(V_m),$$

with $H(V_r)=0, Q(V_r)=V_r$ at a root. Every column is a finite original edge vector. Its original intermediate vertices may be larger than n ; the proof uses the decreasing rule endpoints, not a false monotonicity assertion for all those vertices. Set

$$F = I - Hd_\epsilon \quad \text{on edge vectors.}$$

17.8.1 Theorem 3. Integral chain retraction with all original obstructions retained

On the full original finite-support D-modules,

$$\begin{aligned} d_\epsilon H &= I - Q, & Q^2 &= Q, & HQ &= 0, \\ d_\epsilon F &= Qd_\epsilon, & F^2 &= F, & FH &= 0, & Hd_\epsilon &= I - F. \end{aligned} \quad (24)$$

Thus the image complex $[\text{im } F \rightarrow \text{im } Q]$, with its inclusion and projections F, Q , is a chain-homotopy retract of the original complex over D .

Proof. Equation (23) telescopes under the decreasing recursion to give the first identity. Roots have $H=0$ and $Q=\text{identity}$, so Q is idempotent and $HQ=0$. Applying d to F gives Qd . Next $F^2 = I - 2Hd + HdHd$; substitution of $dH=I-Q$ and $HQ=0$ gives $F^2=F$. The same substitution gives $FH=H-HdH=HQ=0$. The last identity is the definition of F . The projection and inclusion compositions are now identities on the image, and (24) provides the original homotopy on both degrees. All coefficients are integral dual numbers. QED.

The image in vertex degree is the free D-module on retained original roots. In edge degree it is not silently replaced by a free graph with independent projected edge symbols. Its exact presentation is

$$C_{\mathbb{D}}^0 / \text{im } H \xrightarrow{\sim} \text{im } F, \quad [e] \mapsto Fe, \quad (25)$$

with inverse Fe mapped to $[e]$. Indeed $\ker F = \text{im } H$: inclusion follows from $FH=0$, and $Fe=0$ gives $e=Hd e$. This retains the original relations between projected edge columns.

At a declared finite equation set E , include its original target vertices, then add the finite original supports of H on those vertices. Call the result $\text{Sigma}(E)$. Sigma preserves unions and inclusions. The displayed maps are typed on these enlarged supports; compositions are compared in the next such enlargement, without assuming

$\text{Sigma}(E)$ is a finite invariant closure. The global finite-support modules are their union. Thus the support transitions, zero coefficients and all source equations remain available in the Split-Zero reconstruction.

The reduction of every coefficient q^k is 1, and the selected arithmetic rules do not depend on ϵ . Reducing H , Q and F coefficientwise therefore gives the same constructed maps on the original integer incidence complex. The squares with constant reduction commute on each original basis vector and thus on all finite sums. The retraction preserves the full first-jet cohomology and its reduction map, hence both the finite period-index part and the free nonperiodic part of (14). This is not based on dimension alone. The original fixed edge satisfies $F(e_1)=e_1$, since $d_\epsilon e_1 = -\epsilon V_1$ and $H(V_1)=0$. On a longer actual cycle, its period holonomy remains $1 - q^m = -m\epsilon$; no division by m occurs.

17.8.2 Exact old/new comparison for the new shared-future family

Let H_0, Q_0, F_0 use rules 1-2 only, and H_1, Q_1, F_1 also use (22). The earlier rules have priority and are unchanged. The new rule acts only at old roots. Append the old reduction of its endpoint m to its original chain, retaining its q -power. This gives a chain Γ between two old roots, hence $F_0 \Gamma = \Gamma$ by (24). The accumulated correction $K = H_1 - H_0$ is a finite sum of these chains with the old clock factors. Therefore

$$d_\epsilon K = Q_0 - Q_1, \quad F_0 K = K, \quad F_1 F_0 = F_0 F_1 = F_1, \quad Q_1 Q_0 = Q_0 Q_1 = Q_1. \quad (26)$$

For the last identities one can also follow the unchanged old route first and then the new routes at its root; that is exactly the full new route. The edge identities follow by $F_i = I - H_i d$, $F_0 K = K$ and the same root equations. The proof uses actual concatenated chains, not a rank estimate.

17.9 9. The exact remaining arithmetic root set

After rules 1-2, $n > 1$ is a root precisely when

$$n \bmod 36 \in \{3, 7, 15, 19, 27\}, \quad n \not\equiv 8731 \pmod{8748}. \quad (27)$$

After the new family (22), add

$$n \not\equiv 1731 \pmod{2916}. \quad (28)$$

Together with the actual base 1, these are the complete roots of the fixed global retraction.

To verify the residue calculation, rule 1 removes $n=1$ modulo four. On the remaining $n=3$ modulo four, the inverse of (1) removes $n=2$ modulo three. The inverse of (1,2) is $(8n-5)/9$, removing $n=4$ modulo nine, i.e. $n=31$ modulo 36 in that remaining set. The third inverse is $(2048n-2363)/2187$, with $n=-17$ modulo 2187. Intersecting with $n=3$ modulo four gives $n=8731$ modulo 8748. Its smallest remaining source is already positive in the inverse domain, so no small positive exception is omitted. Finally (22) removes exactly the class (28), which lies in the old residue class 3 modulo 36 and is disjoint from the -17 class. There are 1,214 old and 1,211 new nonbase root residues in one period 8,748. These counts describe the **specified reduction rule**, not a density estimate for counterexamples or for all unresolved Collatz sources.

Every nonbase component minimum must survive (27)-(28), because each accepted rule relates it by actual common-future paths to a smaller member of the same component. By Section 3 that minimum must also exceed 5,624,777, and it cannot have a first coefficient crossing with at most 10,280 odd steps. These are necessary restrictions, not a proof that the remaining root set is empty or that its elements are counterexamples.

17.10 10. What remains globally

The new proofs and certificates eliminate a broad class of source questions, but their remaining quantifiers are explicit. The all-depth envelope is established; its finite source verification has the delivered horizon. The global chain retraction is established; its roots are the nonempty explicit set (27)-(28). The all-pair coalescence calculus is established; applicability of lower-source comparisons on every remaining nonbase root is not proved here.

The next useful arithmetic tasks are on those roots: extend the complete source-window certificate at its next record, find further whole shared-future families with exact parameter domains, or control the ordinary-integer support of the surviving infinite noncrossing paths. Formula (20) allows complete non-coprime target synchronization, including order, coefficients and domains, rather than matching only a trajectory word. Any new rule can be attached through (26), preserving its full original kernel.

A finite residual diagram with open frontier is not an infinite nonperiodic witness. A class transported to a smaller-source subgroup is not automatically zero. All bounded source rows here have independent finite certificates; no rule uses a heuristic probability, a discarded affine term, or an assumed finite stopping time.

17.11 11. Implementation and acceptance

`height_control.py` implements (1)-(28), including exact image intersections, both source parameterizations, signed D-coefficients, the two retractions and declared arithmetic roots. `verify.py` checks the formulas on a complete exponent-sum-16 word domain (65,535 words), every row of the 10,280-depth envelope, a complete small pair-word domain, both support/retraction compositions, the full affine-family coefficients and false-certificate rejection. Those are structural regression scopes, not a replacement for the written general proofs.

`scan_sources.py` constructs the complete 2,812,388-row source certificate and an inductive height table. `audit_sources.py` is separately implemented and imports no contribution module. Its replay repeats every original division and checks the source index sequence, shortened coefficient time, original endpoint, descent and induction. The ledger hash records the entire original row stream, not an unspecified statistical summary. Both ordinary and optimized structural and source executions are compared byte for byte. The full proof scan and the named structural checks have separate counts.

The authentic first-jet receiver is pinned by Git blob 97f9ef5ad535f7ec18259d41fab6a3557be17d98. Its source is retained byte for byte. The new finite examples are accepted by its integral validator. No new Lean run or independent external mathematical review is claimed. Repository publication status is in a separate receipt, not inferred from local tests.

17.12 12. Sources and scoped attribution

[WB] KokunoYumeto, collatz-workbench, PR #2 at 9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b. collatz_reconstruction/research_program/intrinsic_zero_firstjet_20260915/note.md, Sections 3-9, and its firstjet.py. These supply the absolute comparison kernel, original integral witness equations and source support conventions. https://github.com/KokunoYumeto/collatz-workbench/tree/9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b/collatz_reconstruction/research_program/intrinsic_zero_firstjet_20260915

[PREV] The supplied local Collatz_Global_Cylinder_Proof.md and its full archived source, global_cylinder_control_20260916. Read in full for continuation. Its generic contracting-word singleton theorem and earlier 18-return audit remain preserved. The new first-crossing envelope is proved directly and does not use its logarithm bound.

[RT] Olivier Rozier and Claude Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948v3 (15 March 2025), Introduction and Section 2; the current v5 (17 May 2026), Introduction, Definitions 1.1-1.2 and Section 2, was also checked. Journal reference: Discrete Mathematics 349 (2026), 115167. It states the original coefficient-stopping-time problem and analyzes the affine remainder of parity vectors. The shortened/odd-return correspondence used here is proved in Section 1 above. <https://arxiv.org/html/2502.00948v3> <https://arxiv.org/html/2502.00948v5>

[INV] Guenther J. Wirsching, *On the problem of positive predecessor density in $3n+1$ dynamics*, Discrete and Continuous Dynamical Systems 9 (2003), 771-787, DOI 10.3934/dcds.2003.9.771. Cited as prior inverse-iteration context, not as a source of a bound or a theorem used in this proof. No exhaustive priority survey was made. <https://www.aims sciences.org/article/doi/10.3934/dcds.2003.9.771>

Research direction and Split-Zero framework: the owner's programme. New derivations, source certificates and implementation: this tool-assisted continuation. An elementary formula or successful test count is not by itself a claim of priority or external review.

Chapter 18

All ternary layers: bounded original-support reductions and a longer crossing window

Source: repository/collatz_reconstruction/research_program/ternary_join_reduction_20260916/notes.md.

16 September 2026. Additive continuation for KokunoYumeto/collatz-workbench, PR #2, read at 9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b. The current receiver is the absolute integral first-jet complex in intrinsic_zero_firstjet_20260915. The local height_window_reduction_20260916 is a source of the preceding envelope and common-future calculus. Its pending files are not claimed to be published by this addition. The definitions and proofs used here are supplied below; execution depends only on the published first-jet validator, whose original hash is checked.

The main new construction has two complete families at every positive ternary valuation, a finite catalogue of all indicated one-step/common-future comparisons, and a whole-source retraction with a quadratic bound on original vertex support. The first-crossing certificate advances from 10,280 to 10,945 odd steps. None of these results asserts that all remaining roots vanish. No logarithm estimate, probabilistic hypothesis, or RH estimate is used.

18.1 1. Original maps and complete affine cylinders

Use the unchanged positive-odd source X and map

$$T(n) = (3n + 1)/2^{a(n)}, \quad a(n) = \nu_2(3n + 1).$$

For a word $w=(a_1, \dots, a_k)$ put $A_j = \sum_{i \leq j} a_i$, $A=A_k$, $U=2^A$, $L=3^k$ and

$$C = \sum_{j=0}^{k-1} 3^{k-1-j} 2^{A_j}, \quad f_w(x) = (Lx + C)/U. \quad (1)$$

The complete source and target are

$$\rho + 2Ut \mapsto \eta + 2Lt, \quad t \geq 0, \quad \rho \equiv (U - C)L^{-1} \pmod{2U}, \quad 0 < \rho < 2U, \quad \eta = (L\rho + C)/U. \quad (2)$$

The parameter inverses are $(n-\rho)/(2U)$ and $(y-\eta)/(2L)$ on these actual images. Ter-

minal oddness in (2) implies each original prefix oddness: reduce the full numerator modulo $2^{\{A_j+1\}}$; its first j terms form the prefix numerator times an odd power of three, its next term has valuation exactly A_j , and all later terms vanish. Consecutive prefix equations then give the exact valuations a_j . Positivity propagates from the positive starting value. The same congruences work for the negative source $\rho-2U$; every one of its original iterates is negative, since $3x+1 < 0$ for a negative odd integer. Therefore

$$0 < \eta < 2L. \quad (3)$$

This sign argument is about the original formulas on a stated additional source; it is not an identification of positive and negative cycles.

18.2 2. Two source families at every original ternary valuation

Fix integers $b \geq 1$ and e in $\{2, 6\}$. Define

$$h_e = (2^e + 2)/3 \in \{2, 22\}, \quad a = a(b, e) = \min\{a \geq 1 : 2^{a+e+2b} < 3^{a+b}\}. \quad (4)$$

This minimum exists with an elementary explicit bound:

$$a(b, e) \leq b + 2e, \quad a(b+1, e) - a(b, e) \in \{0, 1\}. \quad (5)$$

Indeed at $a=b+2e$ the ratio in (4) is $(8/9)^{b+e} < 1$. Increasing b multiplies that ratio by $4/3$, and then increasing a by one multiplies it by $2/3$. The product is $8/9 < 1$, proving the adjacent increment claim along with minimality.

Keep the original right word

$$w_{b,e} = (1)^a (e) (2)^{b-1} (3). \quad (6)$$

Its length is $a+b+1$, and its exponent sum is $a+e+2b+1$. Put

$$Q = 3^{a+b}, \quad J = 2^{e+2b}, \quad K = 2^{a+e+2b} < Q.$$

Define the EXACT larger-source set

$$\mathcal{R}_{b,e} = \{n > 0 : n \equiv 3 \pmod{4}, \quad Jn + h_e 3^b \equiv 0 \pmod{Q}\}. \quad (7)$$

This is one full positive residue class modulo $4Q$. Let n_0 be its representative in $(0, 4Q)$, and set

$$m_0 = \frac{Kn_0 + 2^a h_e 3^b}{Q} - 1.$$

18.2.1 Theorem 1. Complete lower-source joins on every layer

For every b, e above, all original pairs in the designated fibre are

$$\boxed{\begin{aligned} n(t) &= n_0 + 4Qt, \\ m(t) &= m_0 + 4Kt, \\ y(t) &= (3n_0 + 1)/2 + 6Qt, \quad t \in \mathbb{Z}_{\geq 0}, \end{aligned}} \quad (8)$$

and they obey the ORIGINAL forward paths

$$n(t) \xrightarrow{1} y(t) \xleftarrow{w_{b,e}} m(t), \quad 0 < m(t) < n(t). \quad (9)$$

Every $n(t)$ has actual valuation $\nu_3(n(t))=b$. Its ternary unit $n(t)/3^b$ is 1 modulo 3 for $e=2$ and 2 modulo 3 for $e=6$. In particular it has no incoming original odd-return edge. The two families at different (b,e) are disjoint. No claim is made that (7) covers every source with that valuation and unit.

Proof with the original coordinates. Congruence (7) first forces $3^b|n$. With $z=n/3^b$ it is $Jz+h_e=0$ modulo 3^a . Since $a \geq 1$, $J=1$ modulo 3, and h_e is a unit modulo 3, z is a unit of the asserted residue. Thus b is exactly the valuation, not just a divisibility lower bound. The original input is $n=3^b z$ with $n \equiv 3 \pmod{4}$.

Set $w=(Jz+h_e)/3^a$. It is a positive even integer with $\nu_2(w)=1$: J is divisible by 16, $h_e \equiv 2 \pmod{4}$, and division by odd 3^a preserves valuation. Define

$$m = 2^a w - 1 = \frac{Kn + 2^a h_e 3^b}{Q} - 1. \quad (10)$$

After j of the first a odd returns its ORIGINAL value is

$$x_j = 3^j 2^{a-j} w - 1 \quad (0 \leq j \leq a). \quad (11)$$

For $j < a$, $\nu_2(x_{j+1}) \geq 2$, so the next exponent is exactly one. At $j=a$,

$$x_a = 2^e (4/3)^b n + h_e - 1. \quad (12)$$

The next original exponent is exactly e , since $3h_e - 2 = 2^e$ and

$$T(x_a) = 1 + 4^b n / 3^{b-1}.$$

For the subsequent $b-1$ steps, the value after j of those steps is

$$1 + 4^{b-j} n / 3^{b-1-j} \quad (0 \leq j \leq b-1). \quad (13)$$

They have exact exponent two and end at $1+4n$. The last exponent is exactly three: $3(1+4n)+1=4(3n+1)$, and $n \equiv 3 \pmod{4}$ gives $\nu_2(3n+1)=1$. The endpoint is $(3n+1)/2$. Thus every displayed edge is integral, odd at its odd-return endpoints, positive, and has its original prescribed valuation. Conversely equality of these two word endpoints, using (1), gives (10) and hence (7). The first $n \equiv 3 \pmod{4}$ condition is necessary for the left word (1). This proves the full source domain.

For strict height, (10) reads

$$m = (K/Q)n + h_e (2/3)^a - 1.$$

Equation (4) implies $h_e (2/3)^a < (h_e/2^e)(3/4)^b < 1$. Both $K/Q < 1$ and this negative constant term therefore give $m < n$, while $m = 2^a w - 1 > 0$ gives positivity. Since both endpoints are odd, $n-m \geq 2$. CRT between Q and 4 gives (8); subtracting and dividing any of the three original progressions supplies its inverse parameter. An incoming edge to n would give $3x+1=2^c n=0 \pmod{3}$, impossible. QED.

For $e=2, b=1$ this is the previously retained family $1731+2916t$ to $1215+2048t$. For $e=2, b=2$ it gives

$$6219 + 8748t \longrightarrow 9329 + 13122t \longleftarrow 5823 + 8192t. \quad (14)$$

For $e=6, b=2, a=12$, and it gives

$$\boxed{13704327 + 19131876t \longrightarrow 20556491 + 28697814t \longleftarrow 12017663 + 16777216t.} \quad (15)$$

The right word in (15) is $(1)^{12}, (6), (2), (3)$. This is a lower-component relation, not a purported forward descent from the left source to the right source.

18.2.2 Complete treatment of longer leading runs in this template

For $a' \geq a(b, e)$ the same formula with a' gives a subfamily of (7), since its congruence modulo $3^{a'+b}$ implies the original congruence modulo 3^{a+b} . Both source morphisms remain available and generally have different smaller endpoints. The minimal $a(b, e)$ thus covers the complete height-reducing template family with longer leading runs, rather than being a sample of the a -axis.

The membership test at an arbitrary n is finite: compute its ACTUAL $b = \text{nu}_3(n)$, read its ternary unit, select $e=2$ or 6 , compute a by integer comparison, and test (7). The accepted path has at most $2b+2e+1$ odd returns on its right leg. This is a bound on the constructed witness, not on an arbitrary orbit's stopping time.

18.3 3. A complete finite catalogue of further original comparisons

There is also a useful all-length rule for any right word w of length $k \geq 2$ with

$$a_k \text{ odd}, \quad 2^{A-1} < 3^{k-1}. \quad (16)$$

Its target η is 5 modulo 6, because η is odd and $\eta = 2^{-a_k} = -1$ modulo 3. Thus its entire image is inside the original image of the left word (1). Using (2), every pair is

$$\boxed{\begin{aligned} n(t) &= (2\eta - 1)/3 + 4 \cdot 3^{k-1}t, \\ m(t) &= \rho + 2^{A+1}t, \\ y(t) &= \eta + 2 \cdot 3^k t, \quad t \geq 0. \end{aligned}} \quad (17)$$

The common endpoint, either source, and their affine lattice steps have the same inverse parameter. These pairs have $m(t) < n(t)$ throughout. To prove it, note $C \geq 3^k - 2^k$, by termwise comparison $A_j \geq j$. For $k \geq 2$, $3^k - 2^k > 3^{k-1} > 2^{A-1}$. Define $D_0 = (C - 2^{A-1})/3 > 0$; integrality follows from the odd final exponent. The original equation is

$$m = (2^{A-1}n - D_0)/3^{k-1} < n. \quad (18)$$

Its numerator is positive on (17), since m starts at the positive ρ . Equation (3) also proves $0 < (2\eta - 1)/3 < 4 \cdot 3^{k-1}$. Hence this is an ENTIRE residue class without omitted small positive representatives. No bounded search is needed for these statements at arbitrary k .

The finite catalogue takes exactly all (16) with $2 \leq k \leq 12$. For each k the exponent sum has the precise bound $A \leq (3^{k-1}).\text{bit_length}()$. There are 20,397 words. Their independent composition count is

$$\sum_{k=2}^{12} \sum_{A=k}^{\text{bitlength}(3^{k-1})} \sum_{\substack{j \geq 1 \text{ odd} \\ j \leq A-k+1}} \binom{A-j-1}{k-2}. \quad (19)$$

Every original candidate and its active/redundant face label is regenerated and checked. Both original affine coefficient columns are divided by their prescribed 2-powers, retaining the odd constant and even slope at EVERY original intermediate point. This verifies the whole variable-source family, not merely its anchor.

18.4 4. Exact residue cover, without deleting redundant faces

The old retained retraction first uses a forward one-return descent, then inverses on the actual images of (1), (1,2), (1,1,1,2,1,1,4), and then the e=2,b=1 join. Its nonbase roots are exactly

$$n \bmod 36 \in \{3, 7, 15, 19, 27\}, \quad n \not\equiv 8731 \pmod{8748}, \quad n \not\equiv 1731 \pmod{2916}. \quad (20)$$

These follow directly by solving the original expanding affine image congruences; the inverse of (1,2) is (8n-5)/9 and the longer inverse is (2048n-2363)/2187. All their least admitted remaining sources are in the positive inverse domain. The point 1 remains the separate original base root.

At the common period $P=4 \cdot 3^{11}=708588$, (20) has 98,091 nonbase root residues. Visit (17) in the declared order (period, larger-source anchor, exponent sum, word). A family is selected exactly when it covers at least one still-retained old root residue. Its original full family remains its domain; prioritization restricts execution, not its mathematical source. This yields 370 selected families and 93,025 remaining residues. The 5,066 removed residues equal the sum of the disjoint newly covered counts. A second sweep using the UNION OF ALL 20,397 candidates independently reproduces exactly the same remaining set. The signed row/word and all mask hashes are recorded in verification.json's full reconstructed result. These are counts for a specified arithmetic reduction; they are not probabilities or counts of counterexamples.

Let R_{12} denote that explicitly regenerated residue set. The fixed new retraction uses all the old rules, then the selected catalogue, then Theorem 1 on its remaining inputs. Its nonbase roots are exactly

$$\{n > 1 : n \bmod 708588 \in R_{12}\} \setminus \bigcup_{b \geq 1, e \in \{2,6\}} \mathcal{R}_{b,e}. \quad (21)$$

The infinite layer subtraction is an effective single test on the actual valuation of n. It is not approximated by imposing a finite b cap.

The old/catalogue rules already cover the whole e=2 families b=1,2,3,4 and the e=6 family b=1. Every e=2 family b>=5 and e=6 family b>=2 is new relative to the catalogue. For b<=10 these assertions are finite original residue comparisons. For b>=11, every member has $3^{11}|n$ and $n \equiv 3 \pmod{4}$, hence has the single residue 177147 modulo 708588; that residue lies in R_{12} . This proves the comparison at every larger b, not by extrapolation from testing 64 layers. It also proves that no fixed catalogue length of twelve has already supplied these infinite-layer exclusions.

For example the catalogue gives $91+324t \rightarrow 71+256t$ as a height relation, from the original right word (1,1,2,2,1) meeting the left word (1). Cancelling their actual com-

mon last edge recovers the four-edge inverse path; its cancellation is recorded, not interpreted as absence of the source.

18.5 5. Integral first-jet maps and nested retractions

On the absolute original graph use $D=Z[\epsilon]/\epsilon^2$, $q=1+\epsilon$, and

$$d_\epsilon e_n = V_n - qV_{T(n)}, \quad q^j = 1 + j\epsilon \quad (j \in \mathbb{Z}).$$

The inverse $q^{-1}=1-\epsilon$ is integral. For original forward legs $l:n \rightarrow y$ and $r:m \rightarrow y$ of lengths s, t , their weighted chains give

$$\boxed{d_\epsilon(B_l(n) - q^{s-t}B_r(m)) = V_n - q^{s-t}V_m.} \quad (22)$$

Both legs, not only their difference, are retained by the checker. Theorem 1 has $s-t=-(a+b)$. The coefficient kernel receiver sends V_n to $[\epsilon V_n]$. Multiplying (22) by ϵ gives the exact lower-height relation $\xi_n = \xi_m$. It does not assert that either class is zero before a lower-source boundary is supplied.

Each chosen move has $m < n$. Recurse on the original positive integer, leaving roots as $H(V_r)=0$ and $Q(V_r)=V_r$. For a move with chain B and signed clock k , define

$$H(V_n) = B + q^k H(V_m), \quad Q(V_n) = q^k Q(V_m), \quad F = I - H d_\epsilon. \quad (23)$$

The recursion terminates on EVERY original input, independent of Collatz. Its intermediate original path values need not decrease. The identities are

$$d_\epsilon H = I - Q, \quad Q^2 = Q, \quad HQ = 0, \quad d_\epsilon F = Q d_\epsilon, \quad F^2 = F, \quad FH = 0. \quad (24)$$

The first telescopes (22). The next two hold because Q ends at roots and H is zero there. Substituting $dH=I-Q$ and $HQ=0$ in F^2 and FH proves the remaining statements. Thus the projections F, Q and the inclusion of their image complex give an actual chain-homotopy retraction. In edge degree its full presentation is

$$C_D^0 / \text{im } H \xrightarrow{\sim} \text{im } F, \quad [e] \mapsto Fe,$$

with inverse Fe mapped to $[e]$, since $\ker F = \text{im } H$. The projected edges have these relations; they are not declared an independent basis.

The catalogue and all-layer rules act only after the preceding rules fail. The old route is therefore unchanged up to its retained root. Appending the new route there gives the correction $K=H_{\text{new}}-H_{\text{old}}$, whose original boundary is $Q_{\text{old}}-Q_{\text{new}}$ and whose chain lies in the old image: $F_{\text{old}} K=K$. Consequently

$$F_{\text{new}} F_{\text{old}} = F_{\text{old}} F_{\text{new}} = F_{\text{new}}, \quad Q_{\text{new}} Q_{\text{old}} = Q_{\text{old}} Q_{\text{new}} = Q_{\text{new}}. \quad (25)$$

These apply both at the catalogue extension and the infinite-layer extension. They prove preservation through specific maps, not through a comparison of dimensions. All arithmetic choices are independent of ϵ , so constant reduction sends q^k to 1 and commutes with all the maps. The original first-jet comparison kernel and its cycle-index/free component submodules are preserved. In particular e_1 is retained and $F(e_1)=e_1$; no integer cycle period is inverted.

18.6 6. A quantitative bound on the whole-system retraction

18.6.1 Theorem 2. Polynomial original-support control

For every original odd input $n \geq 3$, the construction above uses only original odd vertices at most

$$\boxed{64n^2 + 21.} \quad (26)$$

Before like edge terms are collected, its homotopy column has at most

$$\boxed{K(n) = \frac{n-1}{2} (2\lfloor \log_3 n \rfloor + 14)} \quad (27)$$

signed monomial edge terms. Its constant coefficients have total absolute sum at most $K(n)$; its epsilon coefficients have total absolute sum at most $K(n)^2$. The logarithm in (27) is equivalently the largest b with $3^b \leq n$, so all bounds are integer-computable. This is a bound on the retraction to retained roots, not on all actual stopping times.

Proof. Each selected move drops between positive odd integers by at least two, so there are at most $(n-1)/2$ moves. The finite catalogue has right-leg length at most twelve, and its left leg has length one; old inverse paths are shorter. The all-layer right leg has $a+b+1 \leq 2b+2e+1 \leq 2b+13$ edges, and its left leg adds one. Its actual $b = \nu_3(\text{current source}) \leq \lfloor \log_3 n \rfloor$. Thus (27) bounds the total terms.

On an all-layer move with larger source $x \leq n$, the right leg first rises to the explicit maximum (12), then every subsequent exponent is at least two and the values decrease. Since $(4/3)^b \leq 3^b \leq x$ and $e \leq 6, h_e \leq 22$, (12) is at most $64x^2 + 21 \leq 64n^2 + 21$. The left target is $(3x+1)/2$, smaller than that bound. For every finite catalogue or inverse leg, its smaller source is below x and its length is at most twelve. Each original map is bounded by $g_1(z) = (3z+1)/2$ on the same positive input, hence all intermediate values are less than $(3/2)^{12}(x+1) < 130(x+1) \leq 64x^2 + 21$ for $x \geq 3$. Forward descent is already below x . This proves (26) along every concatenated leg.

All weights in (23) are signed q -powers. Their exponent is a sum of previous signed leg lengths plus the position within the current leg, whose absolute value is at most the total number $K(n)$ of traversed edges. Each term consequently has constant amplitude ± 1 and epsilon amplitude of absolute value at most $K(n)$. Collection cannot increase either absolute-sum bound. QED.

The induced original-height filtration maps satisfy

$$Q(C_{\leq n}^1) \subset C_{\leq n}^1, \quad H(C_{\leq n}^1) \subset C_{\leq 64n^2+21}^0, \quad F(C_{\leq n}^0) \subset C_{\leq 256n^2+21}^0. \quad (28)$$

The last uses $T(x) < 2x$, so each H column in $Hd(e_x)$ obeys the bound at source $2n$. This does not assert that H preserves height without an increase. Under expansion to the original elementary map, intermediate odd-step values $3x+1$ are at most $192n^2 + 64$ in an H column; the divisions do not increase that bound.

For finite declared equation support E , include its original targets and the finite original supports of these H columns. This enlargement preserves unions and inclusions. Equations (24)-(28) describe its actual comparison with larger support stages. The reconstructed supported zero keeps its original fibre and relations. An infinite diagram is being controlled by bounds on every original column, rather than by identi-

fying it with an absent scalar.

18.7 7. Extending the sharp crossing-height window

Retain $b_j = \text{bitlength}(3^j) - 1$, $b_0 = 0$, and the exact original coefficients

$$C_0^* = 0, \quad C_m^* = 3C_{m-1}^* + 2^{b_{m-1}}, \quad D_m^* = 2^{b_m+1} - 3^m, \quad \Theta_m = C_m^*/D_m^*. \quad (29)$$

For a first odd-return coefficient crossing at m , every proper $A_j \leq b_j$ and the final $A_m \geq b_{m+1}$. Thus

$$C_m^* - C_p = \sum_{j < m} 3^{m-1-j} (2^{b_j} - 2^{A_j}) \geq 0, \quad C_p / (2^{A_m} - 3^m) \leq \Theta_m. \quad (30)$$

The maximum is attained at the original word with consecutive prefix sums b_j and final sum b_{m+1} . The differences and prefix sums are inverse operations. This is a sharp bound on the word domain; attainment is not an assertion of an integer non-descending source in that word's cylinder. The original non-descent equation forces $n \leq \text{floor}(\Theta_m)$.

For the shortened map $S(n) = (3n+1)/2$ on odd n and $n/2$ on even n , each original odd edge of exponent a expands to 1 followed by $a-1$ even divisions. The first shortened coefficient crossing with m odd steps occurs at b_{m+1} total shortened steps. Its numerator remains C_p and its denominator is $2^{\{b_{m+1}\}}$; (30) therefore applies at that possibly even endpoint as well. This supplies the exact clock comparison.

Define $H_M = \max_{m \leq M} \text{floor}(\Theta_m)$. All exceptions by horizon M lie among original odd $n \leq H_M$. This proof is valid for every chosen M without a logarithm theorem. The coefficient-stopping-time question itself is the established Terras problem; we do not claim it originated here [RT].

The previous retained window had $M = 10280, H_M = 5624777$. The current exact recurrence gives

$$H_{10945} = 6728242, \quad \text{largest odd candidate } 6728241. \quad (31)$$

The attaining record is $(m, b_{m+1}, H) = (10281, 16295, 6728242)$; the next record is $(10946, 17349, 8129266)$. Thus testing the next original source interval extends the entire horizon through 10945, not just the single next value 10281.

Every original odd source $3 \leq n \leq 6728241$ has been checked, with ledger row $[n, \text{actual odd endpoint}, \text{complete first-crossing exponent word}]$. There are 3,364,120 rows, 551,732 new rows beyond the previous interval. Both the odd endpoint and the earlier shortened crossing endpoint are strictly below n . The latter is computed as $\text{terminal} * 2^{\{A - \text{bitlength}(3^m)\}}$, not inferred from its later odd value.

The independent auditor imports neither `join_control` nor the first-jet code. It replays every individual division, every proper coefficient prefix, and the original consecutive source labels. It checks 11,738,935 original odd-return edges and 23,482,682 divisions. The longest first crossing in this finite source interval is 141 odd returns; this measured statistic does not replace the proved horizon in (31). Inductive endpoint heights give a maximum total 248 odd returns, at source 6649279, on this certified finite interval.

It follows from (30)-(31) and the complete ledger that every original $n \geq 2$ whose first shortened coefficient crossing uses at most 10945 odd steps descends at that cross-

ing. Starting integers and the final valuations are unbounded. Even starting sources descend on their first division. The original fixed loop $n=1$ is retained separately. No source with no crossing is accepted by this theorem.

Each row ends at a smaller positive odd integer. Induction in the original source order supplies actual finite arrival certificates throughout the interval, so the original kernel-height subgroup generated by $x_i = [\epsilon V_n]$, $n \leq 6728241$, is zero with its source rows retained.

18.8 8. Witness extraction and remaining global task

The source first-jet theorem states that a finite integral witness (b,c) obeying

$$db = 0, \quad dc - Pb = V_n \quad (32)$$

extracts an actual path from n to 1. Its proof uses the ORIGINAL integral component relation $-km=1$ and thus forces the actual cycle length $m=1$. It does not allow rational division by a longer period.

From (24), multiply an H column by ϵ . It is a relation ϵV_n minus ϵV_{root} . Add its constant edge coefficients to the linear part of the root's original witness (32); leave the constant cycle untouched. The full result is again (32) at n . The unchanged authenticated validator performs the extraction. Six examples, including $n=13704327$ outside (31), are delivered. Its new arithmetic root is 5341183 inside (31), its signed reduction clock is -16, and its extracted ACTUAL forward path to 1 has 89 odd returns. The root is not misreported as an actual forward iterate at that signed clock. Infinite progressions are proved for lower-source relations; these six point arrival witnesses do not prove arrival of all their variable endpoints.

The complete first-jet component kernel is preserved by (24)-(25). A nonbase cycle retains its $\mathbb{Z}/m\mathbb{Z}$ component and a nonperiodic component its $\mathbb{Z}$. Its least original integer must now exceed 6728241, belong to (21), and have no first coefficient crossing with at most 10945 odd steps. These are necessary original-source restrictions; the remaining root set is not claimed empty.

The next arithmetic work can enlarge either leg of the common-future family, construct additional coefficient templates on (21), or sharpen (30) on remaining source residues. A family must retain its exact source domain and both inverse parameters. The polynomial support bound (26) makes the present whole-system retraction quantitatively available for such further comparisons; it does not supply global applicability of a next root reduction. No unproved well-founded rank, zero measure argument, or infinite intersection assumption is used.

18.9 9. Verification, integration and sources

The general algebraic and arithmetic-family results above have their written proofs. The finite catalogue and original-source window have COMPLETE bounded certificates. Regression checks additionally test 128 layer objects with $b \leq 64$, four source parameters per layer, exact affine coefficients, old/new projection identities, negative controls and original witness extraction. General b coverage is proved by (4)-(13), not extrapolated from that test range. Both ordinary and optimized outputs are compared byte for byte. A typed-cache bug caught by a Boolean-as-integer negative control during development was repaired before the accepted runs; those invalid

inputs remain rejected tests, not silently removed cases.

`verify.py --outputfull.json` reconstructs all 370 selected families, all masks, examples and regression records. `replay.py --full --predecessor` regenerates the complete source ledger in both execution modes and audits every row independently. The source ledger is an external deliverable and a reproducible optional artifact; its hash is pinned in `window_certificate.json`. No large binary need be trusted without this reconstruction. Source/current Git identities and publication scope are recorded separately; an expected or scheduled CI result is not called a pass.

[HW] Owner-directed local continuation, *Global height reduction: sharp crossing windows and original coalescence maps*, 16 September 2026. Supplied `Collatz_Height_Window_Proof.md` and its complete ZIP. The source-height envelope, all-pair common-future formula and unbounded retraction definitions are prior work of this workbench, reproduced where used. Exact input file hashes are in `SOURCE_INTAKE`.

[FJ] KokunoYumeto/collatz-workbench, `intrinsic_zero_firstjet_20260915`, at commit `9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b`. Original `firstjet.py` SHA-256 `a53cef65109285f377e2c1344ae662c4a74574a30c1c40dffffebc9ed6cca06b0`. https://github.com/KokunoYumeto/collatz-workbench/tree/9afcbe2dc9bace77c1ae705b1cd128da22fe8f8b/collatz_reconstruction/research_program/intrinsic_zero_firstjet_20260915

[SZ] Owner's Split-Zero support-fibre and killed-representative construction, traced in [FJ]. Its support fibre and original relation quotient are not defined as chronological weighted-automata histories. No additional Zeta numerical estimate is imported here, and no claim about the complete current RH programme is made.

[RT] Olivier Rozier and Claude Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948, version 5, 17 May 2026; DOI `10.1016/j.disc.2026.115167`. <https://arxiv.org/abs/2502.00948> The coefficient-stopping-time conjecture and paradoxical segments are established literature. No published verification record or novelty priority is claimed here.

Chapter 19

Bilateral original-source reductions, maximal integral lifts, and subquadratic support bounds

Source: repository/collatz_reconstruction/research_program/bilateral_root_bounds_20260916/note.md.

16 September 2026. Research continuation in KokunoYumeto/collatz-workbench. The authenticated base is b2dac4c2cc4a5eedde0af36d6749390c77fa4a62, ternary_join_reduction_20260916. The original integral receiver is intrinsic_zero_firstjet_20260915. All earlier source files remain unchanged.

This note proves two complete new arithmetic families at every positive ternary valuation, their exact disjointness from the preceding layer families, and an all-depth refinement that selects the smallest source within each admitted leading-run template. It also gives a complete finite two-sided catalogue and a subquadratic bound on the original vertex support of the whole-source retraction. The first-crossing window is verified through 11,610 odd steps by a complete original-source certificate and an independent all-row audit.

These are bounds on specified constructions and proved source domains. They do not assert that the residual root set is empty, that every orbit crosses, or that a possible positive nontrivial cycle or nonperiodic component has been found. No estimate from the Zeta programme is imported. The general arithmetic and algebraic arguments below use no theorem on linear forms in logarithms.

19.1 1. Fixed original coordinates and the comparison receiver

Write X for the positive odd integers and retain

$$T(n) = \frac{3n+1}{2^{a(n)}}, \quad a(n) = \nu_2(3n+1). \quad (1)$$

An exponent word $w=(a_1, \dots, a_k)$ has original prefix sums A_j , total A , and

$$L = 3^k, \quad U = 2^A, \quad C = \sum_{j=0}^{k-1} 3^{k-1-j} 2^{A_j}, \quad f_w(n) = (Ln + C)/U. \quad (2)$$

The empty word has $(L,U,C)=(1,1,0)$. Its complete positive odd source is $1+2t$. For every word retain the original cylinder and image

$$\rho + 2Ut \mapsto \eta + 2Lt, \quad t \geq 0, \quad \rho \equiv (U - C)L^{-1} \pmod{2U}, \quad 0 < \rho < 2U, \quad \eta = (L\rho + C)/U. \quad (3)$$

Their inverse parameters are $(n-\rho)/(2U)$ and $(y-\eta)/(2L)$ on the displayed images. The affine numerator satisfies the chronological recurrence $C'=3C+U$. The terminal congruence in (3) gives every prefix oddness: reduce the full numerator modulo $2^{\{A_j+1\}}$, retain the j -prefix term, the next term of exact valuation A_j , and the later terms divisible by $2^{\{A_j+1\}}$. The resulting prefix numerator is $2^{\{A_j\}}$ modulo $2^{\{A_j+1\}}$. The successive original equations then force the exact valuations a_j . This proves the word/cylinder correspondence without replacing exact valuations by divisibility alone.

In particular $0 < \eta < 2L$. Positivity follows from $\rho > 0$. The same congruences at the specified negative source $\rho - 2U$ give the integral odd terminal $\eta - 2L$. Every original odd-return image of a negative odd integer is negative; hence $\eta - 2L < 0$. This proves the image-anchor bound by two evaluations of the same affine map. No identification of the positive and negative source domains is made.

The absolute first-jet complex has original edge and vertex bases E_n, V_n , including E_1, V_1 . Put

$$\mathbb{D} = \mathbb{Z}[\epsilon]/(\epsilon^2), \quad q = 1 + \epsilon, \quad d_\epsilon E_n = V_n - qV_{T(n)}. \quad (4)$$

For every integer k , $q^k = 1 + k\epsilon$; $q^{-1} = 1 - \epsilon$ is an integral unit. Neither a cycle period nor a nonunit integer is inverted. The ordinary incidence is $d = J - P$. The specified comparison kernel is

$$\mathfrak{B}_X = \ker(H^1(K_\epsilon) \rightarrow H^1(K)), \quad \xi_n = [\epsilon V_n]. \quad (5)$$

For an original finite path $l:n \rightarrow y$ of length s let $B_l(n)$ be its weighted chain, with coefficient q^j on its j -th edge. Telescoping gives

$$d_\epsilon B_l(n) = V_n - q^s V_y. \quad (6)$$

Thus two ORIGINAL paths $l:n \rightarrow y$ and $r:m \rightarrow y$ have the exact comparison

$$\boxed{d_\epsilon (B_l(n) - q^{s-t} B_r(m)) = V_n - q^{s-t} V_m, \quad t = |r|.} \quad (7)$$

Multiplication by ϵ gives $\xi_n = \xi_m$. This equality does not by itself make either original class zero. It identifies their images through the original finite integral relation, with both paths and their source labels retained. In particular the integer m below is a smaller source with a common future, not necessarily a forward iterate of n .

The predecessor proves that the full kernel (5) is a direct sum of $\mathbb{Z}/m_C \mathbb{Z}$ for actual nontrivial cycle components and $\mathbb{Z}$ for actual nonperiodic components. Its finite integral witness equations are $db=0$, $dc - Pb = V_n$. They imply an actual path to 1, extracted from the original support, since the cycle-component augmentation would otherwise give $-k \cdot m_C = 1$ with $m_C > 1$ or with no cycle. We retain that exact receiver and independently replay the supplied original validator; no quotient coefficient is silently rationalized.

19.2 2. Two nontrivial legs on every ternary layer

Fix $b \geq 1$ and e in $\{2, 6\}$. Put

$$\begin{aligned} h_e &= (2^e + 2)/3 \in \{2, 22\}, & a &= a(b, e) = \min\{a \geq 1 : 2^{a+e+2b-1} < 3^{a+b}\}, \\ J &= 2^{e+2b-1}, & Q &= 3^{a+b}, & K &= 2^a J < Q. \end{aligned} \quad (8)$$

These definitions involve only integer comparisons. The minimum exists, with

$$a(b, e) \leq b + 2e - 2, \quad a(b + 1, e) - a(b, e) \in \{0, 1\}. \quad (9)$$

At $a = b + 2e - 2$ the ratio K/Q is $(8/9)^{b+e-1} < 1$. Increasing b multiplies the ratio by $4/3$; increasing a once more multiplies it by $2/3$. Their product is $8/9 < 1$, which proves the upper increment, and monotonicity proves the lower increment.

The left ORIGINAL word is $l = (1, 2, 1)$, whose map is $(27n + 23)/16$ and whose source is $n = 27 \bmod 32$. The right word is

$$r_{b,e} = (1)^a, (e), (2)^{b-1}, (1), (1), (3). \quad (10)$$

It has $a + b + 3$ odd returns and exponent sum $a + e + 2b + 3$. Define the exact source set

$$\mathcal{S}_{b,e} = \{n > 0 : n \equiv 27 \pmod{32}, \quad Jn + h_e 3^b \equiv 0 \pmod{Q}\}. \quad (11)$$

The moduli 32 and Q are coprime, so this is one complete positive residue class modulo $32Q$. Let its representative in $(0, 32Q)$ be n_0 and put

$$m_0 = (Kn_0 + 2^a h_e 3^b)/Q - 1, \quad y_0 = (27n_0 + 23)/16.$$

19.2.1 Theorem 1. Complete bilateral joins at every positive ternary valuation

For every b, e above, the entire designated original source fibre is

$$\begin{aligned} n(t) &= n_0 + 32Qt, \\ m(t) &= m_0 + 32Kt, \\ y(t) &= y_0 + 54Qt, \end{aligned} \quad t \in \mathbb{Z}_{\geq 0},$$

(12)

with the original paths

$$n(t) \xrightarrow{(1,2,1)} y(t) \xleftarrow{r_{b,e}} m(t), \quad 0 < m(t) < n(t). \quad (13)$$

Every $n(t)$ has actual $\nu_3(n(t)) = b$. Its unit $n(t)/3^b$ is 2 modulo 3 for $e = 2$ and 1 modulo 3 for $e = 6$. All these sources have no incoming original odd-return edge.

Proof with all intermediate values. Congruence (11) forces $3^b | n$. Write $n = 3^b z$. Then $Jz + h_e = 0 \bmod 3^a$. Since $J = 2 \bmod 3$ and h_e is a ternary unit, z is a unit of the asserted class. Thus b is the exact valuation. Set $W = (Jz + h_e)/3^a$. It is a positive even integer of exact dyadic valuation one: J is divisible by eight and $h_e = 2 \bmod 4$, while the divisor is odd. Define $m = 2^a W - 1$, which is positive odd. Its first a original values are

$$x_j = 3^j 2^{a-j} W - 1 \quad (0 \leq j \leq a). \quad (14)$$

For $j < a$, x_{j+1} is divisible by four, giving exact next exponent one. At $j=a$,

$$x_a = Jz + h_e - 1 = 2^{e-1} (4/3)^b n + h_e - 1. \quad (15)$$

Because $3h_e - 2 = 2^e$, the next exponent is exactly e and its image is

$$v_0 = 1 + 2 \cdot 4^{b-1} n / 3^{b-1}.$$

The following $b-1$ exponents are exactly two, with original values

$$v_j = 1 + 2 \cdot 4^{b-1-j} n / 3^{b-1-j} \quad (0 \leq j \leq b-1). \quad (16)$$

For $j < b-1$ the nonconstant part is divisible by eight, which proves exactness of those valuations. Their last value is $2n+1$. The remaining original values are

$$2n+1 \xrightarrow{1} 3n+2 \xrightarrow{1} (9n+7)/2 \xrightarrow{3} (27n+23)/16. \quad (17)$$

The residue $n=27$ modulo 32 proves all three exact valuations. It also proves the left path $n \rightarrow (3n+1)/2 \rightarrow (9n+5)/8 \rightarrow (27n+23)/16$ with exponents $(1,2,1)$. Conversely equality of the original two endpoint formulas gives $m = (Kn + 2^{ah_e} 3^b)/Q - 1$ and hence (11); the left path requires $n=27$ modulo 32. Thus the source domain is exact, rather than a subset justified only by samples.

The height difference is controlled by

$$m = (K/Q)n + h_e (2/3)^a - 1. \quad (18)$$

The defining comparison $K/Q < 1$ gives

$$h_e (2/3)^a < (2h_e/2^e) (3/4)^b < 1. \quad (19)$$

Both terms in (18) therefore make $m < n$. Positivity follows from $m = 2^a W - 1$. CRT gives (12); subtraction and division in any one of the three displayed progressions supplies the inverse parameter. An incoming edge to n would require $3x+1 = 2^c n = 0$ modulo 3, which is impossible. This proves every assertion. QED.

The smallest displayed case $b=1, e=2$ has $a=3$:

$$\boxed{1275 + 2592t \longrightarrow 2153 + 4374t \longleftarrow 1007 + 2048t.} \quad (20)$$

At $t=0$ the original left path is 1275, 1913, 1435, 2153. The original right path is 1007, 1511, 2267, 3401, 2551, 3827, 5741, 2153. The difference of the two sources is $268 + 544t > 0$. This family can already overlap a finite catalogue; its full integral relation remains valid there.

The case $b=2, e=6$ has $a=10$:

$$n(t) = 8808219 + 17006112t, \quad m(t) = 8689663 + 16777216t, \quad (21)$$

with the common endpoint $(27n(t)+23)/16$. This is an unbounded family of original sources beyond the finite verification interval below, not an extrapolation from that interval.

19.3 3. The old and new layer families have no common source

The preceding published families use

$$J_{b,e}^{\text{old}} = 2^{e+2b}, \quad J_{b,e}^{\text{old}}(n/3^b) + h_e = 0 \pmod{3^{a^{\text{old}}(b,e)}}, \quad n = 3 \pmod{4}, \quad (22)$$

where a^{old} is the least positive a with $2^{\{a+e+2b\}} < 3^{\{a+b\}}$. Their unit classes are 1 modulo 3 for $e=2$ and 2 modulo 3 for $e=6$.

19.3.1 Theorem 2. Exact disjointness, including the first nonreduced lift

The union of the new families (11) is disjoint from the union of all the old families (22). Families with different b are disjoint by the original valuation; at a fixed b the two e values are disjoint by the original unit class.

Proof. The only possible mixed intersections at a fixed b pair new $e=2$ with old $e=6$, or new $e=6$ with old $e=2$. In the first case $J^{\text{old}}_6 = 32J^{\text{new}}_2$. Multiplying the new congruence by 32 and subtracting the old gives the constant

$$32h_2 - h_6 = 64 - 22 = 42, \quad \nu_3(42) = 1. \quad (23)$$

In the second case $J^{\text{new}}_6 = 8J^{\text{old}}_2$, giving

$$h_6 - 8h_2 = 22 - 16 = 6, \quad \nu_3(6) = 1. \quad (24)$$

Every precision in these possible pairs is at least two; this follows either from (8),(22) at $b=1$ and monotonicity, or by checking that $a=1$ cannot contract. Either intersection would therefore force $9|42$ or $9|6$, a contradiction. QED.

The matching congruences can agree modulo 3 and disagree modulo 9. Equations (23)-(24) retain exactly the obstruction at the next lift. Thus matching a unit class alone would not establish an overlap; the original coefficient maps and prime-power precision are essential. This statement concerns the displayed integer source congruences, not an unproved identification with a Zeta conductor.

19.4 4. Maximal leading runs: the whole extra-depth axis in one finite calculation

The original source is preserved while all allowed leading runs are compared. For n in (11), set

$$Z = J(n/3^b) + h_e, \quad A_{\max} = \nu_3(Z). \quad (25)$$

19.4.1 Theorem 3. Exact maximal lift and its full parameter strata

The possible original leading runs in this fixed template have $a' \leq A_{\max}$. Every contracting one has $a \leq a' \leq A_{\max}$, with smaller source

$$m_{a'} = 2^{a'} Z / 3^{a'} - 1. \quad (26)$$

The smallest of these sources is $m_{\{A_{\max}\}}$. For $a' < A_{\max}$, the original edge $T(m_{\{a'+1\}}) = m_{\{a'\}}$ has exact exponent one. The original maximum vertex on

these right paths is independent of a' : their initial rises end at the same $Z-1$ and all subsequent vertices coincide.

Proof. Integrality of the forced endpoint formula requires $3^{\{a'\}}|Z$, hence exactly $a' \leq A_{\max}$. Division by an odd power leaves $\nu_2(Z/3^{\{a'\}})=1$, so (14)-(17) prove the actual words. All $a' \geq a$ contract by the comparison in (8). Equation $m_{\{a'+1\}}+1=(2/3)(m_{\{a'\}}+1)$ proves strict decrease and, since the first odd value has its plus-one valuation at least two, the exact exponent-one edge. The initial rise ends at $Z-1$ for every a' , and is increasing up to that point; the common remaining path proves the maximum assertion. QED.

This theorem exhausts the admitted leading-run axis for the specified template. It makes no claim that every possible right-word shape is one of these templates.

There is an exact all-depth partition of the ORIGINAL family parameter. Write $n=n_0+32Qt$ and

$$W_0 = (J(n_0/3^b) + h_e)/3^a.$$

Then

$$Z = 3^a(W_0 + 32Jt).$$

For an extra depth $r \geq 0$ and a retained unit u in $\{1,2\}$, its entire stratum is

$$\boxed{t = t_0 + 3^{r+1}s, \quad t_0 \equiv (3^r u - W_0)(32J)^{-1} \pmod{3^{r+1}}, \quad 0 \leq t_0 < 3^{r+1}, \quad s \geq 0.} \quad (27)$$

Here $A_{\max}=a+r$. The complete source and smaller-source steps are respectively

$$32Q 3^{r+1}, \quad 2^{a+r}(32J)3. \quad (28)$$

Their constants are obtained from $n(t_0)$ and $2^{\{a+r\}}(W_0+32Jt_0)/3^{r-1}$. Both are original positive integers. The inverse is $s=(t-t_0)/3^{\{r+1\}}$; exact valuation r follows from the displayed nonzero unit u . Every $t \geq 0$ belongs to exactly one such stratum because W_0+32Jt is positive. The same inverse parameters compute its common original endpoint. Thus arbitrary extra depth is retained without a truncation or a generic large-depth flag.

19.4.2 Exact comparison with the minimal-run relation

Let $B_{\min}$ be (7) for the minimal-run pair, with clock $k_{\min}$. Let I be the original r -edge incoming path from $m_{\{A_{\max}\}}$ to m_a . Its weighted chain obeys $d_{\epsilon} I = V_{\{m_{A_{\max}}\}} - q^r V_{\{m_a\}}$. The maximal pair therefore has chain

$$\boxed{B_{\max} = B_{\min} - q^{k_{\min}-r} I.} \quad (29)$$

The two original right words concatenate as 1^r followed by the minimal right word, so this is an equality of original weighted chains, not only of boundaries. Maximization does not supply a different root-exclusion domain: it refines the same admitted family by composing its original relation with r existing incoming ones. That distinction is retained in the implementation.

There is also a source-size bound on the chosen run. Put $k=\text{floor}(\log_3 n)$, computed as the largest integer with $3^k \leq n$. Since $J+h_e < 3^{\{e+2b-1\}}$ and $n/3^b < 3^{\{k+1-b\}}$,

$$Z \leq (J + h_e)n/3^b < 3^{k+b+e}, \quad A_{\max} \leq k + b + e - 1. \quad (30)$$

This is a proved finite search bound and an exact one-step valuation evaluation; it is not an assumption about an arbitrary orbit's stopping time.

19.5 5. Complete two-leg catalogue with both nonempty original words

For original words l, r , keep all five affine data (L, U, C, ρ, η) from (2)-(3). Put $g = \gcd(L_l, L_r)$. Their target images intersect exactly when

$$\eta_l \equiv \eta_r \pmod{2g}. \quad (31)$$

Let y_0 be the least common image value at least $\max(\eta_l, \eta_r)$, and put $u_0 = (y_0 - \eta_l)/(2L_l)$, $v_0 = (y_0 - \eta_r)/(2L_r)$. Then all compatible pairs are

$$\begin{aligned} n(t) &= n_0 + 2U_l(L_r/g)t, & n_0 &= \rho_l + 2U_l u_0, \\ m(t) &= m_0 + 2U_r(L_l/g)t, & m_0 &= \rho_r + 2U_r v_0, \\ y(t) &= y_0 + 2(L_l L_r/g)t, & t &\geq 0. \end{aligned} \quad (32)$$

The common target and its period give the inverse t . The original source parameters are $u = u_0 + (L_r/g)t$ and $v = v_0 + (L_l/g)t$, with the inverse obtained by subtraction and division. The height condition is exactly

$$(n_0 - m_0) + (2U_l L_r/g - 2U_r L_l/g)t > 0. \quad (33)$$

It specifies an integer interval with either slope sign, which is retained even when empty. Formula (31) follows from the congruence compatibility criterion; (32) follows by adding the common target period. Thus both inverse maps and all images are proved. No independent-distribution premise is used.

The new finite domain has these five left words:

$$(1, 1), (1, 2), (1, 1, 1), (1, 1, 2), (1, 2, 1). \quad (34)$$

They are all the indicated length-two and length-three words before coefficient crossing. For each left word and each right length $1 \leq j \leq 12$, include EVERY positive exponent word r with

$$2^{A_r} L_l < U_l 3^j. \quad (35)$$

The total sum A_r has a finite exact ceiling $B(l, j)$, computed by integer powers. There are $\text{binom}(B(l, j), j)$ right words when $B(l, j) \geq j$. This is the positive-composition identity obtained by adjoining the unused budget as a final nonnegative coordinate. Hence the complete domain has 112,050 pairs.

Of these, 24,107 satisfy (31). The complete certificate proves that each of these has domain $t \geq 0$, positive constant and slope in (33), and $0 < n_0 < 2U_l L_r/g$. Both original affine coefficient columns are replayed after EVERY original odd step and division: the endpoint constant is odd, its slope is even, and their divisions have exactly the declared exponent. Thus the certificate proves the entire infinite family at each row, not only its anchor. The other 87,943 pairs are retained incompatible labelled faces.

These whole-family height assertions are finite, computer-assisted statements on the explicitly bounded domain (34)-(35). They are not extended to all word pairs without proof. The general formulas (31)-(33) remain valid for any words.

19.5.1 The exact residue cover and its independent comparison

The published one-left-edge catalogue comprises all right words of lengths 2 through 12, with odd last exponent and $2^{\{A_r-1\}} < 3^{\{j-1\}}$. The new code reconstructs its complete 20,397 words and its stated original source formulas. The old initial arithmetic root condition is

$$n \bmod 36 \in \{3, 7, 15, 19, 27\}, \quad n \not\equiv 8731 \pmod{8748}, \quad n \not\equiv 1731 \pmod{2916}. \quad (36)$$

Before the published infinite layers are removed, the old catalogue leaves 93,025 root residues modulo $P_0=708,588$, using 370 selected original families.

Use the common new period $P=8P_0=5,668,704$. The lifted old set has 744,200 residues. Visit the compatible new pairs in the declared order (source period, first admitted source, total word length, left word, right word). Select a family exactly when it removes a previously retained root residue. This yields 67 selected families and 739,770 residual residues: 4,430 are removed from this periodic set. A separate union of ALL 24,107 compatible families, in reverse order, gives exactly the same resulting set R_{new} .

The complete 112,050-row ledger includes incompatible faces. An independent checker imports no contribution or predecessor module, verifies original coefficient formulas and valuations, checks strict lexical order and the exact binomial count in EVERY domain bucket, and independently rebuilds both residue masks. Completeness is therefore a proved finite-domain count together with all rows, not an observed sample. These counts describe the specified reductions; they are not counts or probabilities of Collatz counterexamples.

The periodic count precedes the old and new infinite layer subtractions. In particular 4,430 is not being called the number of new classes remaining after an uncomputed overlap with the old infinite layers.

19.5.2 Which of the new infinite layers lie beyond the finite catalogue

The new $e=2$ families $b=1,2,3,4$ and $e=6$ family $b=1$ are entirely covered by the finite stage. Every new $e=2$ family $b \geq 5$ and new $e=6$ family $b \geq 2$ lies entirely inside its retained periodic root set and is therefore a further removal.

For $1 \leq b \leq 10$ the complete residue comparison is included in the certificate: list every residue in the image (12) modulo P and test the complete mask, using its exact period divisibility. At $b \geq 11$ every source has $3^{11} | n$ and $n \equiv 27 \pmod{32}$; CRT gives the single residue 177147 modulo P . That residue is retained. This proves the assertion at ALL larger layers, rather than extrapolating from a finite b sample. Theorem 2 also proves these new sources were not removed by any old infinite layer. Thus the cited surviving layers supply genuinely new root reductions at arbitrarily large valuations.

19.6 6. A whole-source integral retraction, with its edge relations retained

Use three ordered rule stages. Stage 0 uses the published rules: one-return forward descent; then the original image inverses of (1), (1,2), and (1,1,1,2,1,1,4); then the old (b,e)=(1,2) join; then the 370-family catalogue; then the old infinite-layer test (22). Stage 1 uses the 67 new families after all stage 0 rules fail. Stage 2 uses (11), with its maximal-run choice (25), after all stage 1 rules fail. The original fixed point1 is left as a root.

Each rule gives a finite original chain B_n and a smaller positive odd source $\pi(n)$, with a signed integer clock k_n and

$$d_\epsilon B_n = V_n - q^{k_n} V_{\pi(n)}. \quad (37)$$

A root has $H(V_r)=0$ and $Q(V_r)=V_r$. At any other source set

$$H(V_n) = B_n + q^{k_n} H(V_{\pi(n)}), \quad Q(V_n) = q^{k_n} Q(V_{\pi(n)}), \quad F = I - Hd_\epsilon. \quad (38)$$

Every selected source decreases by at least two. The recursion therefore terminates at a retained root for EVERY original input, without assuming Collatz. Its intermediate original path vertices need not decrease.

19.6.1 Theorem 4. Nested original chain retractions

For each stage the maps in (38) satisfy

$$d_\epsilon H = I - Q, \quad Q^2 = Q, \quad HQ = 0, \quad d_\epsilon F = Qd_\epsilon, \quad F^2 = F, \quad FH = 0. \quad (39)$$

The inclusion of $[\text{im } F \rightarrow \text{im } Q]$, its projections F, Q , and H are an integral chain-homotopy retraction. In edge degree the complete presentation is

$$C_{\mathbb{D}}^0 / \text{im } H \xrightarrow{\sim} \text{im } F, \quad [e] \mapsto Fe. \quad (40)$$

For any earlier/later pair of the three stages, with $K=H_{\text{new}}-H_{\text{old}}$,

$$F_{\text{old}}K = K, \quad F_{\text{new}}F_{\text{old}} = F_{\text{old}}F_{\text{new}} = F_{\text{new}}, \quad Q_{\text{new}}Q_{\text{old}} = Q_{\text{old}}Q_{\text{new}} = Q_{\text{new}}. \quad (41)$$

Proof. Equation (37) telescopes to $dH=I-Q$. The map Q ends at roots and H is zero there, proving $Q^2=Q, HQ=0$. Substituting these into $F=I-Hd$ gives the last three identities in (39), including $FH=H-H(I-Q)=0$. Also $I-F=Hd$ is the edge-degree homotopy equation. If $Fe=0$, then $e=Hde$ is in $\text{im } H$; conversely $FH=0$ kills $\text{im } H$. Thus $\ker F=\text{im } H$, and the inverse of (40) sends Fe to $[e]$. In particular projected edge columns are not assumed to be an independent new basis.

All earlier rules have priority. A later route therefore follows the earlier route unchanged until its old root, including its terminal q -power, and then continues there. Hence $H_{\text{new}}=H_{\text{old}}+H_{\text{new}}Q_{\text{old}}$ and $Q_{\text{new}}=Q_{\text{new}}Q_{\text{old}}$. The later final root is also an earlier root, so $Q_{\text{old}}Q_{\text{new}}=Q_{\text{new}}$. Applying d and $HQ=0$ yields $dK=Q_{\text{old}}Q_{\text{new}}$, $H_{\text{old}}dK=0$, hence $F_{\text{old}}K=K$. Substitution of H_{new} into both products of $I-Hd$ gives (41). These arguments use the actual paths and coefficients, not only abstract isomorphism types. QED.

The choices depend on original integers, not on epsilon. Constant reduction commutes with the maps and sends every q^k to 1. It therefore induces the same

comparison-kernel equivalence (5) at every stage. E_1 is retained and $FE_1 = E_1$. A longer actual weighted period has boundary $(1-q^m)V_c = -m \epsilon V_c$; no division by m is permitted. The integral cycle-index torsion and the free nonperiodic summands remain represented.

For a finite declared equation support E , include its original targets and the supports of their finite H columns. This enlargement commutes with unions and inclusions. The resulting maps are the literal component maps in the Split-Zero support diagram. They retain the original relation submodule, not an inferred chronological log or an abstract zero without its map.

19.7 7. Subquadratic original-support bounds, including maximal lifts

Let $k(n)$ be the largest integer k with $3^k \leq n$. Define the integer-computable bound

$$R(n) = \max\{130(n+1), \lfloor 64n 4^{k(n)} / 3^{k(n)} \rfloor + 21\}. \quad (42)$$

Also put

$$K(n) = \frac{n-1}{2}(3k(n) + 14) \quad (n \text{ odd}). \quad (43)$$

19.7.1 Theorem 5. Bounds on the complete original homotopy columns

For every original odd $n \geq 3$, every original odd vertex in $H(V_n)$, including vertices in the two uncollected paths of each selected relation, is at most $R(n)$. There are at most $K(n)$ uncollected signed monomial edge terms. The collected constant coefficients have total absolute sum at most $K(n)$; the epsilon coefficients have total absolute sum at most $K(n)^2$.

For $n \geq 27$,

$$R(n) = \lfloor 64n 4^{k(n)} / 3^{k(n)} \rfloor + 21 \leq 64n^{\log_3 4} + 21. \quad (44)$$

Thus the original-support growth exponent is $\log_3 4$, approximately 1.262, rather than the earlier coarse quadratic bound. The exact operational bound is (42), with integer powers; no floating-point exponent is needed by the checker.

Proof. A selected source x in the route is at most n . Every finite-catalogue or inverse leg has length at most 12 and begins below or at x . On positive inputs $g_a(y) \leq g_1(y) = (3y+1)/2$, and $g_1^j(y) = (3/2)^j(y+1) - 1$. Thus these legs stay below $130(n+1)$. The three-edge left leg also obeys that bound.

For an old layer at x with $b = \nu_3(x)$, the explicit largest initial rise is at most $64x(4/3)^b + 21$, and the following falling vertices do not increase it. For a new layer, the initial rise ends at (15), at most $32x(4/3)^b + 21$. The later vertices in (16)-(17) are bounded by this or by $(9x+7)/2$, in turn below the bound with 64. The maximal-lift theorem proves that increasing its leading run changes neither this peak nor the later path. Since $b \leq k(n)$, all these values are bounded by the second term of (42). Integer rounding gives the stated exact bound.

There are at most $(n-1)/2$ moves. The finite pairs have at most 15 total edge terms. Old layers have at most $2b+14$. By (30), new maximal layers have at most $A_{\max} + b +$

$6 \leq k(n) + 2b + e + 5 \leq 3k(n) + 11$. The uniform factor $3k(n) + 14$ therefore bounds all moves. This proves (43). Every term is a signed q -power whose exponent is a sum of earlier signed leg lengths and a position in the current leg. Its absolute value is at most the total uncollected edge count $K(n)$. Expansion $q^j = 1 + j \epsilon$ and the triangle inequality prove the two collected coefficient bounds.

For $k \geq 3$, $64(4/3)^k \geq 4096/27 > 130$. At $n \geq 27$ this gap, with the retained $+21$, dominates $130(n+1)$ even after flooring. Finally $3^k \leq n$ implies $(4/3)^k \leq (n)^{\log_3(4/3)}$, proving (44). QED.

The actual source-height maps consequently obey

$$Q(C_{\leq n}^1) \subset C_{\leq n}^1, \quad H(C_{\leq n}^1) \subset C_{\leq R(n)}^0, \quad F(C_{\leq n}^0) \subset C_{\leq R(2n)}^0. \quad (45)$$

The last uses $T(x) < 2x$. Expanding every odd-return edge into its original odd step and divisions gives intermediate positive integers at most $3R(n) + 1$ for an H column. The factors and clock assignment in that expansion are the original ones.

These bounds apply on the WHOLE original source, including unresolved roots. They are bounds for the chosen retraction, not bounds on all Collatz stopping times. At a retained root its H column is zero with its source and relations still present; no conclusion of convergence is attached to that zero column.

19.8 8. The exact remaining roots

Let R_{new} be the 739,770-residue periodic set from Section 5. The final nonbase roots are exactly

$$\boxed{\{n > 1 : n \bmod 5668704 \in R_{\text{new}}\} \setminus \left(\bigcup_{b,e} \mathcal{R}_{b,e}^{\text{old}} \cup \bigcup_{b,e} \mathcal{S}_{b,e} \right)}. \quad (46)$$

Both unions range over every $b \geq 1$ and e in $\{2,6\}$. Membership is a finite original calculation: compute $nu_3(n)$, its unit, and the matching old/new congruence. The new mask also supplies the $n=27$ modulo 32 restriction. No infinite search over b or cutoff for maximal depth is used. The basepoint 1 is separately retained.

Every least integer of an actual component away from 1 lies in (46): an accepted original shared-future relation would give a smaller original integer in that same component. This follows from the two paths, not from assuming that a lower source is already in the base component. Root residue counts therefore restrict possible component minima; they do not count such components.

19.9 9. The longer sharp crossing window

The source envelope is reproduced from the preceding exact calculation, not claimed as a new theorem in this increment. Put

$$\begin{aligned} b_j &= \text{bitlength}(3^j) - 1, & b_0 &= 0, & C_0^* &= 0, & C_m^* &= 3C_{m-1}^* + 2^{b_{m-1}}, \\ D_m^* &= 2^{b_m+1} - 3^m, & \Theta_m &= C_m^* / D_m^*. \end{aligned} \quad (47)$$

At a first coefficient crossing with m odd returns, all proper $A_j \leq b_j$ and the final $A_m \geq b_{m+1}$. The exact difference

$$C_m^* - C_p = \sum_{j=0}^{m-1} 3^{m-1-j} (2^{b_j} - 2^{A_j}) \geq 0$$

and $D_p \geq D_m^* > 0$ give $C_p/D_p \leq \Theta_m$. Equality occurs at the original word with proper prefix sums b_j and final sum $b_m + 1$. Thus the bound is sharp on its word domain, not necessarily attained at an integer non-descending source. The original non-descent equation gives $n \leq \text{floor}(\Theta_m)$.

For the shortened map, the first crossing with m odd steps occurs after exactly $b_m + 1$ total shortened steps. Its numerator is the same C_p , even when it occurs before the last odd-return edge has finished all its divisions. The envelope therefore bounds exceptions at that original, possibly even, endpoint as well. The exact expansion of an odd exponent a is one shortened odd step followed by $a-1$ shortened even steps; this retains both clocks.

The complete integer recurrence through $m=11610$ gives

$$\max_{1 \leq m \leq 11610} \lfloor \Theta_m \rfloor = 8129266, \quad \text{record at } (m, b_m + 1) = (10946, 17349). \quad (48)$$

The next record is (11611, 18403, 9966964). All 4,064,632 original odd sources from 3 through 8,129,265 have been scanned and independently replayed. Each has a first coefficient crossing, strictly below its source at BOTH the first shortened crossing and the final odd-return endpoint. There are 14,186,680 odd-return relations and 28,378,002 independently repeated divisions in the full audit.

Together with the proved envelope, this finite certificate establishes:

Every positive integer $n \geq 2$ whose first shortened coefficient crossing involves at most 11,610 odd steps strictly descends at that crossing.

Even starting integers descend at the first division; the odd cases above and the envelope prove the rest. The starting integer and final exponent are not bounded in this statement. Sources with no such crossing remain unclaimed.

The two producer runs, ordinary and optimized, are byte-identical. The independent auditor imports no contribution or predecessor module, reads EVERY source row, performs the original divisions separately, reconstructs the envelope with a critical-prefix recurrence, and checks the smaller-source induction table. The largest observed first crossing is 155 odd returns at 8,088,063; the largest inductive total is 248 odd returns at 6,649,279. Those observations are not used as unproved global clock bounds. The largest odd vertex appearing in the audited crossing paths is 20,114,203,639,877, reached from 6,631,675; all large values are retained as exact integers.

The smaller positive odd endpoints supply strong induction to 1 on the entire finite interval. In particular the height-filtered original kernel $F_N = \langle x_i, n: n \leq N \rangle$ has $F_{8129265} = 0$, with its original finite integral witnesses. An arbitrary lower-source relation $x_i, n = x_i, m$ with $m < n$ only kills the associated height increment; it does not erase that retained lower subgroup.

Coefficient stopping time and merging/sufficient-set methods have established literature; see [RT] and [M]. No priority or published verification record is claimed. The new unrestricted family proofs do not depend on the scan or on a literature bound.

19.10 10. What the results do and do not close

Any least vertex of a nonbase component must exceed 8,129,265, lie in (46), and have no first coefficient crossing involving at most 11,610 odd steps. These are simultaneous necessary conditions on original positive integers. The proof has not shown that their intersection is empty.

The new all-layer domains are disjoint from the published ones, and their maximal leading runs now have complete residue strata and bounds. The finite two-sided domain is fully certified, and its exact cover is explicit. The whole-system retraction has a smaller global support-growth bound while keeping its residual component module. None of these facts assumes that a finite open frontier is an infinite orbit or that a zero in a rationalized module vanishes integrally.

The next arithmetic targets are further right-leg templates or genuinely new left-leg patterns on (46), and residue-aware restrictions on the exact canonical crossing sources. The next fixed crossing-record calculation has horizon 11611 and odd candidate ceiling 9,966,963. These are stated remaining calculations, not scheduled tasks or assumed results. A single ever-larger finite certificate does not establish that every original singleton vanishes.

The predecessor bound $\Theta_m > m/6$ follows from $2^{\{b_j\}} > 3^{j/2}$ and $D_m^* < 3^m$, so no fixed finite source interval covers all crossing horizons under that envelope. A global proof needs additional arithmetic on the original remaining source, or a rigorously surviving integral source class for a counterexample. The supplied maps preserve that distinction.

19.11 11. Executed evidence and source lineage

verify.py checks the complete 112,050-pair domain, all 24,107 compatible variable families, all 20,397 old candidate families, both residue unions, the layer formulas and saturation strata, original signed chain maps, the nested retractions, support/length bounds, and four original singleton witness extractions. It rejects seventeen malformed controls. General family statements have the written proofs above; finite symbolic-family rows are complete finite proof components, and point fixtures are regression evidence only.

The structural pair has 1,905,007 named checks per run and byte-identical output. The all-pair auditor independently checks 112,050 rows and 332,467 original affine row equations. The full source scan and its independent audit have the separate scopes in Section 9. Counts are not summed and reported as a number of theorems. The actual command receipts and file hashes identify the executed bytes.

The source programme and Split-Zero framework are attributed to KokunoYumeto. This tool-assisted continuation develops the new arithmetic, proofs, and checks; it is not an independent external mathematical review. No new Lean execution is claimed. The complete earlier contribution is neither relicensed nor rewritten.

[WB] KokunoYumeto, Collatz workbench, ternary_join_reduction_20260916/note.md, commit b2dac4c2cc4a5eedde0af36d6749390c77fa4a62. Full relevant source text and implementation priority were read through the authenticated GitHub connector. The new code reconstructs its arithmetic rules; it does not claim that a full remote checkout was replayed. The local preceding height-window source is also retained as lineage, not silently reported as newly published.

[IJ] `intrinsic_zero_firstjet_20260915/firstjet.py`, Git blob `97f9ef5ad535f7ec18259d41fab6a3557be17d98`, is the unchanged executable receiver. Its byte identity is checked before every imported witness verification.

[RT] Olivier Rozier and Claude Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948v5 (17 May 2026), definitions 1.1–1.2 and discussion of Terras’ co-efficient stopping-time conjecture. Source for attribution, not an input to the present arithmetic proofs. <https://arxiv.org/html/2502.00948v5>

[M] Keenan Monks, Kenneth G. Monks, Kenneth M. Monks and Maria Monks, *Strongly sufficient sets and the distribution of arithmetic sequences in the $3x+1$ graph*, arXiv:1204.3904. Prior merging/sufficient-set context; no result from that paper is used without an additional map or invoked as a premise here. <https://arxiv.org/abs/1204.3904>

Chapter 20

All even interior exponents: complete source templates and budget-preserving root reduction

Source: repository/collatz_reconstruction/research_program/all_even_join_extension_20260916/note.md.

16 September 2026. Continuation from the original bilateral-root calculation, whose source has now been integrated into main at 9453b5306b3a897389dece309f9c119dffbcc489 in KokunoYumeto/collatz-workbench. The earlier restriction to interior exponents 2 and 6 is removed on both displayed template types. All positive even exponents are admitted. The induced ternary order, exact integer source, and two original paths remain specified.

This note proves a larger family of comparisons, not global Collatz convergence. A shared-future comparison identifies two original first-jet classes; it is not necessarily forward iteration from the larger source to the smaller source. The complete even-exponent test is finite on every original input. One version preserves the preceding subquadratic support budget by testing actual path support; another admits the entire enlarged template with its own proved budget. The old support bound is not silently assigned to the unrestricted version.

20.1 1. Original source and first-jet receiver

Let $X = \{1, 3, 5, \dots\}$, and use

$$T(n) = \frac{3n + 1}{2^{\nu_2(3n+1)}}.$$

An exponent word $w = (a_1, \dots, a_j)$ retains all its ordered exponents, prefix sums A_i , and original affine data

$$L_w = 3^j, \quad U_w = 2^{A_j}, \quad C_w = \sum_{i=0}^{j-1} 3^{j-1-i} 2^{A_i}, \quad f_w(n) = \frac{L_w n + C_w}{U_w}.$$

Its complete positive source is the odd-terminal congruence

$$L_w \rho_w + C_w \equiv U_w \pmod{2U_w}, \quad 0 < \rho_w < 2U_w,$$

with progression and image

$$\rho_w + 2U_w t \mapsto \eta_w + 2L_w t, \quad \eta_w = (L_w \rho_w + C_w)/U_w, \quad t \geq 0.$$

The original predecessor supplies the exact word/cylinder proof. All valuations in the new families are also derived independently below.

Keep the absolute original graph, including the fixed vertex and loop at 1. Over $\mathbb{D} = \mathbb{Z}[\epsilon]/(\epsilon^2)$, put $q = 1 + \epsilon$ and

$$d_\epsilon E_n = V_n - qV_{T(n)}, \quad q^k = 1 + k\epsilon \quad (k \in \mathbb{Z}).$$

The last identity follows for nonnegative k by the binomial formula and for negative k from the integral inverse $q^{-1} = 1 - \epsilon$. No integer period is inverted. For actual finite paths $l : n \rightarrow y$ and $r : m \rightarrow y$, of lengths s, t , retain the weighted edge columns B_l, B_r . Their literal boundary identity is

$$d_\epsilon(B_l(n) - q^{s-t}B_r(m)) = V_n - q^{s-t}V_m. \quad (1)$$

Multiplying by epsilon identifies the original reduction-kernel classes $\xi_n = [\epsilon V_n]$ and ξ_m . It does not certify their vanishing. Every support label below is the set of original equations used, with its full coefficient module and incoming relation map. It is not an automata word label.

20.2 2. The previously omitted ternary orders

For every even integer $e \geq 2$ define

$$h_e = \frac{2^e + 2}{3}, \quad t_e = \nu_3(h_e).$$

The numerator is divisible by 3, h_e is positive, and $h_e \equiv 2 \pmod{4}$.

Lemma 1. For every such e ,

$$\boxed{t_e = \nu_3(e - 1)}. \quad (2)$$

Proof. Set $k=e-1$, which is odd. Then $2^e + 2 = 2(1 - (-2)^k)$. For k coprime to 3, expansion of $(1 - 3)^k$ modulo 9 gives $1 - (-2)^k = 3k$ modulo 9, with valuation one. For any integer $x \equiv 1 \pmod{3}$, write $x = 1 + 3z$. Then $x^2 + x + 1 = 3(1 + 3z + 3z^2)$ has valuation one. Factoring $x^3 - 1 = (x - 1)(x^2 + x + 1)$ therefore increases the valuation by exactly one on cubing. Write $k = 3^r k_0$ with 3 not dividing k_0 and iterate the factorization. It gives $\nu_3(1 - (-2)^k) = 1 + r$. Division by 3 in h_e proves (2). QED.

In particular $e=4$ contributes $t_e=1$ and $e=10$ contributes $t_e=2$. Treating their h_e as ternary units would put the family on the wrong original source layer. The earlier values $e=2, 6$ both have $t_e=0$ and are recovered by (2).

20.3 3. One exact family for every even exponent and tail length

Take $b \geq 1$, even $e \geq 2$, and a selector $\sigma \in \{0, 1\}$. Define

$$a = a(b, e, \sigma) = \min\{a \geq 1 : 2^{a+e+2b-\sigma} < 3^{a+b}\}, \quad J = 2^{e+2b-\sigma}, \quad Q = 3^{a+b}, \quad K = 2^a J < Q. \quad (3)$$

The minimum exists. At $a = b + 2e - 2\sigma$, the ratio K/Q equals $(8/9)^{b+e-\sigma} < 1$, so

$$a \leq b + 2e - 2\sigma. \quad (4)$$

Also $a \geq e$. Indeed $a \leq e-1$ would give $(3/2)^a < 2^{e-1}$, whereas (3) requires $(3/2)^a > 2^{e-\sigma}(4/3)^b > 2^{e-1}$. Since $h_e < 3^e$, it follows that $a > t_e$.

For $\sigma=0$ retain left word (1) and the residue $n=3$ modulo 4. For $\sigma=1$ retain left word (1,2,1) and the residue $n=27$ modulo 32. Write these dyadic data as $(d_\sigma, r_\sigma) = (4, 3)$ or $(32, 27)$. The complete larger-source domain is

$$\mathcal{E}_{b,e,\sigma} = \{n > 0 : n \equiv r_\sigma \pmod{d_\sigma}, \quad Jn + h_e 3^b \equiv 0 \pmod{Q}\}. \quad (5)$$

Because J is a unit modulo Q , the second condition is one residue modulo Q . CRT with the dyadic condition has one representative n_0 in $(0, d_\sigma Q)$.

Put

$$m_0 = \frac{Kn_0 + 2^a h_e 3^b}{Q} - 1, \quad y_0 = f_{l_\sigma}(n_0).$$

Then the complete pair and its common image are

$$\boxed{\begin{aligned} n(v) &= n_0 + d_\sigma Qv, \\ m(v) &= m_0 + d_\sigma Kv, \\ y(v) &= y_0 + \frac{L_{l_\sigma} d_\sigma Q}{U_{l_\sigma}} v, \end{aligned} \quad v \in \mathbb{Z}_{\geq 0}. \quad (6)$$

The original right word is

$$w_0 = 1^a, e, 2^{b-1}, 3, \quad w_1 = 1^a, e, 2^{b-1}, 1, 1, 3. \quad (7)$$

Here repeated symbols denote the literal ordered exponents, not powers of a map after discarding its original intermediate positions.

Theorem 2. All pairs (6) satisfy

$$n(v) \xrightarrow{l_\sigma} y(v) \xleftarrow{w_\sigma} m(v), \quad 0 < m(v) < n(v), \quad \boxed{\nu_3(n(v)) = b + t_e}. \quad (8)$$

Both paths have exactly their displayed valuations. Conversely the equality of these two original word endpoints, on their positive sources, gives (5)-(6). The parameter inverse is $(n - n_0)/(d_\sigma Q)$, with the same value recovered from m or y using (6).

Proof. From (5), reduction modulo 3^b first gives 3^b dividing n . Write $z = n/3^b$. Then $Jz + h_e = 0$ modulo 3^a . Since $a > t_e$ and J is a ternary unit, $\nu_3(z) = t_e$, proving the actual layer in (8). The integer

$$W = \frac{Jz + h_e}{3^a}$$

is positive and has dyadic valuation one: J is divisible by 8, $h_e = 2$ modulo 4, and division by 3^a preserves valuation one. Set $m = 2^a W - 1$. After j of the first a original returns the value is

$$x_j = 3^j 2^{a-j} W - 1 \quad (0 \leq j \leq a). \quad (9)$$

For $j < a$ the valuation of x_{j+1} is at least two; therefore the next exponent is exactly one. At $j=a$, the value is

$$x_a = \frac{J}{3^b}n + h_e - 1.$$

The identity $3h_e - 2 = 2^e$ gives the exact next exponent e and endpoint

$$1 + \frac{2^{2b-\sigma_n}}{3^{b-1}}.$$

After j of the next $b-1$ steps the value is

$$1 + \frac{2^{2(b-j)-\sigma_n}}{3^{b-1-j}} \quad (0 \leq j \leq b-1). \quad (10)$$

At every nonterminal position of (10), its numerator correction is divisible by 8; hence the next exponent is exactly two. The last value is $4n+1$ at $\sigma=0$, and $2n+1$ at $\sigma=1$.

For $\sigma=0$ the final exponent three gives $(3n+1)/2$, the left endpoint. For $\sigma=1$ the remaining exact path is

$$2n+1 \xrightarrow{1} 3n+2 \xrightarrow{1} (9n+7)/2 \xrightarrow{3} (27n+23)/16.$$

Its valuations follow from $n=27$ modulo 32. This is the left word's endpoint. All values are positive. Reversing the original affine equality gives $3^{a+b}W = Jn + h_e 3^b$ and hence (5); the original left valuations give the dyadic residue. This proves completeness and both parameter inverses.

For height, retain the full affine identity

$$m = (K/Q)n + h_e(2/3)^a - 1. \quad (11)$$

The coefficient K/Q is less than one. Also (3) implies

$$h_e(2/3)^a < 2^{\sigma-e} h_e(3/4)^b \leq \frac{2+2^{2-e}}{3} (3/4)^b < 1.$$

Thus the constant in (11) is negative and $m < n$. Positivity follows from $m = 2^a W - 1$ with $W \geq 2$. QED.

No assertion that the different e -families are disjoint is made. Their overlaps are calculated by the original congruences or by the full list at each source. All eligible labels remain in that list, even when only one is selected.

20.4 4. Removing the apparent infinite search

For a fixed admitted source n , the largest leading run in the same template is

$$A_{\max} = \nu_3(J(n/3^b) + h_e). \quad (12)$$

Every original a' in the exact interval $a \leq a' \leq A_{\max}$ gives a smaller positive source

$$m_{a'} = 2^{a'} \frac{J(n/3^b) + h_e}{3^{a'}} - 1. \quad (13)$$

For adjacent values,

$$m_{a'+1} + 1 = \frac{2}{3}(m_{a'} + 1), \quad T(m_{a'+1}) = m_{a'}.$$

The shared-future endpoint is unchanged. The rise endpoint is the same integer $J(n/3^b) + h_e - 1$; the later right-hand path is unchanged as well. Thus (12) selects the smallest source within this entire admitted leading-run template. The exact signed chain correction is obtained by adjoining the original incoming ones, as in the preceding bilateral note.

Theorem 3. The complete set of admitted coefficient-contracting templates at any positive odd source n is determined by finitely many explicit tests. For $n > 1$ put

$$k(n) = \lfloor \log_2(n-1) \rfloor - 1, \quad B = \nu_3(n).$$

Only even e with

$$2 \leq e \leq k(n) \quad (14)$$

need be tested. For each e , the only possible tail layer is

$$b = B - \nu_3(e-1) \geq 1. \quad (15)$$

There are at most two support selectors to test at that (b, e) . The tests are the original dyadic residue, $a \leq k(n)$, and (5). Thus the number of template labels examined is at most $2 \lfloor k(n)/2 \rfloor$, regardless of how large n is.

Proof. Every admitted smaller source has $m = 2^{\{a'\}W-1}$ with $W \geq 2$ and $m < n$. Both sources are odd, so $m+1 \leq n-1$. Hence $2^{a'+1} \leq n-1$ and $a' \leq k(n)$. Since $a' \geq a \geq e$, (14) follows. Equation (8) and Lemma 1 give (15), uniquely. Every admitted template is therefore in the finite list. Conversely, a successful test of its defining congruences is in Theorem 2's complete source domain; (12) then gives the maximal run and smallest template source. These implications prove the exact finite search, rather than merely a termination heuristic. QED.

This proof handles all even exponents; it does not supply a bound on arbitrary Collatz stopping times. It eliminates an unnecessary infinite parameter search within the stated construction.

20.5 5. A whole new root progression

Take $b=2, e=8, \sigma=0$. Then $h=86, t_e=0, a=16$, and the original common-future pair is

$$\begin{array}{c} 20\,241\,207 + 1\,549\,681\,956v \xrightarrow{1} 30\,361\,811 + 2\,324\,522\,934v \\ \xleftarrow{1^{16}, 8, 2, 3} 14\,024\,703 + 1\,073\,741\,824v, \quad v \geq 0. \end{array} \quad (16)$$

The exact height difference is

$$6\,216\,504 + 475\,940\,132v > 0.$$

The left sources have actual ternary valuation two and no incoming original edge:

$3x+1=2^n$ with 3 dividing n is impossible modulo 3. Both original paths are needed. At $v=0$ the right path rises to 9,211,998,293 and then reaches the displayed common endpoint. The comparison's signed odd-return clock is $1-19=-18$.

The complete previous reduction leaves every source of (16) as a root. This is a finite exact residue certificate, not an extrapolation from its anchor. The old finite mask has period 5,668,704. Since the new progression has fixed ternary valuation, the four possible old $e=2,6$ layer congruences have fixed moduli. Their least common multiple with the finite-mask period makes the membership of $n_0+n_{\text{step}}v$ periodic in v with period 8. The checker tests all eight parameter classes against the full previous mask and all four original congruences. All eight are roots. Periodicity proves the assertion for every $v \geq 0$.

Consequently (16) is additional source coverage beyond both the preceding finite catalogue and its infinite $e=2,6$ families. At its anchor the new comparison ends at the still-retained root 14,024,703. No arrival-at-1 conclusion is inferred from that class equality.

The other supported source orders are retained as well. For $b=1, e=4, \sigma=1$ the actual layer is two, not one, and the entire pair is

$$45\,243 + 69\,984v \longrightarrow 76\,349 + 118\,098v \longleftarrow 42\,367 + 65\,536v.$$

This illustrates the importance of (2); no claim that this smaller family is new relative to the old finite catalogue is made.

20.6 6. Budget-preserving and unrestricted global retractions

The primary new retraction first applies every old rule in its original priority. Only at a retained root does it examine the complete finite list in Theorem 3. For each candidate it constructs BOTH original paths, their integral chain (1), and their maximum original vertex. It admits the candidate only when that maximum is at most the predecessor's proved budget

$$R_0(n) = \max\{130(n+1), \lfloor 64n(4/3)^{\lfloor \log_3 n \rfloor} \rfloor + 21\}. \quad (17)$$

Among the admitted candidates it selects the smallest target, with the explicit word-length and label tie-breakers used in `all_even.py`. Rejected budget faces are not erased from the full arithmetic candidate list. A source without an accepted move remains a root.

All selected targets are strictly smaller odd integers. Therefore the procedure terminates on every original input. For a selected move with chain B_n , target m , and signed clock j , define

$$H(V_n) = B_n + q^j H(V_m), \quad Q(V_n) = q^j Q(V_m), \quad F = I - H d_\epsilon.$$

At roots use $H=0$ and $Q=I$. The exact identities are

$$d_\epsilon H = I - Q, \quad Q^2 = Q, \quad HQ = 0, \quad d_\epsilon F = Q d_\epsilon, \quad F^2 = F, \quad FH = 0. \quad (18)$$

The first telescopes (1). The next two follow at roots. Substitution in F proves the remaining identities. Thus these are an actual finite-column chain-homotopy retraction, not merely an equality of module ranks.

Theorem 4. The budget-preserving extension still uses only original vertices at most

$R_0(n)$ in its H column. For $n \geq 27$ the inherited bound

$$R_0(n) \leq 64n^{\log_3 4} + 21$$

therefore remains valid.

Proof. Every chosen reduction source x is at most the original n . Old moves have the original bound. The new acceptance test verifies that bound for both complete original paths at x . The integer function R_0 is nondecreasing, so their union stays below $R_0(n)$. This also handles cancellation: support is checked on both paths before equal edge coefficients are collected. QED.

The whole new progression (16) passes this budget. For every n there, $n \geq 20,241,207 > 3^{15}$. Its largest right-hand value is

$$(4096/9)n + 85.$$

The inequality

$$(4096/9)n + 85 \leq 64(4/3)^{15}n + 21$$

holds at $n=20,241,207$ by integer multiplication and has strictly positive margin slope. It holds throughout the progression. The floor in (17) does not change this integer endpoint conclusion. The short left path has smaller values. This proves the budget acceptance on the entire family, not only at the eight residue representatives used to test novelty.

For completeness, an unrestricted mode admits every successful template from Theorem 3. For positive odd $n > 1$ its larger support budget is separately proved:

$$R_*(n) = \max\{130(n+1), \lfloor (n-1)3^{k(n)}/2^{k(n)} \rfloor - 1\}. \quad (19)$$

At the fixed root $n=1$, set $R_*(1)=1$; the homotopy column $H(V_1)$ is zero. During the initial rising run, $m+1 \leq n-1$ and $a' \leq k(n)$, so $3^{a'}W - 1 = (3/2)^{a'}(m+1) - 1$ is bounded by the second term. The later falling steps decrease; the bilateral final three-edge segment stays below $5(n+1)$, covered by the first term. The finite old paths also obey that first term. Hence (19) holds along the entire decreasing-source recursion. In real exponents its second term is at most $(2/3)(n-1)^{\log_2 3} - 1$. This is not the smaller budget (17), and it is not assigned that smaller exponent without the explicit acceptance test.

Both versions have the integer uncollected-term bound, for positive odd $n > 1$,

$$K_*(n) = \frac{n-1}{2} \max\{24, k(n) + \lfloor \log_3 n \rfloor + 6\}. \quad (20)$$

Set $K_*(1)=0$ at the fixed root. There are at most $(n-1)/2$ strict source reductions. A new pair uses at most $k(n) + \text{floor}(\log_3 n) + 6$ edges in total, and the finite older pairs use at most 24. Every signed q -exponent is bounded in absolute value by that total. Therefore collected constant coefficients have total absolute sum at most $K_*(n)$, and first-order coefficients at most $K_*(n)^2$.

The old route is unchanged until its retained root. Its new tail gives the comparison $K = H_{\text{new}} - H_{\text{old}}$, with

$$d_\epsilon K = Q_{\text{old}} - Q_{\text{new}}, \quad F_{\text{old}} K = K, \quad F_{\text{new}} F_{\text{old}} = F_{\text{old}} F_{\text{new}} = F_{\text{new}}. \quad (21)$$

The correction lies in the old projected edge module: its boundary endpoints are old roots, on which H_{old} is zero. This proves the displayed F_{old} identity; substitution gives both projection compositions. The analogous Q compositions follow from the fixed old route and the new root endpoints, including all signed q -factors.

All arithmetic rule choices and budget tests are independent of ϵ . Hence constant reduction commutes with these maps and transports the integral first-jet comparison kernel through the actual homotopy equivalence. The fixed loop stays present. Original periods are not divided out, and a still-open component is not called absent because a particular comparison sends an amplitude to supported zero.

20.7 7. Verification and the still-global task

The exact checker covers both selectors, $b=1$ through 16, and every even e from 2 through 32. It verifies both original affine coefficient columns at every step, then checks actual path instances, maximal leading-run recovery, ternary layers, first-jet boundaries, and support bounds. These are regression fixtures for the written all-parameter proofs. The finite-search implementation is compared with a larger direct template search on all odd inputs below 4,000. Retraction tests verify (18) and (21) on original integral vectors, including newly reduced roots. The separate affine auditor imports neither the new module nor its predecessor.

The next arithmetic task is the exact residual root set after these additional families and their budget faces. The finite algorithm for template membership does not prove that one of its templates applies to every retained source. The unchanged first-crossing horizon remains 11,610 odd steps in this increment; no extra crossing-window scan or numerical stopping-time record is claimed.

A complete original singleton primitive would still require a route to the fixed component, or another proved arithmetic exhaustion of the surviving comparison kernel. New height relations improve its presented source; they do not erase it. The cumulative archive keeps the earlier source files and execution receipts separately from the checks executed for this note.

20.8 Sources and exact provenance

Research direction and Split-Zero framework: the owner's workbench. This is a continuation developed in the present tool-assisted session, not an external independent mathematical review or a priority claim.

The mathematical predecessor is `bilateral_root_bounds_20260916/note.md`, with the earlier original source and first-jet conventions preserved. The executable predecessor is the unchanged `bilateral_control.py`, whose SHA-256 is pinned in `SOURCE_INTAKE.json`. Current GitHub integration and PR state were read on 16 September 2026 at main 9453b5306b3a897389dece309f9c119dffbcc489; PR #2 is now merged. No remote write is inferred from generating this local addition.

For historical terminology, the coefficient-stopping-time question is distinct from the present template-exhaustion problem. The original expansion of an odd-return exponent a to one odd shortened step followed by $a-1$ divisions is the clock comparison used in earlier notes. Literature attribution remains in those notes and in Rozier-Terracol, *Paradoxical behavior in Collatz sequences*, <https://arxiv.org/abs/2502.00948>. That paper supplies no premise of Theorems 2-4.